NTE 1010.1 - Introduction to Finance for Engineers
Spring 2020

week 1

Introduction

NTE 1010 – Introduction to Finance for Engineers

Spring 2020 - Course Contents & Weekly Schedule

Spring 2020		NTE 1010.1 - Introduction to Finance for Engineers	
wk	Wed	subject	textbook chapters
1	19-Feb	Financial Accounting	Atrill&McLaney Ch.2-3
2	26-Feb	Financial Statements	Atrill&McLaney Ch.4-5
3	4-Mar	Analysis of Fin Statements	Atrill&McLaney Ch.6
4	11-Mar	Management Accounting HW-1	Atrill&McLaney Ch.7-9
5	18-Mar	Fin System & Money Quiz-1	Mishkin Ch.2-3
6	25-Mar	Interest Rates	Mishkin Ch.4-6
7	1-Apr	Review before MidTerm	
	8-Apr	MidTerm	
8	15-Apr	Stock Market	Mishkin Ch.7
9	22-Apr	Financial Crises	Mishkin Ch.9-10
10	29-Apr	Banking HW-2	Mishkin Ch.11
11	6-May	Central Banks & Monetary Policy	Mishkin Ch.14-17
12	13-May	FX & Intl Fin System Quiz-2	Mishkin Ch.18-19
13	20-May	Financial Management	Atrill&McLaney Ch.10-12
14	27-May	Review before Final	

NTE 1010 – Introduction to Finance for Engineers

Exams & Recommended Reading

At the exams,
you will need to have a calculator which can calculate power of a number.

You may not use computers, telephones or any other electronic equipment of that sort.

You may refer to an A4-sized sheet with formulas & notes on both sides
and to the A3-sized table of Time Value Factors.

All other notes and books are not to be referred to.

Every week, after the lecture, lecture notes will be uploaded to internet, DISCORD
you may reach to them via the following link: https://discord.gg/J3gVqr3

Recommended Reading:	
Accounting and Finance for Non-Specialists Fifth Edition, Peter Atrill & Eddie McLaney Prentice Hall, ISBN: 978-0-273-70244-3	The Economics of Money, Banking, and Financial Markets Tenth Edition, Frederic s. Mishkin Pearson, ISBN: 978-0-273-76573-8

NTE 1010 – Introduction to Finance for Engineers

Spring 2020 - Course Grading

Spring 2020	NTE 1010.1 - Introduction to Finance for Engineers				
No of Questions	Exam Time (minutes)	Exam	Q'ty	Weight in Course Grade	
				Each (%)	Total (%)
6	60	Final Exam	1		40
		Semester Evaluation			60
4	45	MidTerm	1		24
2	25	Quiz	2	12	24
		Homework	2	6	12

AA	90 - 100
BA	80 - 89
BB	70 - 79
CB	60 - 69
CC	50 - 59
DC	45 - 49
DD	40 - 44

Half of the exam questions will be numerical; the other half will be verbal.

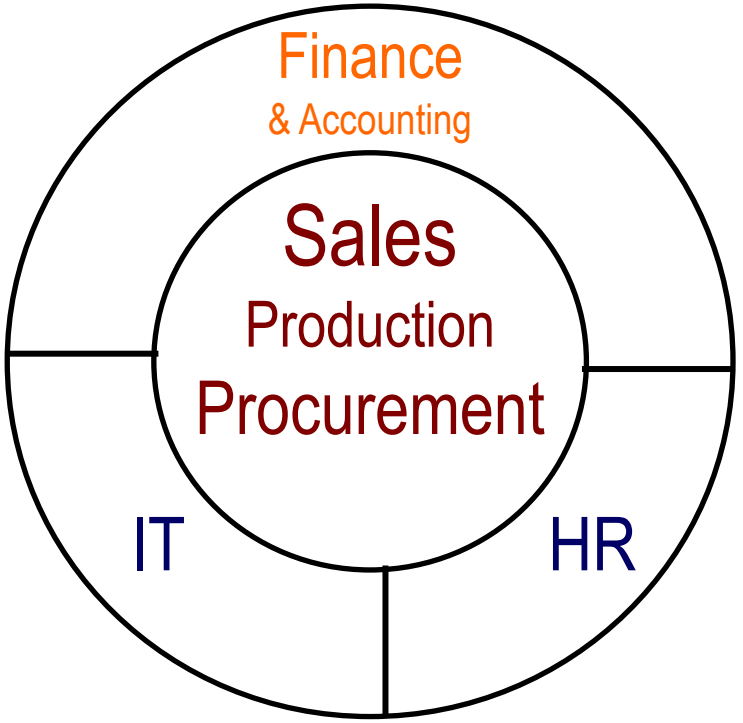
Finance for Engineers

Finance for Engineers

Functions in a Company

Basic functions in a company are the following:

1. Procurement (for production and trading companies)
2. Production (for production companies)
3. Sales
4. Information Technologies, IT
5. Human Resources, HR
6. Accounting
7. Finance



Finance for Engineers

Finance Function

Finance uses
internal information (basically accounting data such as Balance Sheet, Income Statement, Cash Flow)

and external information to perform its job.

Basic external info neded (actual as well as forecasts) are
interest rates,
FX rates & devaluation,
inflation,
growth rates of economy and markets.

Financial Management takes care of
the "money side" of the business.

NTE 1010

Introduction to Finance for Engineers

Financial Accounting

Introduction

Financial Accounting

Accounting is concerned with
collecting, analysing and communicating
financial information.

This information is useful for
those who need to make decisions and plans about businesses,
and for those who need to control those businesses.

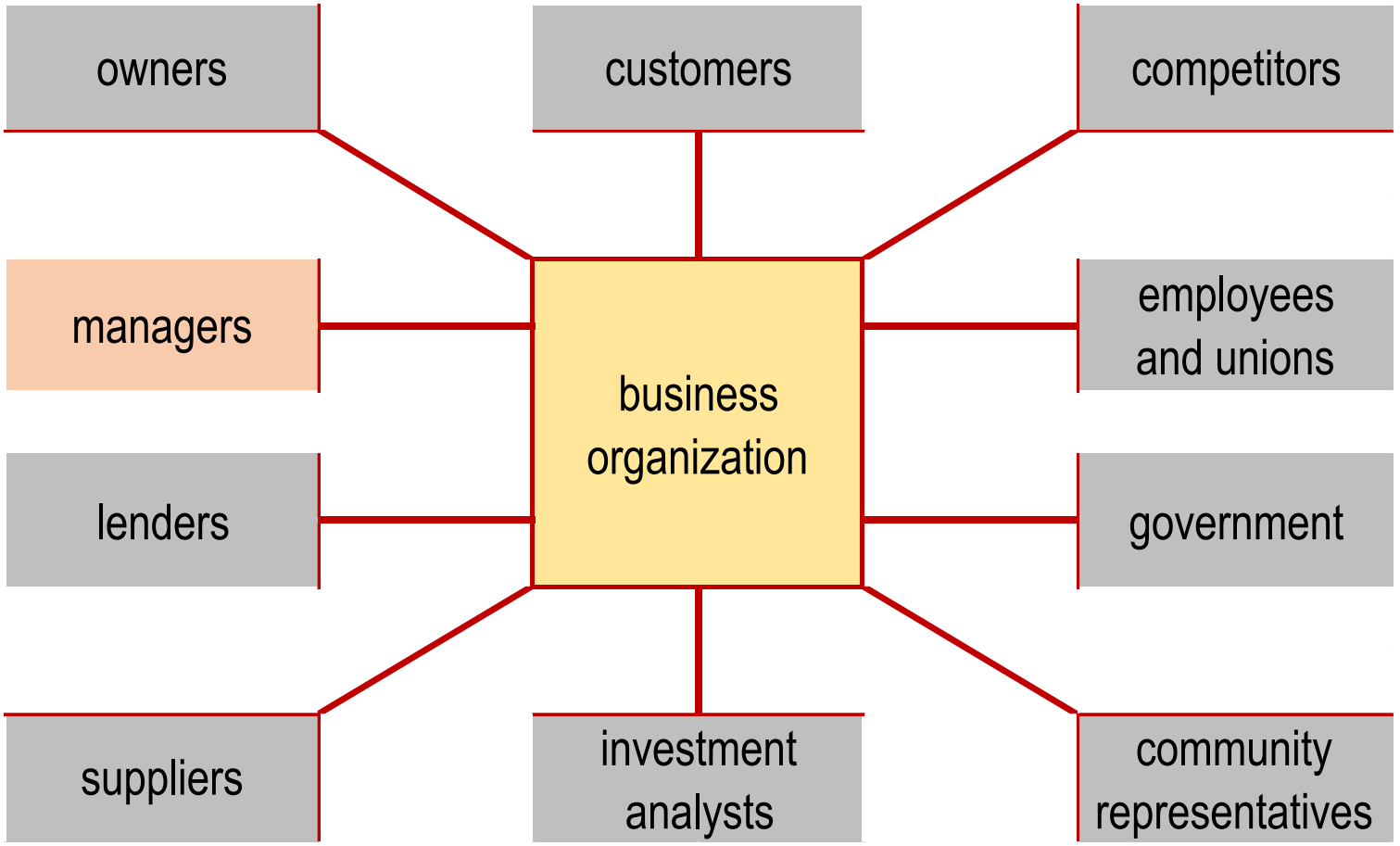
Finance is concerned with the ways in which funds for a business are raised and invested.

Finance, like accounting, exists to help decision makers.

A high-level of understanding of accounting and finance
is required by most engineers within a company in order to be able to
communicate better with the managers and accounting staff of the company.

Financial Accounting

Main users of financial information relating to a business



Financial Accounting versus Management Accounting

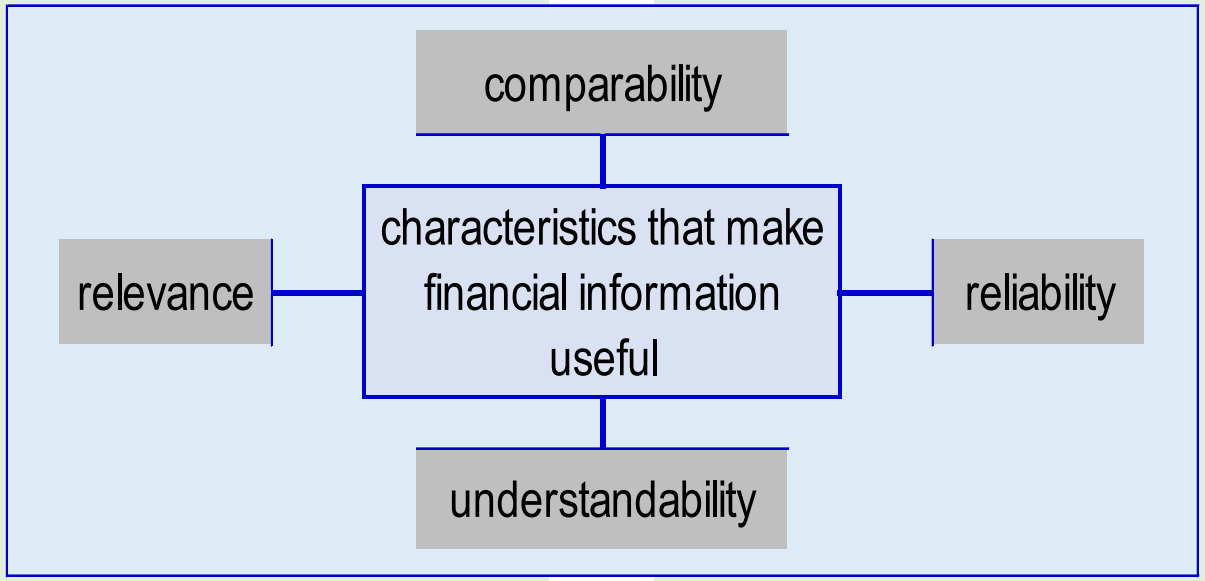
	Accounting	
	Financial	Management
seeks to meet the needs of	all other users	managers
nature of reports produced	general purpose	specific purpose; for decision or for business unit
level of detail	information is aggregated	considerable details, needed for operational decisions
regulations	subject to accounting regulations, mostly called GAAP "Generally Accepted Accounting Principles"	for internal use only, designed to meet needs of managers
reporting interval	mostly annual, also some half-yearly or quarterly	as required by managers; monthly, weekly or even daily
time horizon	for the past period; backward looking	past (actual) + generally forecast
range and quality of information	in monetary terms and verifiable	monetary + non-monetary figures; need overweights verifiability

Financial Accounting

Usefulness of financial information

Materiality

To be included in the financial statements, information needs to be "material".
If omission or misinterpretation of an information in the financial reports would really alter the decisions that users make, that the information is "material".

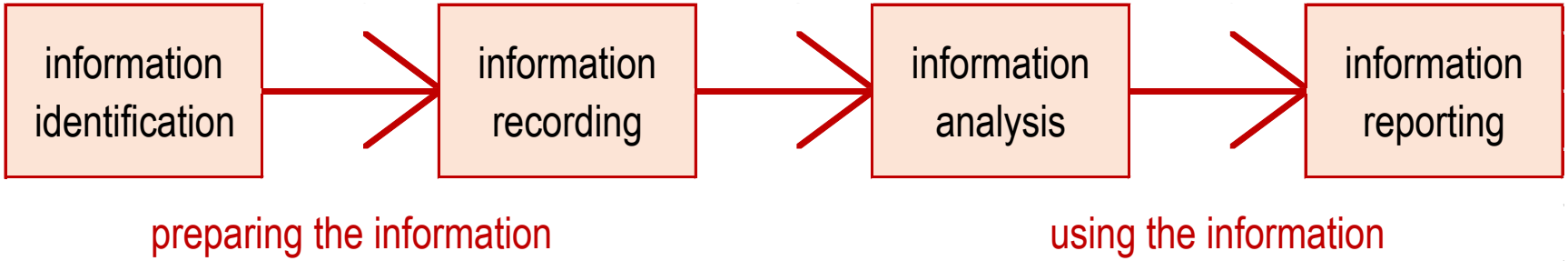


Cost / Benefit

To be included in the financial statements, the benefits of providing it should outweigh the costs of acquiring it.

Financial Accounting

Accounting as Information System



Financial Accounting

Accounting Principles and Conventions

Each country has a different variation of "Generally Accepted Accounting Principles".
For example, GAAP accepted in Turkey are called TR GAAP, in USA as US GAAP.

A version that is harmonized internationaly is called IFRS,
"International Financial Reporting Standards".

Although it is not necessary to have a detailed understanding of GAAP or IFRS
in order to understand the financial statements,
it is necessary to understand
some of the basic accounting principles and conventions:

- Business Entity

The business and its owners are treated separately for the purposes of accounting;
this differs from the legal treatment.

- Accrual Accounting
- Historical Cost

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Basic Accounting Principles and Conventions

Business Entity

The business and its owners are treated separately for the purposes of accounting; this differs from the legal treatment.

The three main forms of legal structures of a business are

- Sole propriertorship (şahıs şirketi)

where owner is personally responsible for the debts of the business,

- Partnership (adi ortaklık)

where partners are "jointly and severally responsible"
meaning that creditors can attempt to recover bad debts
from all the owners, or each one separately, in their personal capacities,

- Corporation (şirket)

where the most that the owners can lose is their equity investment in the company.

Basic Accounting Principles and Conventions

Accrual Accounting

Accrual accounting recognises a sale when it occurs, that is, when the sale documents are signed, not when the customer finally pays the cash in settlement.

If the transaction occurs before the cash transfer, the tranaction is accrued,
and if the transaction is recognised after the cash transfer, it is deferred.
For example, deferred taxation is the recognition of taxes owed but not paid yet.

Because of the difference in timing between the recognition of a transaction and cash receipt or payment, there is the need to match the expenses to the correct period in which the revenue is recognised. Matching expenses and revenues allows for
the correct calculation of the profit that a company has made in a particular period.

On the other hand, the cost of an asset with a useful life of longer than a year
is spread out over its expected useful life.
This procedure is called depreciation.

Basic Accounting Principles and Conventions

Accrual Accounting

Accrual accounting recognises a sale when it occurs,
that is, when the sale documents are signed,
not when the customer finally pays the cash in settlement.

If the transaction occurs before the cash transfer,
the tranaction is accrued,
and
if the transaction is recognised after the cash transfer,
it is deferred.

Basic Accounting Principles and Conventions

Historical Cost

The figures in the accounts represent historical values,
i.e. the cost of acquiring the assets or goods at the date of acquisition.
The value of an asset in the accounts, that is, its "book value" is unrelated to its value either within the business (replacement value) or to its value to others (market value).
Although this seems unrealistic, historical cost reporting removes the requirement for valuation based on subjective opinion,
which could result in manipulation of the financial position.

This convention also removes the need to adjust the figures for inflation.
Nevertheless, in high inflation environments (where cumulative inflation rate over three years exceeds 100%), "inflationary accounting" is allowed.

This is also in accord with the conservative position adopted by accounting. For example, if losses are probable, and can be reasonably estimated, they are recorded. However, if gains ocur, these are only recorded when they occur.

Basic Accounting Principles and Conventions

Historical Cost

The figures in the accounts represent historical values,
i.e. the cost of acquiring the assets or goods
at the date of acquisition.

The value of an asset in the accounts,
that is, its "book value" is unrelated to its value
either within the business (replacement value)
or to its value to others (market value).

Major Financial Statements

Additional Accounting Conventions

- **Money Measurement**

  Resources not capable of being measured in monetary terms will be excluded.

  Accounting normally deals with only those items that are capable of being expressed in monetary terms. Money has the advantage that it is a useful common denominator with which to express the wide variety of resources held by a business.

- **Going Concern**

  Financial statements are to be prepared on the assumption that the business will continue operations for the foreseeable future, unless this is known not to be true.

  In other words, it is assumed that there is no intention, or need, to sell off the non-current assets of the business. Based on this assumption, it is prudent to continue to show the value of non-current assets at their recorded values (that is, based on historic cost).

Major Financial Statements

Additional Accounting Conventions

- Dual Aspect

  Each transaction has two aspects, both of which will affect the balance sheet.

  The purchase of a car for cash results in an increase in one asset (vehicles) and a decrease in another (cash). The repayment of a bank loan results in the decrease in a liability (loans) and the decrease in an asset (cash / bank).

- Prudence

  Financial statements should err on the side of caution.

  The convention represents an attempt to deal with the uncertainty surrounding many events reported in the financial statements, and evolved to counteract the excessive optimism of some managers and owners, which resulted in an overstatement of financial position.

- Objectivity convention seeks to reduce personal bias in financial statements.

  Financial statements should be based on objective verifiable evidence rather than on matters of opinion.

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Spring 2020

week 2

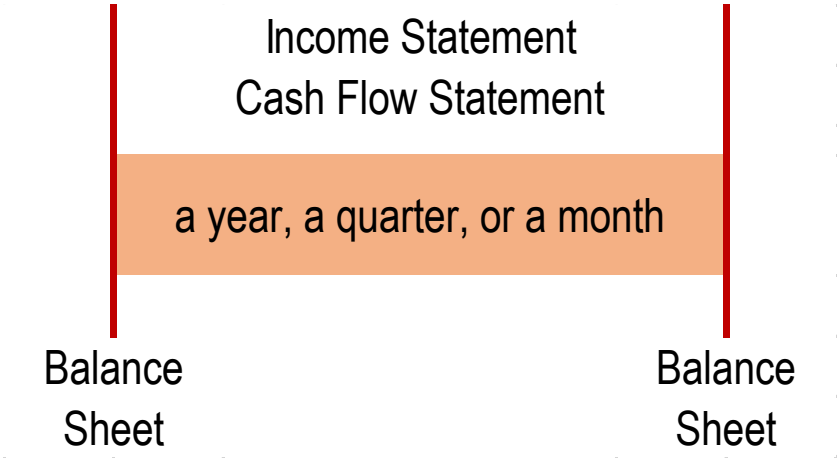
Financial Statements

Reports of Financial Accounting

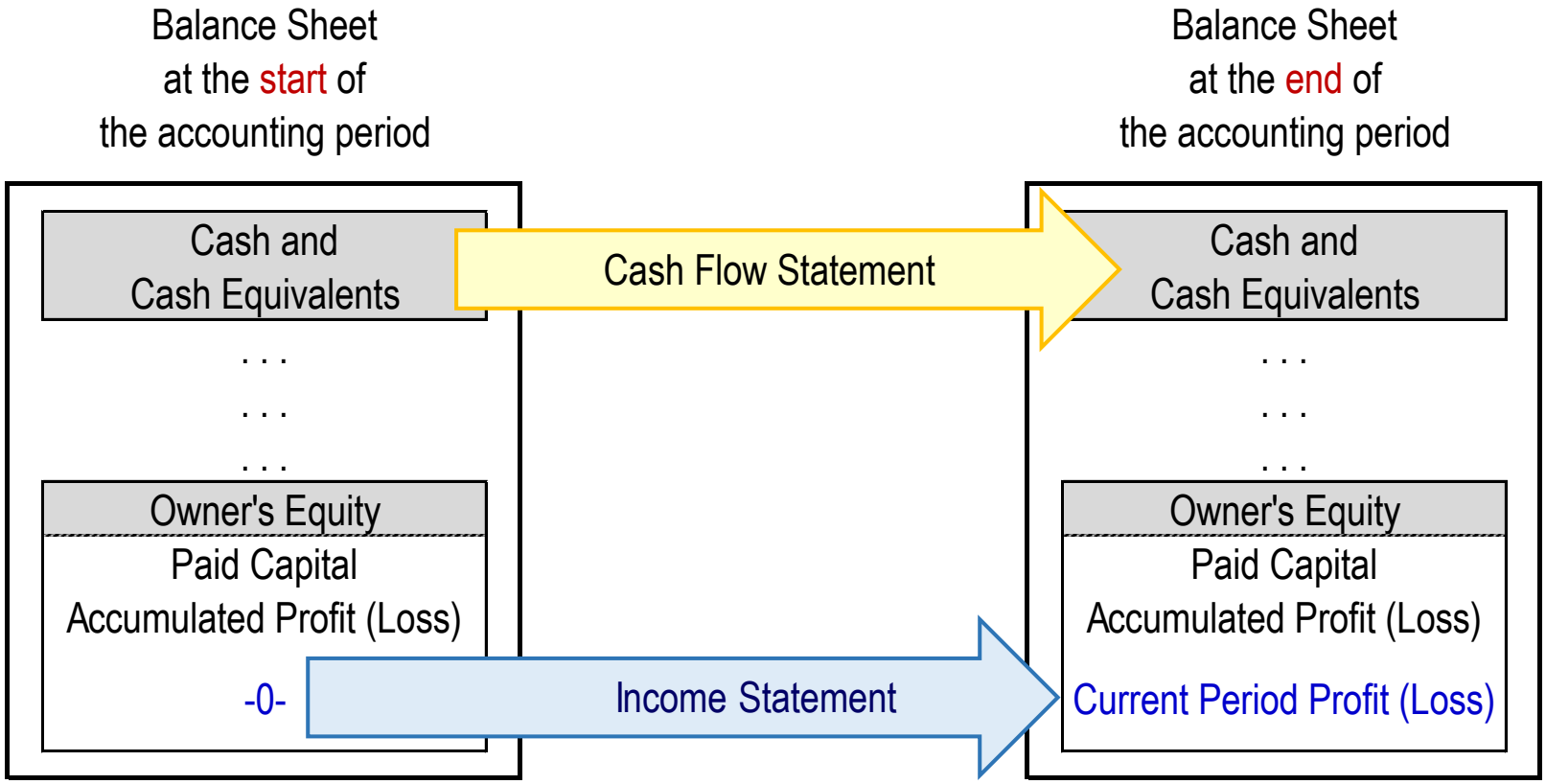
Three major financial statements that form the core of financial accounting:

- Balance Sheet
  shows the accumulated wealth of the business at the end of the period,
- Income Statement (also known as Profit and Loss Account)
  shows the wealth (profit) generated or lost by the business during the period,
- Cash Flow Statement
  shows the cash in and cash out during the period.

Taken together, they provide an overall picture of the financial health of the business.



Relationship among Primary Financial Statements



Balance Sheet

Income Statement

Cash Flow Statement

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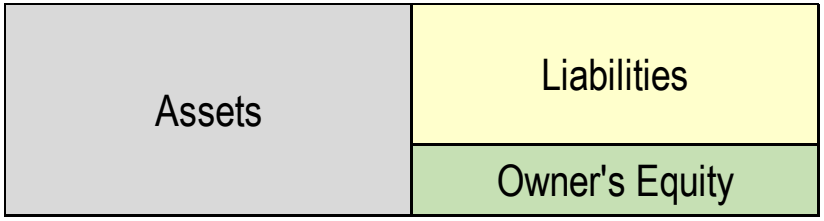
Major Financial Statements

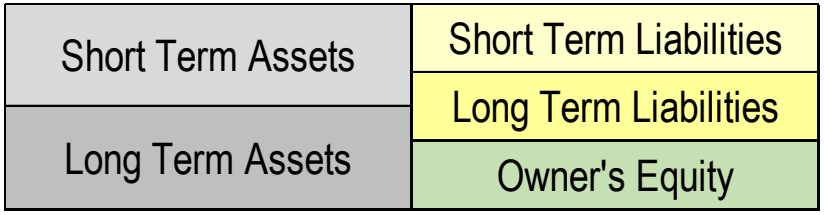
Balance Sheet

what is the accumulated wealth of the business (Owners' Equity)
at the end of a particular period

Basic Structure of a Balance Sheet

+ Assets (resources held by the business)
- Liabilities (obligation of the business to provide cash, or some other form of benefit to other individuals and organizations apart from the owners)
= Owner's Equity (accumulated wealth of the business)





Short Term ≤ 1 year, Long Term > 1 year

My Shop Ltd		
Balance Sheet	31-12-2017	($m)
Current Assets		
Cash, Cash Equivalents & Bank Deposits		
Trade Receivables		
Inventories		
Prepaid Expenses		
Other Current Assets		
Non-Current Assets		
Plant, Machinery & Equipment, gross		
less: Accumulated Depreciation		
equals: Plant, Machine & Equipm, net		
Land & Buildings		
Other Non-Current Assets		
TOTAL ASSETS		
Current Liabilities		
Short-Term Bank Loans		
Trade Payables		
Interest Payable		
Corporate Tax Payable		
Accrued Expenses		
Dividends Payable (Unpaid Dividends)		
Other Current Liabilities		
Long-Term Liabilities		
Long-Term Trade Payables		
Long-Term Bank Loans		
Other Long-Term Liabilities		
Long-Term Bonds		
Total Liabilities		
Owner's Equity		
Paid-in Capital		
Equity Reserves		
Retained Earnings		
Current Period Profit (Profit after Tax)		
Declared Dividends		
TOTAL LIABILITIES & OWNER'S EQUITY		

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Balance Sheet

Basic Structure

Name of the Company
BALANCE SHEET (currency)
date

CURRENT ASSETS	CA=CA1+CA2+CA3+CA4+CA5	CURRENT LIABILITIES	CL=CL1+CL2+CL3+CL4
Cash & Bank Deposits	CA1	Bank Loans	CL1
Receivables	CA2	Payables	CL2
Inventories	CA3	Short Term Portion of Long Term Liabilities	CL3
Prepaid Expenses	CA4	Other Current Liabilities	CL4
Other Current Assets	CA5	LONG TERM LIABILITIES	LTL=LTL1+LTL2
FIXED ASSETS	FA=FA1+FA2+FA3	Long Term Bank Loans	LTL1
Plant & Equipment, net of Depreciation	FA1=P&Eg-Depr	Other Long Term Liabilities	LTL2
Plant & Equipment, gross	P&Eg	OWNER'S EQUITY	OE=OE1+OE2+OE3
less: Accumulated Depreciation	Depr	Paid Capital	OE1
Intangible Assets	FA2	Accumulated Profit (Loss)	OE2
Other Non-current Assets	FA3	Current Period's Profit (Loss)	OE3
TOTAL ASSETS	TA=CA+FA	TOTAL LIABILITIES & OWNER'S EQUITY	TL&OE=CL+LTL+OE

TA = TL&OE

Balance Sheet

Assets

For a particular item to be treated as an asset for accounting purposes, it should have the following characteristics.

- A probable future benefit must exist.
- The business must have an exclusive right to control the benefit.
- The benefit must arise from some past transaction or event.
- The asset must be capable of measurement in monetary terms.

Note that all four of these conditions must apply.

Still, some resources not satisfying all four may well be assets in a broader sense, but not for accounting purposes.

Once an asset has been acquired by a business, it will continue to be considered an asset until the benefits are exhausted or the business disposes of it in some way.

Assets that have a physical substance and can be touched are referred to as *tangible assets*.

Assets that have no physical substance but which, nevertheless, provide expected future benefits (such as patents) are referred to as *intangible assets*.

My Shop Ltd		
Balance Sheet	31-12-2017	($m)
Current Assets		
Cash, Cash Equivalents & Bank Deposits		
Trade Receivables		
Inventories		
Prepaid Expenses		
Other Current Assets		
Non-Current Assets		
Plant, Machinery & Equipment, gross		
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Long-Term Bank Loans		
Other Long-Term Liabilities		
Long-Term Bonds		
Total Liabilities		
Owner's Equity		
Paid-in Capital		
Equity Reserves		
Retained Earnings		
Current Period Profit (Profit after Tax)		
Declared Dividends		
TOTAL LIABILITIES & OWNER'S EQUITY		

Balance Sheet

Assets

Current assets are basically assets that are held for the short term.
They have to meet any one of the following four criteria:

- held for sale or consumption in the normal course of a business' operating cycle
- are for the short term (i.e. to be sold within a year)
- are held primarily for trading
- are cash, or near cash such as easily marketable, short-term investments

The most common current assets are inventories (or stock), customers who owe money for goods or services supplied on credit (known as trade receivables or debtors), and cash.

Non-current assets (also called fixed assets) are assets that do not meet the above criteria.
They are held for the long-term operations of the business.
Essentially, they are "tools" of the business and are held with the objective of generating wealth.

My Shop Ltd		
Balance Sheet	31-12-2017	($m)
Current Assets		
Cash, Cash Equivalents & Bank Deposits		
Trade Receivables		
Inventories		
Prepaid Expenses		
Other Current Assets		
Non-Current Assets		
Plant, Machinery & Equipment, gross		
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Declared Dividends		
TOTAL LIABILITIES & OWNER'S EQUITY		

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Financial Statements

Arçelik A.Ş.

30 June 2019

Assets

http://www.arcelikas.com/UserFiles/file/Arcelik%202019%202%C3%87%20Finansal%20Rapor.pdf

ARÇELİK ANONİM ŞİRKETİ

30 HAZİRAN 2019 VE 31 ARALIK 2018 TARİHLERİ İTİBARIYLA KONSOLİDE FİNANSAL DURUM TABLOSU

(Tutarlar aksi belirtilmedikçe bin Türk Lirası ("TL") olarak ifade edilmiştir.)

		Sınırlı denetimden geçmiş	Bağımsız denetimden
	Notlar	30 Haziran 2019	31 Aralık 2018
VARLIKLAR			
Dönen varlıklar:			
Nakit ve nakit benzerleri	5	5.787.781	5.341.524
Ticari alacaklar			
-İlişkili taraflardan ticari alacaklar	28	31.157	12.059
-İlişkili olmayan taraflardan ticari alacaklar	9	9.454.764	7.744.252
Türev araçlar	8	237.989	28.405
Stoklar	10	6.081.641	5.087.676
Peşin ödenmiş giderler	17	335.535	181.987
Cari dönem vergisiyle ilgili varlıklar	18	102.716	118.220
Diğer dönen varlıklar	20	720.259	681.903
Toplam dönen varlıklar		**22.751.842**	**19.196.026**
Duran varlıklar:			
Finansal yatırımlar	6	2.033	1.856
Ticari alacaklar			
-İlişkili olmayan taraflardan ticari alacaklar	9	28.053	35.690
Türev araçlar	8	3.808	166.813
Özkaynak yöntemiyle değerlenen yatırımlar	12	475.289	346.605
Maddi duran varlıklar	13	5.735.721	4.534.276
Maddi olmayan duran varlıklar			
-Şerefiye		820.274	507.966
-Diğer maddi olmayan duran varlıklar	14	2.724.856	2.600.849
Peşin ödenmiş giderler		55.355	100.033
Ertelenmiş vergi varlıkları	26	978.682	878.247
Toplam duran varlıklar		**10.824.071**	**9.172.335**
Toplam varlıklar		**33.575.913**	**28.368.361**

Balance Sheet

Liabilities

Liabilities are obligations of the business to provide cash, or some other form of benefit to other individuals and organizations apart from the owners.

Once a liability has been incurred by a business, it will remain as an obligation until it is settled.

Liabilities are classified into two groups:

Current liabilities are basically amounts due for settlement in the short term.

They have to meet any one of the following four criteria:

- expected to be settled within the normal course of the business' operating cycle
- due to be settled within 12 months of the balance sheet date
- held primarily for trading purposes
- Business does not have the right to defer settlement beyond 12 months after the balance sheet date

Non-current liabilities represent those amounts due to outside parties that are not current liabilities.

My Shop Ltd		
Balance Sheet	31-12-2017	($m)
Current Assets		
Cash, Cash Equivalents & Bank Deposits		
Trade Receivables		
Inventories		
Prepaid Expenses		
Other Current Assets		
Non-Current Assets		
Plant, Machinery & Equipment, gross		
less: Accumulated Depreciation		
equals: Plant, Machine & Equipm, net		
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Interest Payable		
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Long-Term Trade Payables		
Long-Term Bank Loans		
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Owner's Equity		
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Equity Reserves		
Retained Earnings		
Current Period Profit (Profit after Tax)		
Declared Dividends		
TOTAL LIABILITIES & OWNER'S EQUITY		

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Financial Statements

Arçelik A.Ş.

30 June 2019

Liabilities

http://www.arcelikas.com/UserFiles/file/Arcelik%202019%202%C3%87%20Finansal%20Rapor.pdf

ARÇELİK ANONİM ŞİRKETİ

30 HAZİRAN 2019 VE 31 ARALIK 2018 TARİHLERİ İTİBARIYLA KONSOLİDE FİNANSAL DURUM TABLOSU

(Tutarlar aksi belirtilmedikçe bin Türk Lirası ("TL") olarak ifade edilmiştir.)

		Sınırlı denetimden geçmiş	Bağımsız denetimden
	Notlar	30 Haziran 2019	31 Aralık 2018
YÜKÜMLÜLÜKLER			
Kısa vadeli yükümlülükler:			
Kısa vadeli borçlanmalar	7	4.255.535	4.022.086
Uzun vadeli borçlanmaların kısa vadeli kısımları	7	1.545.620	1.494.640
Ticari borçlar			
-İlişkili taraflara ticari borçlar	28	701.818	441.597
-İlişkili olmayan taraflara ticari borçlar	9	5.502.030	4.292.847
Türev araçlardan borçlar	8	280.218	251.095
Çalışanlara sağlanan faydalar kapsamında borçlar	19	333.559	344.171
Diğer borçlar			
-İlişkili taraflara diğer borçlar	28	30.612	27.794
-İlişkili olmayan taraflara diğer borçlar	11	328.315	241.069
Dönem karı vergi yükümlülüğü	26	68.285	42.978
Kısa vadeli karşılıklar			
-Diğer kısa vadeli karşılıklar	16	742.135	582.339
Diğer kısa vadeli yükümlülükler	20	1.083.550	756.340
Toplam kısa vadeli yükümlülükler		**14.871.677**	**12.496.956**
Uzun vadeli yükümlülükler:			
Uzun vadeli borçlanmalar	7	8.276.942	6.431.552
Diğer borçlar			
-İlişkili taraflara diğer borçlar	28	15.758	28.180
Uzun vadeli karşılıklar			
-Çalışanlara sağlanan faydalara ilişkin uzun vadeli karşılıklar		316.901	289.560
-Diğer uzun vadeli karşılıklar	16	160.159	155.868
Ertelenmiş vergi yükümlülüğü	26	523.958	511.386
Diğer uzun vadeli yükümlülükler	20	245.229	235.697
Toplam uzun vadeli yükümlülükler		**9.538.947**	**7.652.243**
Toplam yükümlülükler		**24.410.624**	**20.149.199**

Balance Sheet

Owner's Equity

Represents the claim of the owners against the business.

For accounting purposes, a clear distinction is made
between the business and the owner(s).
Viewed from this perspective,
any funds contributed by the owners will be seen as
coming from outside the business
and will appear as a claim against the business in its balance sheet.

My Shop Ltd		
Balance Sheet	31-12-2017	($m)
Current Assets		
Cash, Cash Equivalents & Bank Deposits		
Trade Receivables		
Inventories		
Prepaid Expenses		
Other Current Assets		
Non-Current Assets		
Plant, Machinery & Equipment, gross		
less: Accumulated Depreciation		
equals: Plant, Machine & Equipm, net		
Land & Buildings		
Other Non-Current Assets		
TOTAL ASSETS		
Current Liabilities		
Short-Term Bank Loans		
Trade Payables		
Interest Payable		
Corporate Tax Payable		
Accrued Expenses		
Dividends Payable (Unpaid Dividends)		
Other Current Liabilities		
Long-Term Liabilities		
Long-Term Trade Payables		
Long-Term Bank Loans		
Other Long-Term Liabilities		
Long-Term Bonds		
Total Liabilities		
Owner's Equity		
Paid-in Capital		
Equity Reserves		
Retained Earnings		
Current Period Profit (Profit after Tax)		
Declared Dividends		
TOTAL LIABILITIES & OWNER'S EQUITY		

NTE 1010 – Introduction to Finance for Engineers

Financial Statements

Arçelik A.Ş. 30 June 2019 Owners' Equity

http://www.arcelikas.com/UserFiles/file/Arcelik%202019%202%C3%87%20Finansal%20Rapor.pdf

ARÇELİK ANONİM ŞİRKETİ

30 HAZİRAN 2019 VE 31 ARALIK 2018 TARİHLERİ İTİBARIYLA KONSOLİDE FİNANSAL DURUM TABLOSU

(Tutarlar aksi belirtilmedikçe bin Türk Lirası ("TL") olarak ifade edilmiştir.)

		Sınırlı denetimden geçmiş	Bağımsız denetimden
	Notlar	30 Haziran 2019	31 Aralık 2018
Özkaynaklar:			
Ödenmiş sermaye	21	675.728	675.728
Sermaye düzeltme farkları	21	468.811	468.811
Paylara ilişkin primler/ iskontolar		889	889
Kar veya zararda yeniden sınıflandırılmayacak birikmiş diğer kapsamlı gelirler ve giderler			
Yeniden değerleme ve ölçüm kazanç/ kayıpları			
-Tanımlanmış fayda planları yeniden ölçüm kazanç/ kayıpları		(195.503)	(177.981)
- Gerçeğe uygun değer farkı diğer kapsamlı gelire yansıtılan finansal varlıklardan kazançlar / kayıplar		1.516	1.348
Kar veya zararda yeniden sınıflandırılacak birikmiş diğer kapsamlı gelirler ve giderler			
Yabancı para çevirim farkları		3.378.316	2.871.675
Riskten korunma kazanç/ kayıpları			
-Yurt dışındaki işletmeye ilişkin yatırım riskinden korunma kazanç/ kayıpları		(663.939)	(601.215)
-Nakit akış riskinden korunma kazanç/ kayıpları		1.658	5.405
Birleşme denkleştirme hesabı		14.507	14.507
Kardan ayrılan kısıtlanmış yedekler	21	409.115	409.115
Geçmiş yıllar karları		4.515.065	3.663.309
Net dönem karı		444.484	851.756
Ana ortaklığa ait özkaynaklar		9.050.647	8.183.347
Kontrol gücü olmayan paylar		114.642	35.815
Toplam özkaynaklar		**9.165.289**	**8.219.162**
Toplam kaynaklar		**33.575.913**	**28.368.361**
Taahhütler,koşullu varlık ve yükümlülükler	15		

Financial Statements – Balance Sheet

Akbank T.A.Ş. 30 June 2019

Balance Sheet Highlights

Consolidated (TL mn)	2018	1H19
Cash and Due from Banks	49,618	56,345
Securities	56,782	64,951
Loans (net) (1)	201,332	211,353
Other	46,950	43,178
Total Assets	**354,682**	**375,827**
Deposits	208,630	229,503
Funds Borrowed and Bonds Issued	61,506	59,181
Repo	14,275	11,375
- TL Repo	3,918	910
- FX Repo	10,357	10,465
Other	26,484	26,427
Equity	43,787	49,341
Total Liabilities and S/H Equity	**354,682**	**375,827**

Major Financial Statements

Balance Sheet: Asset Valuation

- Property, plant and equipment are shown at historic cost
  less any amounts written off for depreciation.
  However, fair values, if below depreciated cost, should rather be used.

- Where the amount that can be recovered from the tangible non-current assets
  is below the net book value,
  this lower amount should be reflected in the balance sheet.

- Intangible non-current assets broadly follow the same valuation rules as described above.

- Current assets are shown at the lower of cost or net realisable value.

TDHP - Tek Düzen Hesap Planı

	HESAP KODU	HESAP ADI
100 CARİ VARLIKLAR	100	KASA: Nakit, Çek, Banka
	110	HİSSE SENEDİ, TAHVİL, BONO: Özel, Kamu
	120	KISA VADELİ TİCARİ ALACAKLAR
	130	KV ORTAK, İŞTİRAK, PERSONEL ALACAK
	150	STOKLAR
	170	YILLARA YAYILAN İNŞAAT, ONARIM
	180	GELECEK AY GİDER, GELİR TAHAKKUK
	190	KISA VADELİ KDV, AVANS, DİĞER
200 DURAN VARLIKLAR	220	UZUN VADELİ ALACAKLAR
	230	UV ORTAK, İŞTİRAK, PERSONEL ALACAK
	240	BAĞLI MENKUL KIYMET, İŞTİRAK
	250	ARAZİ, BİNA, MAKİNA, TAŞIT, DEMİRBAŞ
	260	HAKLAR, ŞEREFİYE, KURULUŞ GİDER
	270	ARAMA, HAZIRLIK, GELİŞTİRME GİDER
	280	GELECEK YIL GİDER, GELİR TAHAKKUK
	290	UZUN VADELİ KDV, AVANS, DİĞER

	HESAP KODU	HESAP ADI
300 CARİ BORÇLAR	300	KV BANKA KREDİ, UV KREDİ TAHVİL TAKSİT
	320	SATICILARA BORÇ, DEPOZİTO
	330	KV ORTAK, İŞTİRAK, PERSONEL BORÇ
	340	ALINAN KISA VADELİ SİPARİŞ AVANSLARI
	350	YILLARA YAYILAN İNŞAAT, ONARIM
	360	ÖDENECEK VERGİ VE FONLAR
	370	VERGİ, KIDEM TAZMİNATI KARŞILIĞI
	380	GELECEK AY GELİR, GİDER TAHAKKUK
	390	KISA VADELİ KDV, AVANS, DİĞER
400 UZUN VADELİ BORÇLAR	400	UV BANKA KREDİLERİ, TAHVİL
	420	SATICILAR
	430	UV ORTAK, İŞTİRAK, KAMU BORÇ
	440	ALINAN UZUN VADELİ SİPARİŞ AVANSLARI
	472	KIDEM TAZMİNATI KARŞILIĞI
	479	DİĞER BORÇ VE GİDER KARŞILIK.
	480	GELECEK YIL GELİR, GİDER TAHAKKUK
	490	DİĞER UZUN VADELİ YABANCI KAYNAKLAR
500 ÖZKAYNAKLAR	500	SERMAYE, SERMAYE YEDEKLERİ
	520	HS İHRAÇ PRİM, DEĞER ARTIŞLARI
	540	YASAL, STATÜ YEDEKLER, ÖZEL FONLAR
	570	GEÇMİŞ YILLAR KARLARI
	580	GEÇMİŞ YILLAR ZARARLARI (-)
	590	DÖNEM NET KARI (ZARARI)

	HESAP KODU	HESAP ADI
600 KAR / ZARAR	600	SATIŞLAR
	610	SATIŞTAN İADE, İSKONTO (-)
	620	SATILANLARIN MALİYETİ (-)
	630	ARGE, PAZ, SATIŞ, GENEL YÖNT GİDER (-)
	640	TEMETTÜ, FAİZ, KOM, MENK K SATIŞ K, FX K
	650	FAİZ, KOM, KARŞ, MENK K SATIŞ Z, FX Z (-)
	660	FAİZ (-)
	670	ÖNCEKİ DÖNEM, OLAĞANÜSTÜ K/Z
	680	DİĞER K/Z
	690	DÖNEM KARI VEYA ZARARI

700	MALİYET MUHASEBESİ DETAYLARI

NTE 1010 – Introduction to Finance for Engineers

Major Financial Statements

Balance Sheet exercise

Jerry & Co deposits TL 20,000 in a bank account on March 1st in order to commence business. The cash is supplied by the owner (TL 6,000) and by a lender (TL 14,000) and paid into the business bank account.

Jerry & Co Balance Sheet on March 1st

Current Assets		**Liabilities**	
Cash at Bank	20.000	Loans	14.000
Fixed Assets		**Owner's Equity**	
		Capital	6.000
Total Assets	20.000	T Liabilities & OE	20.000

On March 2nd, Jerry & Co purchased a motor van for TL 5,000 paying by check.

Jerry & Co Balance Sheet on March 2nd

Current Assets		**Liabilities**	
Cash at Bank	15.000	Loans	14.000
Fixed Assets		**Owner's Equity**	
Motor Van	5.000	Capital	6.000
Total Assets	20.000	T Liabilities & OE	20.000

On March 3rd, Jerry & Co purchased inventories (i.e. goods to be sold) on one month's credit for TL 3,000.

Jerry & Co Balance Sheet on March 3rd

Current Assets		**Liabilities**	
Cash at Bank	15.000	Loans	14.000
Inventories	3.000	Trade Payable	3.000
Fixed Assets		**Owner's Equity**	
Motor Van	5.000	Capital	6.000
Total Assets	23.000	T Liabilities & OE	23.000

NTE 1010 – Introduction to Finance for Engineers

Major Financial Statements

Balance Sheet exercise

On March 4th, Jerry & Co repaid TL 2,000 of the loan from the lender.

Jerry & Co Balance Sheet on March 7th

Current Assets		**Liabilities**	
Cash at Bank	13.000	Loans	12.000
Inventories	3.000	Trade Payable	3.000
Fixed Assets		**Owner's Equity**	
Motor Van	5.000	Capital	6.000
Total Assets	21.000	T Liabilities & OE	21.000

On March 6th, owner introduced another TL 4,000 into the business bank account.

Jerry & Co Balance Sheet on March 7th

Current Assets		**Liabilities**	
Cash at Bank	17.000	Loans	12.000
Inventories	3.000	Trade Payable	3.000
Fixed Assets		**Owner's Equity**	
Motor Van	5.000	Capital	10.000
Total Assets	25.000	T Liabilities & OE	25.000

On March 7th, the business managed to sell all of the inventories for TL 5,000 and received a check immediately from the customer for this amount.

Jerry & Co Balance Sheet on March 7th

Current Assets		**Liabilities**	
Cash at Bank	22.000	Loans	12.000
Inventories	0	Trade Payable	3.000
Fixed Assets		**Owner's Equity**	
Motor Van	5.000	Capital	10.000
		Profit	2.000
Total Assets	27.000	T Liabilities & OE	27.000

Balance Sheet

Income Statement

Cash Flow Statement

NTE 1010 – Introduction to Finance for Engineers

Major Financial Statements

Income Statement

The purpose of the Income Statement,
also called Profit and Loss Account,
is to measure and report how much profit (wealth)
the business has generated
over a period
(most commonly a year, a quarter, a month).

My Shop Ltd Income Statement 1/1 - 31/12 /2017 ($m)
Sales Revenue
- Cost of Goods Sold
Gross Profit
- Selling, General and Admin Costs
- Other Operating Costs
- Depreciation Expense
Profit before Interest & Tax
- Interest Expense
- Uncollectible Receivables
+ Other Operational Income
- Other Operatinal Expense
+/- Net Investment Income (Expense)
Profit before Tax
- Corporate Tax on Profit
Current Period Profit (Profit after Tax)

Income Statement

Revenue

Recognition of sales revenue
is independent of cash collection.
This is *Accrual Accounting*.

For credit sales,
revenue is recognized before the cash collected.

Advance payment received
can not be recognized as revenue.

Sales revenue is recognized
when ownership and control of goods sold
passes to the buyer.

My Shop Ltd Income Statement 1/1 - 31/12 /2017 ($m)
Sales Revenue
- Cost of Goods Sold
Gross Profit
- Selling, General and Admin Costs
- Other Operating Costs
- Depreciation Expense
Profit before Interest & Tax
- Interest Expense
- Uncollectible Receivables
+ Other Operational Income
- Other Operatinal Expense
+/- Net Investment Income (Expense)
Profit before Tax
- Corporate Tax on Profit
Current Period Profit (Profit after Tax)

Income Statement

Cost of Goods Sold

Value of Inventories held at the beginning of the Period
+ Additions to the Inventory during the Period
- Cost of Goods Sold
= Value of Inventories held at the end of the Period

Commonly used methods of are the following:

- First In, First Out (FIFO)
  the earliest inventories held are the first to be sold.
- Last In, First Out (LIFO)
  the latest inventories held are the first to be sold.
- Weighted Average Cost (AVCO)

My Shop Ltd Income Statement 1/1 - 31/12 /2017 ($m)
Sales Revenue
- Cost of Goods Sold
Gross Profit
- Selling, General and Admin Costs
- Other Operating Costs
- Depreciation Expense
Profit before Interest & Tax
- Interest Expense
- Uncollectible Receivables
+ Other Operational Income
- Other Operatinal Expense
+/- Net Investment Income (Expense)
Profit before Tax
- Corporate Tax on Profit
Current Period Profit (Profit after Tax)

Income Statement

Expenses

Expenses should be matched
to the revenue
that they helped to generate.

Recognition of expenses is also
independent of cash payment
as per *Accrual Accounting*.

Nevertheless, Accrual Accounting is subject to
the *Materiality Convention* of accounting.

My Shop Ltd Income Statement 1/1 - 31/12 /2017 ($m)
Sales Revenue
- Cost of Goods Sold
Gross Profit
- Selling, General and Admin Costs
- Other Operating Costs
- Depreciation Expense
Profit before Interest & Tax
- Interest Expense
- Uncollectible Receivables
+ Other Operational Income
- Other Operatinal Expense
+/- Net Investment Income (Expense)
Profit before Tax
- Corporate Tax on Profit
Current Period Profit (Profit after Tax)

Income Statement

Depreciation

Depreciation is an attempt to measure
that portion of the cost of a non-current asset
that is used to generate the revenue
recognized during a particular period.

The following have to be considered
to calculate *Depreciation Expense*:

- Cost

(all costs incurred to to make the asset ready for use),

- Useful life

(shorter of the physical and economic life),

- Residual (salvage or disposal) value

of the asset (needs to be estimated), and

- Depreciation method.

My Shop Ltd Income Statement 1/1 - 31/12 /2017 ($m)
Sales Revenue
- Cost of Goods Sold
Gross Profit
- Selling, General and Admin Costs
- Other Operating Costs
- Depreciation Expense
Profit before Interest & Tax
- Interest Expense
- Uncollectible Receivables
+ Other Operational Income
- Other Operatinal Expense
+/- Net Investment Income (Expense)
Profit before Tax
- Corporate Tax on Profit
Current Period Profit (Profit after Tax)

Income Statement

Depreciation

- Depreciation method (most commonly used are the following):

	Straight-Line	
End of Year	Book Value	Annual Depreciation
0	100.000	
1	70.000	-30.000
2	40.000	-30.000
3	10.000	-30.000

Estimated Salvage Value

	Reducing-Balance	
End of Year	Book Value	Annual Depreciation
0	100.000	
1	46.416	-53.584
2	21.544	-24.872
3	10.000	-11.544

Estimated Salvage Value

Depreciation Percentage

53,6%

$1-(BV_3/BV_0)^{\wedge 1/3}$



Income Statement

"Bad" Receivables

Many businesses sell goods on credit.

When it becomes reasonably certain that the customer will never pay, the receivable is considered to be "bad".

To provide a more realistic picture of the financial performance and position, "bad" receivables must be "written off" by

- reducing the receivables and
- increasing expenses.

Income Statement 1/1 - 31/12 /2017 ($m)
Sales Revenue
- Cost of Goods Sold
Gross Profit
- Selling, General and Admin Costs
- Other Operating Costs
- Depreciation Expense
Profit before Interest & Tax
- Interest Expense
- Uncollectible Receivables
+ Other Operational Income
- Other Operatinal Expense
+/- Net Investment Income (Expense)
Profit before Tax
- Corporate Tax on Profit
Current Period Profit (Profit after Tax)

My Shop Ltd Balance Sheet 31-12-2017 ($m)
Current Assets
Cash, Cash Equivalents & Bank Deposits
Trade Receivables
Inventories
Prepaid Expenses
Other Current Assets
Non-Current Assets
Plant, Machinery & Equipment, gross
less: Accumulated Depreciation
equals: Plant, Machine & Equipm, net
Land & Buildings
Other Non-Current Assets
TOTAL ASSETS
Current Liabilities
Short-Term Bank Loans
Trade Payables
Interest Payable
Corporate Tax Payable
Accrued Expenses
Dividends Payable (Unpaid Dividends)
Other Current Liabilities
Long-Term Liabilities
Long-Term Trade Payables
Long-Term Bank Loans
Other Long-Term Liabilities
Long-Term Bonds
Total Liabilities
Owner's Equity
Paid-in Capital
Equity Reserves
Retained Earnings
Current Period Profit (Profit after Tax)
Declared Dividends
TOTAL LIABILITIES & OWNER'S EQUITY

BS & Income St exercise

Shop Co has the next Balance Sheet on 31-12-2016. The following is a summary of transactions that occurred during the year 2017: Prepare 2017 year-end Balance Sheet and 2017 Income Statement of Shop Co.

1 - Inventories totalling $550,000 were bought on credit.

2 - Sales revenue on credit totalled $700,000 (cost of goods sold was $450,000).

3 - Receipts from trade receivables totalled $600,000. At year-end, it was clear that a trade debtor who owed $2,000 would not be able to pay any part of the debt.

4 - Payments to trade payables totalled $500,000.

5 - Wages totalling $76,500 were paid during the year. At year-end, $1,500 was owed for the last week of the year.

6 - Corporate tax of 2017 will be paid in 2018.

Shop Co ($)	BS 31-12-16	1	2	3 collection	3 write-off	4	5 wages		6 payable	BS 31-12-17
Current Assets	220.000									441.500
Cash	60.000			600.000		-500.000	-76.500			83.500
Trade Receivables	90.000		700.000	-600.000	-2.000					188.000
Inventories	70.000	550.000	-450.000							170.000
Non-Current Assets	80.000									80.000
Total Assets	300.000									521.500
Current Liabilities	115.000									200.500
Short-Term Payables	110.000	550.000				-500.000				160.000
Wages Payable	5.000							1.500		6.500
Corporate Tax Payable									34.000	34.000
Non-Current Liabilities	45.000									45.000
Owner's Equity	140.000									276.000
Capital	100.000									100.000
Retained Earnings	40.000									40.000
Current Period Profit	0		250.000		-2.000		-76.500	-1.500	-34.000	136.000
Tot Liabilities & OE	300.000									521.500

Shop Co ($)	1	2	3 collection	3 write-off	4	5 wages		6 payable	Inc St yr 2017
Sales Revenue		700.000							700.000
- Cost of Goods Sold		-450.000							-450.000
Gross Profit									250.000
- Wages						-76.500	-1.500		-78.000
- Uncollectible Receivables				-2.000					-2.000
Profit Before Tax									170.000
- Corporate Tax								-34.000	-34.000
Current Period Profit		250.000		-2.000		-76.500	-1.500	-34.000	136.000

NTE 1010 – Introduction to Finance for Engineers

Financial Statements

Arçelik A.Ş. 1 January - 30 June 2019

Income Statement

http://www.arcelikas.com/UserFiles/file/Arcelik%20 2019 %202%C3%87%20 Finansal%20 Rapor.pdf

ARÇELİK ANONİM ŞİRKETİ

30 HAZİRAN 2019 TARİHİNDE SONA EREN ARA DÖNEME AİT KONSOLİDE KAR VEYA ZARAR TABLOSU

(Tutarlar aksi belirtilmedikçe bin Türk Lirası ("TL") olarak ifade edilmiştir.)

		Sınırlı denetimden geçmiş		Denetimden geçmemiş	
	Notlar	1 Ocak - 30 Haziran 2019	1 Ocak - 30 Haziran 2018	1 Nisan - 30 Haziran 2019	1 Nisan - 30 Haziran 2018
Hasılat	4	15.329.757	11.794.111	8.427.600	6.511.893
Satışların maliyeti		(10.447.728)	(8.169.419)	(5.784.268)	(4.537.873)
Brüt kar	**4**	**4.882.029**	**3.624.692**	**2.643.332**	**1.974.020**
Genel yönetim giderleri		(784.704)	(537.899)	(430.451)	(278.671)
Pazarlama giderleri		(2.880.673)	(2.242.539)	(1.500.715)	(1.262.253)
Araştırma ve geliştirme giderleri		(121.543)	(90.046)	(61.998)	(46.571)
Esas faaliyetlerden diğer gelirler	22	515.615	668.135	167.282	397.237
Esas faaliyetlerden diğer giderler	22	(307.014)	(310.978)	(153.679)	(176.824)
Esas faaliyet karı		**1.303.710**	**1.111.365**	**663.771**	**606.938**
Yatırım faaliyetlerinden gelirler	23	1.474	6.747	924	4.277
Yatırım faaliyetlerinden giderler	23	(2.153)	(3.746)	(1.131)	(1.306)
Özkaynak yöntemiyle değerlenen yatırımların kar/zararlarından paylar (net)	12	16.372	34.278	1.404	23.587
Finansman gelir/(gideri) öncesi faaliyet karı		**1.319.403**	**1.148.644**	**664.968**	**633.496**
Finansman gelirleri	24	1.367.748	898.573	648.009	563.946
Finansman giderleri	25	(2.124.705)	(1.699.951)	(1.011.373)	(1.007.997)
Sürdürülen faaliyetler vergi öncesi kar		**562.446**	**347.266**	**301.604**	**189.445**
Sürdürülen faaliyetler vergi gelir/(gideri)					
- Dönem vergi gideri	26	(182.721)	(81.313)	(122.017)	(50.966)
- Ertelenmiş vergi geliri	26	78.364	56.224	52.408	6.111
Dönem karı		**458.089**	**322.177**	**231.995**	**144.590**

Financial Statements – Income Statement

Akbank T.A.Ş. 1 January - 30 June 2019

Income Statement Highlights

Consolidated (TL mn)	1H18	1H19
Interest Income	15,203	**18,724**
Interest Expense	(7,992)	(11,014)
Net Interest Income	**7,212**	**7,710**
Trading Gain (Loss)	(272)	(198)
- Securities	(10)	79
- Other	(263)	(277)
Provision for Loan Losses, net of collections	(2,034)	(2,223)
Fees and Commissions (Net)	1,737	2,403
Operating Expense	(2,743)	(3,368)
Other Income	328	92
Other Provisions (1)	(66)	(961)
Income Before Tax	**4,162**	**3,456**
Tax	(882)	(789)
Net Income	**3,280**	**2,667**

Balance Sheet

Income Statement

Cash Flow Statement

NTE 1010 – Introduction to Finance for Engineers

Major Financial Statements

Cash Flow Statement

Income Statement sets out the revenue and expenses,
rather than the cash receipts and cash payments, for the period.

Thus, Income Statement may have little or no relation
to the cash generated for the period.

Why is cash so important?
Because, people and organizations will not normally accept other than cash
in settlement of their claims against the business.
When businesses fail, it is their inability to find the cash
to pay the amounts owed that really pushes them under.

Analysts tend to watch most carefully when trying to assess
the ability of businesses to survive
and/or take advantage of commercial opportunities as they arise.

NTE 1010 – Introduction to Finance for Engineers

Balance Sheet

Cash and Cash Equivalents

Cash are banknotes and coins in hand
and deposits in banks and similar institutions
that are accessible to the business on demand.

Cash equivalents are short-term, highly liquid investments
that are readily convertible to known amounts of cash
and which are subject to insignificant risk of changes of value.

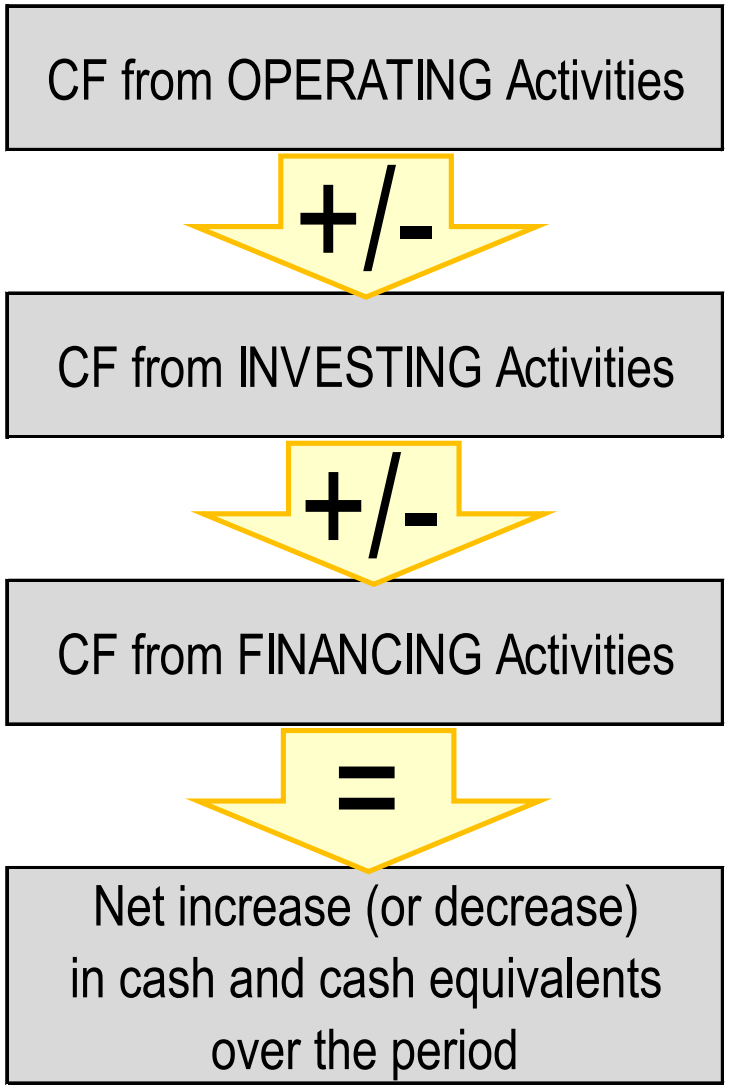
Cash equivalents are held for
the purpose of meeting short-term cash commitments
rather than for investment or other purposes.

My Shop Ltd		
Balance Sheet	31-12-2017	($m)
Current Assets		
Cash, Cash Equivalents & Bank Deposits		
Trade Receivables		
Inventories		
Prepaid Expenses		
Other Current Assets		
Non-Current Assets		
Plant, Machinery & Equipment, gross		
less: Accumulated Depreciation		
equals: Plant, Machine & Equipm, net		
Land & Buildings		
Other Non-Current Assets		
TOTAL ASSETS		
Current Liabilities		
Short-Term Bank Loans		
Trade Payables		
Interest Payable		
Corporate Tax Payable		
Accrued Expenses		
Dividends Payable (Unpaid Dividends)		
Other Current Liabilities		
Long-Term Liabilities		
Long-Term Trade Payables		
Long-Term Bank Loans		
Other Long-Term Liabilities		
Long-Term Bonds		
Total Liabilities		
Owner's Equity		
Paid-in Capital		
Equity Reserves		
Retained Earnings		
Current Period Profit (Profit after Tax)		
Declared Dividends		
TOTAL LIABILITIES & OWNER'S EQUITY		

NTE 1010 - Introduction to Finance for Engineers

Cash Flow Statement

Standard Layout



Cash Flow from OPERATING Activities
cash receipts from trade receivables and cash receipts from cash sales less the sums paid to buy inventories, to pay rent, to pay wages and so on interest paid on short-term business borrowings, taxes paid net cash flows of normal day-to-day trading activities with the tax that has to be paid on them and the interest paid on loans that are needed to support them
Cash Flow from INVESTING Activities
cash payments made to acquire additional non-current assets such as buildings, machinery and so on, shares of other businesses interest received on loans made by the business dividends received from shares in other businesses that are owned by the business cash receipts from disposal of non-current assets net cash flows from making new investments and/or disposing of existing ones
Cash Flow from FINANCING Activities
repayment/redemption of long-term financing (borrowings, share issues) as well as raising it dividends paid to business owners net cash flows from raising and/or paying back long-term finance
Net increase (or decrease) in cash and cash equivalents over the period

Cash Flow Statement

Normal Direction of Cash Flows

Normally "operating activities" provide positive cash flows.

Investing activities typically cause net negative cash flows.

This is because most types of non-current asset wear out,
and because businesses tend to seek to expand their asset base.

When a business sells some non-current assets,
this will give rise to positive cash flows,
but in net terms the cash flows are normally negative
with cash spent on new assets outweighing that received from disposal of old ones.

Financing can go in either direction, depending on the financing strategy at the time.

Since businesses seek to expand, there is a general tendency for this area to lead to cash coming into the business rather than leaving it.

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Introduction to Finance for Engineers

Cash Flow Statement

exercise

ABC Co $m year-end	2016	2017
Current Assets	112	235
Cash	10	81
Trade Receivables	53	93
Inventories	49	61
Non-current Assets	309	325
Plant and Machinery [*]	309	325
Total Assets	421	560
Current Liabilities	75	107
Trade Payables	54	74
Interest Payable	5	10
CorporateTaxes Payable	16	23
Long-Term Loans	160	130
Owner's Equity	186	323
Capital	160	200
Retained Earnings	26	-24
Current Period Profit		147
Total Liabilities & OE	421	560

ABC Co $m year	2017
Sales Revenue	576
- Cost of Goods Sold	-216
Gross Profit	360
- Distribution Costs	-65
- Administrative Expenses	-26
+ Other Operating Income	21
- Depreciation Expense	-79
- Interest Expense	-26
- Investment Income	8
Profit Before Tax	193
- Corporate Tax Expense	-46
Current Period Profit After Tax	147

[*] In 2017, ABC Co acquired additional plant and machinery; there were no disposals.

Net Profit, after Interest, before Tax i.e. Profit Before Tax	193
+ Depreciation expense	79
+ Interest EXPENSE	26
+/- (Increase) Decrease in Inventories	-12
+/- (Increase) Decrease in Receivables	-40
+/- Increase (Decrease) in Payables	20
- Interest PAID [a]	-21
- Corporate Tax PAID [b]	-39
- Investment Income	-8
= Net CF from OPERATING Activities	198
- Plant & Machinery acquired [c]	-95
+ Investment Income	8
= Net CF from INVESTING Activities	-87
- Long-Term Loans paid-back	-30
+ New Capital Raised	40
- Dividends Paid	-50
= Net CF from FINANCING Activities	-40
Net Inc (Dec) in Cash & Cash Equivalents	71

[a] Interest PAID = Int EXP -Inc(Dec) Int Payable

[b] Tax PAID = Tax EXP -Inc(Dec) Tax Payable

[c] Plant&Machinery acquired =
Increase in Plant&Machinery +Depreciation Exp
(+325 -309) +79 =95

NTE 1010 – Introduction to Finance for Engineers

Financial Statements

Arçelik A.Ş. 1 January - 30 June 2019

Cash Flow Statement

http://www.arcelikas.com/UserFiles/file/Arcelik%202019%202%C3%87%20Finansal%20Rapor.pdf

ARÇELİK ANONİM ŞİRKETİ

30 HAZİRAN 2019 TARİHİNDE SONA EREN ARA DÖNEME AİT KONSOLİDE NAKİT AKIŞ TABLOSU

(Tutarlar aksi belirtilmedikçe bin Türk Lirası ("TL") olarak ifade edilmiştir.)

		Sınırlı denetimden geçmiş	
	Notlar	1 Ocak - 30 Haziran 2019	1 Ocak - 30 Haziran 2018
İşletme faaliyetlerinden nakit akışları:			
Dönem karı		**458.089**	**322.177**
İşletme faaliyetlerinden sağlanan net nakdin vergi sonrası net dönem karı ile mutabakatına yönelik düzeltmeler:			
Vergi (geliri) gideri ile ilgili düzeltmeler	26	104.357	25.089
Amortisman ve itfa gideri ile ilgili düzeltmeler		507.083	310.575
Değer düşüklüğü ile ilgili düzeltmeler	31	42.855	20.201
Karşılıklar ile ilgili düzeltmeler	31	451.405	250.389
Faiz giderleri ile ilgili düzeltmeler	25	779.478	401.587
Faiz gelirleri ile ilgili düzeltmeler	24	(70.082)	(18.212)
Özkaynak yöntemiyle değerlenen yatırımların dağıtılmamış karları ile ilgili düzeltmeler	12	(16.372)	(34.278)
Türev finansal araçların gerçeğe uygun değer kayıpları (kazançları) ile ilgili düzeltmeler	24,25	(54.410)	23.212
Gerçekleşmemiş yabancı para çevrim farkı ile ilgili düzeltmeler	24,25	96.865	389.451
Kar (zarar) mutabakatı ile ilgili diğer düzeltmeler	24,25	5.106	5.340
Devlet teşviklerinden elde edilen gelirler ile ilgili düzeltmeler	22	(20.909)	(27.335)
Kar payı (geliri) gideri ile ilgili düzeltmeler	23	-	(94)
Duran varlıkların elden çıkarılmasından kaynaklanan kayıplar (kazançlar) ile ilgili düzeltmeler	23	679	(2.907)
İşletme varlık ve yükümlülüklerindeki değişiklik öncesi işletme faaliyetlerinden sağlanan net nakit		**2.284.144**	**1.665.195**
İşletme sermayesinde gerçekleşen değişimler:			
Ticari alacaklardaki azalış (artış) ile ilgili düzeltmeler		(1.610.777)	(1.466.685)
Stoklardaki azalışlar (artışlar) ile ilgili düzeltmeler		(715.923)	(984.552)
Peşin ödenmiş giderlerdeki azalış (artış)		(153.548)	13.707
Ticari borçlardaki artış (azalış) ile ilgili düzeltmeler		1.481.376	1.033.780
Çalışanlara sağlanan faydalar kapsamında borçlardaki artış (azalış)		(85.291)	(81.793)
Faaliyetler ile ilgili diğer borçlardaki artış (azalış) ile ilgili düzeltmeler		(14.597)	29.055
Devlet teşvik ve yardımlarındaki artış (azalış)		29.010	6.799
İşletme sermayesinde gerçekleşen diğer artış (azalış) ile ilgili düzeltmeler		105.217	(70.604)
Vergi iadeleri (ödemeleri)		(66.955)	(56.206)
İşletme faaliyetlerinden nakit akışları		**1.252.656**	**88.696**
Yatırım faaliyetleri:			
Bağlı ortaklıkların kontrolünün elde edilmesine yönelik alışlara ilişkin nakit çıkışları	3	(406.321)	-
Maddi ve maddi olmayan duran varlıkların alımından kaynaklanan nakit çıkışları		(587.333)	(628.916)
Maddi ve maddi olmayan duran varlıkların satışından kaynaklanan nakit girişleri		6.509	7.488
Verilen nakit avans ve borçlar		-	(54.639)
Alınan temettüler		-	26.265
İştiraklar ve/veya iş ortaklıkları pay alımı veya sermaye artırımı sebebiyle oluşan nakit çıkışları	12	(112.480)	(57.237)
Yatırım faaliyetlerinden kaynaklanan nakit akışları		**(1.099.625)**	**(707.039)**
Finansman faaliyetleri:			
Borçlanmadan kaynaklanan nakit girişleri		2.996.566	3.231.983
Borç ödemelerine ilişkin nakit çıkışları		(3.008.987)	(1.796.089)
Ödenen temettüler		-	(435.000)
Kira sözleşmelerinden kaynaklanan borç ödemelerine ilişkin nakit çıkışları	7	(132.990)	-
Ödenen faiz		(506.950)	(345.096)
Alınan faiz		69.506	18.009
İhraç edilen tahviller		500.000	-
Türev araçlardan nakit girişleri (net)		40.668	(49.757)
Diğer nakit girişleri (çıkışları)	24,25	(5.106)	(5.340)
Finansman faaliyetlerinden nakit akışları		**(47.293)**	**618.710**
Yabancı para çevirim farklarının etkisinden önce nakit ve nakit benzerlerindeki net artış/(azalış)		**105.738**	**367**
Yabancı para çevirim farklarının nakit ve nakit benzerleri üzerindeki etkisi		339.943	373.440
Nakit ve nakit benzerlerindeki net artış/(azalış)		**445.681**	**373.807**
1 Ocak itibarıyla nakit ve nakit benzeri değerler	5	**5.337.278**	**2.580.893**
30 Haziran itibarıyla nakit ve nakit benzeri değerler	5	**5.782.959**	**2.954.700**

NTE 1010
Introduction to Finance for Engineers
Spring 2020

Week 3

Analysing & Interpreting Financial Statements

Ratio Analysis

Vertical & Horizontal Analysis

Ratio Analysis

NTE 1010 – Introduction to Finance for Engineers

Financial Ratios

Financial ratios provide a quick and relatively simple means of assessing the financial health of a business. A ratio simply relates one figure appearing in the financial statements to some other figure appearing there.

Ratios can be very helpful when comparing the financial health of different businesses. Nevertheless, differences in scale of operations need to be taken into acount in such comparisons.

Though ratios are not difficult to calculate, they can be difficult to interpret and so it is important to appreciate that they are really only the starting point for further analysis. Ratios help to highlight the financial strengths and weaknesses of a business, but they cannot, by themselves, explain why certain strengths and weaknesses exit, or why certain changes have occurred. Only a detailed investigation will reveal these underlying reasons.

There is no generally accepted list of ratios that can be applied to the financial statements, nor is there a standard method of calculating many ratios.

Financial Ratios

Classifications

1. Profitability
   businesses exist with the primary purpose of creating wealth for their owners
2. Efficiency
   also known as Activity Ratios
3. Liquidity
   it is vital to the survival of a business for there to be sufficient liquid resources available to meet maturing obligations
4. Financial leverage
   highlight the extent to which the business uses loan finance
5. Investment
   Return on Investment, etc

Financial Ratios

Comparison

Merely calculating a ratio will not tell us very much
about the position of performance of a business.
It is only when we compare the ratio with some "benchmark"
that the information can be interpreted and evaluated.

1. Past periods

   By comparing the ratio calculated with the same ratio, but for a previous period, it is possible to detect whether there has been an improvement or deterioration in performance. Nevertheless, when comparing the performance of a single business over time, operating inefficiencies may not be clearly exposed.

2. Similar businesses

   In a competitive environment, a business must consider its performance in relation to that of other businesses operating in the same industry.

3. Planned performance

   Ratios may be compared with the targets that management developed before the start of the period under review.

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Financial Ratios & Financial Statement Analysis

Alex & Co ($m)	Balance Sheet 31-12 2016	31-12 2017
Current Assets	544	679
Cash, Cash Equivalents & Bank Deposits	4	
Trade Receivables	240	273
Inventories	300	406
Prepaid Expenses		
Other Current Assets		
Non-Current Assets	510	587
Plant, Machinery & Equipment, gross		
less: Accumulated Depreciation		
equals: Plant, Machine & Equipm, net [f]	129	160
Land & Buildings	381	427
Other Non-Current Assets		
TOTAL ASSETS	**1.054**	**1.266**
Current Liabilities	291	432
Short-Term Bank Loans		76
Trade Payables	221	314
Interest Payable		
Corporate Tax Payable	30	2
Accrued Expenses		
Dividends Payable (Unpaid Dividends) [d]	40	40
Other Current Liabilities		
Long-Term Liabilities	200	300
Long-Term Trade Payables		
Long-Term Bank Loans		
Other Long-Term Liabilities		
Long-Term Bonds [g]	200	300
Total Liabilities	**491**	**732**
Owner's Equity [g]	**563**	**534**
Paid-in Capital [a]	300	300
Equity Reserves		
Retained Earnings	138	263
Current Period Profit (Profit after Tax)	165	11
Declared Dividends [d]	-40	-40
TOTAL LIABILITIES & OWNER'S EQUITY	**1.054**	**1.266**

Alex & Co ($m)	Income Statement year 2016	year 2017
Sales Revenue [b]	2.240	2.681
- Cost of Goods Sold [c]	-1.745	-2.272
Gross Profit	495	409
- Selling, General and Admin Costs		
- Other Operating Costs	-252	-362
- Depreciation Expense		
Profit before Interest & Tax	243	47
- Interest Expense	-18	-32
- Uncollectible Receivables		
+ Other Operational Income		
- Other Operatinal Expense		
+/- Net Investment Income (Expense)		
Profit before Tax	225	15
- Corporate Tax on Profit	-60	-4
Current Period Profit (Profit after Tax)	165	11

[a] Face value of each share is $0.50. Market value at 2016 year-end was $2.50 and $1.50 at 2017 year-end.

[b] All sales and purchases are made on credit.

[c] Cost of Goods Sold	2016	2017
Opening inventories	241	300
+ Purchases	1.804	2.378
- Closing inventories	-300	-406
= Cost of Goods Sold	1.745	2.272

[d] Declared dividends are paid in the following year.

[e] Staff size at 2016 year-end was 13,995 and 18,623 at 2017 year-end.

[f] The business expanded its capacity in 2017 by setting up a new warehouse and distribution center.

[g] On 1-1-2016, Owner's Equity was $438m; sum of OE and Long-Term Bonds was $638m.

Profitability

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Financial Ratios – Profitability

Return on Equity (RoE)

Compares the amount of profit for the period available to the owners, with the owners' average stake in the business during that same period.

$$RoE = \frac{\text{Current Period Profit}}{\text{average Owner's Equity}}$$

$$RoE_{2016} = \frac{165}{(\ 438\ +\ 563\ )\ /\ 2} = 33\%$$

$$RoE_{2017} = \frac{11}{(\ 563\ +\ 534\ )\ /\ 2} = 2\%$$

In general, businesses seek to generate as high a value as possible for this ratio, provided that it is not achieved at the expense of potential future returns by, for example, taking on more risky activities.

The RoE ratio is widely used by businesses when establishing targets for profitability. These targets are sometimes made public by companies shares of which are traded in stock exchanges.

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NTE 1010 – Introduction to Finance for Engineers

Financial Ratios – Profitability

Net Profit Margin

Relates Profit before Interest & Tax for the period to the Sales Revenue during that period.

This is often regarded as the most appropriate measure of operational performance, when used as a basis of comparison, because differences arising from the way in which the business is financed will not influence the measure.

$$\text{Net Profit Margin} = \frac{\text{Profit before Interest \& Tax}}{\text{Sales Revenue}}$$

$$\text{Net Profit Margin}_{2016} = \frac{243}{2.240} = 11\%$$

$$\text{Net Profit Margin}_{2017} = \frac{47}{2.681} = 2\%$$

This ratio can vary considerably between types of business. For example, supermarkets thend to operate on low margins. Jewellers, on the other hand, tend to have high net profit margins.

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NTE 1010 – Introduction to Finance for Engineers

Financial Ratios – Profitability

Gross Profit Margin

Relates Gross Profit for the period
to the Sales Revenue during that period.
It is a measure of profitability
in buying (or producing) and selling goods
before any other expenses are taken into account.

$$\text{Gross Profit Margin} = \frac{\text{Gross Profit}}{\text{Sales Revenue}}$$

$$\text{Gross Profit Margin}_{2016} = \frac{495}{2.240} = 22\%$$

$$\text{Gross Profit Margin}_{2017} = \frac{409}{2.681} = 15\%$$

The decline in this ratio could mean that
sales prices were lower and/or
the purchase cost of goods sold increased.
It is possible that both were reduced,
but sales prices at a greater rate.
Or, both incresed,
but cost of goods sold at a greater rate.
Further analysis is needed
to discover the causes.

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+/- Net Investment Income (Expense)		
Profit before Tax	225	15
- Corporate Tax on Profit	-60	-4
Current Period Profit (Profit after Tax)	165	11

[a] Face value of each share is $0.50.
Market value at 2016 year-end was $2.50
and $1.50 at 2017 year-end.

[b] All sales and purchases are made on credit.

[c] Cost of Goods Sold	2016	2017
Opening inventories	241	300
+ Purchases	1.804	2.378
- Closing inventories	-300	-406
= Cost of Goods Sold	1.745	2.272

[d] Declared dividends are paid in the following year.

[e] Staff size at 2016 year-end was 13,995
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Efficiency

NTE 1010 – Introduction to Finance for Engineers

Financial Ratios – Efficiency

Average Inventories Turnover Period

Inventories often represent a significant investment for a business.

This ratio measures the average period (in days) for which the inventories are being held.

In other words, it measures "how fast" the inventories are being sold ("turned over").

$$\text{Inventory Turnover Period (in days)} = \frac{\text{average Inventories held}}{\text{Cost of Goods Sold}} \times 365$$

$$\text{Inventory Turnover Period}_{2016} = \frac{(241 + 300)/2}{1.745} \times 365 = 57 \text{ days}$$

$$\text{Inventory Turnover Period}_{2017} = \frac{(300 + 406)/2}{2.272} \times 365 = 57 \text{ days}$$

In case of a highly seasonal business, where inventory levels may vary considerably over the year, a monthly average may be more appropriate.

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Current Assets	544	679
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Trade Receivables	240	273
Inventories	300	406
Prepaid Expenses		
Other Current Assets		
Non-Current Assets	510	587
Plant, Machinery & Equipment, gross		
less: Accumulated Depreciation		
equals: Plant, Machine & Equipm, net [f]	129	160
Land & Buildings	381	427
Other Non-Current Assets		
TOTAL ASSETS	**1.054**	**1.266**
Current Liabilities	291	432
Short-Term Bank Loans		76
Trade Payables	221	314
Interest Payable		
Corporate Tax Payable	30	2
Accrued Expenses		
Dividends Payable (Unpaid Dividends) [d]	40	40
Other Current Liabilities		
Long-Term Liabilities	200	300
Long-Term Trade Payables		
Long-Term Bank Loans		
Other Long-Term Liabilities		
Long-Term Bonds [g]	200	300
Total Liabilities	**491**	**732**
Owner's Equity [g]	**563**	**534**
Paid-in Capital [a]	300	300
Equity Reserves		
Retained Earnings	138	263
Current Period Profit (Profit after Tax)	165	11
Declared Dividends [d]	-40	-40
TOTAL LIABILITIES & OWNER'S EQUITY	**1.054**	**1.266**

Alex & Co ($m)	Income Statement year 2016	year 2017
Sales Revenue [b]	2.240	2.681
- Cost of Goods Sold [c]	-1.745	-2.272
Gross Profit	495	409
- Selling, General and Admin Costs		
- Other Operating Costs	-252	-362
- Depreciation Expense		
Profit before Interest & Tax	243	47
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NTE 1010 – Introduction to Finance for Engineers

Financial Ratios – Efficiency

Avg Settlement Period for Receivables

Speed of payment by customers can have a significant effect on business' cash flow.
This ratio measures the average period (in days) credit customers to pay the amounts that they owe to the business.

$$\text{Avg Settlement Period of Receivables (in days)} = \frac{\text{Average Trade Receivables}}{\text{Credit Sales Revenue}} \times 365$$

$$\text{Avg Settlement Period of Receivables}_{2016} = \frac{(215 + 240)/2}{2{,}240} \times 365 = 37 \text{ days}$$

On 1-1-2016, Trade Receivables was $215m.
When opening balances are available, averaging Trade Receivables should be preferred.

$$\text{Avg Settlem Period of Receivables}_{2017} = \frac{(240 + 273)/2}{2.681} \times 365 = 35 \text{ d}$$

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Financial Ratios – Efficiency

Avg Settlement Period for Payables

Measures how long, on the average, the business takes to pay its trade payables.

As trade payables provide a free source of finance for the business, most businesses attempt to increase their average settlement period. However, such a policy, if taken too far, would result in a loss of goodwill of suppliers.

$$\text{Avg Settlement Period for Payables (in days)} = \frac{\text{Average Trade Payables}}{\text{Credit Purchases}} \times 365$$

$$\text{Avg Settlement Period for Payables}_{2016} = \frac{(263 + 221)/2}{1{,}804} \times 365 = 49 \text{ days}$$

On 1-1-2016, Trade Payables was $263m.

$$\text{Avg Settlem Period for Payables}_{2017} = \frac{(221 + 314)/2}{2.378} \times 365 = 41 \text{ d}$$

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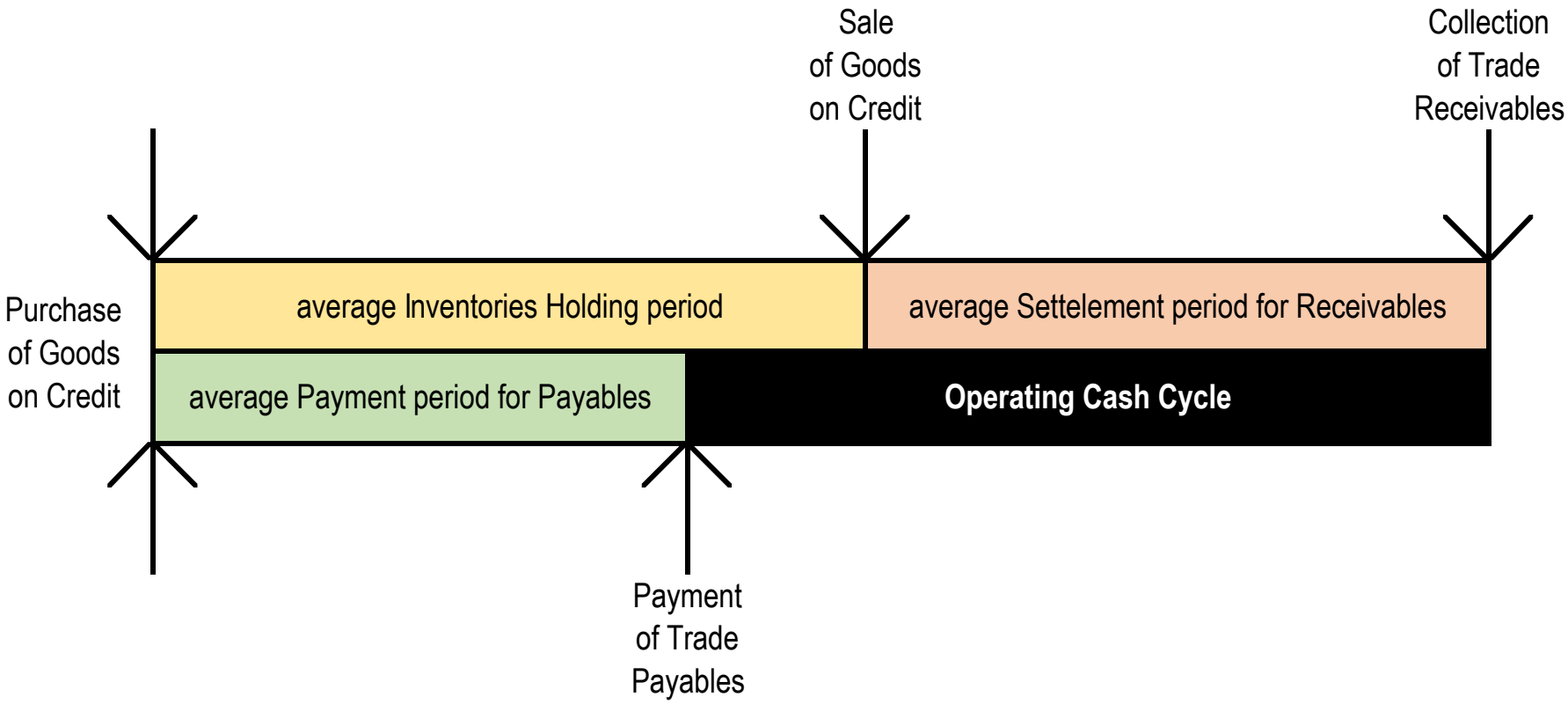
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Financial Ratios - Efficiency

Operating Cash Cycle



Financial Ratios - Efficiency

Measuring Operating Cash Cycle

average Inventories Holding period (days)

$$\frac{\text{average Inventories}}{\text{Cost of Goods Sold}} \times 365$$

avg Settelement period for Receivables (days)

$$\frac{\text{average Trade Receivables}}{\text{Credit Sales}} \times 365$$

avg Payment period for Payables (days)

$$\frac{\text{average Trade Payables}}{\text{Credit Purchases}} \times 365$$

Operating Cash Cycle (days)

Arçelik	31-12 2018	30-6 2019
Inventories	5.1	6.1
Trade Receivables	7.7	9.5
Trade Payables	4.7	6.2
(Credit) Sales		15.3
Cost of Goods Sold = Credit Purchases		10.5

Migros	31-12 2018	30-6 2019
Inventories	2.25	2.53
Trade Receivables	0.12	0.19
Trade Payables	4.9	5.2
(Credit) Sales		10.7
Cost of Goods Sold = Credit Purchases		7.8

Liquidity

NTE 1010 – Introduction to Finance for Engineers

Financial Ratios – Liquidity

Current Ratio

Compares the "liquid" assets
(cash and assets to be turned into cash soon)
of the business with the current liabilities.

There is no "ideal" Curent Ratio as diiferent types of business require different current ratios.

As liquidity is vital to the survival of a business, a higher current ratio is preferable.

$$\text{Current Ratio} = \frac{\text{Current Assets}}{\text{Current Liabilities}}$$

$$\text{Current Ratio}_{2016} = \frac{544}{291} = 1{,}9 \text{ times}$$

$$\text{Current Ratio}_{2017} = \frac{679}{432} = 1{,}6 \text{ times}$$

Though there is a decline from 2016 to 2017, it is not necessarily a matter of concern.

The next ratio may provide a clue as to whether there seems to be a problem.

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Financial Ratios – Liquidity

Acit Test Ratio

Very similar to the Current Ratio,
but it represents a more stringent test of liquidity.
It excludes inventories,
as it can be argued that, for many businesses,
inventories cannot be converted into cash quickly.

$$\text{Acid Test Ratio} = \frac{\text{Current Assets excluding Inventories, Prepaid Exp, Other CA}}{\text{Current Liabilities}}$$

$$\text{Acid Test Ratio}_{2016} = \frac{544 - 300}{291} = 0{,}8 \text{ times}$$

$$\text{Acid Test Ratio}_{2017} = \frac{679 - 406}{432} = 0{,}6 \text{ times}$$

We can see now that the "liquid" Current Assets
do not quite cover the current liabilities,
so the business may be experiencing
some liquidity problems.

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Leverage

Financial Ratios

Financial Leverage

Occurs when a business is financed, at least in part, by borrowing,
instead of by finance provided by the owners (the shareholders).

Level of leverage of a business is an important factor in assessing its risk.
Where a business borrows, it takes on a commitment
to pay interest charges and make capital repayments.

Where the borrowing is heavy, this can be a significant financial burden;
it can increase the risk of the business becoming insolvent.

Nevertheless, most businesses are leveraged to some extent.

This may be so, because the owners do not sufficient funds,
so the only way to finance the business adequately is to borrow from others.

Another reason is that leveraging can be used to increase the returns to owners.
This is possible provided the returns generated from borrowed funds
exceed the cost of paying interest.

NTE 1010 – Introduction to Finance for Engineers

Financial Ratios

Financial Leverage

$	Lee &Co	Nova &Co
Paid-in Capital	100.000	200.000
10% Long-Term Loan	200.000	100.000
Capital Employed	300.000	300.000

$	Lee &Co	Nova &Co
Profit before Int & Tax	50.000	50.000
- Interest Expense	-20.000	-10.000
Profit before Tax	30.000	40.000
20% Corporate Tax	-6.000	-8.000
Current Period Profit	24.000	32.000

	Lee &Co	Nova &Co
$\frac{\text{Current Period Profit}}{\text{Paid-in Capital}}$	24%	16%

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Financial Ratios – Leverage

Leverage Ratio

Measures debt level of a company relative to its owner's equity.

$$\text{Leverage Ratio} = \frac{\text{Total Liabilities}}{\text{Owner's Equity}}$$

$$\text{Leverage Ratio}_{2016} = \frac{291 + 200}{563} = 0{,}9 \text{ times}$$

$$\text{Leverage Ratio}_{2017} = \frac{432 + 300}{534} = 1{,}4 \text{ times}$$

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Financial Ratios – Leverage

Interest Cover Ratio

Measures the amount of profit available to cover interest payable.

$$\text{Interest Cover Ratio} = \frac{\text{Profit before Interest \& Tax}}{\text{Interest Expense}}$$

$$\text{Interest Cover Ratio}_{2016} = \frac{243}{18} = 13{,}5 \text{ times}$$

$$\text{Interest Cover Ratio}_{2017} = \frac{47}{32} = 1{,}5 \text{ times}$$

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- Selling, General and Admin Costs		
- Other Operating Costs	-252	-362
- Depreciation Expense		
Profit before Interest & Tax	243	47
- Interest Expense	-18	-32
- Uncollectible Receivables		
+ Other Operational Income		
- Other Operatinal Expense		
+/- Net Investment Income (Expense)		
Profit before Tax	225	15
- Corporate Tax on Profit	-60	-4
Current Period Profit (Profit after Tax)	165	11

[a] Face value of each share is $0.50. Market value at 2016 year-end was $2.50 and $1.50 at 2017 year-end.

[b] All sales and purchases are made on credit.

[c] Cost of Goods Sold	2016	2017
Opening inventories	241	300
+ Purchases	1.804	2.378
- Closing inventories	-300	-406
= Cost of Goods Sold	1.745	2.272

[d] Declared dividends are paid in the following year.

[e] Staff size at 2016 year-end was 13,995 and 18,623 at 2017 year-end.

[f] The business expanded its capacity in 2017 by setting up a new warehouse and distribution center.

[g] On 1-1-2016, Owner's Equity was $438m; sum of OE and Long-Term Bonds was $638m.

Investment

NTE 1010 – Introduction to Finance for Engineers

Financial Ratios – Investment

Dividend Payout Ratio

Measures the proportion of earnings that a business pays out to shareholders in the form of dividends.

$$\text{Dividend Payout Ratio} = \frac{\text{Dividends Declared}}{\text{Current Period Earnings available for Dividends*}}$$

* normally current period profit after tax

$$\text{Dividend Payout Ratio}_{2016} = \frac{40}{165} = 24\%$$

$$\text{Dividend Payout Ratio}_{2017} = \frac{40}{11} = 364\% \ *$$

* dividends to be paid out of earnings of previous years (i.e. Retained Earnings)

Alex & Co ($m)	Balance Sheet 31-12 2016	31-12 2017
Current Assets	544	679
Cash, Cash Equivalents & Bank Deposits	4	
Trade Receivables	240	273
Inventories	300	406
Prepaid Expenses		
Other Current Assets		
Non-Current Assets	510	587
Plant, Machinery & Equipment, gross		
less: Accumulated Depreciation		
equals: Plant, Machine & Equipm, net [f]	129	160
Land & Buildings	381	427
Other Non-Current Assets		
TOTAL ASSETS	**1.054**	**1.266**
Current Liabilities	291	432
Short-Term Bank Loans		76
Trade Payables	221	314
Interest Payable		
Corporate Tax Payable	30	2
Accrued Expenses		
Dividends Payable (Unpaid Dividends) [d]	40	40
Other Current Liabilities		
Long-Term Liabilities	200	300
Long-Term Trade Payables		
Long-Term Bank Loans		
Other Long-Term Liabilities		
Long-Term Bonds [g]	200	300
Total Liabilities	**491**	**732**
Owner's Equity [g]	**563**	**534**
Paid-in Capital [a]	300	300
Equity Reserves		
Retained Earnings	138	263
Current Period Profit (Profit after Tax)	165	11
Declared Dividends [d]	-40	-40
TOTAL LIABILITIES & OWNER'S EQUITY	**1.054**	**1.266**

Alex & Co ($m)	Income Statement year 2016	year 2017
Sales Revenue [b]	2.240	2.681
- Cost of Goods Sold [c]	-1.745	-2.272
Gross Profit	495	409
- Selling, General and Admin Costs		
- Other Operating Costs	-252	-362
- Depreciation Expense		
Profit before Interest & Tax	243	47
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NTE 1010 – Introduction to Finance for Engineers

Financial Ratios – Investment

Dividend Yield Ratio

Relates the cash return from a share to its current market value.

$$\text{Dividend Yield Ratio} = \frac{\text{Dividend per Share}}{\text{Market Value per Share}}$$

$$\text{Dividend Yield Ratio}_{2016} = \frac{\text{dividend}}{\text{no of shares}} = \frac{40\ \text{m\$} / (\ 300\ \text{m\$} / \ 0{,}50\ \$)}{2{,}50\ \$} = \frac{0{,}067}{2{,}50} = 2{,}7\%$$

$$\text{Dividend Yield Ratio}_{2017} = \frac{\text{dividend}}{\text{no of shares}} = \frac{40\ \text{m\$} / (\ 300\ \text{m\$} / \ 0{,}50\ \$)}{1{,}50\ \$} = \frac{0{,}067}{1{,}50} = 4{,}4\%$$

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Prepaid Expenses		
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Plant, Machinery & Equipment, gross		
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NTE 1010 – Introduction to Finance for Engineers

Financial Ratios – Investment

Earnings per Share (EPS)

Relates the earnings generated by the business, and available to shareholders, during a period to the number of shares in issue.

$$EPS = \frac{\text{Current Period Earnings available to Shareholders *}}{\text{no of Shares}}$$

* normally current period profit after tax

Many investment analysts regard the EPS ratio as a fundamental measure of share performance. Though it is possible to make total profits rise through shareholders investing more in the business, this will not necessarily result in profitability per share rising.

$$EPS_{2016} = \frac{165 \text{ m\$}}{300 \text{ m\$} / 0{,}50 \text{ \$}} = 0{,}28 \text{ \$}$$

$$EPS_{2017} = \frac{11 \text{ m\$}}{300 \text{ m\$} / 0{,}50 \text{ \$}} = 0{,}02 \text{ \$}$$

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Financial Ratios – Investment

Price/Earnings Ratio (P/E)

Relates market value of a share
to the earnings per share.
This ratio is a measure of market confidence
in the future of a business.

$$\text{P/E Ratio} = \frac{\text{Market Value per Share}}{\text{EPS}}$$

$$EPS_{2016} = \frac{165\ m\$}{300\ m\$\ /\ 0{,}50\ \$} = 0{,}28\ \$$$

$$\text{P/E Ratio}_{2016} = \frac{2{,}50}{0{,}28} = 9{,}1\ \text{times}$$

$$EPS_{2016} = \frac{11\ m\$}{300\ m\$\ /\ 0{,}50\ \$} = 0{,}02\ \$$$

$$\text{P/E Ratio}_{2016} = \frac{1{,}50}{0{,}02} = 82\ \text{times}$$

Alex & Co ($m)	Balance Sheet 31-12 2016	31-12 2017
Current Assets	544	679
Cash, Cash Equivalents & Bank Deposits	4	
Trade Receivables	240	273
Inventories	300	406
Prepaid Expenses		
Other Current Assets		
Non-Current Assets	510	587
Plant, Machinery & Equipment, gross		
less: Accumulated Depreciation		
equals: Plant, Machine & Equipm, net [f]	129	160
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Other Non-Current Assets		
TOTAL ASSETS	**1.054**	**1.266**
Current Liabilities	291	432
Short-Term Bank Loans		76
Trade Payables	221	314
Interest Payable		
Corporate Tax Payable	30	2
Accrued Expenses		
Dividends Payable (Unpaid Dividends) [d]	40	40
Other Current Liabilities		
Long-Term Liabilities	200	300
Long-Term Trade Payables		
Long-Term Bank Loans		
Other Long-Term Liabilities		
Long-Term Bonds [g]	200	300
Total Liabilities	**491**	**732**
Owner's Equity [g]	**563**	**534**
Paid-in Capital [a]	300	300
Equity Reserves		
Retained Earnings	138	263
Current Period Profit (Profit after Tax)	165	11
Declared Dividends [d]	-40	-40
TOTAL LIABILITIES & OWNER'S EQUITY	**1.054**	**1.266**

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+/- Net Investment Income (Expense)		
Profit before Tax	225	15
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[a] Face value of each share is $0.50.
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and $1.50 at 2017 year-end.

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sum of OE and Long-Term Bonds was $638m.

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Financial Ratios

Summary of Major Ratios

Profitability

$$\text{RoE} = \frac{\text{Current Period Profit}}{\text{average Owner's Equity}}$$

$$\text{Net Profit Margin} = \frac{\text{Profit before Interest \& Tax}}{\text{Sales Revenue}}$$

$$\text{Gross Profit Margin} = \frac{\text{Gross Profit}}{\text{Sales Revenue}}$$

$$\text{P/E Ratio} = \frac{\text{Market Value per Share}}{\text{EPS}}$$

Leverage

$$\text{Leverage Ratio} = \frac{\text{Total Liabilities}}{\text{Owner's Equity}}$$

$$\text{Interest Cover Ratio} = \frac{\text{Profit before Interest \& Tax}}{\text{Interest Expense}}$$

$$\text{Dividend Payout Ratio} = \frac{\text{Dividends Declared}}{\text{Current Period Earnings available for Dividends}}$$

$$\text{Dividend Yield Ratio} = \frac{\text{Dividend per Share}}{\text{Market Value per Share}}$$

$$\text{EPS} = \frac{\text{Current Period Earnings available to Shareholders}}{\text{no of Shares}}$$

Efficiency

$$\text{Inventory Turnover Period (in days)} = \frac{\text{average Inventories held}}{\text{Cost of Goods Sold}} \times 365$$

$$\text{Avg Settlement Period of Receivables (in days)} = \frac{\text{average Trade Receivables}}{\text{Credit Sales Revenue}} \times 365$$

$$\text{Avg Settlement Period for Payables (in days)} = \frac{\text{average Trade Payables}}{\text{Credit Purchases}} \times 365$$

Liquidity

$$\text{Current Ratio} = \frac{\text{Current Assets}}{\text{Current Liabilities}}$$

$$\text{Acid Test Ratio} = \frac{\text{Current Assets excluding Inventories, Prepaid Exp, Other CA}}{\text{Current Liabilities}}$$

Investment

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Introduction to Finance for Engineers
Spring 2020

Week 3

Analysing & Interpreting Financial Statements

Ratio Analysis

Vertical & Horizontal Analysis

NTE 1010 – Introduction to Finance for Engineers

Financial Ratios

Horizontal Analysis

Alex & Co ($m)	Balance Sheet 31-12 2015	31-12 2016	31-12 2017	Horizontal Analysis 31-12 2015	31-12 2016	31-12 2017
Current Assets	479	544	679	100	114	142
Cash, Cash Equivalents & Bank Deposits	30	4		100	13	0
Trade Receivables	208	240	273	100	115	131
Inventories	241	300	406	100	124	168
Prepaid Expenses						
Other Current Assets						
Non-Current Assets	361	510	587	100	141	163
Plant, Machinery & Equipment, gross						
less: Accumulated Depreciation						
equals: Plant, Machine & Equipm, net [f]	80	129	160	100	161	200
Land & Buildings	281	381	427	100	136	152
Other Non-Current Assets						
TOTAL ASSETS	**840**	**1.054**	**1.266**	**100**	**125**	**151**
Current Liabilities	288	291	432	100	101	150
Short-Term Bank Loans			76			
Trade Payables	240	221	314	100	92,1	131
Interest Payable						
Corporate Tax Payable	30	30	2	100	100	7
Accrued Expenses						
Dividends Payable (Unpaid Dividends) [d]	18	40	40	100	222	222
Other Current Liabilities						
Long-Term Liabilities	200	200	300	100	100	150
Long-Term Trade Payables						
Long-Term Bank Loans						
Other Long-Term Liabilities						
Long-Term Bonds [g]	200	200	300	100	100	150
Total Liabilities	**488**	**491**	**732**	**100**	**101**	**150**
Owner's Equity [g]	**352**	**563**	**534**	**100**	**160**	**152**
Paid-in Capital [a]	250	300	300	100	120	120
Equity Reserves						
Retained Earnings	60	138	263	100	230	438
Current Period Profit (Profit after Tax)	60	165	11	100	275	18
Declared Dividends [d]	-18	-40	-40	100	222	222
TOTAL LIABILITIES & OWNER'S EQUITY	**840**	**1.054**	**1.266**	**100**	**125**	**151**

NTE 1010 – Introduction to Finance for Engineers

Financial Ratios

Vertical Analysis

Alex & Co ($m)	Income Statement year 2016	Income Statement year 2017	Vertical Analysis year 2016	Vertical Analysis year 2017
Sales Revenue [b]	2.240	2.681	100	100
- Cost of Goods Sold [c]	-1.745	-2.272	-78	-85
Gross Profit	495	409	22	15
- Selling, General and Admin Costs				
- Other Operating Costs	-252	-362	-11	-14
- Depreciation Expense				
Profit before Interest & Tax	243	47	11	2
- Interest Expense	-18	-32	-1	-1
- Uncollectible Receivables				
+ Other Operational Income				
- Other Operatinal Expense				
+/- Net Investment Income (Expense)				
Profit before Tax	225	15	10	1
- Corporate Tax on Profit	-60	-4	-3	0
Current Period Profit (Profit after Tax)	165	11	7	0

Alex & Co ($m)	Balance Sheet 31-12 2016	Balance Sheet 31-12 2017	Vertical Analysis 31-12 2016		Vertical Analysis 31-12 2017	
Current Assets	544	679	52	100	54	100
Cash, Cash Equivalents & Bank Deposits	4		0	1	0	0
Trade Receivables	240	273	23	44	22	40
Inventories	300	406	28	55	32	60
Prepaid Expenses						
Other Current Assets						
Non-Current Assets	510	587	48	100	46	100
Plant, Machinery & Equipment, gross						
less: Accumulated Depreciation						
equals: Plant, Machine & Equipm, net [f]	129	160	12	25	13	27
Land & Buildings	381	427	36	75	34	73
Other Non-Current Assets						
TOTAL ASSETS	**1.054**	**1.266**	**100**		**100**	
Current Liabilities	291	432	28	100	34	100
Short-Term Bank Loans		76				
Trade Payables	221	314	21	76	25	73
Interest Payable						
Corporate Tax Payable	30	2	3	10	0	0
Accrued Expenses						
Dividends Payable (Unpaid Dividends) [d]	40	40	4	14	3	9
Other Current Liabilities						
Long-Term Liabilities	200	300	19	100	24	100
Long-Term Trade Payables						
Long-Term Bank Loans						
Other Long-Term Liabilities						
Long-Term Bonds [g]	200	300	19	100	24	100
Total Liabilities	**491**	**732**	**47**		**58**	
Owner's Equity [g]	**563**	**534**	**53**	**100**	**42**	**100**
Paid-in Capital [a]	300	300	28	53	24	56
Equity Reserves						
Retained Earnings	138	263	13	25	21	49
Current Period Profit (Profit after Tax)	165	11	16	29	1	2
Declared Dividends [d]	-40	-40	-4	-7	-3	-7
TOTAL LIABILITIES & OWNER'S EQUITY	**1.054**	**1.266**	**100**		**100**	

NTE 1010
Introduction to Finance for Engineers
Spring 2020

Week 3

Analysing & Interpreting Financial Statements

Ratio Analysis

Vertical & Horizontal Analysis

Ratio Analysis exercise

NTE 1010.1 Intro to Finance for Engr

Financial Ratios

In-Class Exercise

My Shop Ltd Income Statement ($m)	year 2017
Sales Revenue [a]	2.000
- Cost of Goods Sold [b]	-1.100
Gross Profit	900
- Selling, General and Admin Costs	-225
- Other Operating Costs	-40
- Depreciation Expense	-300
Profit before Interest & Tax	335
- Interest Expense	-50
- Uncollectible Receivables	-5
+ Other Operational Income	6
- Other Operatinal Expense	-3
+/- Net Investment Income (Expense)	5
Profit before Tax	288
- Corporate Tax on Profit	-58
Current Period Profit (Profit after Tax)	230

[a] 85% of sales are on credit.
[b] 40% of Cost of Goods Sold were credit purchases.
[c] Face value of each share is $1.
Market value at 2017 year-end was $4.25.

My Shop Ltd Balance Sheet ($m)	31-12 2016	31-12 2017
Current Assets	301	496
Cash, Cash Equivalents & Bank Deposits	10	25
Trade Receivables	165	140
Inventories	100	250
Prepaid Expenses	20	64
Other Current Assets	6	17
Non-Current Assets	734	1.086
Plant, Machinery & Equipment, gross	1.000	1.430
less: Accumulated Depreciation	-350	-550
equals: Plant, Machine & Equipm, net	650	880
Land & Buildings	80	200
Other Non-Current Assets	4	6
TOTAL ASSETS	**1.035**	**1.582**
Current Liabilities	304	428
Short-Term Bank Loans	80	88
Trade Payables	81	138
Interest Payable	35	42
Corporate Tax Payable	35	58
Accrued Expenses	18	21
Dividends Payable (Unpaid Dividends)	40	70
Other Current Liabilities	15	11
Long-Term Liabilities	146	189
Long-Term Trade Payables	4	6
Long-Term Bank Loans	40	30
Other Long-Term Liabilities	2	3
Long-Term Bonds	100	150
Total Liabilities	**450**	**617**
Owner's Equity	**585**	**965**
Paid-in Capital [c]	400	600
Equity Reserves	40	60
Retained Earnings	50	145
Current Period Profit (Profit after Tax)	135	230
Declared Dividends	-40	-70
TOTAL LIABILITIES & OWNER'S EQUITY	**1.035**	**1.582**

$$\text{RoE} = \frac{\text{Current Period Profit}}{\text{average Owner's Equity}}$$

$$\text{Leverage Ratio} = \frac{\text{Total Liabilities}}{\text{Owner's Equity}}$$

$$\text{RoE}_{2017} = \frac{230}{(\ 585\ +\ 965\ )\ /\ 2} = 30\%$$

$$\text{Leverage Ratio}_{2017} = \frac{617}{965} = 0{,}6 \text{ times}$$

NTE 1010.1 Intro to Finance for Engr

Financial Ratios

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Short-Term Bank Loans	80	88
Trade Payables	81	138
Interest Payable	35	42
Corporate Tax Payable	35	58
Accrued Expenses	18	21
Dividends Payable (Unpaid Dividends)	40	70
Other Current Liabilities	15	11
Long-Term Liabilities	146	189
Long-Term Trade Payables	4	6
Long-Term Bank Loans	40	30
Other Long-Term Liabilities	2	3
Long-Term Bonds	100	150
Total Liabilities	**450**	**617**
Owner's Equity	**585**	**965**
Paid-in Capital [c]	400	600
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Current Period Profit (Profit after Tax)	135	230
Declared Dividends	-40	-70
TOTAL LIABILITIES & OWNER'S EQUITY	**1.035**	**1.582**

$$\text{Inventory Turnover Period (in days)} = \frac{\text{average Inventories held}}{\text{Cost of Goods Sold}} \times 365$$

$$\text{Avg Settlement Period of Receivables (in days)} = \frac{\text{average Trade Receivables}}{\text{Credit Sales Revenue}} \times 365$$

$$\text{Avg Settlement Period for Payables (in days)} = \frac{\text{average Trade Payables}}{\text{Credit Purchases}} \times 365$$

$$\text{Inventory Turnover Period}_{2017} = \frac{(100 + 250)/2}{1.100} \times 365 = 58 \text{ days}$$

$$\text{Avg Settlement Period of Receivables}_{2017} = \frac{(165 + 140)/2}{1.700} \times 365 = 33 \text{ days}$$

$$\text{Avg Settlement Period for Payables}_{2017} = \frac{(81 + 138)/2}{440} \times 365 = 91 \text{ days}$$

NTE 1010.1 Intro to Finance for Engr

Financial Ratios

In-Class Exercise

My Shop Ltd Income Statement ($m)	year 2017
Sales Revenue [a]	2.000
- Cost of Goods Sold [b]	-1.100
Gross Profit	900
- Selling, General and Admin Costs	-225
- Other Operating Costs	-40
- Depreciation Expense	-300
Profit before Interest & Tax	335
- Interest Expense	-50
- Uncollectible Receivables	-5
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+/- Net Investment Income (Expense)	5
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[a] 85% of sales are on credit.
[b] 40% of Cost of Goods Sold were credit purchases.
[c] Face value of each share is $1. Market value at 2017 year-end was $4.25.

My Shop Ltd Balance Sheet ($m)	31-12 2016	31-12 2017
Current Assets	301	496
Cash, Cash Equivalents & Bank Deposits	10	25
Trade Receivables	165	140
Inventories	100	250
Prepaid Expenses	20	64
Other Current Assets	6	17
Non-Current Assets	734	1.086
Plant, Machinery & Equipment, gross	1.000	1.430
less: Accumulated Depreciation	-350	-550
equals: Plant, Machine & Equipm, net	650	880
Land & Buildings	80	200
Other Non-Current Assets	4	6
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$$\text{Net Profit Margin} = \frac{\text{Profit before Interest \& Tax}}{\text{Sales Revenue}}$$

$$\text{Gross Profit Margin} = \frac{\text{Gross Profit}}{\text{Sales Revenue}}$$

$$\text{Interest Cover Ratio} = \frac{\text{Profit before Interest \& Tax}}{\text{Interest Expense}}$$

$$\text{Net Profit Margin}_{2017} = \frac{335}{2.000} = 17\%$$

$$\text{Gross Profit Margin}_{2017} = \frac{900}{2.000} = 45\%$$

$$\text{Interest Cover Ratio}_{2017} = \frac{335}{50} = 6{,}7 \text{ times}$$

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Financial Ratios

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$$\text{Current Ratio} = \frac{\text{Current Assets}}{\text{Current Liabilities}}$$

$$\text{Acid Test Ratio} = \frac{\text{Current Assets excl Inv, Prepaid Exp, Other CA}}{\text{Current Liabilities}}$$

$$\text{Current Ratio}_{2017} = \frac{496}{428} = 1.16 \text{ times}$$

$$\text{Acid Test Ratio}_{2017} = \frac{496 - 331}{428} = 0.39 \text{ times}$$

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Financial Ratios

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$$\text{Dividend Payout Ratio} = \frac{\text{Dividends Declared}}{\text{Current Period Earnings available for Dividends}}$$

$$\text{Dividend Yield Ratio} = \frac{\text{Dividend per Share}}{\text{Market Value per Share}}$$

$$\text{Dividend Payout Ratio}_{2017} = \frac{70}{230} = 30\%$$

$$\text{Dividend Yield Ratio}_{2017} = \frac{\text{dividend}}{\text{4,25 \$}} = \frac{\text{70 m\$ / (600 m\$ / 1 \$)}}{\text{4,25 \$}} = \frac{\text{0,117 \$}}{\text{4,25 \$}} = 2{,}7\%$$

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Financial Ratios

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$$\text{EPS} = \frac{\text{Current Period Earnings available to Shareholders *}}{\text{no of Shares}}$$

* normally current period profit after tax

$$\text{P/E Ratio} = \frac{\text{Market Value per Share}}{\text{EPS}}$$

$$EPS_{2017} = \frac{230 \text{ m\$}}{(600 \text{ m\$} / 1 \$)} = 0.38 \$$$

$$\text{P/E Ratio}_{2017} = \frac{4.25 \$}{0.38 \$} = 11.1 \text{ times}$$

NTE 1010

Introduction to Finance for Engineers

week 4

Management Accounting

Cost – Volume – Profit

Break-Even

Management Accounting – Cost, Volume, Profit

Costs

Costs represent the resources that have to be sacrificed to achieve a business objective;
in other words, it is spending money for business.
The objective may be to make a particular product, to provide a particular service and so on.

(If the benefit of spending money is realized immediately or up to within a year,
then it is a cost.
On the other hand,
if the benefit is realized over a period longer than one year,
then it is an investment.
The cost of investments are spread out over their useful lives
by a method called depreciation.)

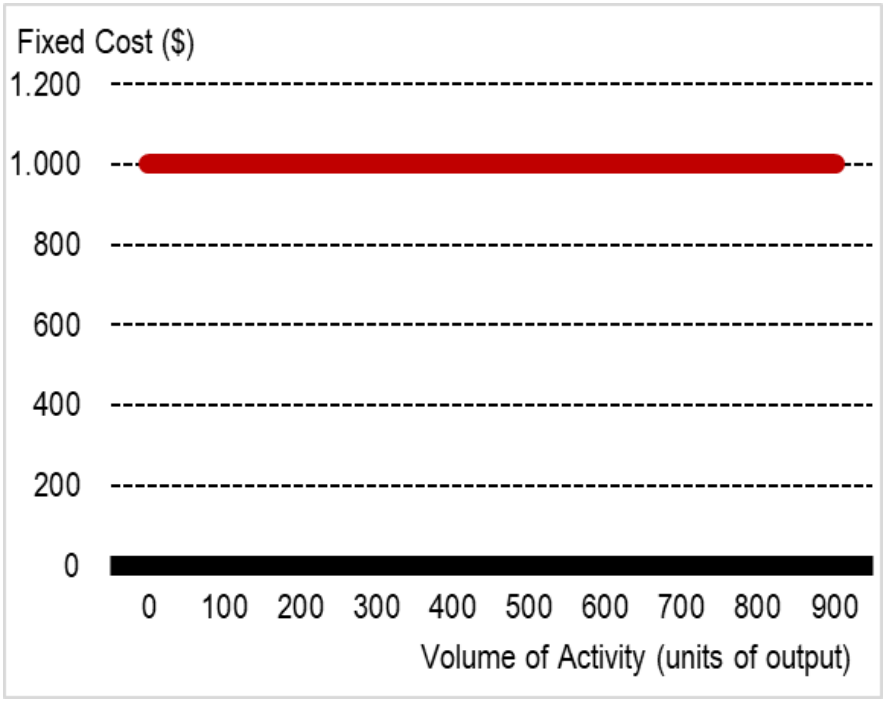
Costs may be classified in many ways;
and, one important way is according to how they behave in relation to
changes in the volume of activity.

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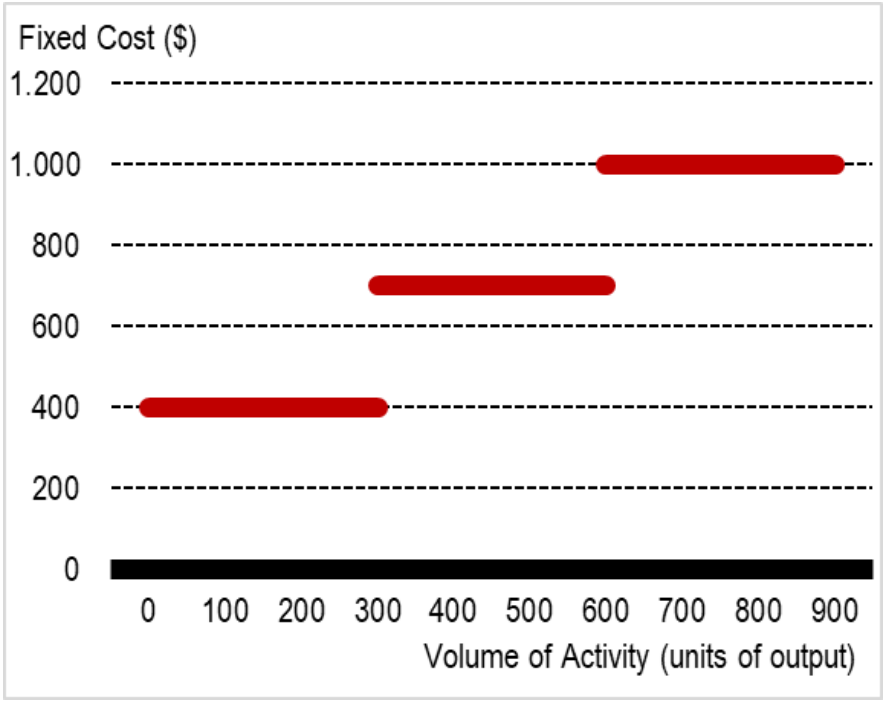
Management Accounting

Costs

Fixed Costs are
those that are independent of
the level of activity
(for example, rent).



Nevertheless,
as the volume of activity increases,
at a particular point,
the volume of activity cannot increase further
without additional fixed cost
such as renting additional space.
Fixed costs that behave like this are often
referred to as Stepped Fixed Costs.

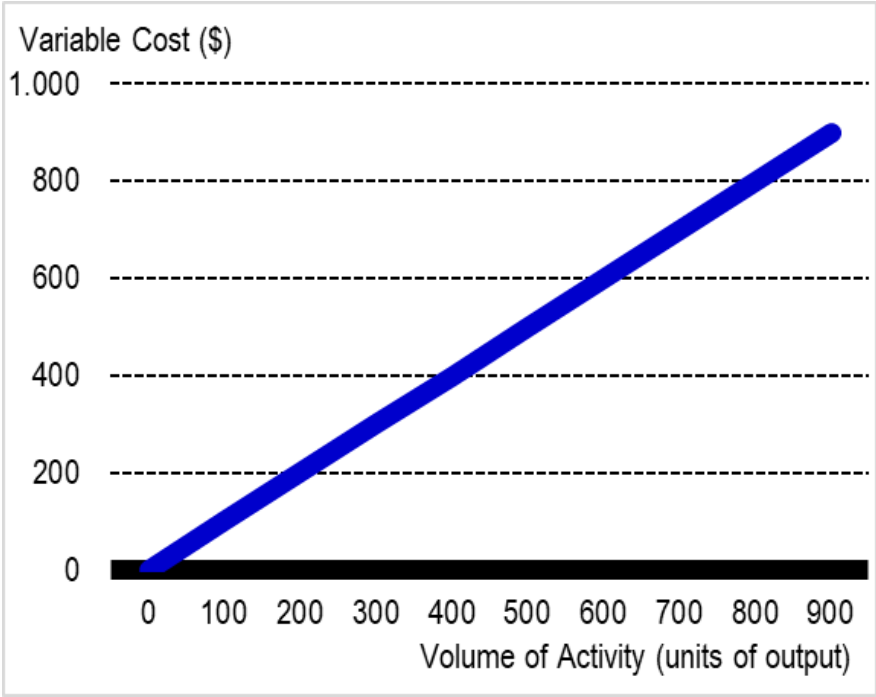


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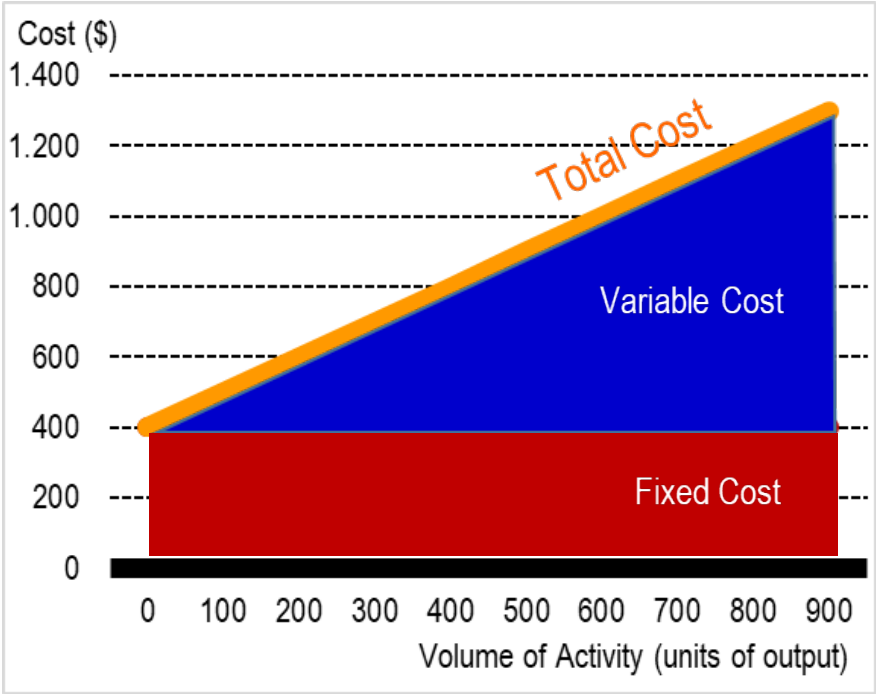
Management Accounting

Costs

Variable Costs are
those that vary with
the level of activity
(for example, raw materials).



Total Costs = Fixed Costs + Variable Costs



Semi-fixed (semi-variable) costs are
a mixture of the two (for example, electricity).

Management Accounting – Cost, Volume, Profit

Break-Even Analysis

The break-even point (BEP) is the level of activity (in units of output or sales revenue) at which total costs (fixed +variable) = total sales revenue.

At BEP,

Total Sales Revenue = Total Costs.

BEP Volume in Units * Price = Fixed Costs +(BEP Volume in Units * Unit Variable Costs)

BEP Volume in Units = Fixed Costs / (Price - Unit Variable Costs)

BEP Volume in Units = Fixed Costs / Contribution per Unit

(Price - Unit Variable Costs) is called "contribution", because it contributes to meeting the fixed costs and, if there is any excess, it also contributes to profit.

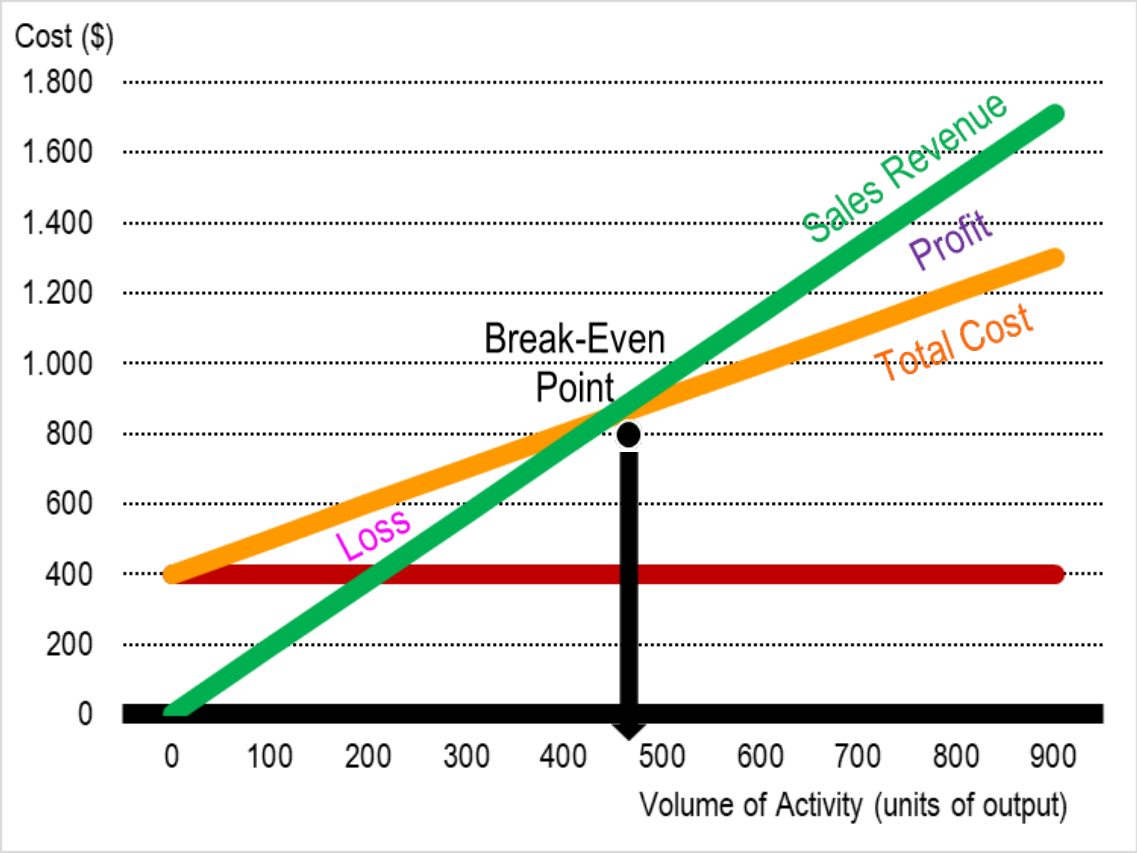
Knowledge of BEP for a particular activity can be used as a basis for risk assessment. Planning to operate only just above the Break-Even Volume of Activity may indicate that it is a risky venture, since only a small fall from the planned / expected volume of activity could lead to a loss.

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Management Accounting – Cost, Volume, Profit

Break-Even Analysis

The break-even point (BEP) is
the level of activity (in units of output or sales revenue)
at which total costs (fixed +variable) = total sales revenue.



Management Accounting – Cost, Volume, Profit

Break-Even Analysis

Cottage Industries Ltd makes baskets.
The fixed costs of operating the workshop for a month total $500.
Each basket requires materials that cost $2.
Each basket takes one hour to make, and the business pays the basket makers $10 an hour.
The basket makers are all on contracts such that
if they do not work for any reason, they are not paid.
The baskets are sold to a wholesaler for $14 each.

What is the Break-Even Point (BEP) for basket making for the business?

BEP (in number of baskets) = Fixed Costs / (Unit Price - Variable Costs per unit)
BEP = $500 per month / ($14 -$2 -$10) per basket
BEP = 500 / 2 = 250 baskets per month

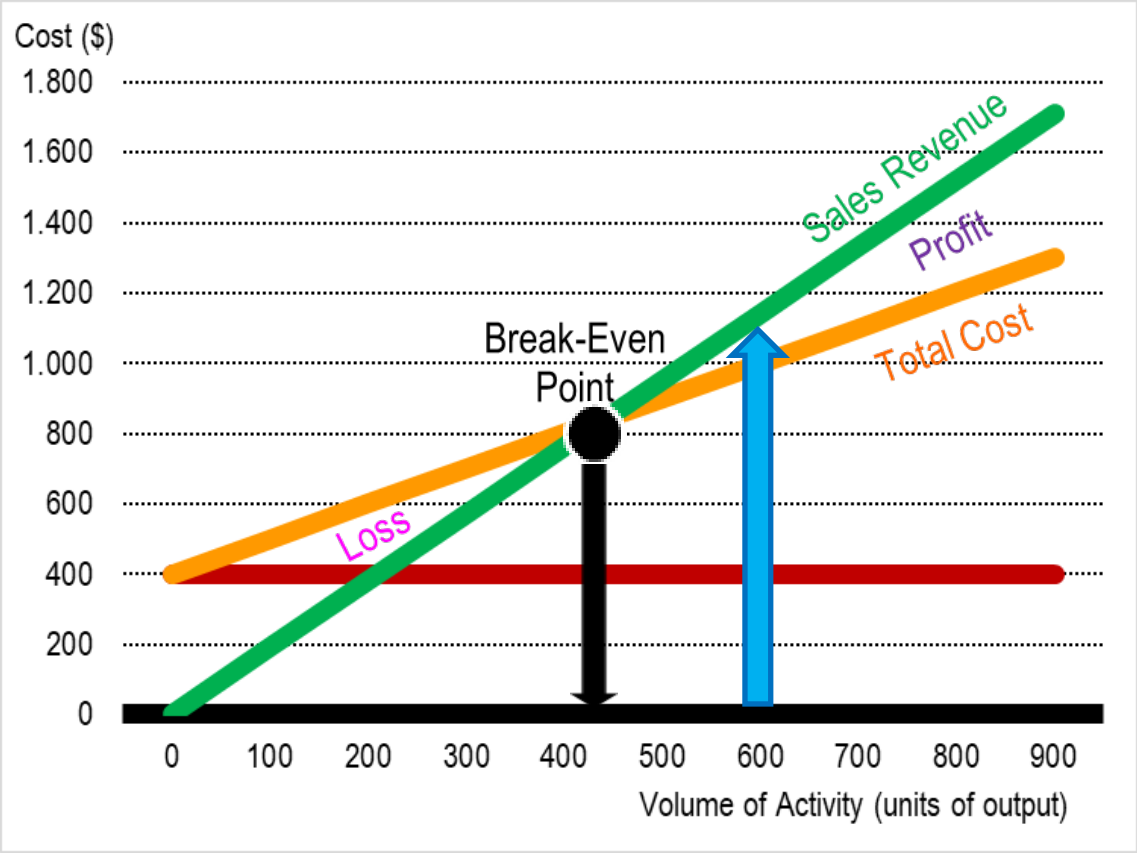
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Management Accounting – Cost, Volume, Profit

Break-Even Analysis: Margin of Safety

Margin of Safety = excess of planned volume of activity over BEP / planned volume of activity

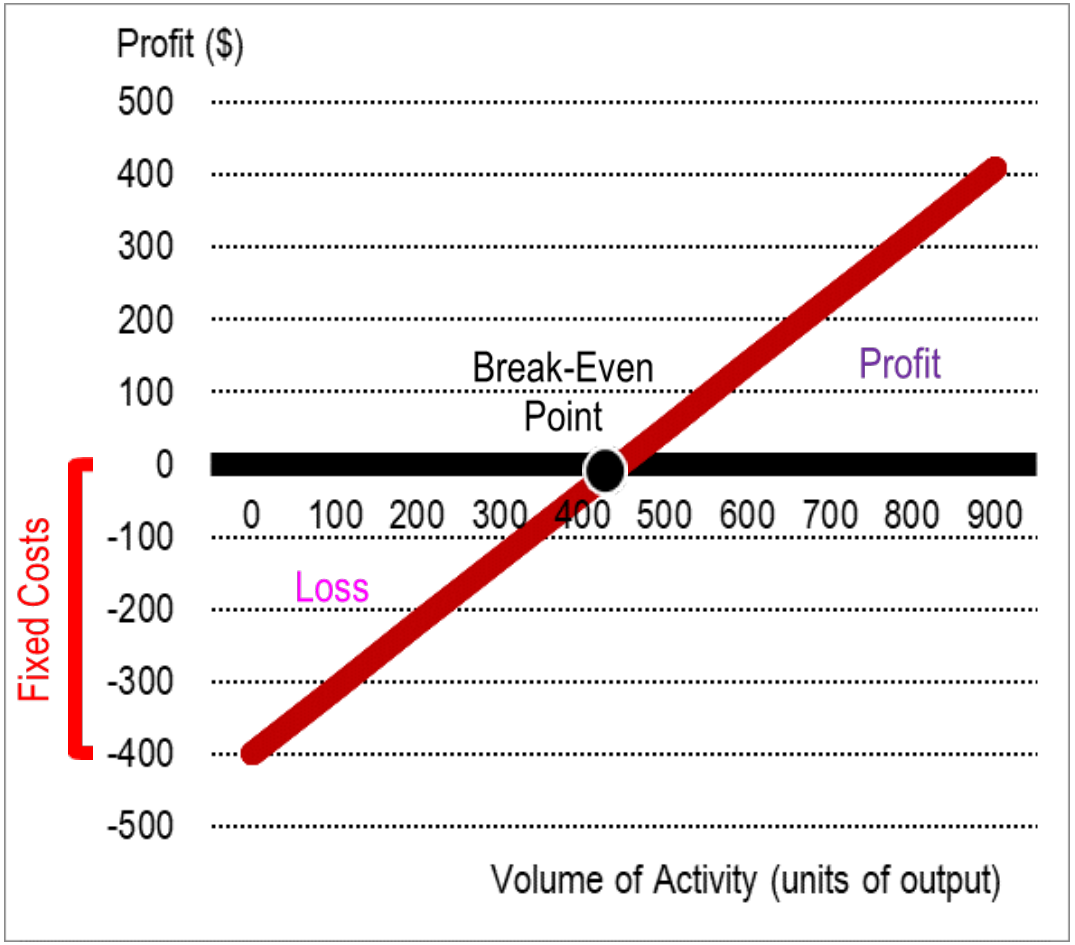
Margin of Safety = (600-420) / 600 = 30%



Management Accounting – Cost, Volume, Profit

Break-Even Analysis

Profit-volume (PV) chart is an alternative approach to the BE chart.



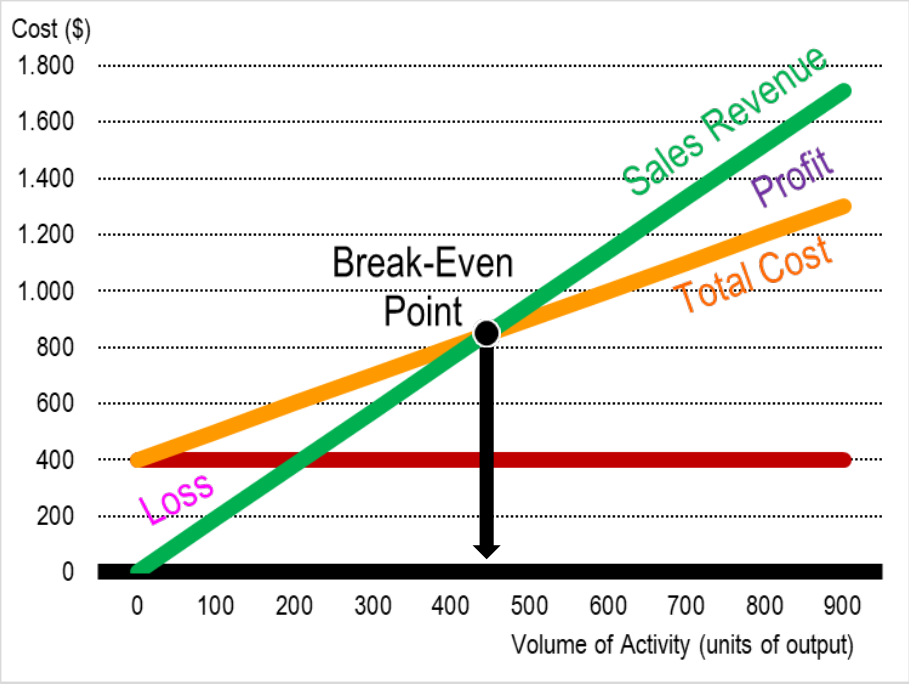
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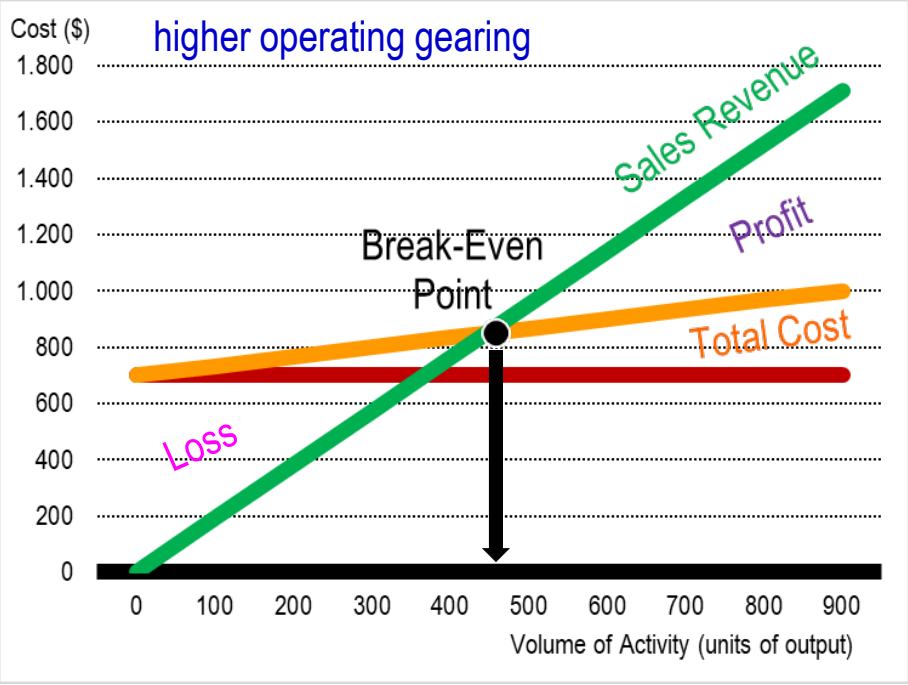
Management Accounting – Cost, Volume, Profit

Break-Even Analysis

Operating gearing = the extent to which the total costs of some activity are fixed rather than variable.

An activity with relatively high fixed costs compared with its variable costs is said to have higher operating gearing.



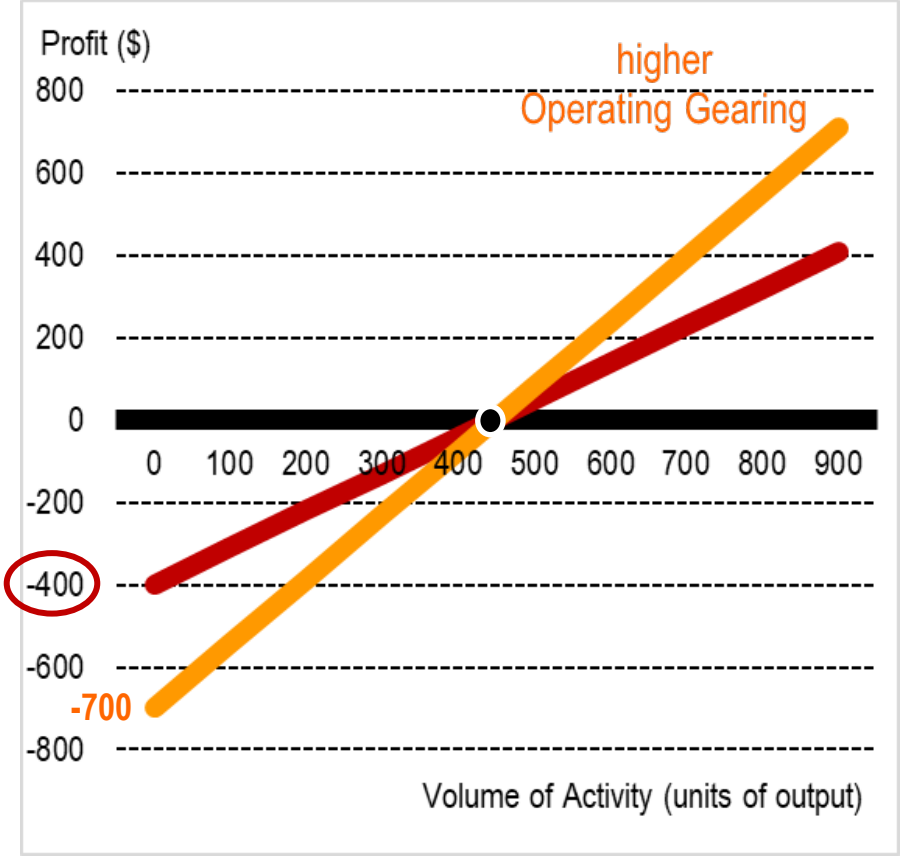


Management Accounting – Cost, Volume, Profit

Break-Even Analysis

Operating gearing = the extent to which the total costs of some activity are fixed rather than variable.

Increasing the level of operating gearing tends to make profits more sensitive to changes in the volume of activity.



Management Accounting – Cost, Volume, Profit

Weaknesses of Break-Even Analysis

- Non-linear relationships:

  The basic Break-Even analysis assumes that the relationships between sales revenues, variable costs and volume are linear ones; in real life, this unlikely to be so.

- Stepped fixed costs:

  Most fixed costs such as rent, supervisory salaries, and administration costs are not fixed over all volumes of activity in real life.

- Multi-product businesses:

  Fixed costs tend to relate more than one activity; while there are ways of dividing fixed costs between activities, these tend to be arbitrary.

Management Accounting – Cost, Volume, Profit

Marginal Analysis

(ignores fixed costs where these are not affected by the decision):

- Accepting/rejecting special contracts
  - consider only the effect on contributions.

  Since the fixed costs will be incurred in any case, they are not relevant. We need to see whether the price offered by the special contract will yield a contribution, i.e. Price - Unit Variable Cost.
- Using scarce resources
  - the limiting factor is most effectively used by maximising its contribution per unit.

  We tend to think in terms of the size of the market being the brake on output.

  This is to say that the ability of a business to sell will limit production, rather than the ability to produce will limit sales. In some cases, however, it is a limit on what can be produced that limits sales.

  Limited production might stem from a shortage of any factor of production
    - labour, raw materials, space, machinery and so on.
- Make-or-buy decisions (outsourcing / subcontracting)
  - take the action that leads to the highest total contributions.

  Compare the variable cost of subcontracting to variable cost of internal manufacture.
- Closing/continuing an activity
  - should be assessed by net effect on total contributions.

  For a department that is making loss, if it is making positive contribution before allocated fixed costs, it should not be closed.

  Because the allocated fixed costs would continue whether the department were closed or not.

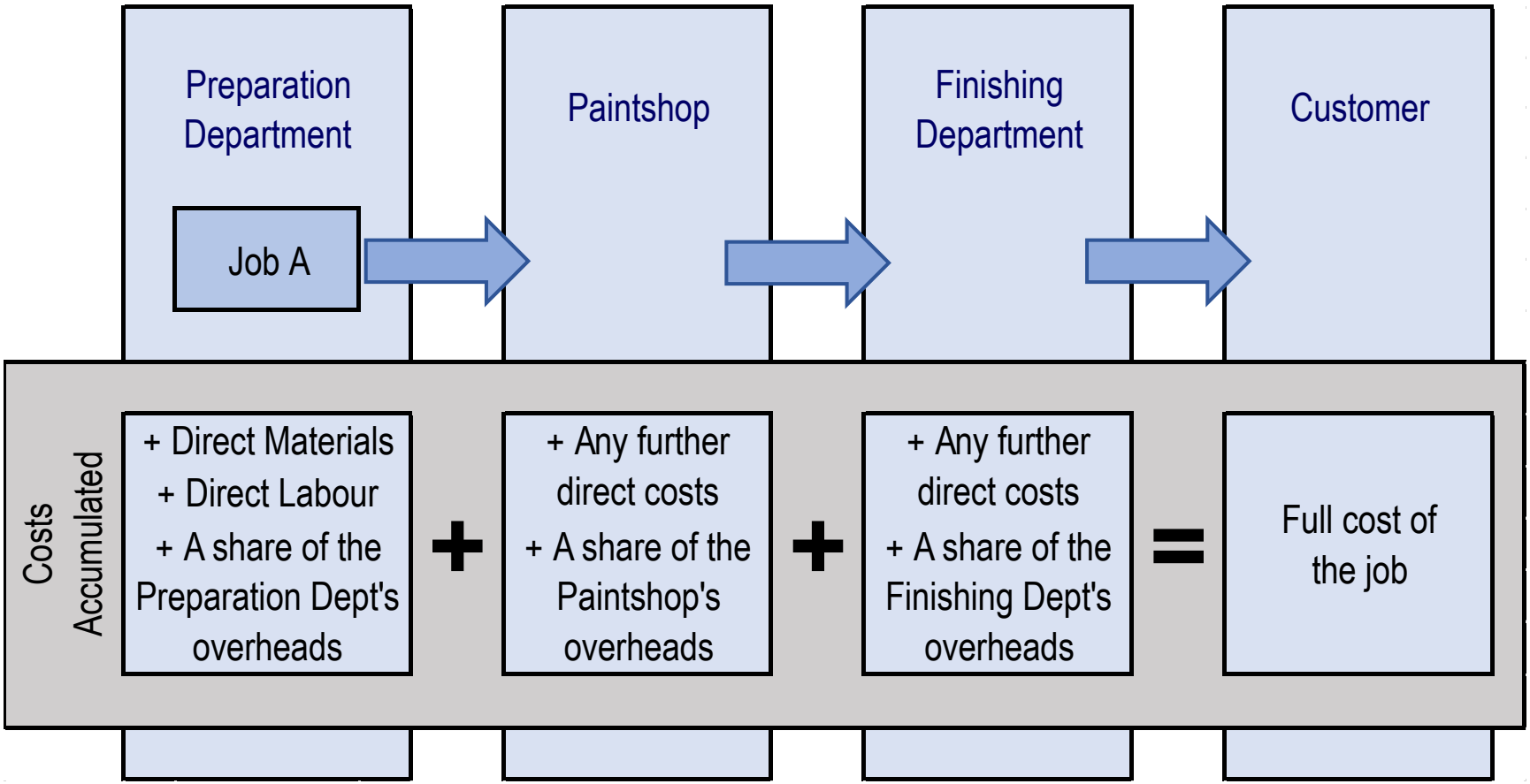
Full Costing

Management Accounting – Full Costing

Full cost = the total amount of resources sacrificed to achieve a particular objective

To derive the full cost figure,
we must accumulate the costs incurred and then assign them to the particular product or service.

A cost unit (Job A) passing through Autosparkle Ltd's process



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Management Accounting – Full Costing

Single Product Operations

Full cost = the total amount of resources sacrificed to achieve a particular objective

To derive the full cost figure,
we must accumulate the costs incurred and then assign them to the particular product or service.

Single-product operations:
Where all the units of output are identical,
the full cost can be calculated as follows:

Cost per unit = Total cost of output / Number of units produced

NTE 1010 – Introduction to Finance for Engineers

Management Accounting – Full Costing

Single Product Operations

Fruitjuice Ltd has just one product, a sparkling orange drink that is marketed as "Orange Fizz".
During last month, the business produced 7,300 litres of the drink.
The costs incurred were as follows:

"Orange Fizz" Costs	$
Ingredients (oranges and so on)	390
Fuel	85
Rent of premises	350
Depreciation of equipment	75
Labour	880

What is the full cost per litre of producing "Orange Fizz"?

This figure is found by simply adding together all of the costs incurred and then dividing by the number of litres produced.

Total Cost	1.780	[a]
Litres produced	7.300	[b]
Full Cost per litre	0,24	[a/b]

Management Accounting – Full Costing

Multi-Product Operations

Where units of output are not identical,
it is necessary to divide the costs into two categories: direct costs and indirect costs.

Direct costs = costs that can be identified with specific cost units
(for example, labour of a garage mechanic in relation to a particular car repair).

Indirect costs (overheads) = costs that cannot be directly measured in respect of a particular job
(for example, rent of a garage)

Full cost = direct cost +indirect cost.

Direct/indirect is not linked to variable/fixed.

Indirect costs are difficult to relate to individul cost units - arbitrary bases are used
and there is no single correct method.

Traditionally, indirect costs are seen as the costs of providing a "service" to cost units.

Direct labour hour basis of applying indirect costs to cost units
is the most popular in practice.

Management Accounting – Full Costing

Overhead Costs

Dealing with overheads on a departmental basis:

Indirect costs can be segmented
usually on a departmental basis - each department has its own overhead recovery rate.

Each department is a separate cost centre
that is an area, activity or function for which costs are separately collected.

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Management Accounting – Full Costing

Batch Costing / Activity-Based Costing

Batch Costing: A variation of job costing
where each job consists of a number of identical (or near identical) cost units:
Cost per unit = Cost of the batch (direct +indirect) / Number of units in the batch
If the full cost is charged as the sales price and things go according to plan,
the business will break even.

Activity-based costing (ABC) is an approach to dealing with overheads (in full costing)
that treats all costs as being caused or "driven" by activities.
While in traditional approach, overhead costs are allocated on the basis of a parameter,
in ABC approach, overhead costs are charged on the basis of activities.

Advocates argue that it is more relevant to the modern commercial environment
than is the traditional approach.
Identification of the cost drivers can lead to more relevant indirect cost treatment in full costing.
Identification of the cost drivers can lead to better control of overheads.

Critics argue that ABC is time-consuming and expensive to apply,
not justified by the possible improvement in the quality of information.

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Management Accounting – Full Costing

Activity-Based Costing

Modern Producers Ltd runs, like virtually all manufacturers, finished goods stores.
The business has two product lines, A and B.
Product A tends to be made in small batches and so low levels of finished goods inventories are held.
Product B is supllied in relatively large quantities instantly. As a consequence,
much of the finished goods stores is filled with Product B ready to be dispatched as an order is received.

Traditionally, the cost of of operationg the stores (an overhead cost) are allocated to products A and B
according to the number of direct labour hours worked on each one.
And, that did not seem fair to the new accountant Modern Producers Ltd.

He estimated that the costs of running the finished goods stores for the next year will be 90,000$.

It is also estimated that each unit of Product A will spend an average of 1 week in the stores before being sold.
With Product B, the equivalent period is 4 weeks.
Both products are of roughly similar size and have very similar storage needs.

It is estimated that next year 50,000 units of Product A and 25,000 units of Product B will pass through the stores.

The new accountant wants to allocate the overhead cost of running the finished goods stores to products A and B
according to Activity-Based Costing
with the quantity of products and the time spent in the stores as the cost drivers.

Activity-Based Costing

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The total number of "product weeks" in the stores will be:

Product		estimated annual units for next year		average weeks spent in the stores before being sold		annual "product weeks"
A	:	50.000	x	1	=	50.000
B	:	25.000	x	4	=	100.000
						150.000

The finished goods store cost per "product week" is given by 90,000$ / 150,000 product weeks = 0.60$
Therefore each unit of product A and product B will be charged as follows:

Product		average weeks spent in the stores before being sold		cost ($) per "product week"		finished goods stores cost per unit ($)
A	:	1	x	0,60	=	0,60
B	:	4	x	0,60	=	2,40

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Management Accounting – Full Costing

Pricing (full cost) on a cost-plus basis.

In some industries and circumstances, full costs are used as the basis of pricing.
Here, the full cost is deduced and a percentage is added for profit.
This is known as cost-plus pricing.
Vehicle repair shops typically operate this way.
Lawyers and accountants doing work for clients often use this approach as well.

Income measurement.

Where a service is partially rendered in one accounting period
but the revenue is realised in the next,
Full-Costing information may be useful in assessing profitability.

Full cost information is seen by some as not very useful
because it can be backward looking.

Budgeting

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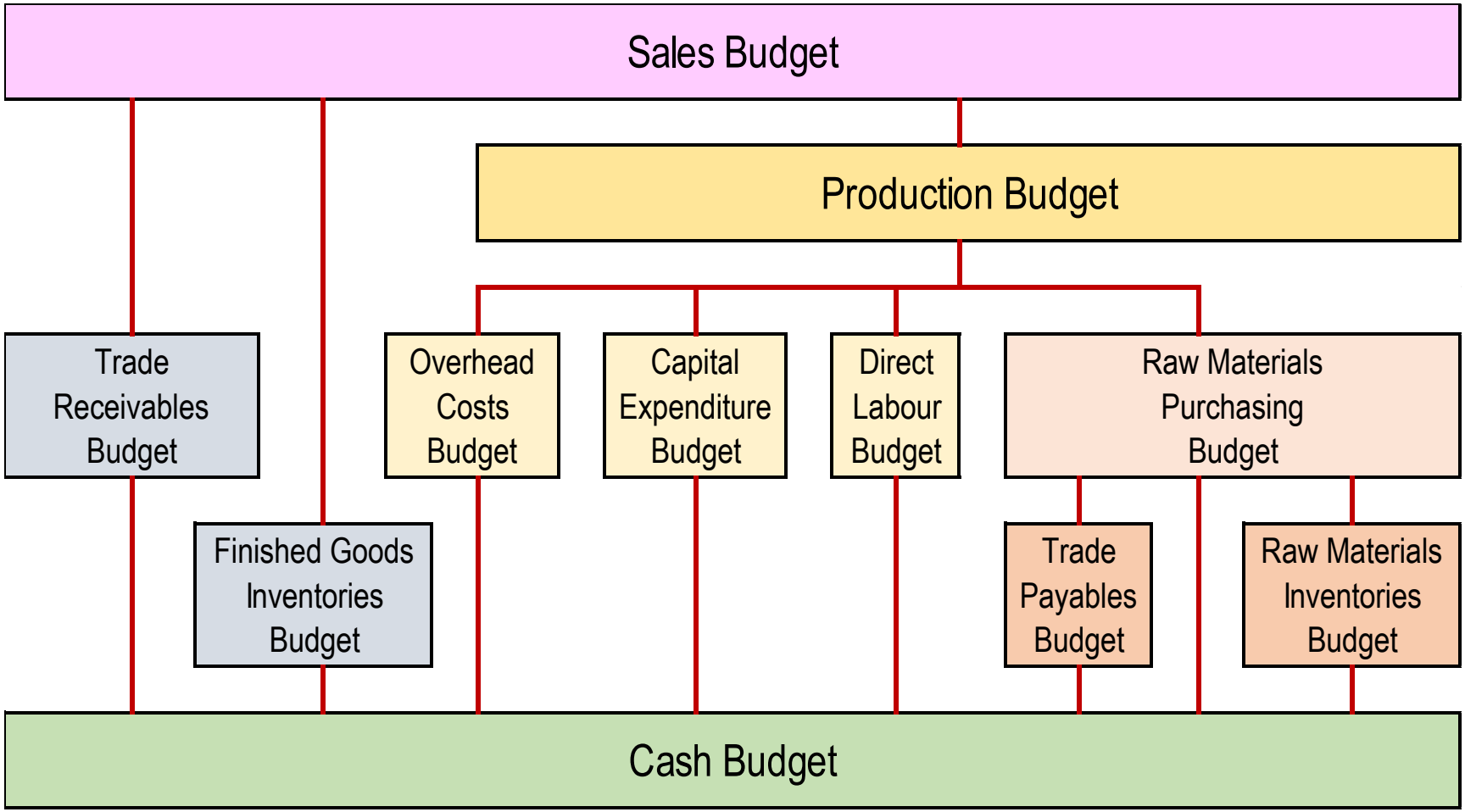
Management Accounting – Budgeting

Budget = a short-term business plan, mainly expressed in financial terms.
Budgets are the short-term means of working towards the business's objectives.
Usually for 12 months with subperiods of a month.
Uses are to:

- promote forward thinking
- help coordinate the various aspects
- motivate performance
- provide the basis of a system of control
- provide a system for authorisation

Preparing budgets:
There is no standard style - practicality and usefulness are the key issues.
They are usually prepared in columnar form,
with a column for each month (or similarly short period).
Each budget must link (coordinate) with others.

Management Accounting – Budgeting



interrelationship of budgets for a manufacturing business

starting point is usually the sales budget which defines the overall level of activity

Vierra Popova Ltd is a wholesale business.
The budgeted income statements for each of the net six months are as follows:
The business allows all of its customers one month's credit. Sales revenue during December totalled 60,000$.
The business plans to maintain inventories at their existing level until some time in March, when they are to be reduced by 5,000$. Inventories will remain at this lower level indefinitely.
Inventories purchases are made on one month's credit. December purchases totalled 30,000$.
Salaries, wages and "other overhead costs" are paid in the month concerned.
Electricity is paid quarterly in arrears in March and June.
The business plans to buy and pay for a new delivery van in March. This will cost a total of 15,000$, but an existing van will be traded in for 4,000$ as part of the deal.
The business expects to have 12,000$ in cash at the beginning of January.
Please prepare the cash budget for the six months ending in June.

Income Statement Budget ($ 000)	Jan	Feb	Mar	Apr	May	Jun
Sales revenue	52	55	55	60	55	53
Cost of goods sold	-30	-31	-31	-35	-31	-32
Salaries and wages	-10	-10	-10	-10	-10	-10
Electricity	-5	-5	-4	-3	-3	-3
Depreciation	-3	-3	-3	-3	-3	-3
Other overhead costs	-2	-2	-2	-2	-2	-2
Total expenses	-50	-51	-50	-53	-49	-50
Net profit	2	4	5	7	6	3

Cash Budget ($ 000)	Jan	Feb	Mar	Apr	May	Jun
Receivables [1]	60	52	55	55	60	55
Payables [2]	-30	-30	-31	-26	-35	-31
Salaries and wages	-10	-10	-10	-10	-10	-10
Electricity			-14			-9
Other overhead costs	-2	-2	-2	-2	-2	-2
Van purchase			-11			
Total payments	-42	-42	-68	-38	-47	-52
Cash surplus or (deficit)	18	10	-13	17	13	3
Balance at the beginning of the month [3]	12	30	40	27	44	57
Balance at the end of the month	30	40	27	44	57	60

- [1] The cash receipts from trade receivables lag a month behind sales, because customers are given a month in which to pay for their purchases. So, December sales will be paid for in January and so on.
- [2] In most months, purchases of inventories will equal the cost of goods sold. This is because the business maintains a constant level of inventories. For inventories to remain constant at the end of each month, the business must replace eactly the amount that has been used.
- During March, however, the business plans to reduce its inventories by 5,000$. this means that inventories purchases will be lower than inventories usage that month. The payments for inventories purchases lag a month behind purchases because the business expects to be allowed a month to pay for what it buys.
- [3] Each month's cash balance is the previous month's figure plus the cash surplus (or minus the cash deficit) for the current month. Th ebalance at the start of January is 12,000$ according to the information provided earlier.
- [4] Depreciation does not give rise to a cash payment.

Management Accounting – Budgeting

Trade Receivables Budget:
Balance at the beginning of the month + Sales revenue - Cash receipts
= Balance at the end of the month

Trade Payables Budget:
Balance at the beginning of the month + Purchases - Cash payments
= Balance at the end of the month

Inventories Budget:
Balance at the beginning of the month + Purchases - Inventories used
= Balance at the end of the month

Management Accounting – Budgeting

Controlling through Budgeting

Flexing the budget to match actual volume of output.
Variance = increase (favourable) or decrease (adverse) in profit, relative to the budgeted profit, as a result of some aspect of the business's activities taken alone.
Budgeted profit plus all favourable variances less all adverse variances equals actual profit.

Commonly calculated variances:

- Sales volume variance = difference between budgeted and actual volume (in units) multiplied by the contribution per unit
- Sales price variance = difference between actual sales revenue and actual volume at the budgeted sales price per unit
- direct materials usage variance = difference between actual usage and budgeted usage, for the actual volume of output, multiplied by the budgeted material cost per unit of material
- direct material price variance = difference between the actual material cost and the actual usage multiplied by the budgeted cost per unit of material
- direct labour efficiency variance = difference between actual labour time and budgeted time, for the actual volume of output, multiplied by the budgeted labour rate
- direct labour rate variance = difference between the actual labour cost and the actual labour time multiplied by the budgeted labour rate
- fixed overhead spending variance = difference between the actual and budgeted spending on fixed overheads

Management Accounting – Budgeting

Controlling through Budgeting

- Standards = budgeted physical quantities and financial values for one unit of inputs and outputs.
  Standards are useful in providing data for decision making.
- Significant and/or persistent variances need to be investigated to establish their cause.
- Good budgetary control requires establishing systems and routines to ensure such things as clear distinction between individual managers' areas of responsibility, prompt, frequent and relevant variance reporting and senior management commitment.
- Not all activities can usefully be controlled through traditional variance analysis.
  Expenses such training and advertising are discretionary and not linked to the level of output in a direct way.
  Standards can quickly become out of date as a result of technological change.
  Sometimes factors that are outside the control of the manager concerned can affect the calculation of the variance for which the manager is held accountable; this should be taken into account.
  In practice, creating clear lines of demarcation between the areas of responsibility of various managers may be difficult.
- There is a behavioural aspect of control, which should be taken into accountnby senior managers.
  Examples, performance improvement, motivation, participation improves performance and motivation.

NTE 1010
Introduction to Finance for Engineers

Spring 2020 - Week 4

Homework 1

NTE 1010 – Introduction to Finance for Engineers

Homework-1

Analysis of Financial Statements

1. Find 2017, 2018 and 2019 year-end Financial Statements of a company
   shares of which are being traded in İstanbul Stock Exchange.
   Attach copies of those statements to your homework and state the source of the information.
   Afterwards, choose 3 ratio classifications out of
   [1] Profitability, [2] Efficiency, [3] Liquidity, [4] Financial leverage, [5] Investment.
   And then, pick a ratio in each one; state the formulas of each ratio that you picked.
2. Calculate 2018 and 2019 values for these 3 ratios that you picked.
3. Comment for each ratio whether the company performed better or worse in 2019 compared to 2018
   and explain why so.

- You may do your homework in groups (of max 3 students) if you wish.
- The homework should either be a computer printout or a legible handwriting.
- At the top, it must have the following heading:
  "NTE 1010 Introduction to Finance for Engineers, Spring 2020, Homework-1"
  followed by your name(s), family name(s), and student ID no(s).

Homework-1 is to be handed-in two weeks later, on Wednesday, March 25th in the class.
Each homework is worth 6% of your course grade.
Do not copy the work of somebody else.

NTE 1010

Introduction to Finance for Engineers

Spring 2020

week 5

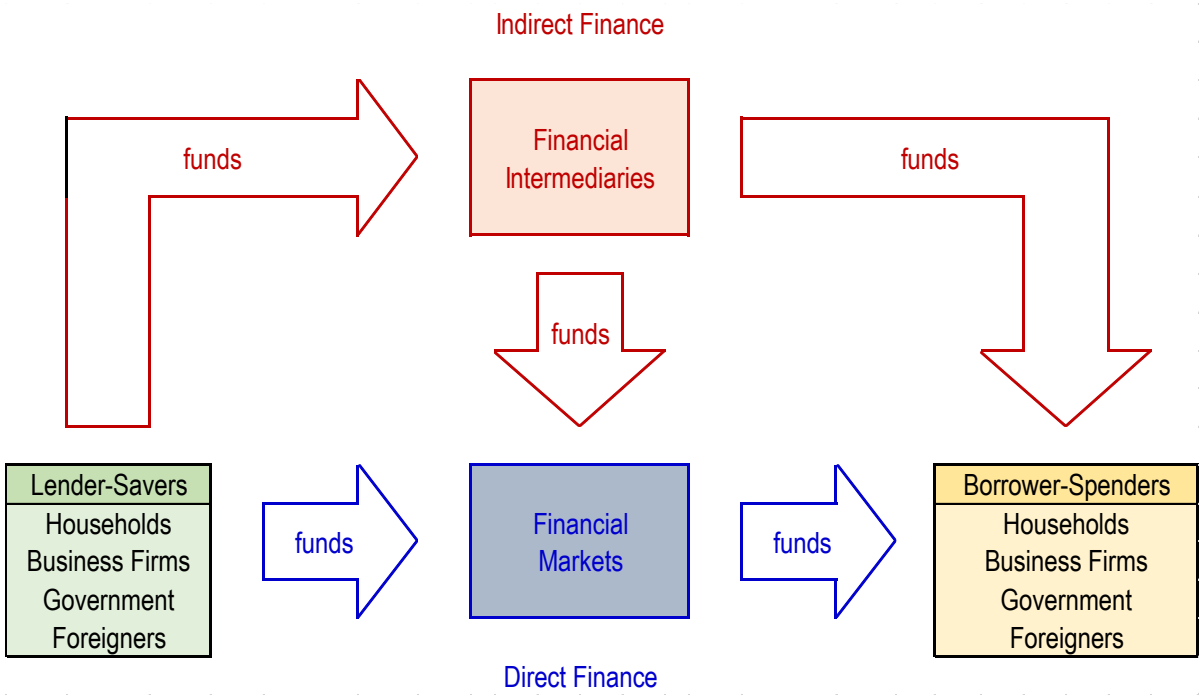
Wednesday, March 18, 2020

Financial System & Money

Financial System

Flow of Funds through the Financial System

In Indirect Finance, borrowers borrow funds directly from Lenders in Financial Markets
by selling them securities (financial instruments)
which are claims on the borrower's future income or assets.



Financial markets perform the essential economic function of channeling funds
from households, firms, and governments
that have saved surplus funds by spending less than their income
to those that have a shortage of funds because they wish to spend more than their income.

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Financial Markets (Direct Finance)

Primary and Secondary Markets

Primary Market

is a financial market in which new issues of a Financial Security, such as a bond or stock, are sold to initial buyers by the corporation or the government agency borrowing the funds.

Secondary Market

is a financial market in which securities that have been previously issued can be resold.

An important financial institution that assists in the initial sale of securities in the primary market is the investment bank.
It does this by underwriting securities:
It guarantees a price for a corporation's securities and then sells them to the public.

Securities brokers and dealers are crucial to a well-functioning secondary market.
Brokers are agents of investors who match buyers with sellers of securities;
dealers link buyers and sellers by buying and selling securities at stated prices.

Secondary markets serve two important functions:
First, they make it easier and quicker to sell financial instruments;
that is, they make the financial instruments more liquid.
Second, they determine the price of the security that the issuing firm sells in the primary market.

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Financial Markets (Direct Finance)

Types of Secondary Markets

Secondary markets can be organized in two ways:

One method is to organize Exchanges, where buyers and sellers of securities
(or their agents or brokers) meet in one central location to conduct trades.

The New York Stock Exchange for stocks
and the Chicago Board of Trade for commodities (wheat, corn, silver, and other raw materials)
are examples of organized exchanges.

As well as Borsa Istanbul and
Izmir Ticaret Borsası (commodity trade).

The other is to have an Over-the-Counter (OTC) market,
in which dealers at different locations who have an inventory of securities stand ready
to buy and sell securities "over the counter"
to anyone who comes to them and is willing to accept their prices.

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Financial Markets (Direct Finance)

Money Markets and Capital Markets

Another way of distinguishing between markets is
on the basis of the maturity of the securities traded in each market.

A. Money Market

is a financial market in which only short-term debt instruments
(generally those with original maturity of less than one year) are traded;

B. Capital Market

is the market in which longer-term debt
(generally those with original maturity of one year or greater)
and equity instruments are traded.

Financial Markets (Direct Finance)

Further Categories

A. Money Markets

B. Capital Markets

C. Futures and Derivatives Markets

Futures are agreements to purchase an asset at a date in the future at a price that is agreed now.

A swap is an agreement to exchange the rights to one asset for the rights to another.

For example, a fixed interest rate loan may be swaped with a variable rate loan.

An options contract provides the owner of the contract
with the right to buy or sell within a period at a known price.

(Spot markets are those in which payment and delivery occur within a few days.
Unless otherwise indicated, financial markets are spot markets.)

D. Foreign Exchange (FX) Markets

E. Commodities Markets

(for petroleum, for valuable metals such as gold and silver, for industrial metals such as copper and aluminum, for basic food such as wheat and corn)

Most of the time physical settlements do not take place.

Investors hedge (reduce risk exposure) or speculate (willingly take risk).

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Financial Markets (Direct Finance)

Equity - Debt

- Equity Markets
- Debt Markets

A firm or an individual can obtain funds in a financial market
by issuing a debt instrument, such as a bond or a mortgage,
which is a contractual agreement by the borrower to pay the holder of the instrument
fixed dollar amounts at regular intervals (interest and principal payments)
until a specified date (the maturity date),
when a final payment is made.

The maturity of a debt instrument is the number of years (term)
until the instrument's expiration date.

A debt instrument is short-term if its maturity is less than a year
and long-term if its maturity is ten years or longer.
Debt instruments with a maturity between one and ten years are said to be intermediate-term.

Financial Securities

Debt Instruments

Secured debt is secured over one or more of the assets of the company,
which means that in the case of liquidation,
the proceeds from the sale of secured assets
would be used to first meet the claims of the debt-holders.

The secured assets are called collateral.

In case of liquidation,
there is a ranking of the claims on the assets of the company;
senior debt has first claim
and subordinated debt has a subsequent claim.

Unsecured debt does not lay claim to any of the assets.

Financial Securities

Debt Instruments

Fixed interest debt is that in which the interest rate is fixed for the period of the loan.
Bonds are often fixed interest instruments.

Variable interest or floating interest debt is that in which
the interest rate is not fixed but varies according to the market rate.
Mortgages are a common form of floating interest debt.

Financial Securities

Types of Loans

- Regular loan is one in which the interest is paid at regular periods during the term of the loan and the principal is paid at the end.

- Discount loan is structured so that the entire interest amount is paid in advance.
  The borrower receives the loan amount less the interest payment.
  At the end of the loan period, the borrower repays the principal amount.

- Instalment loan is one which is paid back in equal instalments.
  An instalment is composed of interest for the current period and partial principal repayment.

- A line of credit or revolving credit is a flexible form of short-term financing
  that allows the borrower to access funds up to a specified limit as needed.
  The company pays interest on the amount borrowed;
  however, it may be required to pay a (small) charge on the unused balance as well.

Financial Securities

Additional Debt Instruments

- A bond is a long-term loan agreement which sets terms for the principal, the coupon rate, the coupon frequency and the maturity.
  In case the coupon rate is variable, it is indexed to a specific rate such as prime rate, LIBOR, or inflation.
  The coupon payments are usually made semi-annually or annually.

- Leasing provides funding by allowing the company to use the leased assets without incurring the initial cost of purchase.
  - Operational lease: e.g. cars where leasing company covers the operational expenses except for gasoline, highway tolls, etc. Asset is returned to the leasing company once the leasing period is over.
  - Capital lease is used for investment equipment. The operating company covers the costs and generally buys the asset for a predetermined price once the leasing period is over.

- Factoring: Discounting of business' receivables by a factoring company.
  Discount rate is higher if the factoring company accepts the non-payment risk.

NTE 1010 – Introduction to Finance for Engineers

Financial Markets (Direct Finance)

Equity - Debt

- Debt Markets
- **Equity Markets**

A firm or an individual can obtain funds in a financial market
by issuing equities, such as common stock,
which are claims to share in the net income (income after expenses and taxes)
and the assets of a business.

Equities often make periodic payments (dividends) to their holders
and considered long-term because they have no maturity date.

In addition, owning stock means that you own a portion of the firm
and thus have the right to vote on issues important to the firm and to elect its directors.

The main disadvantage of owning equity is that an equity holder is a residual claimant; that is,
the corporation must pay all its debt holders before it pays its euity holders in case of liquidation.

Its advantage is that equity holders benefit directly from
any increases in the corporation's profitability or asset value
because equities confer ownership rights on the equity holders.

Financial Securities

Equity Types – Ordinary Shares

There are two forms of equity in a business;

1. Ordinary shares (common stock),
2. Preferred stock.

1. Ordinary Shares:

The investors provide the share capital for partial ownership of the business.

The return to the holders of ordinary shares
is directly related to the performance of the business in two forms;

- The growth in the value of the company,
- The dividends paid by the company.

The holders of ordinary shares are exposed to the entire risk of the business.

Financial Securities

Equity Types – Ordinary Shares

The sale of new shares in an existing company can be in the form of a rights offer or a follow-on offer.

If the company sells new shares to existing shareholders who have pre-emptive right to subscribe to new shares, it is called a rights offer.
This allows the shareholders to maintain their proportional share of the ownership of the company.

Current shareholders may also waive their right to subscribe to new shares. In such a case, the company can raise equity capital by selling new shares to new shareholders for cash; this called follow-on offer.

Companies can retain more of the profits by paying a smaller dividend; this increases the equity capital of the company without selling additional shares.
The company may offer additional shares instead of a cash dividend; this is known as stock dividend.

Financial Securities

Equity Types – Preferred Stock

1. Ordinary shares (common stock)

2. Preferred stock

Preferred Stock has the characteristics of both equity and debt.

Dividend offered by Preferred Stock is fixed, a percent of the face value of the shares.
These payments are similar to the interest payments on debt.

However, if there are insufficient funds, no payments are made
and this does not throw the company into bancruptcy.
Usually, these unpaid dividends accumulate and holders of Preferred Stock are paid in arrears.

Preference shares usually have no maturity date.

Unlike debt, the dividend payment to Preferred Stock is not tax-deductible.

In case of liquidation of the company,
the holders of Preferred Stock have a claim to to the assets
prior to that of the ordinary shareholders.

Financial Markets (Direct Finance)

Borsa İstanbul

Established as İstanbul Menkul Kıymetler Borsası (İMKB)
at the end of 1985.

Converted to Borsa İstanbul A.Ş. at the end of 2012.

Stock Markets in the World

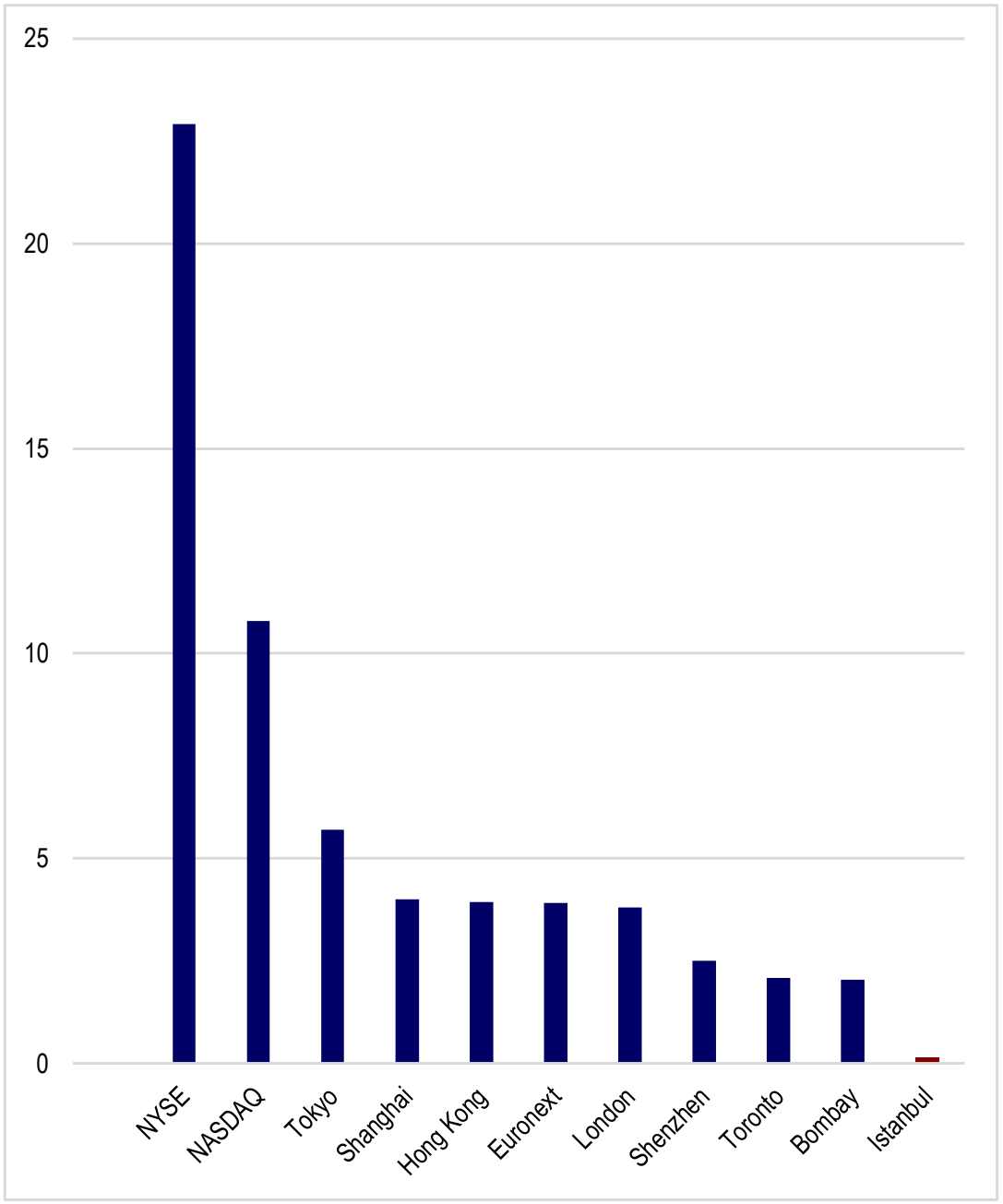
1- New York Stock Exchange, United States: Founded in 1792, the New York Stock Exchange has been the world's largest stock exchange since the end of World War I, when it overtook the London Stock Exchange. It has a market capitalization of $22.9 trillion and about 2,400 listed companies.
2- NASDAQ, United States: The NASDAQ Stock Market was founded in 1971 in New York City. NASDAQ had a market capitalization of $10.8 trillion with an average monthly trading volume of $1.26 trillion.
3- Tokyo Stock Exchange, Japan: Founded in 1878, the Tokyo Stock Exchange has close to 2,300 listed companies with a combined market capitalization of $5.67 trillion.
4- Shanghai Stock Exchange, China: It has a market capitalization of $4.02 trillion. It has more than 1,000 listed companies. Though its origins date back to 1866, it was suspended following the Chinese Revolution in 1949. The Shanghai Exchange in its modern avatar was founded in 1990.
5- Hong Kong Stock Exchange, Hong Kong: The Hong Kong Stock Exchange was founded in 1891. It has close to 2,000 listed companies, about half of which are from mainland China. It has a market capitalization of $3.93 trillion.
6- Euronext, Eurozone: Headquartered in Amsterdam, the Netherlands, Euronext is a pan-European stock exchange with a presence in France, Belgium, Ireland, and Portugal. It has approximately 1,300 listed companies with a combined market capitalization of $3.92 trillion.
7- London Stock Exchange, United Kingdom: The London Stock Exchange was founded in 1698. It has more than 3,000 listed companies with a combined market capitalization of $3.76 trillion.
8- Shenzhen Stock Exchange, China: Formally established in 1990, the Shenzhen Stock Exchange has a market capitalization of $2.5 trillion as of November 2018.
9- Toronto Stock Exchange, Canada: Toronto Stock Exchange (TSX) has 2,207 listed companies with a combined market capitalization of $2.1 trillion.
10- Bombay Stock Exchange, India: Founded in 1875, the Bombay Stock Exchange was the first stock exchange in Asia. As of November 2018, it had a market capitalization of $2.05 trillion. BSE has 5,749 listed companies.

Istanbul Stock Exchange as a Market Capitalization of $163.58 Billion.

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Stock Markets in the World

	Market Capitalization (trn $)
NYSE	22.9
NASDAQ	10.8
Tokyo	5.7
Shanghai	4.0
Hong Kong	3.9
Euronext	3.9
London	3.8
Shenzhen	2.5
Toronto	2.1
Bombay	2.1
Istanbul	0.16



Borsa İstanbul

Debt Securities Market

Debt securities, securitized asset and income backed debt securities, lease certificates, liquidity bills issued by the Central Bank of Turkey are traded.

Trading is conducted electronically via the automated multiple price-continous auction system.

Settlement operations are realized by İstanbul Settlement and Custody Bank Inc. (Takasbank).

Sub-markets are:

- Outright Purchases and Sales Market of fixed income securities
- Repo – Reverse Repo Market of fixed income securities
- Interbank Repo – Reverse Repo Market of fixed income securities
- Repo Market for Specified Securities
- Equity Repo Market
- Negotiated Repo Deals Market
- Offering Market for Qualified Investors
- International Bonds Market of "Eurobonds" issued by the Turkish government

Borsa İstanbul

Equity Market

Equities, preemptive rights, exchange traded funds, warrants and certificates are traded.

Trading is carried out based on price and priority, with "continous auction", "continous auction and market maker" and "single price" trading methods with a fully automated electronic trading system.

Sub-markets are:

- BIST STARS: market value of free float > 100 million TL or included in BIST 100 index
- BIST MAIN: 100 million TL > market value of free float > 25 million TL
- BIST EMERGING CO's: 25 million TL > market value of free float
- WATCHLIST: Co's subject to monitoring due to financial problems
- COLLECTIVE & STRUCTURED PRODUCTS: Exchange traded funds, warrants, certificates.
- QUALIFIED INVESTORS: Shares sold directly to qualified investors; no public offering.
- PRE-MARKET TRADING: Publicly held co's (100+ shareholders) not listed in Borsa İstanbul.

Borsa İstanbul

Derivatives Market

Trading volume in 2017 was 868 billion TL, 12% of total BIST volume.

Sub-markets:

- FX
- Electric
- Commodity
- Index
- Percious Metals
- Equity
- Foreign Indices
- Metal
- ETF
- Interest Rates

Borsa İstanbul

Precious Metals & Diamond Market

Trading volume in 2017 was 68 billion TL, 1% of total BIST volume.

Sub-markets:

- Precious Metals – includes the spot trade of standard and non-standard gold, silver, platinum, and palldium.
- Precious Metal Lending – provides lending and certificate transactions.
- Diamond and Precious Stone

Borsa İstanbul

Other Services

In addition to Trading:

- Listing
- Settlement – Takasbank
- Custody – Central Registry Agency: securities depository
- Index and Data

Financial System

Indirect Finance / Financial Intermediaries

Indirect Finance involves
Financial Intermediaries that stand between the lender-savers and the borrower-spenders
and helps transfer funds from one to the other.
This is called financial intermediation.

Financial Intermediaries can substantially reduce transaction costs
because they have developed expertise in lowering them
and because their large size allows them to take advantage of Economies of Scale,
i.e. the reduction in transaction costs per dollar of transactions
as the size (scale) of transactions increase.

Financial Intermediaries may also provide liquidity services
such as banks providing depositors with checking accounts that enable them to pay their bills easily.

Financial Intermediaries help reduce the exposure of investors to risk.
They create and sell assets with risk characteristics that people are comfortable with,
and then use these funds to purchase other assets that may have far more risk.
This process is called risk sharing or asset transformation.

Financial System

Types of Financial Intermediaries

- Banks accept deposits from individuals and institutions and make loans.

- Life Insurance / Pension Companies insure people against financial hazards following a death and sell annuities (annual income payments upon retirement). They acquire funds from the premiums that people pay to keep their policies in force and use them to invest in financial markets.

- Mutual Funds acquire funds by selling their own shares to many individuals and use the proceeds to purchase diversified portfolios of stocks and bonds.

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Financial System

Regulation of the System

Financial systems are among the most heavily regulated sectors in the world.

Governments regulate financial systems,

- to increase the information available to investors, and
- to ensure the soundness of the system.

Bankacılık Düzenleme ve Denetleme Kurulu (BDDK)
regulates and supervises
banks.

Before Hazine Müsteşarlığı (Undersecretariat of the Treasury)
used to regulate and supervise
insurance companies;
now, a separate supervisory board similar to BDDK has been established.

Sermaye Piyasası Kurulu (SPK)
regulates and supervises
capital markets.

Financial System

Regulation of the System

Governments regulate financial systems via the following:

- Restrictions on Entry – Those who want to establish a financial intermediary must obtain a charter from the government.Only those with impeccable credentials and a large amount of initial funds will be given a charter.
- Disclosure - Bookkeeping of financial intermediaries must follow certain strict principles, their books are subject to periodic inspection, and they must make certain information available to public.
- Restrictions on Assets and Activities – Financial Intermediaries are restricted from holding certain risky assets and/or holding them in unreasonably high quantities.
- Deposit Insurance – The government insures people's deposits so that they do not suffer great financial loss if the financial intermediary that holds these deposits should fail.
  In Turkey, deposits of individuals are insured up to TL 100,000 separately for each bank.
- Limits on Competition – Opening a bank branch is subject to regulatory approval.
- Restrictions on Interest Rates – Regulator may restrict interest paid on deposits and other financial instruments which may be unrealistic in economic terms for the sake of overall good.
  Also, Central Bank may guide banks by determining the interest of the funds that it provides to banks.

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week 5

Financial System & Money

Money

Economy

The term "economy" originates from the Greek Word of "oikonomia"
which means "household management".
Nowadays, "economy" very simply covers topics of
"producing" and "sharing".

"First human" is estimated to have appeared on earth 200,000 years ago.
Agriculture and trading started 10,000 years ago in Mesopothamia;
until then humans were "gatherers" and "hunters".
With agriculture, they became "producers";
surplus production made them "traders".
"Trading" introduced "sharing".

So appeared the "economy".

Money

Trading of goods was cumbersome;
it was not easy to find some who had what you needed
and at the same time who needed what you had.

"Money" as the common denominator of trade made things easier.
First Money, in metal form, was used by the Lydians 2,700 years ago
who had lived in and around Teke Yarımadası of today.
Most precious metal money was in gold,
secondarily in silver.

As the production and trade increased in history,
supply of precious metals became insufficient in supporting world trade.
So, "paper money" appeared as a promise to give gold on demand.

About 45 years ago, "gold standard" in paper money was abandoned.

Money

Primary Functions

- Money in the form of currency, checks or electronic cards is a Medium of Exchange; it is used to pay for goods and services.

This promotes economic efficiency by minimizing the time spent in exchanging goods and services.

- Money provides a Unit of Account; it is used to measure value in the economy.

We measure the value of goods and services in terms of money, just as we measure weight in terms of grams or distance in meters.

- Money also functions as a Store of Value; it is a repository of purchasing power over time.

A store of value is used to save purchasing power from the time income is received until the time it is spent.

Money

Measurement

There are expanding levels of money measurement:

M0 = physical money
(paper, coin, deposits of banks at Central Bank which the Central Bank has to provide cash on demand)

M1 = M0 +current deposits

M2 = M1 +deposits with maturity of less than 30 days

M3 = M2 +time deposits

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week 6

Interest Rates

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Interest Rates

Present Value

Different debt instruments have
very different streams of cash payments to the holder (known as cash flows)
with very different timing.

We need to understand how we can compare
the value of one kind of debt instrument with another.

The concept of present value (or present discounted value) is based on
the commonsense notion that
a dollar paid to you one year from now is less valuable
than a dollar paid to you today.

This notion is true because
you can deposit a dollar today in a bank to earn interest
and have more than a dollar in one year.

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Interest Rates

Present Value

If the annual interest rate (i) is 10%,
then you would be just as happy having \$100 today as having \$110 a year from now.

You would also just as happy having \$100 today as having \$121 two years from now,
or \$133 three years from now,
or \$100 x (1+10%)^n n years from now.

So, \$100 x (1+10%)^n n years from now is worth \$100 today.

Today's value of so much \$ n years from now (i.e. Cash Flow, CF) is called Present Value (PV).
This yields to the following formula:

$$PV = \frac{CF}{(1+i)^n}$$

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Interest Rates

Yield to Maturity

Present Value (PV) of the Debt Instrument may be formulated as follows:

$$PV = \sum_{t=1}^{n} \frac{CF_t}{(1+i)^t}$$

Yield to Maturity is the interest rate (i) that equates the present value of cash flow payments received from a debt instrument with its value today.
Yield to Maturity is also known as Internal Rate of Return (IRR).

If we know CF_t and the PV of the debt instrument,
then we can calculate i, that is, its yield to maturity.

If we know CF_t and the market interest rate relevant for the debt instrument,
then we can calculate its PV.

The effective cost of a corporate bond is determined by calculating the Yield to Maturity (also known as Internal Rate of Return (IRR)) of the cash flows arising from the issuance of bond.

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Yield to Maturity

Effective Cost of a Corporate Bond

A company issues a bond repayable after 10 years.
The bond is issued with a face value of $ 100.000 and carries 7% coupon.
The lender provides the cash to the issuer by buying a corporate bond certificate.
Over the years, the lender receives coupon payments
(called coupon because the certificate contained coupons for each payment
which the lender have to submit in order to redeem that payment).
The coupon is a fixed proportion of the principal, in this case 7% .
At the end of the period, the lender is repaid the total face value of the bond by the issuer.
The company issued bonds worth $ 20 million.
The investment banking costs for arranging the bond amounted to 2,4% of the amount raised.
The bond was offered at a discount 1% to the face value to attract investors.
The tax rate of the company is 22% .
What is the before-tax and after-tax effective cost of this corporate bond?

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Yield to Maturity

Effective Cost of a Corporate Bond

(m $) time (years)	principal (face value)	2,4% investment banking fees	1,0% discount offered	7,0% coupon (interest) payments	principal payment	corporate bond cash flow	effective before-tax cost of the corporate bond (IRR) 7,50% discounting factors	discounted cash flows
0	20,0	-0,5	-0,2			19,3	1,0000	19,3
1				-1,4		-1,4	0,9302	-1,3
2				-1,4		-1,4	0,8653	-1,2
3				-1,4		-1,4	0,8050	-1,1
4				-1,4		-1,4	0,7488	-1,0
5				-1,4		-1,4	0,6966	-1,0
6				-1,4		-1,4	0,6480	-0,9
7				-1,4		-1,4	0,6028	-0,8
8				-1,4		-1,4	0,5607	-0,8
9				-1,4		-1,4	0,5216	-0,7
10				-1,4	-20,0	-21,4	0,4852	-10,4
								0,0 sum of disc CF

effective after-tax cost of the corporate bond = (1 -22% * 7,5%) = 5,9%

Compound Interest

Periods with which interest is compounded may vary.

12%	interest compounded		yearly
12%	interest for the compounding period (i.e. yearly)		
Period No	Amount Owed at Beginning of Period	Interest Amount for Period	Amount Owed at End of Period
1	1.000	120,00	1.120,00

effective interest rate = 12,00%

12%	interest compounded		quarterly
3%	interest for the compounding period (i.e. quarterly)		
Period No	Amount Owed at Beginning of Period	Interest Amount for Period	Amount Owed at End of Period
1	1.000	30,00	1.030,00
2	1.030	30,90	1.060,90
3	1.061	31,83	1.092,73
4	1.093	32,78	1.125,51

effective interest rate = 12,55%

12%	interest compounded		monthly
1%	interest for the compounding period (i.e. monthly)		
Period No	Amount Owed at Beginning of Period	Interest Amount for Period	Amount Owed at End of Period
1	1.000	10,00	1.010,00
2	1.010	10,10	1.020,10
3	1.020	10,20	1.030,30
4	1.030	10,30	1.040,60
5	1.041	10,41	1.051,01
6	1.051	10,51	1.061,52
7	1.062	10,62	1.072,14
8	1.072	10,72	1.082,86
9	1.083	10,83	1.093,69
10	1.094	10,94	1.104,62
11	1.105	11,05	1.115,67
12	1.116	11,16	1.126,83

effective interest rate = 12,68%

[12% compounded quarterly] is the same as [3% per quarter]; where effective interest rate is 12.55%.

[12% compounded monthly] is the same as [1% per month]; where effective interest rate is 12.68%.

Note that for [12% per year] in other words [12% compounded yearly] effective interest rate is 12%.

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Interest Rates

Real versus Nominal Interest Rate

When the effect of inflation is deducted from nominal interest rate, real interest rate is obtained.

Nominal Interest Rate	25%
Inflation	19%

	Today	A Year Later
Price of an Apple (TL)	1.00	1.19
Money that I have (TL)	1,000	1,250
No of Apples that I can Buy	1,000	1,050

Real Interest Rate	5%

The above relationship may be formulated as follows:

$$1 + i_{real} = \frac{1 + i_{nominal}}{1 + inflation_{expected}} \qquad 1+5\% = \frac{1+25\%}{1+19\%}$$

Note that the expected inflation is used.

	Oct 2017	Oct 2018	Oct 2019	Oct 2020
[1-year deposit rate] nominal interest rate	10%	20%	11,5%	
[Δ CPI Last 12 Months] actual inflation	25%	9%	?	
real interest	-12%	10%	?	

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Effective Cost of a Bank Borrowing which is paid back in equal installments

(m TL) time (months)	amount borrowed	5% initial fees deducted	repayment installments	cash flow of the bank loan	effective monthly cost of the bank loan (IRR) 2,0305% discounting factors	discounted cash flows
0	100	-5		95	1,0000	95,000
1			-9	-9	0,9801	-8,821
2			-9	-9	0,9606	-8,645
3			-9	-9	0,9415	-8,473
4			-9	-9	0,9227	-8,305
5			-9	-9	0,9044	-8,139
6			-9	-9	0,8864	-7,977
7			-9	-9	0,8687	-7,819
8			-9	-9	0,8515	-7,663
9			-9	-9	0,8345	-7,511
10			-9	-9	0,8179	-7,361
11			-9	-9	0,8016	-7,215
12			-9	-9	0,7857	-7,071
						0,000 sum of disc CF

$$(1 + i_m)^{12} - 1 = i_a$$

effective annual cost of the bank loan (IRR) 27,3% nominal

annual expected inflation
20%

$$1 + i_{real} = \frac{1 + i_{nominal}}{1 + inflation_{expected}}$$

effective annual cost of the bank loan
6,1% real

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Interest Rates

Interest-Rate Risk

Present Value (PV) of the Debt Instrument = $\sum_{t=1}^{n} \frac{CF_t}{(1+i)^t}$

For a given set of CF_t ,
when market interest rates (i) go up,
the value (PV), that is, the price of the debt instrument drops.
And, vice versa.

Such volatility of prices of debt instruments is called Interest-Rate Risk.

Prices of longer-maturity bonds respond more drastically to changes in interest rates.

Dealing withinterest-rate risk is a major concern of
managers of financial institutions and investors.

Interest Rates

Interest-Rate Risk

(m $) time (years)	principal (face value)	7,0% coupon (interest) payments	principal payment
0	20,0		
1		-1,4	
2		-1,4	
3		-1,4	
4		-1,4	
5		-1,4	
6		-1,4	
7		-1,4	
8		-1,4	
9		-1,4	
10		-1,4	-20,0

(m $) time (years)	market interest rate 10% discounting factors	discounted cash flows	
0			
1	0,9091	1,3	
2	0,8264	1,2	
3	0,7513	1,1	
4	0,6830	1,0	
5	0,6209	0,9	
6	0,5645	0,8	
7	0,5132	0,7	10,3
		17,1 m $ sum of disc CF	

market value of the corporate bond
at market interest rate of 10%

market interest rate 11% discounting factors	discounted cash flows	
0,9009	1,3	
0,8116	1,1	
0,7312	1,0	
0,6587	0,9	
0,5935	0,8	
0,5346	0,7	
0,4817	0,7	9,6
	16,2 m $ sum of disc CF	

market value of the corporate bond
at market interest rate of 11%

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week 6

Behavior of Interest Rates

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Behavior of Interest Rates

Asset Price – Yield to Maturity Relationship

Interest rate, i (in other words, yield to maturity of a security) is linked to the price of that security (in other words, the debt instrument) as follows:

$$\text{(price) PV} = \sum_{t=1}^{n} \frac{CF_t}{(1+i)^t}$$

For a given set of CF_t,
when market interest rates (i) go up,
the value (PV), that is, the price of the debt instrument drops.
And, vice versa.

On the other hand,
when price of a debt instrument (e.g. bond) drops,
its yield to maturity (IRR, interest rate) goes up.
And, vice versa.

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Behavior of Interest Rates

Determinants of Asset Demand

And, the basic economic theory tells us that
price is determined by demand and supply.

Demand for financial securities (in other words, assets) is affected by the following factors:

- **Wealth**

Ceteris paribus, an increase in wealth raises the quantity demanded of an asset.

- **Expected Return**

An increase in an asset's expected return relative to that of an alternative asset, raises the demand.

- **Risk**

If an asset's risk rises relative to that of an alternative asset, its quantity demanded will fall.

- **Liquidity**

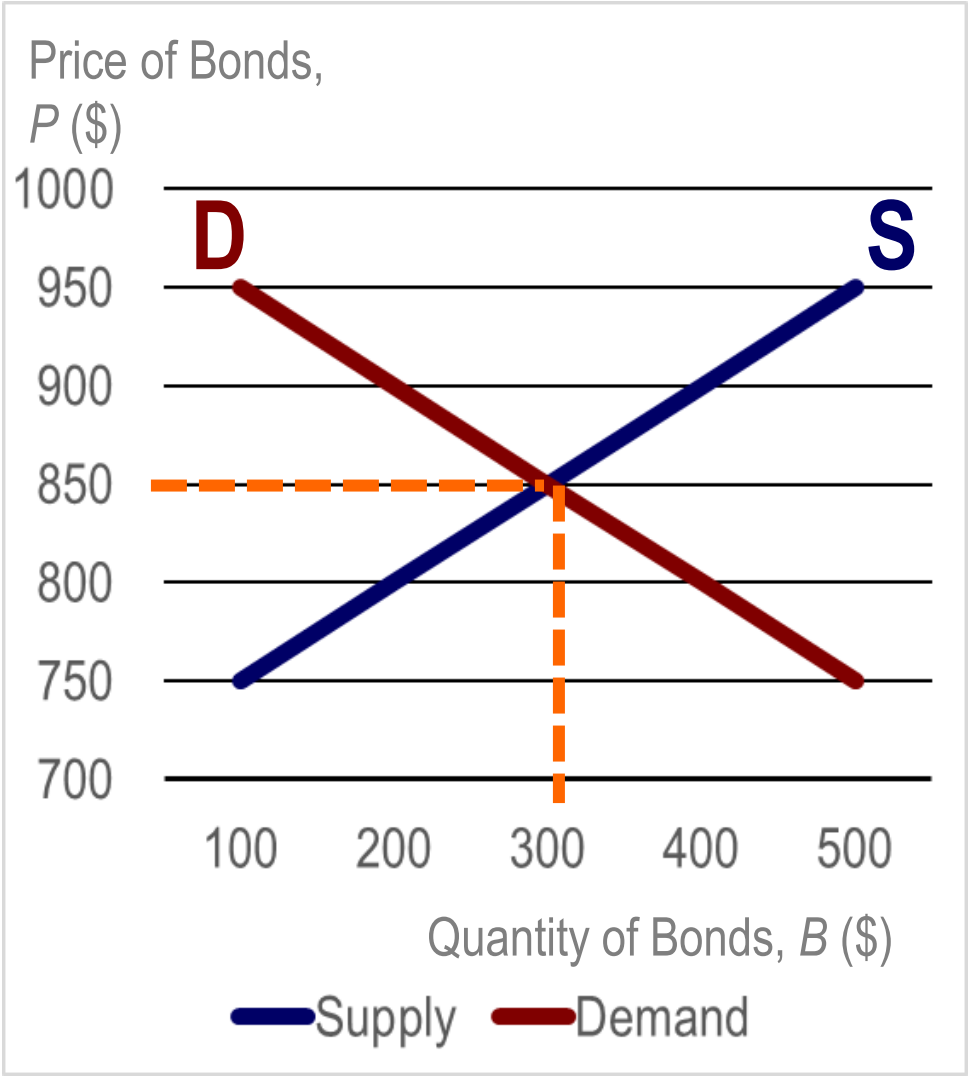
The more liquid an asset is relative to alternative assets, the more desirable it is,
and the greater will be the quantity demanded.

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Behavior of Interest Rates

Supply and Demand in the Bond Market

In economics,
market equilibrium occurs
when the amount
that people are willing to buy (demand)
equals
the amount
that people are willing to sell (supply)
at a given price.



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Changes in Equilibrium Interest Rates

Shifts in Demand for Bonds

When we find out that our wealth has increased, we have more resources available with which to purchase assets, and so, not surprisingly, the quantity of assets we demand increases. When the economy is growing rapidly in a business cycle expansion, wealth increases.

Another factor that affects wealth is the public's propensity to save. If households save more, wealth increases.

In a business cycle expansion (with growing wealth), the demand for bonds rises and the demand curve for bonds shifts to the right.

Price of bonds go up, interest rates drop.

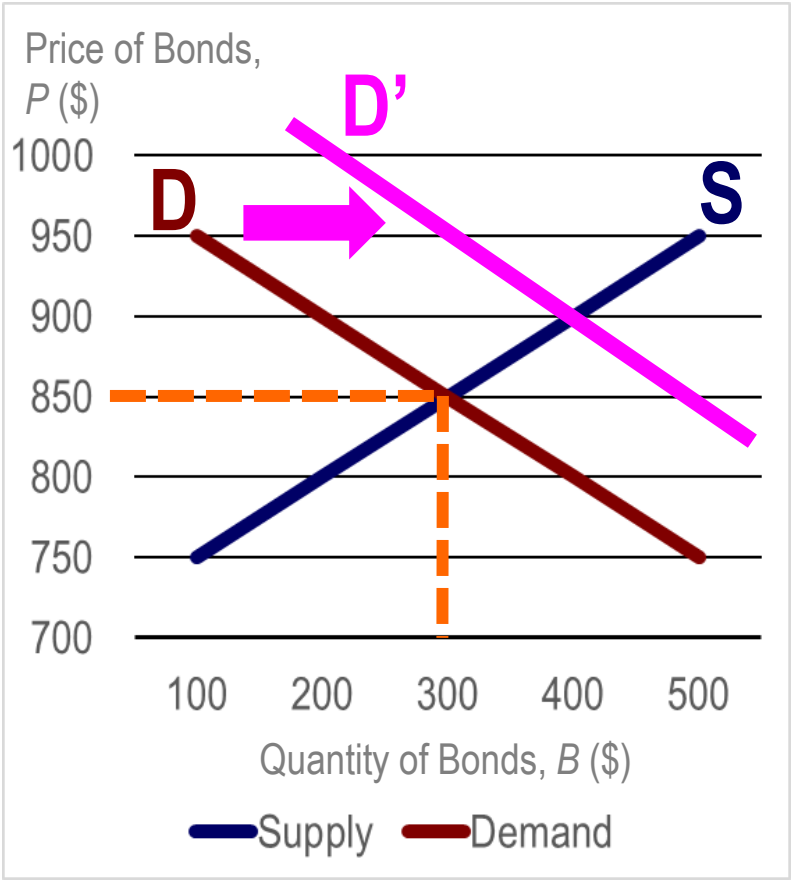
- PV = $\sum_{t=1}^{n} \frac{CF_t}{(1+i)^t}$

If more people started trading in the bond market, and as a result it became easier to sell bonds quickly, the increase in their liquidity would cause the quantity of bonds demanded at each interest rate to rise.

Increased liquidity of bonds relative to alternative assets results in an increased demand for bonds, and the demand curve shifts to the right.

Price of bonds go up, interest rates drop.

On the other hand, the reduction of brokarage commissions for trading common stocks that occured when the fixed-rate commission structure was abolished in 1975 in the US, for example, increased the liquidity of stocks relative to bonds, and the resulting lower demand for bonds shifted the demand curve to the left.



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Changes in Equilibrium Interest Rates

Shifts in Demand for Bonds

When we make a decision to buy an asset, we are influenced by what we expect the return on the asset to be. Higher expected interest rates in the future lower the expected return for long-term bonds today, decrease the demand, and shift the demand curve to the left. Changes in expected returns on other assets can also shift the demand curve for bonds.

If people suddenly became more optimistic about the stock market and began to expect higher stock prices in the future, both expected capital gains and expected returns on stocks would rise. With the expected return on bonds held constant, the expected return on bonds today relative to stocks would fall, lowering the demand for bonds and shifting the demand curve to the left.

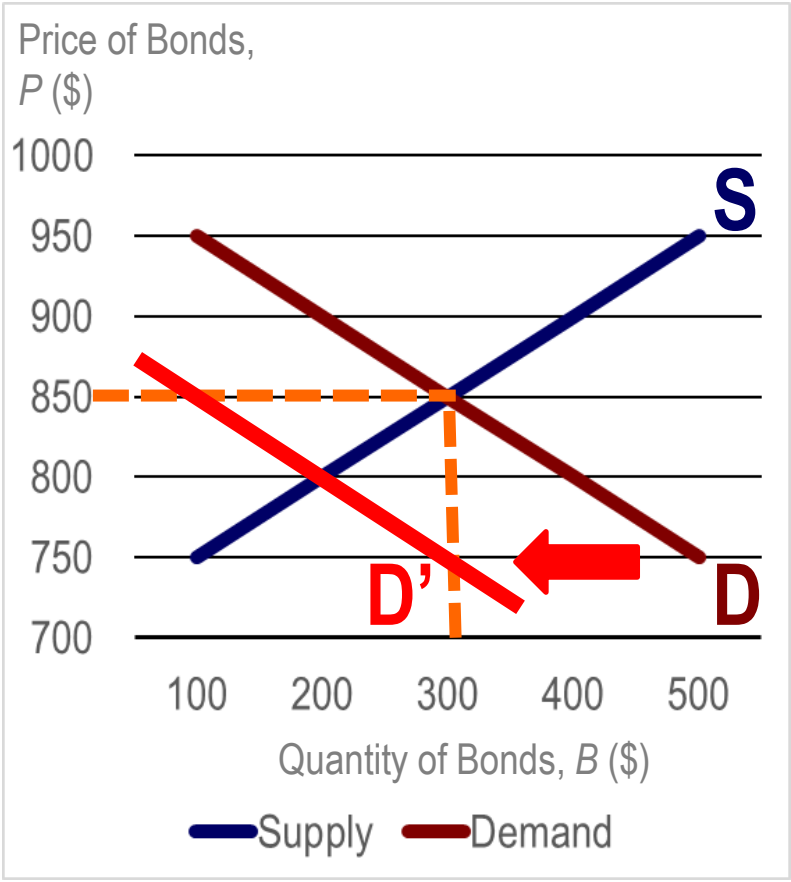
Price of bonds come down, interest rates go up.

- PV = $\sum_{t=1}^{n} \frac{CF_t}{(1+i)^t}$

An increase in expected inflation will lead to higher prices on physical assets (also called real assets) such as cars and houses in the future and hence higher nominal capital gains. The resulting rise in the expected returns today on these real assets will lead to a fall in the expected return on bonds relative to the expected return on real assets today and thus cause the demand for bonds to fall.

An increase in the expected rate of inflation lowers the expected return for bonds, causing their demand to decline and the demand curve to shift to the left. Price of bonds come down, interest rates go up.

If prices in the bond market become more volatile, the risk associated with bonds increases, and bonds become a less attractive asset. An increase in the riskiness of bonds relative to alternative assets causes the demand for bonds to fall and the demand curve to shift to the left. Price of bonds come down, interest rates go up.



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Changes in Equilibrium Interest Rates

Shifts in Supply of Bonds

The more profitable plant and equipment investments
that a firm expects it can make,
the more willing it will be to borrow for financing these investments.
When the economy is growing rapidly,
investment opportunities that are expected to be profitable abound,
the supply of bonds increases, and the supply curve shifts to the right.

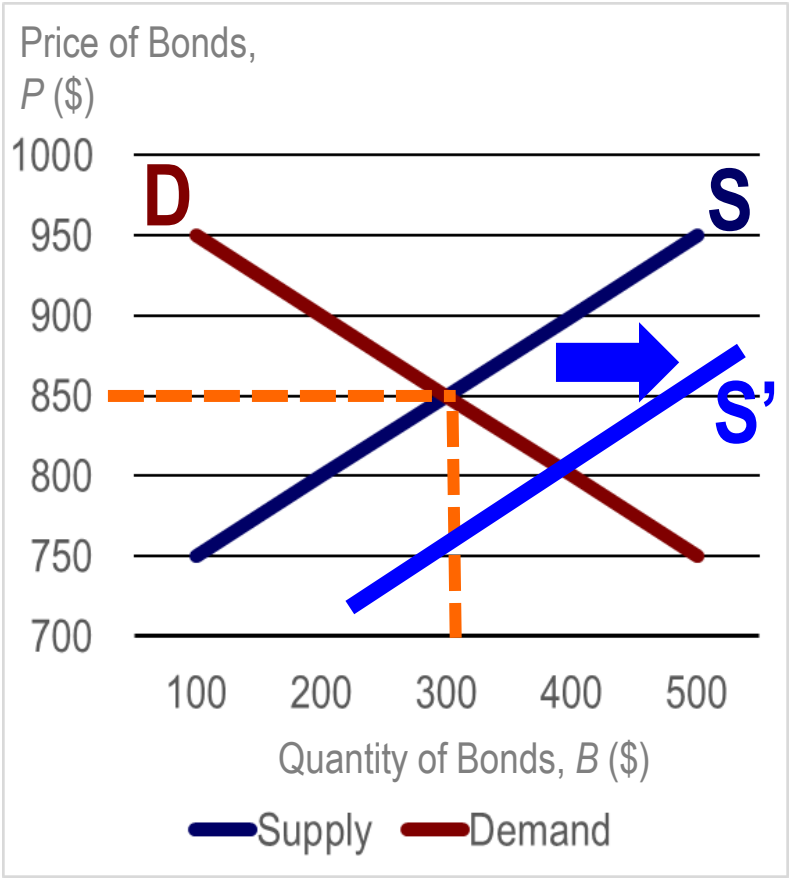
Price of bonds come down,
interest rates go up.

- PV = $\sum_{t=1}^{n} \frac{CF_t}{(1+i)^t}$

The real cost of borrowing is more accurately measured by
the real interest rate,
which equals the (nominal) interest rate *less** the expected inflation rate.
When expected inflation increases,
the real cost of borrowing falls;
hence the quantity of bonds supplied increase at any given bond price,
i.e. the supply curve shifts to the right.
Price of bonds come down, interest rates go up.

$$1 + i_{real} = \frac{1 + i_{nominal}}{1 + inflation_{expected}}$$

Government issues bonds to finance government deficits,
the gap between the government's expenditures and its revenues.
When these deficits are large, the government sells more bonds,
and the quantity of bonds supplied at each bond price increases.
Higher government deficits
increase the supply of bonds and shift the supply curve to the right.
Price of bonds come down, interest rates go up.



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week 6

The Risk and Term Structure of Interest Rates

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Risk Structure of Interest Rates

Default Risk

One attribute of a bond that influences its interest rate is its risk of default,
which occurs when the issuer of the bond is unable or unwilling to
make interest payments when promised
or pay off the face value when the bond matures.

Government bonds have usually been considered to have no default risk
because the government can always increase taxes or print money to pay off its obligations;
they are called default-free bonds.

The spread between the interest rates on bonds with default risk and default-free bonds,
both of the same maturity,
called the risk premium,
indicates how much additional interest people must earn
to be willing to hold that risky bond.

Purchasers of bonds need to know whether a Corporation is likely to default on its bonds;
this information is provided by credit-rating agencies.

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Risk Structure of Interest Rates

Bond Ratings

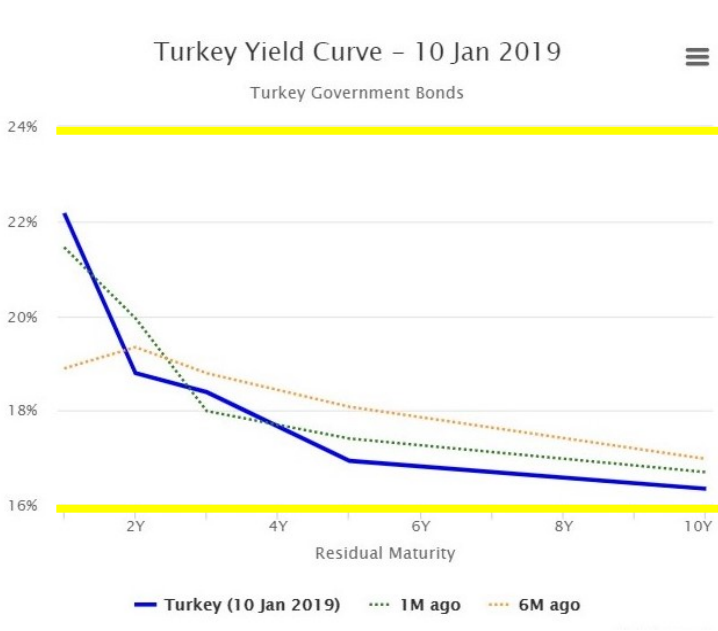
	Moody's	S&P	Fitch	Definitions
	Aaa	AAA	AAA	Prime Maximum Safety
	Aa1	AA+	AA+	High Grade High Quality
	Aa2	AA	AA	
	Aa3	AA-	AA-	
	A1	A+	A+	Upper Medium Grade
	A2	A+	A+	
	A3	A-	A-	
	Baa1	BBB+	BBB+	Lower Medium Grade
	Baa2	BBB	BBB	
	Baa3	BBB-	BBB-	
	Ba1	BB+	BB+	Noninvestment Grade
	Ba2	BB	BB	Speculative
	Ba3	BB-	BB-	
TR	B1	B+	B+	Highly Speculative
	B2	B	B	
	B3	B-	B-	
	Caa1	CCC+	CCC	Substantial Risk
	Caa2	CCC		In Poor Standing
	Caa3	CCC-		
	Ca	--	--	Extremely Speculative
	C	--	--	May Be in Default
	--	--	DDD	Default
	--	--	DD	
	--	D	D	

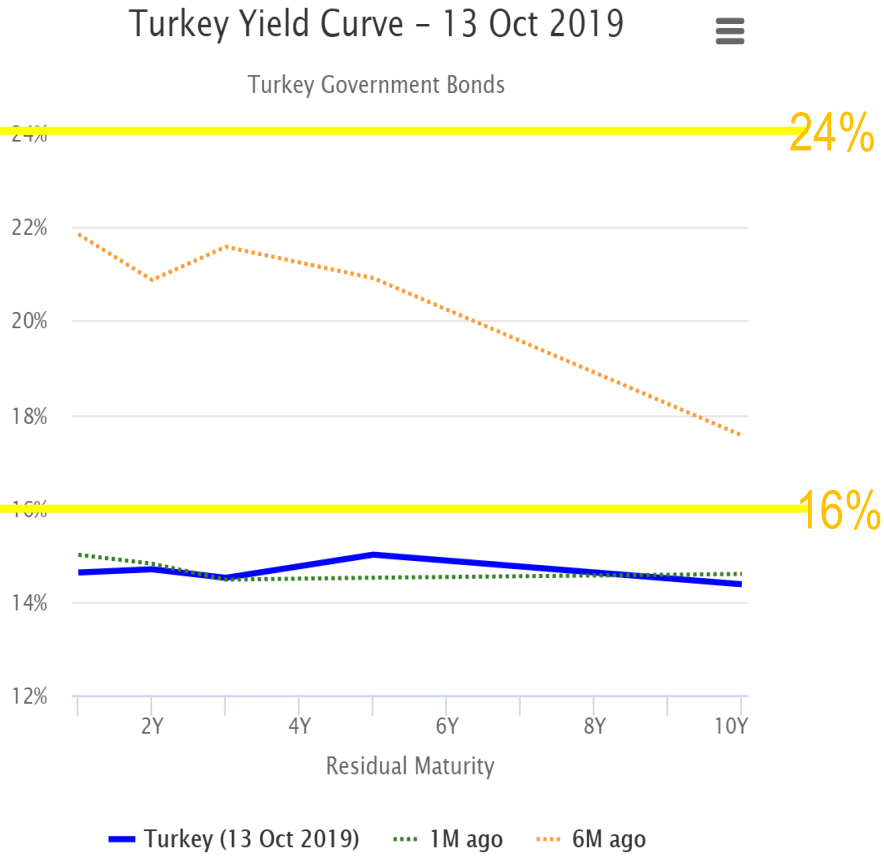
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Term Structure of Interest Rates

Yield Curve

A plot of the yields on bonds with differing terms to maturity
but the same risk, liquidity, and tax considerations is called a yield curve,
and it describes the term structure of interest rates for particular types of bonds.



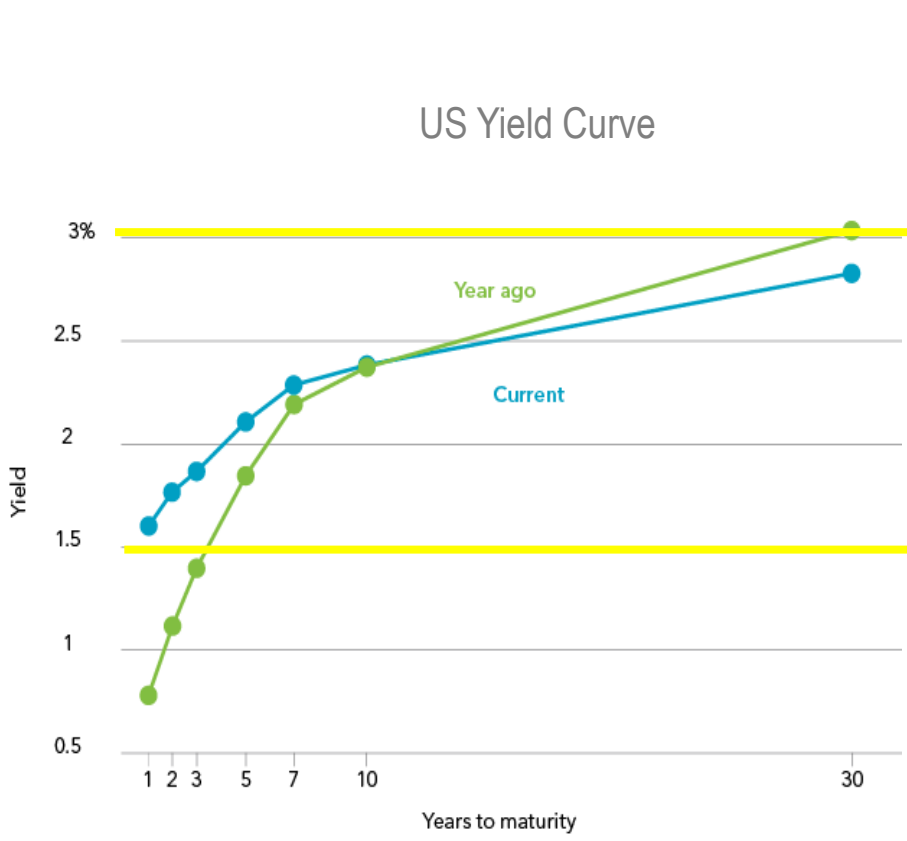


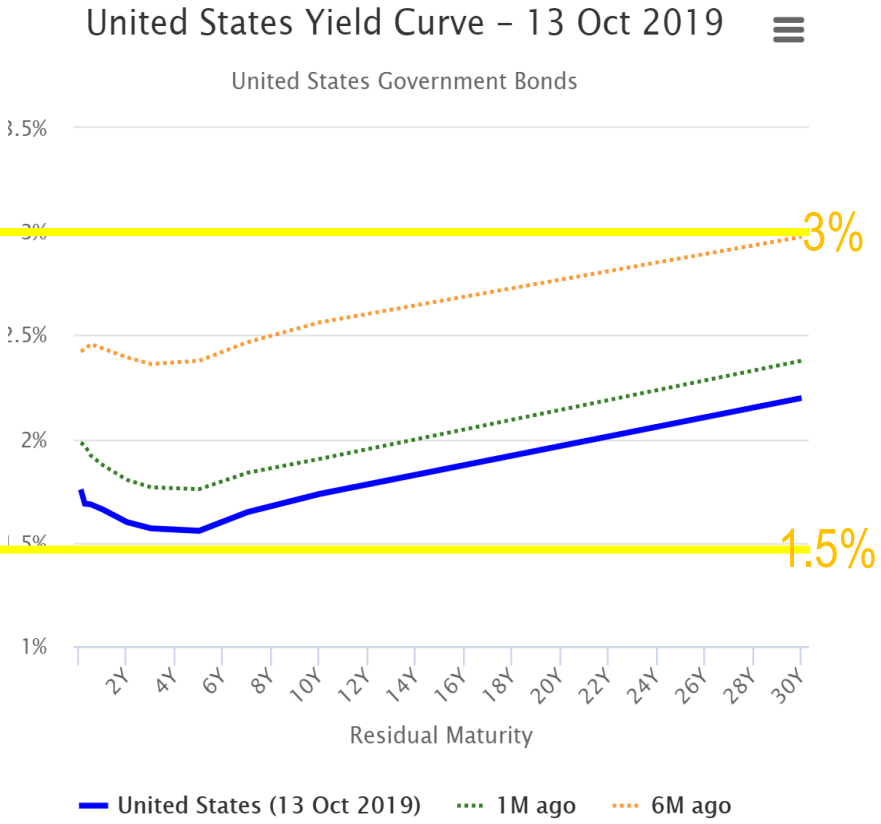
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Term Structure of Interest Rates

Yield Curve

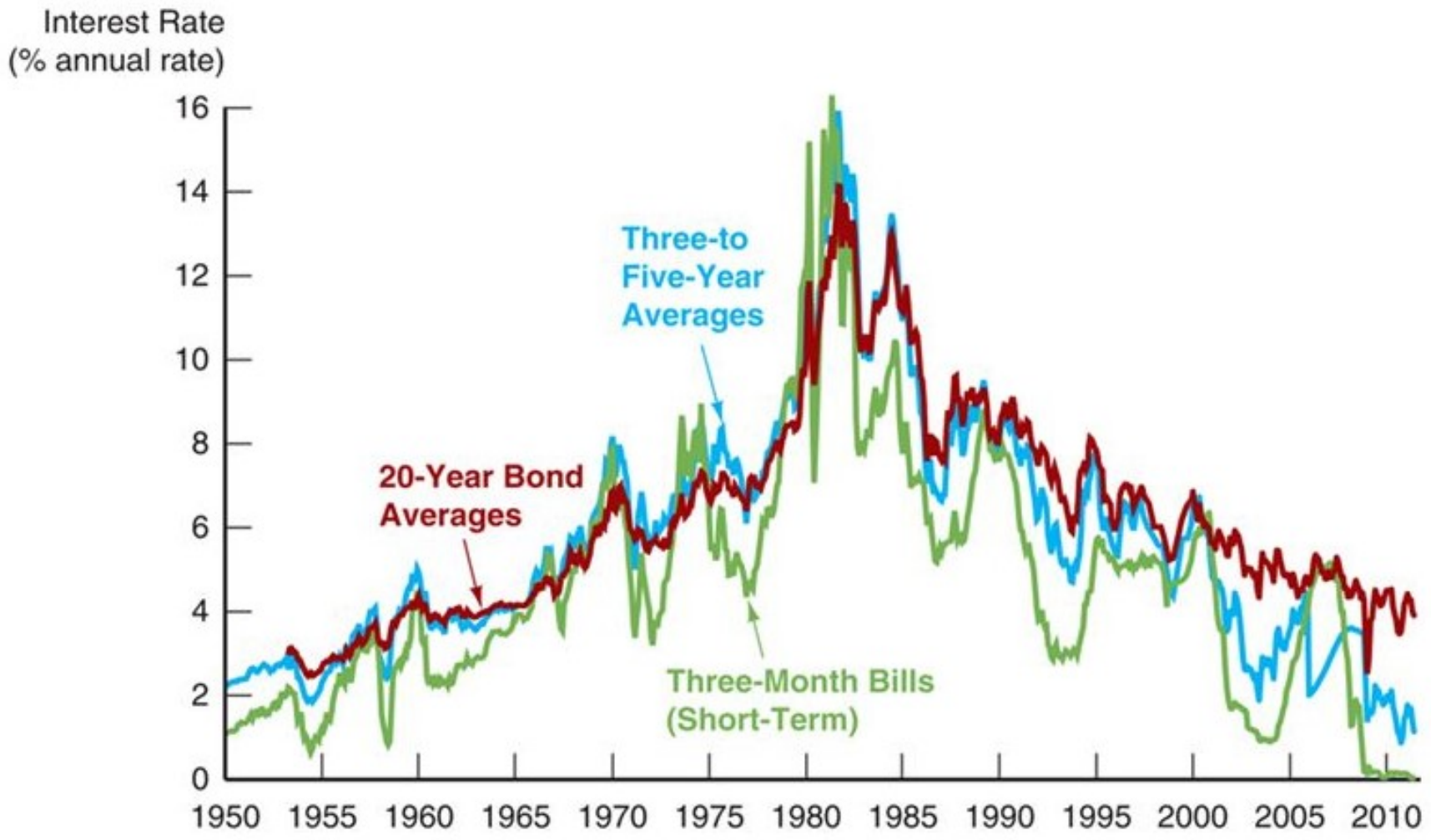
A plot of the yields on bonds with differing terms to maturity
but the same risk, liquidity, and tax considerations is called a yield curve,
and it describes the term structure of interest rates for particular types of bonds.





Term Structure of Interest Rates

Movements over Time of Interest Rates on US Government Bonds with Different Maturities



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Term Structure of Interest Rates

Yield Curve

A plot of the yields on bonds with differing terms to maturity
but the same risk, liquidity, and tax considerations is called a yield curve,
and it describes the term structure of interest rates for particular types of bonds.

Three empirical facts about term structure of interest rates are:

- Interest rates on bonds of different maturities move together over time.
- When short-term interest rates are low, yield curves are more likely to have an upward slope; when short-term interest rates are high, yield curves are more likely to slope downward (i.e. inverted).
- Yield curves almost always slope upward.

Three theories have been put forward to explain the term structure of interest rates,
i.e. shapes of yield curves:

- The Expectations theory
- The Segmented Markets theory
- The Liquidity Premium theory

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Term Structure of Interest Rates

Expectations Theory

"The interest rate on a long-term bond will equal
an average of the short-term interest rates
that people expect to occur over the life of the long-term bond."

For a 5-year bond, the yield formulation would be as follows:

$$(1 + i_{5yr})^5 = \prod_{t=1}^{5}(1 + i_t)$$

When the yield curve is upward-sloping,
the expectations theory suggests that
short-term interest rates are expected to rise in the future.

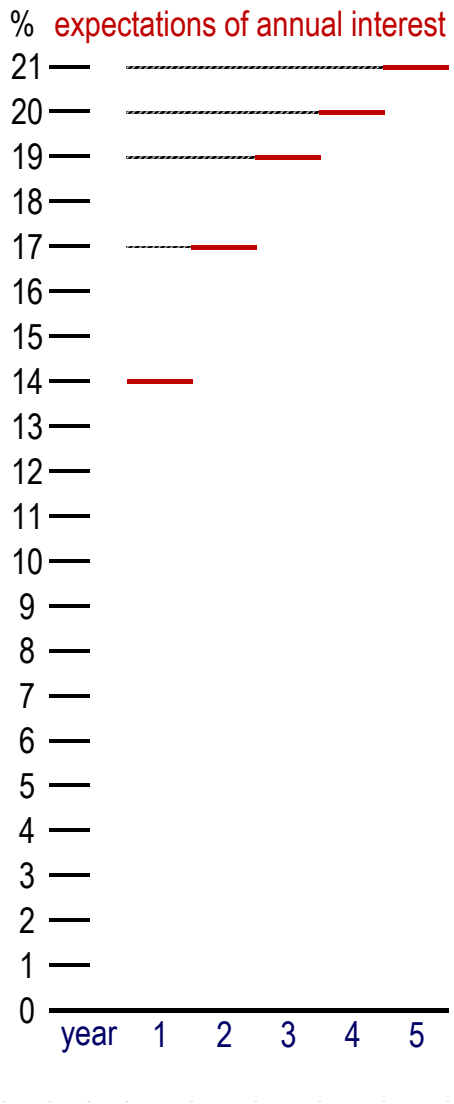
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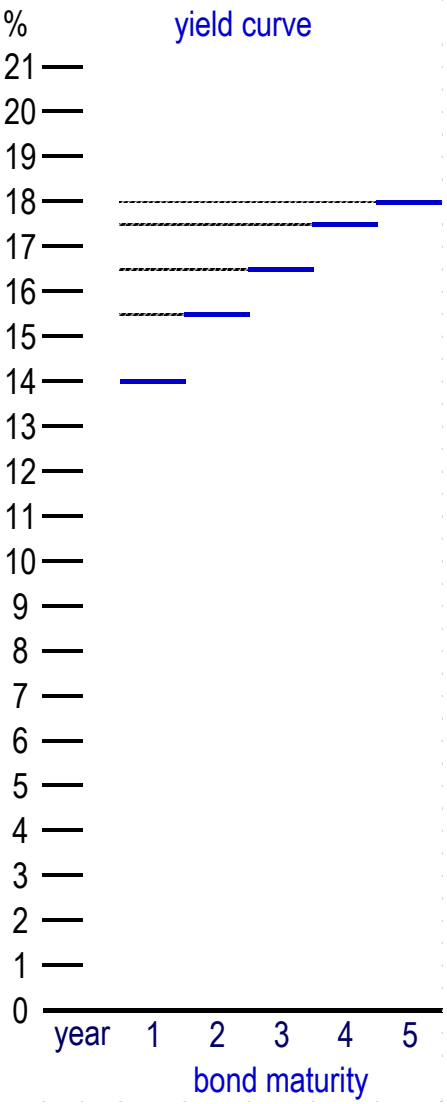
Term Structure of Interest Rates

Expectations Theory

If expectation of annual interest rates
for the next 5 years
is as follows,
what would be the yield curve
according to the expectations theory?

year, t	interest rate, i	$1+i$	Π	$\sqrt[t]{}$	interest rate	
1	14%	1,14	1,14	0,140	14,0%	on 1-year bond
2	17%	1,17	1,33	0,155	15,5%	on 2-year bond
3	19%	1,19	1,59	0,166	16,6%	on 3-year bond
4	20%	1,20	1,90	0,175	17,5%	on 4-year bond
5	21%	1,21	2,30	0,182	18,2%	on 5-year bond





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Term Structure of Interest Rates

Segmented Markets Theory

The segmented markets theory of the term structure
sees markets for different-maturity bonds as completely separate and segmented.

The interest rate for each bond with a different maturity
is then determined by the supply and demand for that bond,
with no effects from expected returns on other bonds with other maturities.

This theory of the term structure is at the opposite extreme to
the expectations theory, which assumes that bonds of different maturities are perfect substitutes.

According to the segmented market theory,
as risk averse investors generally prefer bonds with shorter maturities that have less interest-rate risk,
the demand for long-term bonds is relatively lower than that for short-term bonds;
long-term bonds will have lower prices and higher interest rates,
and hence the yield curve will typically slope upwards.

However, the segmented market theory fails to explain the other two empirical facts about yield curves:

- Interest rates on bonds of different maturities move together over time.
- When short-term interest rates are low, yield curves are more likely to have an upward slope;
when short-term interest rates are high, yield curves are more likely to slope downward (i.e. inverted).

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Term Structure of Interest Rates

Liquidity Premium Theory

This theory states that the interest rate on a long-term bond will equal
an average of short-term interest rates expected to occur over the life of the long-term bond
(as in the Expectations theory)
plus a liquidity premium (also referred to as term premium)
that responds to supply and demand conditions for that bond.

Bonds of different maturities are assumed to be substitutes
but not perfect substitutes.

Investors tend to prefer short-term bonds
because these bonds bear less interest-rate risk.

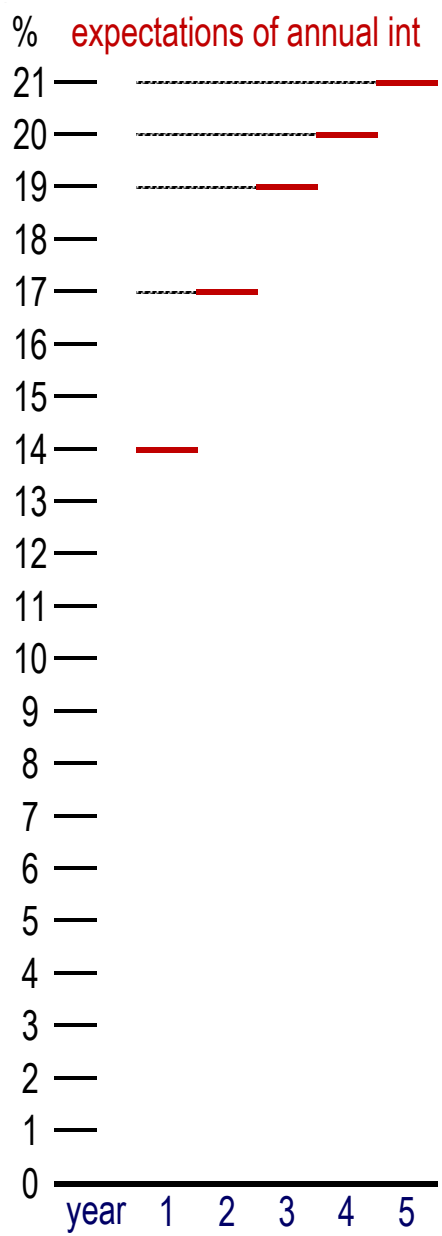
For these reasons,
investors must be offered a positive liquidity premium
to induce them to hold longer-term bonds.

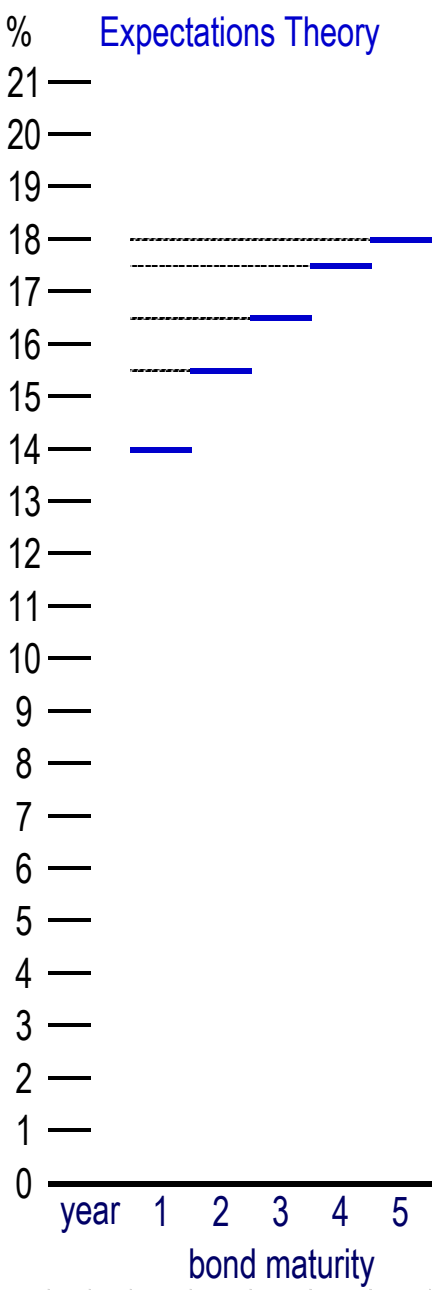
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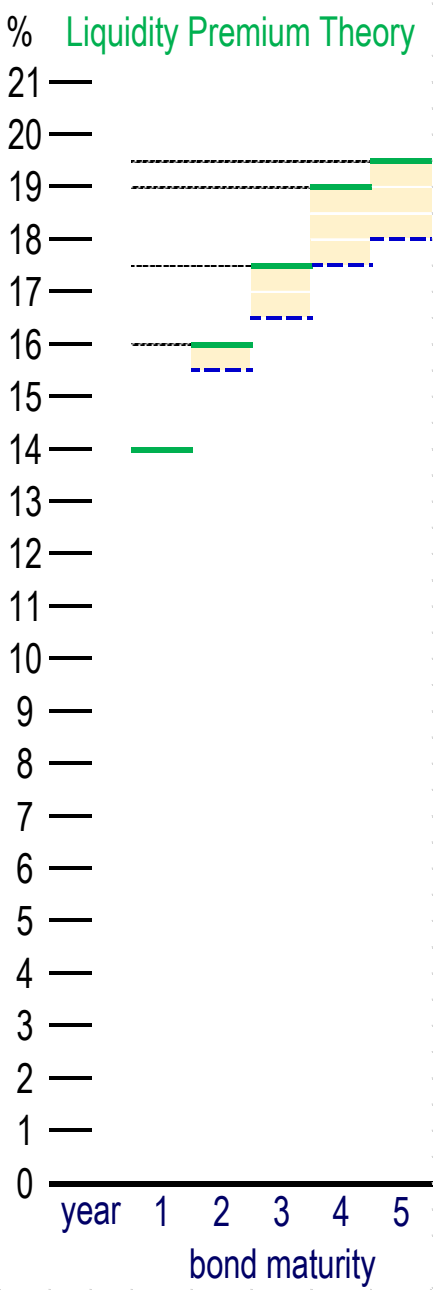
Term Structure of Interest Rates

Liquidity Premium Theory

Because
the liquidity premium
is always positive
and grows
as the term to maturity
increases,
the yield curve implied
by the liquidity
premium theory
is always above
the yield curve implied
by the expectations
theory and
has a steeper slope.





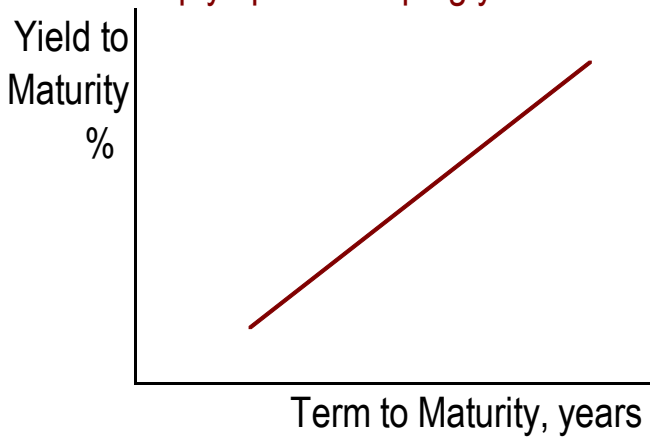


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Term Structure of Interest Rates

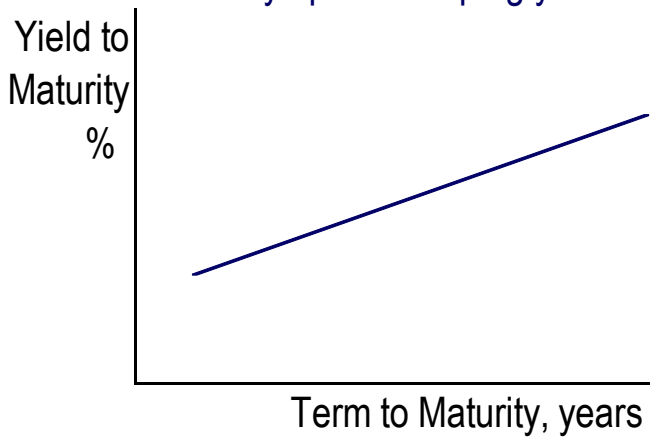
Liquidity Premium Theory

steeply upward-sloping yield curve



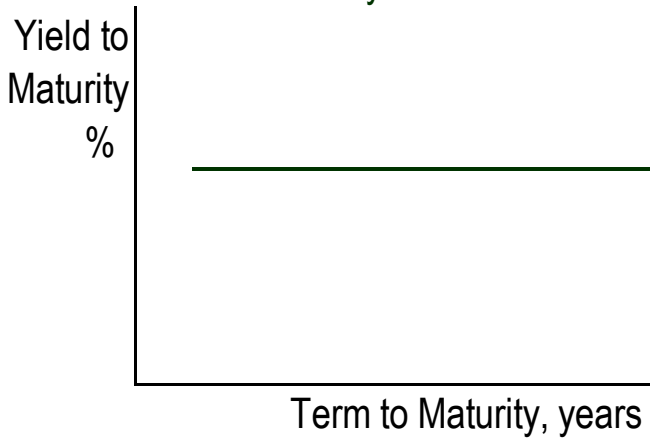
Short-term interest rates are expected to rise in the future.

mildly upward-sloping yield curve



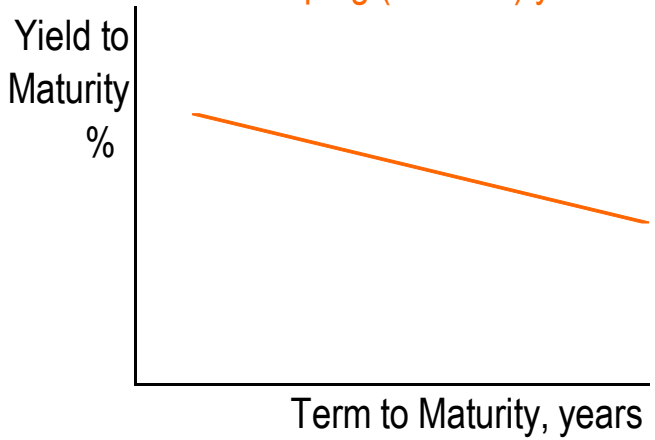
Short-term interest rates are not expected to rise or fall much in the future.

flat yield curve



Short-term interest rates are expected to fall moderately in the future.

downward-sloping (inverted) yield curve



Short-term interest rates are expected to fall sharply in the future.

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Introduction to Finance for Engineers

week 7
Stock Markets

Stock Markets in the World

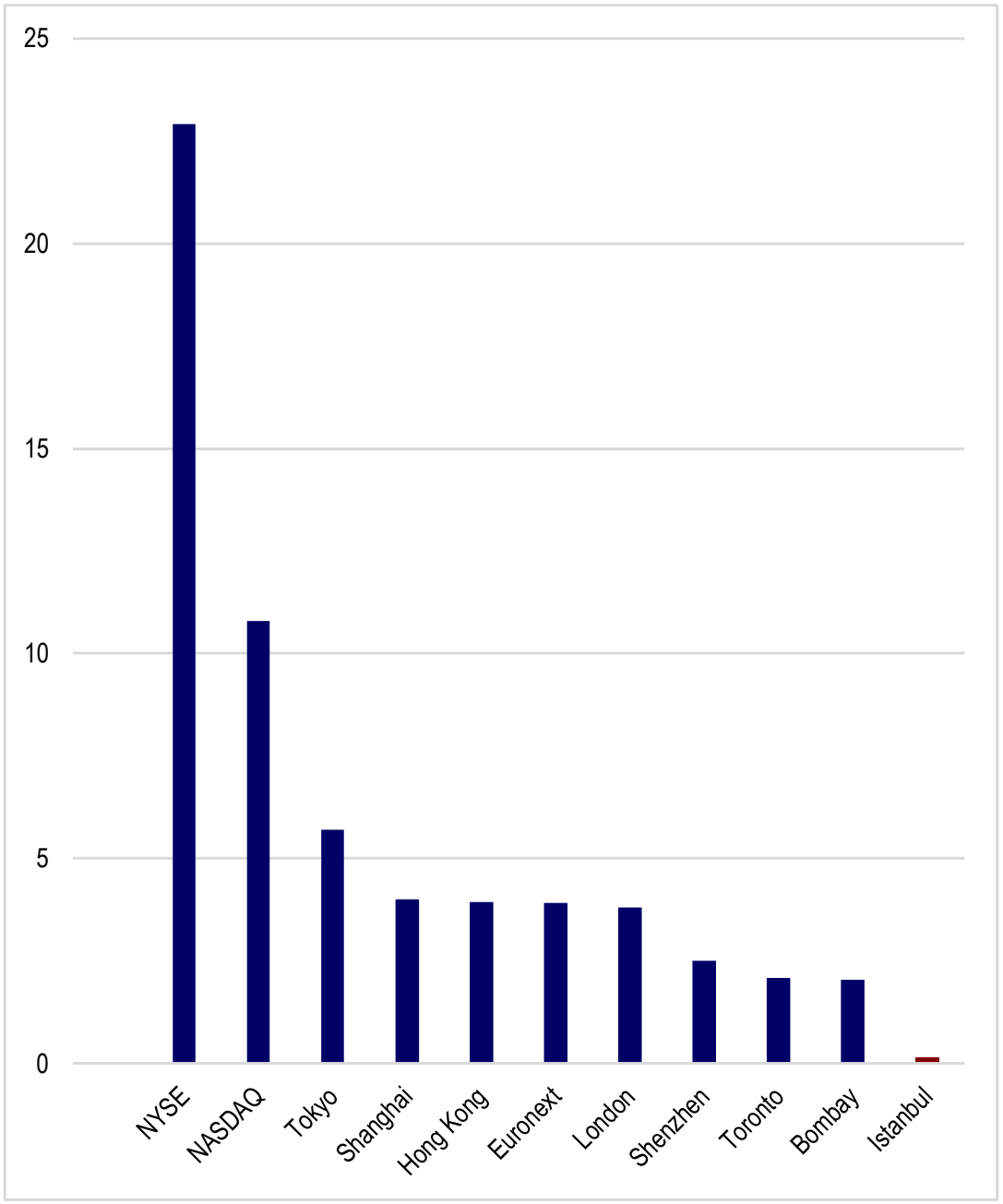
1- New York Stock Exchange, United States: Founded in 1792, the New York Stock Exchange has been the world's largest stock exchange since the end of World War I, when it overtook the London Stock Exchange. It has a market capitalization of $22.9 trillion and about 2,400 listed companies.
2- NASDAQ, United States: The NASDAQ Stock Market was founded in 1971 in New York City. NASDAQ had a market capitalization of $10.8 trillion with an average monthly trading volume of $1.26 trillion.
3- Tokyo Stock Exchange, Japan: Founded in 1878, the Tokyo Stock Exchange has close to 2,300 listed companies with a combined market capitalization of $5.67 trillion.
4- Shanghai Stock Exchange, China: It has a market capitalization of $4.02 trillion. It has more than 1,000 listed companies. Though its origins date back to 1866, it was suspended following the Chinese Revolution in 1949. The Shanghai Exchange in its modern avatar was founded in 1990.
5- Hong Kong Stock Exchange, Hong Kong: The Hong Kong Stock Exchange was founded in 1891. It has close to 2,000 listed companies, about half of which are from mainland China. It has a market capitalization of $3.93 trillion.
6- Euronext, Eurozone: Headquartered in Amsterdam, the Netherlands, Euronext is a pan-European stock exchange with a presence in France, Belgium, Ireland, and Portugal. It has approximately 1,300 listed companies with a combined market capitalization of $3.92 trillion.
7- London Stock Exchange, United Kingdom: The London Stock Exchange was founded in 1698. It has more than 3,000 listed companies with a combined market capitalization of $3.76 trillion.
8- Shenzhen Stock Exchange, China: Formally established in 1990, the Shenzhen Stock Exchange has a market capitalization of $2.5 trillion as of November 2018.
9- Toronto Stock Exchange, Canada: Toronto Stock Exchange (TSX) has 2,207 listed companies with a combined market capitalization of $2.1 trillion.
10- Bombay Stock Exchange, India: Founded in 1875, the Bombay Stock Exchange was the first stock exchange in Asia. As of November 2018, it had a market capitalization of $2.05 trillion. BSE has 5,749 listed companies.

Istanbul Stock Exchange as a Market Capitalization of $163.58 Billion.

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Stock Markets in the World

	Market Capitalization (trn $)
NYSE	22.9
NASDAQ	10.8
Tokyo	5.7
Shanghai	4.0
Hong Kong	3.9
Euronext	3.9
London	3.8
Shenzhen	2.5
Toronto	2.1
Bombay	2.1
Istanbul	0.16



Stock Markets

Computing the Price of Common Stock

Common stock is the principal way that corporations raise equity capital.
Stockholders – those who hold stock in a Corporation –
own an interest in the Corporation proportional to the percentage outstanding shares they own.
This ownership interest gives them the right to vote
and to be the residual claimant of all funds flowing into the firm (known as cash flows),
meaning that the stockholder receives whatever remains
after all other claims against the firm's assets have been satisfied.

Stockholders may receive dividends from the net earnings of the Corporation.
Dividends are payments made periodically,
usually every quarter in the U.S. and annually in Turkey, to stockholders.

In addition, the stocholder has the right to sell the stock.
One basic principle of finance is that the value of any investment is found by computing
the present value of all cash flows the investment will generate over its life.

The cash flows a stockholder might earn from stock are dividends, the sales price, or both.

Stock Markets

The One-Period Valuation Model

To value the stock today,
you need to find the present discounted value of the expected cash flows (future payments).
The cash flows consist of one dividend payment plus a final sales price.

P_0 = the curent price of the stock
(the zero subscript refers to time period zero, or the present)

Div_1 = the dividend paid at the end of year 1

k_e = the required return on investments in equity

P_1 = the price at the end of the first year; predicted sales price of the stock

$$P_0 = \frac{Div_1}{(1+k_e)} + \frac{P_1}{(1+k_e)}$$

Stock Markets

The Generalized Dividend Valuation Model

Using the present value concept,
we can extend the one-period dividend valuation model to any number of periods:
The value of a stock today is the present value of all future cash flows.

P_0 = the curent price of the stock

D_t = the dividend paid at the end of year t

k_e = the required return on investments in equity

P_N = the price at the end of year N; predicted sales price of the stock

$$P_0 = \frac{D_1}{(1+k_e)^1} + \frac{D_2}{(1+k_e)^2} + \ldots + \frac{D_t}{(1+k_e)^t} + \ldots + \frac{D_N}{(1+k_e)^N} + \frac{P_N}{(1+k_e)^N}$$

Stock Markets

The Generalized Dividend Valuation Model

The generalized dividend model may be rewritten without the final sales price.

P_0 = the curent price of the stock
D_t = the dividend paid at the end of year t
k_e = the required return on investments in equity

$$P_0 = \sum_{t=1}^{\infty} \frac{D_t}{(1+k_e)^t}$$

This tells us that the price of stock is determined only by the present value of the dividends and nothing else matters.
In other words, buyers of the stock expect that the firm will pay dividends someday.
Most of the time a firm institutes dividends
as soon as it has completed the rapid growth phase of its life cycle.
However, this model requires that we compute the present value of an infinite stream of dividends, a process that could be difficult, to say the least.

Stock Markets

Valuation - Gordon Growth Model

The Gordon growth model is a simplified method of computing stock value that depends on the assumption that the dividends are growing at a constant rate forever. Given our uncertainty regarding future dividends, this assumption is often the best we can do. Also, the growth rate is assumed to be less than the required return on equity which is a reasonable assumption.

D_0 = the most recent paid dividend
g = the expected constant growth rate in dividends
k_e = the required return on investments in equity

$$P_0 = \frac{D_0 \times (1+g)^1}{(1+k_e)^1} + \frac{D_0 \times (1+g)^2}{(1+k_e)^2} + \ldots + \frac{D_0 \times (1+g)^t}{(1+k_e)^t} + \ldots + \frac{D_0 \times (1+g)^\infty}{(1+k_e)^\infty}$$

$$P_0 = \frac{D_0 \times (1+g)}{(k_e - g)}$$

$$P_0 = \frac{D_1}{(k_e - g)}$$

Stock Markets

Volatility

The interaction among traders in the market is what actually sets prices on a day-to-day basis.
The trader who values the security the most
(either because of less uncertainty about the cash flows
or because of greater estimated cash flows)
will be willing to pay the most.
As the new information is released,
investors will revise their estimates of the true value of the security and will either buy or sell it,
depending on how the market price compares with their estimated valuation.
Because small changes in estimated growth rates or required return
result in large changes in price,
it is not surprising that the markets are often volatile.

Stock Markets

Efficient Market Hypothesis

The efficient market hypothesis states that
current security prices will fully reflect all available information,
because in an efficient market, all unexploited profit opportunities are eliminated.

The elimination of unexploited profit opportunities necessary for a financial market to be efficient
does not require that all market participants be well informed.

The efficient market hypothesis indicates that
hot tips and investment advisors' published recommendations
cannot help an investor outperform the market.

The prescription for investors is to pursue a buy-and-hold strategy
– purchase stocks and hold them for long periods of time.

Empirical evidence generally supports
these implications of the efficient market hypothesis in the stock market.

Stock Markets

Market Crashes and Bubles

The existence of market crashes and bubles has convince many economists that
the stronger version of market efficiency,
which states that asset prices reflect the true fundemental (intrinsic) value of securities,
is not correct.

It is far less clear that these crashes show that
the efficient market hypothesis is wrong.

Even if the stock market was driven by factors other than fundementals,
these crashes do not clearly demonstrate that
many basic lessons of the efficient market hypothesis are no longer valid,
as long as these crashes could not have been predicted.

Stock Markets

Behavioral Finance

The new field of behavioral finance applies concepts from other social sciences such as anthropology, sociology, and psychology to understand the behavior of security prices. Loss aversion, overconfidence, and social contagion can explain why trading volume is so high, stock prices become overvalued, and speculative bubbles occur.

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Financial Crises

Stock Markets

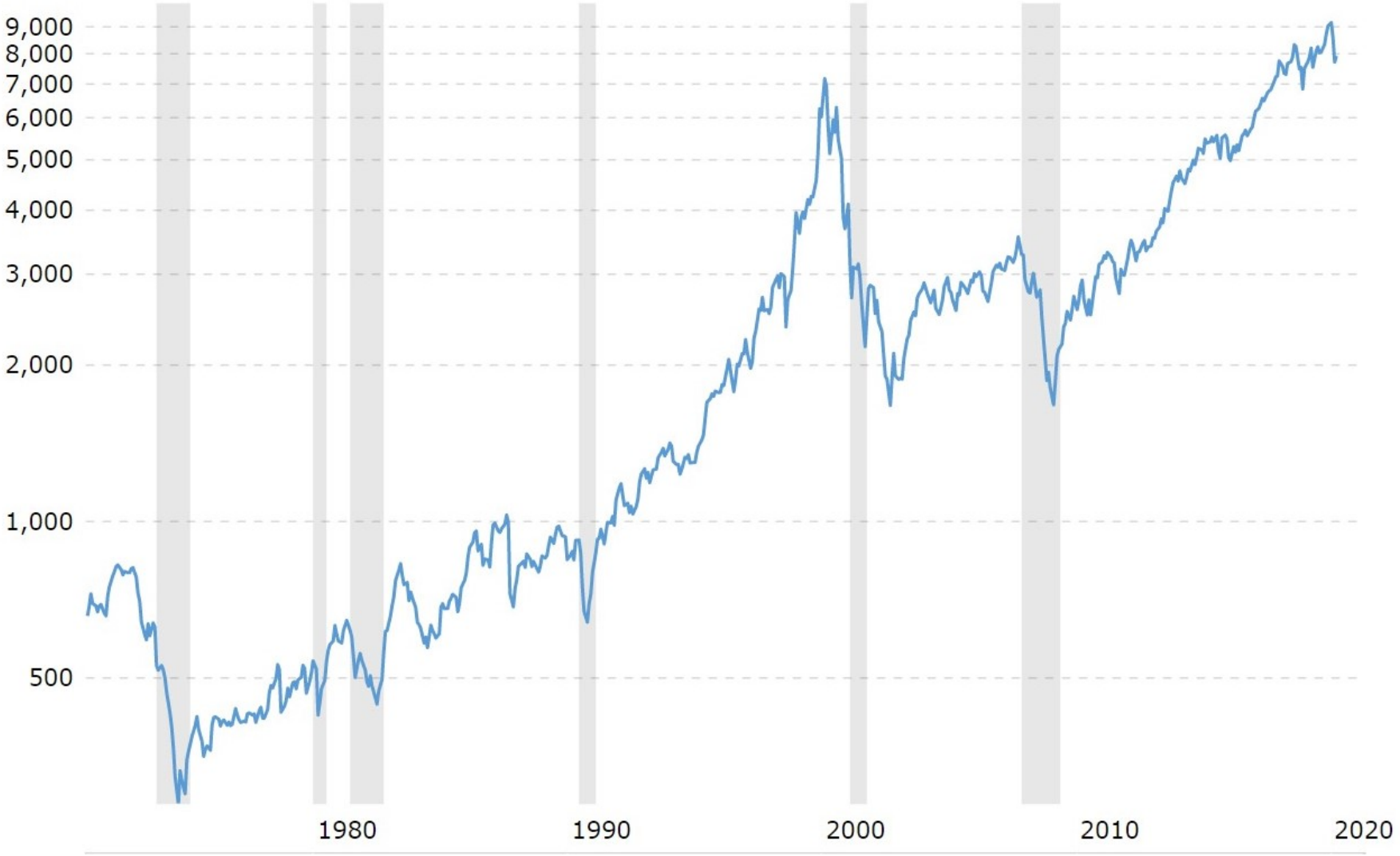
Huge Swings in the Stock Market in Recent Years

The 1990s were an extraordinary decade for stocks:
The Dow Jones and S&P500 indexes increased more than 400%,
while the tech-laden NASDAQ index rose more than 1,000%.
By early 2000, all indexes had reached record highs.
Unfortunately, the good times did not last.
Starting in early 2000, the stock market began to decline and many investors lost their shirts.
The NASDAQ crashed, falling by more than 50%,
while the Dow Jones and S&P500 indexes fell by 30% through January 2003.
After subseuently rising over 30%,
the stock market crashed again during the global financial crisis,
falling over 50% from its peak in the fall of 2007.
Starting in 2009, the stock market recovered quickly, rising by more than 50% by 2011.

Stock Markets

Dow Jones Industrial Average Stock Market Index

Inflation-adjusted, gray bands: recession periods



Stock Markets

Recession of 1990-1991

The recession lasted from July 1990 to March 1991.

From November 1982 to July 1990 the U.S. economy experienced
robust growth, modest unemployment, and low inflation.
The "Reagan boom" rested on shaky foundations, however,
and as the 1980s progressed signs of trouble began to mount.

On October 19, 1987 stock markets around the world crashed (started in the Far East).
In the U.S. the Dow Jones Industrial Average lost over 22% of its value.
Although the causes of "Black Monday" were complex, many saw the crash as a sign that
investors were worried about the inflation that might result from large U.S. budget deficits.
The American housing market presented another sign of weakness,
as in the second half of the 1980s a large number of savings and loan associations (private banks that specialized in home mortgages) went bankrupt.
The collapse of the S&L industry negatively impacted the welfare of many American households and precipitated a large government bailout that placed further strain on the budget.

Other causes of the early 1990s recession included
moves by the U.S. Federal Reserve to raise interest rates in the late 1980s

and Iraq's invasion of Kuwait in the summer of 1990 which drove up the world price of oil,
decreased consumer confidence, and exacerbated the downturn that was already underway.

Stock Markets

Recession of 2001

The 2001 recession resulted from the Y2K scare (i.e. Year 2000).
In 1999, there was an economic boom in computer and software sales
as many companies and individuals bought new computer systems
to make sure their software was Y2K compliant.
As a result, the stock price of many high-tech companies started to increase.
Investors' began buying stock in any high tech company, whether they were showing profits or not.
The exuberance for dot.com companies became irrational. The boom led to a bust in dot-com businesses.
As a result, the stock market dropped in March 2000.
As stock prices declined, so did the value of the dot.com companies and many went bankrupt.

The Federal Reserve ignored the markets and continued raising interest rates.
Interest rates remained high when the economy needed low rates for cheap credit.

In 2001, the 9/11 attack worsened the downturn.
The New York Stock Exchange closed for four trading days after the attacks.
When the stock market reopened, the Dow Jones Industrial Average fell 7 percent in one day.

The Bush administartion helped to end the recession
with expansionary fiscal policy (e.g. tax cuts, and more government spending).

Stock Markets

Financial Crisis of 2007-2008

It began in 2007 with a crisis in the subprime mortgage market in the United States,
and developed into a full-blown international banking crisis
with the collapse of the investment bank Lehman Brothers on September 15, 2008.
Excessive risk-taking by banks such as Lehman Brothers
helped to magnify the financial impact globally.
Massive bail-outs of financial institutions
and other palliative monetary ("quantitative easing")
and fiscal policies
were employed to prevent a possible collapse of the world financial system.

The crisis was nonetheless followed by a global economic downturn, the Great Recession.

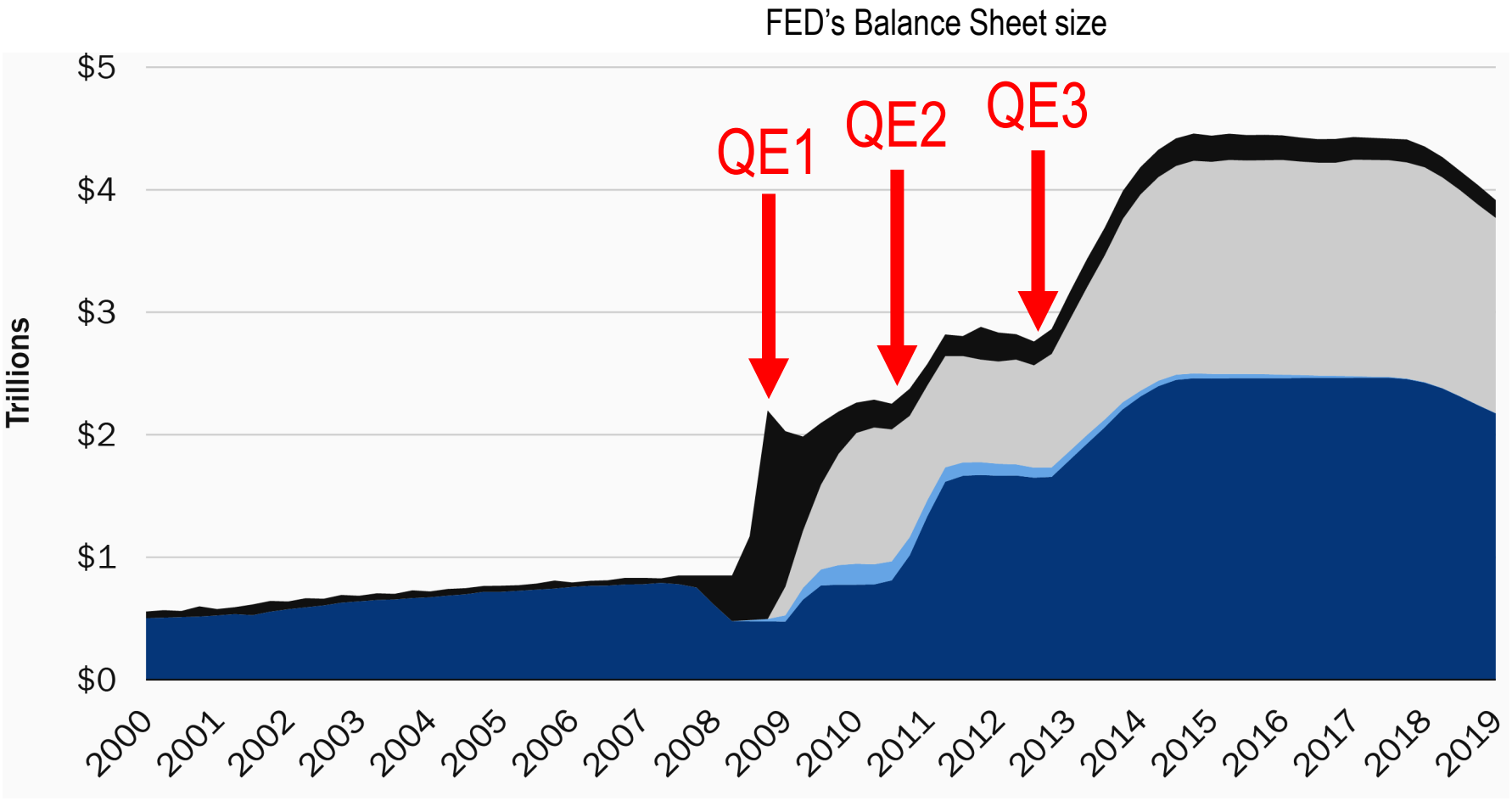
The Europen debt crisis,
a crisis in the banking system of the European countries using the euro, followed later.

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Stock Markets

Quantitative Easing

Quantitative easing (QE) is a monetary policy, implemented by the Central Bank, primarily to energize the economy.
The Central Bank will create money to buy government securities from the market in order to lower interest rates and increase the money supply.
These economic conditions will then trigger financial institutions to promote increased lending and to make money supply more liquid.



Financial Crises

Basics of the World Financial Structure

Stocks are not the most important source of external financing for businesses.
Issuing marketable debt and equity securities is not the primary way
in which businesses finance their operations either.
(Bank Loans, Non-Bank Loans, Bonds, Stocks)

Indirect finance, which involves the activities of financial intermediaries,
is many times more important than direct finance,
in which businesses raise funds directly from lenders in financial markets.

Financial intermediaries, particularly banks, are the most important source of external funds
used to finance businesses.

The financial system is among the most heavily regulated sectors of the economy.

Only large, well-established corporations
have easy access to securities markets to finance their activities.

Collateral is a prevalent feature of debt contracts for both households and businesses.
Collateral is a property that is pledged to a lender to guarantee payment
in the event that the borrower is unable to make deby payments.

Debt instruments typically are extremely complicated legal documents
that place substantial restrictions on the behavior of the borrower.

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Financial Crises

Dynamics in Advanced Economies

Stage One:
Initiation of Financial Crisis

a - Mismanagement of

Financial Innovation

introduction of new types of loans or other financial products

and/or Liberalization

elimination of restrictions on financial markets and institutions:
which can promt financial institutions to go on a lending spree, called a credit boom
even with proper management
credit booms eventually outstrip the ability of institutions – and government regulators –
to screen and monitor credit risks,
leading to overly risky lending

eventually, losses on loans begin to mount,
and the value of the loans (on the asset side of the balance sheet) falls relative to liabilities,
thereby driving down the net worth (capital) of banks and other financial institutions

with less capital, these financial institutions cut back on their lending to borrower-spenders

Financial Crises

Dynamics in Advanced Economies

Stage One:
Initiation of Financial Crisis

b – Asset-Price Boom and Bust

Prices of assets such as equity shares and real estate
can be driven by investor psychology well above thier fundemental economic values,
that is, their values based on realistic expectations of the assets' future income streams.
The rise of asset prices above their fundemental economic values is an asset-price bubble.
Asset-price bubbles are often also driven by credit booms,
in which the large increase in credit is used to fund purchases of assets,
thereby driving up their price.

c – Increase in Uncertainty

With information hard to come by in a period of high uncertainty,
financial frictions increase, reducing lending and economic activity.

Financial Crises

Dynamics in Advanced Economies

Stage One:
Initiation of Financial Crisis

Stage Two:
Banking Crisis

Deteriorating balance sheets and tougher business conditions lead some financial institutions into insolvency, when net worth becomes negative.

Unable to pay depositors or other creditors, some banks go out of business.

If severe enough, these factors can lead to a bank panic, in which multiple banks fail simultaneously.

Eventually, public and private authorities shut down insolvent firms and sell them off or liquidate them.

Uncertainty in financial markets declines, the stock market recovers, and balance sheets improve.

Financial frictions diminish and the financial crisis subsides.

With the financial markets able to operate well again,
the stage is set for an economic recovery.

Financial Crises

Dynamics in Advanced Economies

Stage One: Initiation of Financial Crisis

Stage Two: Banking Crisis

Stage Three:
Debt Deflation

If, however, the economic downturn leads to a sharp decline in the price level,
the recovery process can be short-circuited.
In economies with moderate inflation, which characterizes most advanced economies, many debt contracts
with fixed interest rates are typically of fairly long maturity, ten years or more.
Because debt payments are contractually fixed in nominal terms,
an unanticipated decline in the price level raises the value of borrowing firms' liabilities in real terms
(increases the burden of debt) but does not raise the value of borrowing firms' assets.
The borrowing firms' net worth in real terms
(the difference between asets and liabilities in real terms) thus declines.

And, this causes borrowing / lending apetite to decline,
thus the economy cools down.

Financial Crises

Causes of 2007-2009 Financial Crisis

a – Financial innovations in the mortgage markets

Advances in computer technology and new statistical techniques, known as data mining, led to enhanced, quantitative evaluation of the credit risk for a new class of risky residential mortgages.
By lowering transaction costs, computer technology enabled the bundling together of smaller loans (like mortgages) into standard debt securities, a process known as securitization.
These factors made it possible for banks to offer subprime mortgages to borrowers with less-than-stellar credit records.

b – Agency problems in the mortagage markets

The mortgage brokers that originated the loans often did not make a strong effort to evaluate whether the borrower could pay off the loan, since they would quickly sell (distribute) the loans to investors in the form of mortgage-backed securities.
Once the mortgage broker earns his fee, why should he care if the borrower makes good on his payment?
Risk-loving investors lined up to obtain loans to acquire houses that would be very profitable if housing prices went up, knowing they could "walk away" if housing prices went down.

c – Asymmetric information and Credit-Rating Agencies

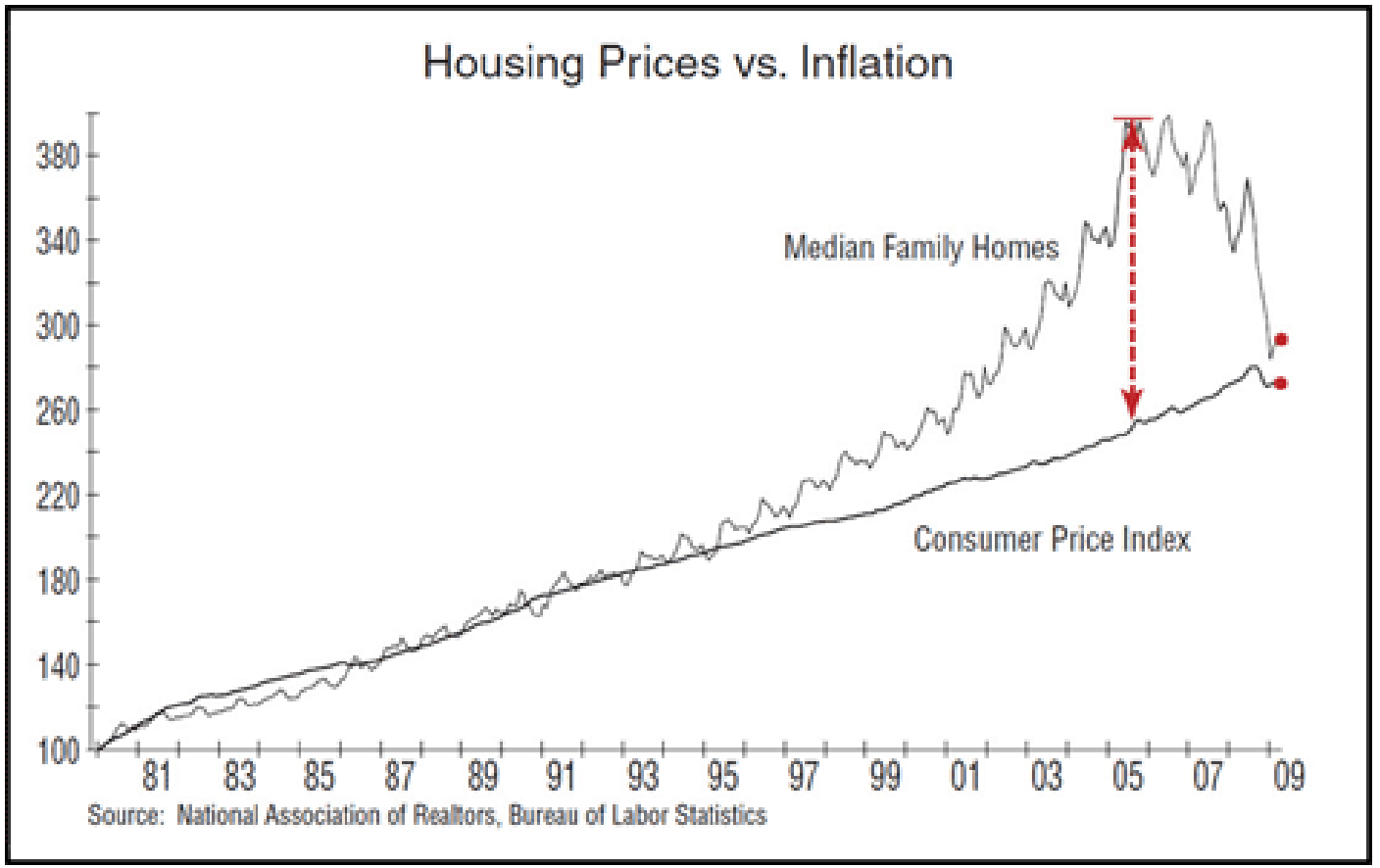
The rating agencies advised clients on how to structure complex financial instruments, at the same time they were rating these identical products.
They were thus subject to conflicts of interest because the large fees they earned from advising clients on how to structure products they were rating meant that they did not have sufficient incentives to make sure that their rating were accurate.

Financial Crises

Effects of 2007-2009 Financial Crisis

Aided by liquidity from huge cash inflows into the United States from countries like China and India,
and low interest rates on residential mortgages,
the subprime mortgage market took off after the recession ended in 2001.
It was lauded by economists and politians alike because it led to a "democratization of credit"
and helped raise US homeownership rates to the highest levels in history.

Housing priced boomed from 2002 to 2006, fueling the market for subprime mortgages and forming an asset-price buble. Housing prices began declining in 2006, falling by more than 30% subsequently, which led to defaults by subprime mortgage holders.

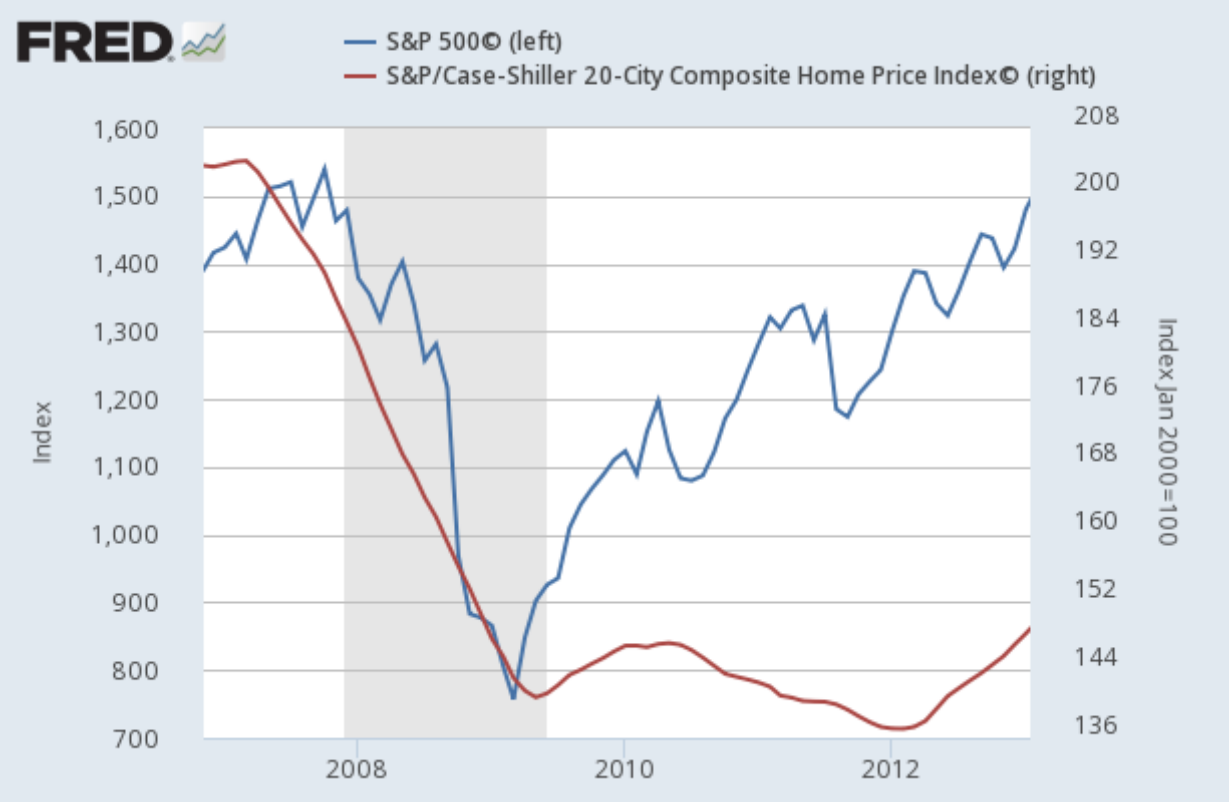


Financial Crises

Effects of 2007-2009 Financial Crisis

The sharp decline in the value of mortgages and other financial assets triggered a run on the shadow banking system, composed of hedge funds, investment banks, and other nondepository financial firms, which are not as tightly regulated as banks.

To raise funds, financial institutions had to engage in fire sales and sell off their assets very rapidly. Because selling assets quickly requires lowering their price, the fire sales led to a further decline in financial institutions' asset values.



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Financial Crises

Dynamics in Emerging Market Economies

Stage One:
Initiation of Financial Crisis

a - Mismanagement of Financial Liberalization and/or Globalization

opening up economies to flows of capital and financial firms from other countries

b – Asset-Price Boom and Bust

c – Increase in Uncertainty

d – Severe fiscal imbalances

When governments in emerging market countries face large fiscal imbalances and cannot finance their debt, they often cajole or force banks to purchase government debt.
If investors lose confidence in the ability of the government to repay its debt, the banking system weakens.
Also, with less free resources, banks will lend less.

e – Increase in interest rates

When interest rate rise, high-risk firms are most willing to pay the high interest rates, so the asset quality of banks deteriorates.

Financial Crises

Dynamics in Emerging Market Economies

Stage One: Initiation of Financial Crisis

Stage Two: Currency Crisis

As the effects of any or all of the factors mentioned previously build on each other,
participants in the foreign exchange market sense an opportunity:
they can make huge profits if they bet on a depreciation of the currency.
Speculators engage in massive sales of the currency.
As the currency sales flood the market, supply far outstrips demand,
the value of the currency collapses, and a currency crisis ensues.
High interest rates abroad, increases in uncertainty, and falling asset prices all play a role.

Stage Three: Full-Fledged Financial Crisis

In contrats to most advanced economies that typically denominate debt in domestic currency,
emerging market economies denominate many debt contracts in foreign currency
leading to what is referred to as currency mismatch.
An unanticipated depreciation or devaluation of the domestic currency in emerging market countries
increases the debt burden of domestic firms in terms of domestic currency.
The collapse of a currency also can lead to higher inflation as prices of imported goods go up.

NTE 1010
Introduction to Finance for Engineers

week 9

Banking

Banks

Balance Sheet

The balance sheet of commercial banks
can be thought of as
a list of the sources and uses of bank funds.

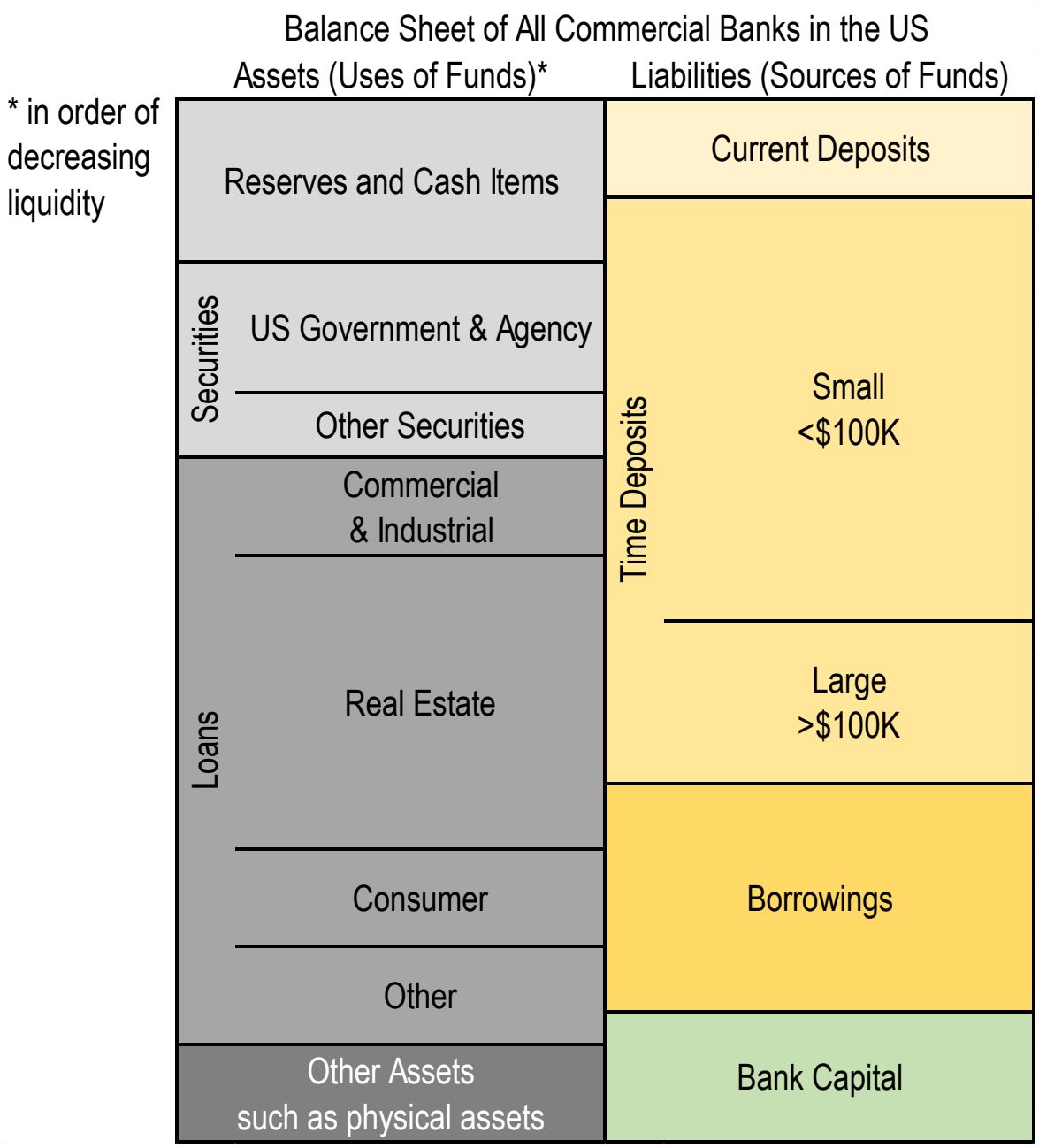
A bank's liabilities are its sources of funds,
which include checkable deposits, time deposits, discount loans from the Fed, borrowings from other banks and corporations, and bank capital.

A bank's assets are its uses of funds,
which include reserves, cash items in process of collection, deposits at other banks, securities, loans, and other assets (mostly physical capital).

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Banks

Balance Sheet



Banks

Asset Transformation

Banks make profits through the process of asset transformation:
They borrow short (accept short-term deposits)
and lend long (make long-term loans).

When a bank takes in additional deposits, it gains an equal amount of reserves;
when it pays out deposits, it loses an equal amount of reserves.

Banks

Bank Management

The bank manager has four primary concerns.

1. To make sure that the bank has enough ready cash to pay its depositors when there are deposit outflows

– that is, when deposits are lost because depositors make withdrawels and demand payment.
To keep enough cash on hand, the bank must engage in Liquidity Management,
the acquisition of sufficiently liquid assets to meet the bank's obligations to depositors.

2. Pursuing an acceptably low level of risk
by acquiring assets that have a low rate default
and by diversifying asset holdings (Asset Management).

3. Acquring funds at low cost (Liability Management).

4. Deciding the amount of capital the bank should maintain
and then acquiring the needed capital (Capital Adequacy Management).

Banks

Banking Risks

Primary risks in banking are as follows.

1. Credit Risk

The risk arising because borrowers may default.

2. Interest-rate Risk

The riskiness of earnings and returns on bank assets that results from interest-rate changes.

Banks

Bank Management

Although more-liquid assets tend to earn lower returns, banks still desire to hold them.

Specifically, banks hold excess and secondary reserves
because they provide insurance against the costs of a deposit outflow.

Banks manage their assets to maximize profits
by seeking the highest returns possible on loans and securities
while at the same time trying to lower risk and making adequate provisions for liquidity.

Although liability management was once a staid affair,
large (money center) banks now actively seek out sources of funds
by issuing liabilities such as negotiable CDs
or by actively borrowing from other banks and corporations.

Banks manage the amount of capital they hold
to prevent bank failure
and to meet bank capital requirements set by the regulatory authorities.
However, they do not want to hold too much capital
because by so doing they will lower the returns to equity holders.

Banks

Equity Multiplier

$$\frac{\text{net profit after taxes}}{\text{equity capital}} = \frac{\text{net profit after taxes}}{\text{assets}} \times \frac{\text{assets}}{\text{equity capital}}$$

Return on Equity		Return on Assets		Equity Multiplier
RoE	=	RoA	x	EM

Higher bank capital benefits the owners of a bank
in that it makes their investment safer by reducing the likelyhood of bankruptcy.

But bank capital is costly
because the higher it is, the lower will be the return on equity
for a given return on assets.

The bank management needs to aim for an optimal capital level
balancing the avoidance of bankruptcy and the maximization of return on equity.

Banks

Credit Risk Management

Credit risk management principles
involving loan activities
are the following:

1. screening and monitoring
2. long-term customer relationships
3. loan commitments
4. collateral and compensating balances
5. credit rationing

Banks

Credit Risk Management

Screening and Monitoring:

Asymmetric information is present in loan markets because lenders have less information about the investment opportunities and activities of borrowers than borrowers do.
This situation leads to two information-producing activities by banks and other financial institutions – screening and monitoring.

For effective screening, lenders must collect reliable information from prospective borrowers.
Bank collects information about the company's profits and losses (income) and about its assets and liabilities.
The lender also has to evaluate the likely future success of the business.
So, in addition to obtaining information on such items as sales figures,
a loan officer might ask questions about
the company's future plans, the purpose of the loan, and the competition in the industry.
The officer may even visit the compant to obtain a firsthand look at its operations.

A lender should write provisions (restrictive covenants) into loan contracts
that restrict borrowers from engaging in risky activities.
By monitoring borowers' activities to see whether they are complying with the restrictive covenants
and by enforcing covenants if they are not,
lenders can make sure that borrowers are not taking risks at their expense.

Banks

Credit Risk Management

Long-term Customer Relationships:

- Reduce the costs of information collection and make it easier to screen out bad credit risks.
  - The costs of monitoring long-term customers are lower.
  - Long-term relationships benefit the customer as well as the bank.
- The long-term customer has the incentive to avoid risky activities that would upset the bank, even if restrictions on these risky activities are not specified in the loan contract.

Banks

Credit Risk Management

Loan Commitments:

A loan commitment is a bank's commitment (for a specified future period of time)
to provide a firm with loans up to a given amount
at an interest rate that is tied to some market interest rate.

The advantage for the firm is that it has a source of credit when it needs it.

The advantage for the bank is that the loan commitment promotes a long-term relationship,
which in turn facilitates information collection.
In addition, provisions in the loan commitment agreement require that
the firm continually supply the bank with information
about the firm's income, asset and liability position, business activities, and so on.

Banks

Credit Risk Management

Collateral and Compensating Balances:

Collateral, which is property promised to the lender as compensation if the borrower defaults
lessens the adverse consequences
because it reduces the lender's losses in the case of a loan default.

One particular form is called Compensating Balances:
A firm receiving a loan is asked to keep
a required minimum (on the average) amount of funds in a checking account at the bank.

The bank can observe the firm's check payment practices,
which may yield a great deal of information about the borrower's financial condition.

Banks

Credit Risk Management

Credit Rationing:

Credit rationing takes two forms.

- A lender refuses to make a loan of any amount to a borrower, even if the borrower is willing to pay a higher interest rate. Individuals and firms with the riskiest investment projects are exactly those that are willing to pay the highest interest rates.

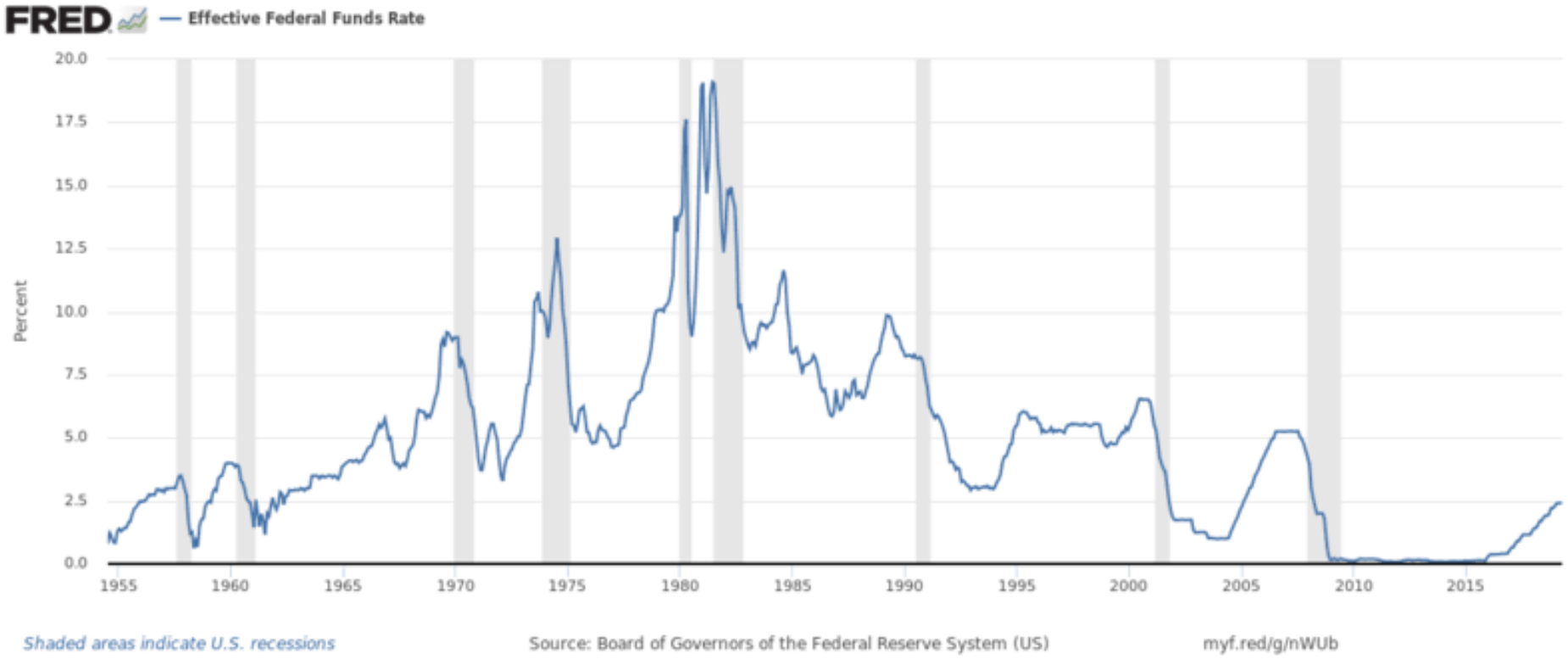
- A lender is willing to make a loan but restricts the size of the loan to less than the borrower would like. Because more borrowers repay their loans if the loan amounts are small, financial institutions ration credit by providing borrowers with smaller loans than they seek.

Banks

Interest-Rate Risk

With the increased volatility of interest rates that occured in the 1980s,
the financial institutions became more concerned about their exposure to interest-rate risk.

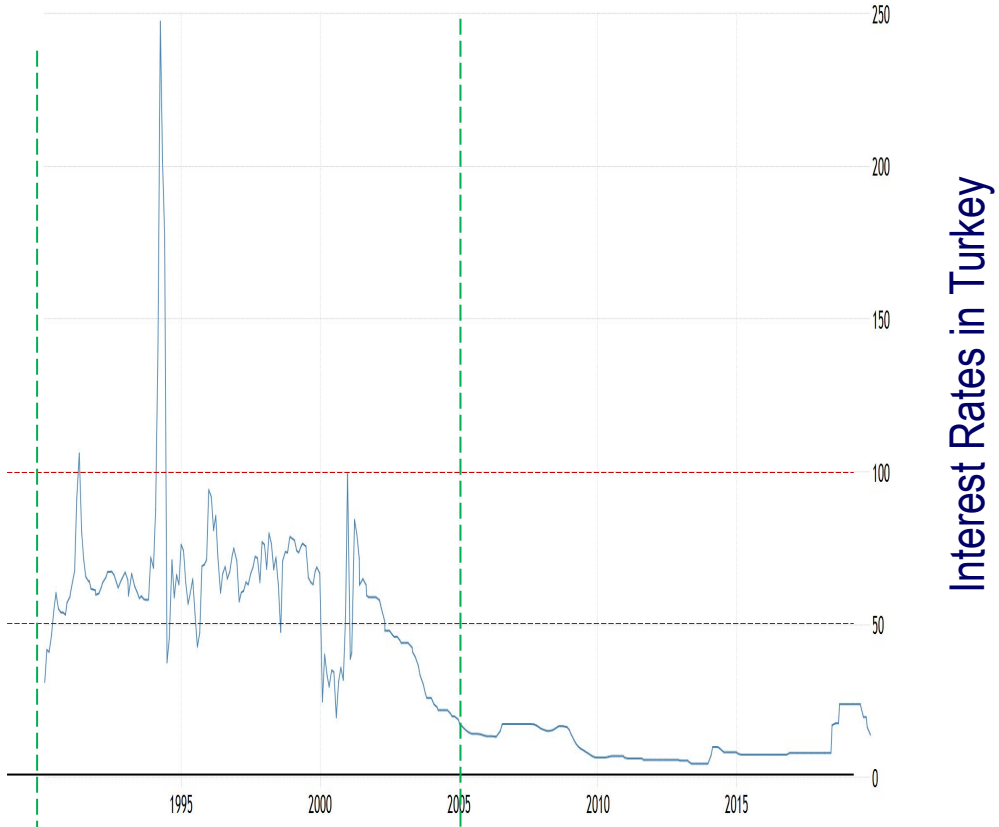
US Interest Rates

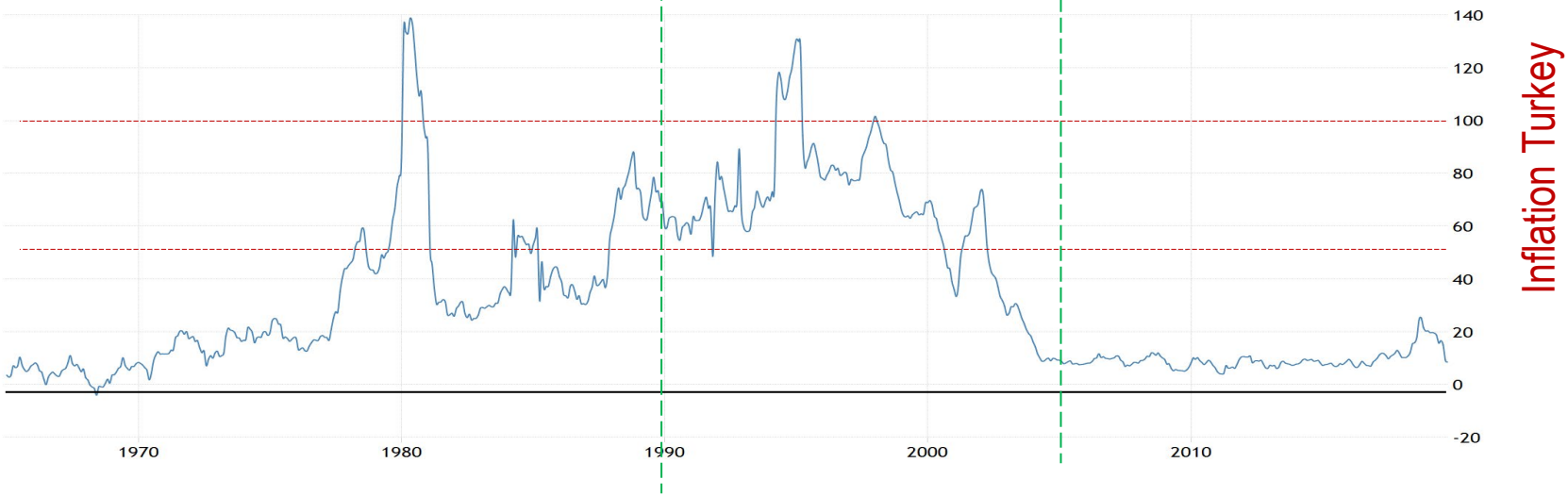


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Interest-Rate Risk





Banks

Estimating the Size of the Interest-Rate Risk

Gap and duration analyses tell a financial institution
if it has more rate-sensitive liabilities than assets
["rate-sensitive" = with interest rates that change frequently (at least once a year)
"fixed-rate" = with interest rates that remain unchanged for a long period (over a year)]
in which case a rise in interest rates will reduce profits
and a fall in interest rates will raise profits.

Gap = rate-sensitive Liabilities – rate-sensitive Assets
By multiplying the Gap times change in interest rate,
we can immediately obtain the effect on bank profits.

Duration = the average lifetime of a security's stream of payments
By multiplying the weighted average duration of a bank's assets and of its liabilities
times the amount of assets and of liabilities
times change in interest rate,
we can immediately obtain the approximate effect on bank's total asset value.

Financial institutions manage their interest-rate risk by modifying their balance sheets
but can also use strategies involving financial derivatives.

Banks

Estimating the Size of the Interest-Rate Risk

Gap Analysis

The Bank

Assets	m$	Liabilities	m$
Rate-sensitive Assets	20	Rate-sensitive Liabilities	50
Variable-rate and short-term loans		Variable-rate Certificates of Deposits	
Short-term securities		Money market deposit accounts	
Fixed-rate Assets	80	Fixed-rate Liabilities	50
Reserves		Current account deposits	
Long-term loans		Savings deposits	
Long-term securities		Equity Capital	

Gap = rate-sensitive Liabilities – rate-sensitive Assets

Gap = 50 -20 = 30 million $

For example, if interest rates fall by 5%, then the bank's profits increase by 30 x 5% = 1.5 million $

Banks

Estimating the Size of the Interest-Rate Risk

Duration Analysis

Duration = the average lifetime of a security's stream of payments

"Duration" concept approximates
the sensitivity of a security's market value (*MV*) to a change in its interest rate (i).

% Δ in *MV* of security ~ -% Δ in i * duration in years

average weighted duration (years)	The Bank	m$		m$	years
3	Assets	100	Liabilities	90	2
			Equity Capital	10	

For example, if interest rates fall by 5%, then the change in bank's net worth would be

100 m$ * 5% * 3 years = 15 m$
-90 m$ * 5% * 2 years = -9 m$

15 -9 = 6 m$ increase.

Banks

Off-Balance-Sheet Activities

Off-balance-sheet activities consist of trading financial instruments
and generating income from fees and loan sales,
all of which affect bank profits but are not visible on bank balance sheets.

Because these off-balance-sheet activities expose banks to increased risk,
bank management must pay particular attention to
risk assessment procedures and internal controls
to restrict employees from taking on too much risk.

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Central Banks

Central Banks

Among the most important players in financial markets throughout the world are central banks, the government authorities in charge of monetary policy.
(Governments are in charge in charge fiscal policy.)

Central banks' actions affect interest rates, the amount of credit, and the money supply, all of which have direct impacts not only on financial markets, but also on aggregate output and inflation.

Central Banks

US Federal Reserve System (Fed)

The Federal Reserve System was created in 1913 to lessen the frequency of bank panics.
Because of public hostility to central banks and the centralization of power,
the system was created with many checks and balances to diffuse power.
It consists of 12 regional Federal Reserve banks,
around 2,500 member commercial banks,
the Board of Governors, the Federal Open Market Committee (FOMC),
and the Federal Advisory Council.

The case for an independent Fed rests on the view that
curtailing its independence and subjecting it to more political pressures
would impart an inflationary bias to monetary policy.
An independent Fed can afford to take the long view and not respond to short-term problems
that would result in expansionary monetary policy and a political business cycle.

The case against an independent Fed holds that it is undemocratic to have a monetary policy (so important to the public) controlled by an elite that is not accountable to the public.
An independent Fed also makes the coordination of monetary and fiscal policy dificult.

Central Banks

EU

The European System of Central Banks has
a similar structure to the Federal Reserve System of US,
with each member country having a National Central Bank,
and an Executive Board of the European Central Bank
being located in Frankfurt, Germany.

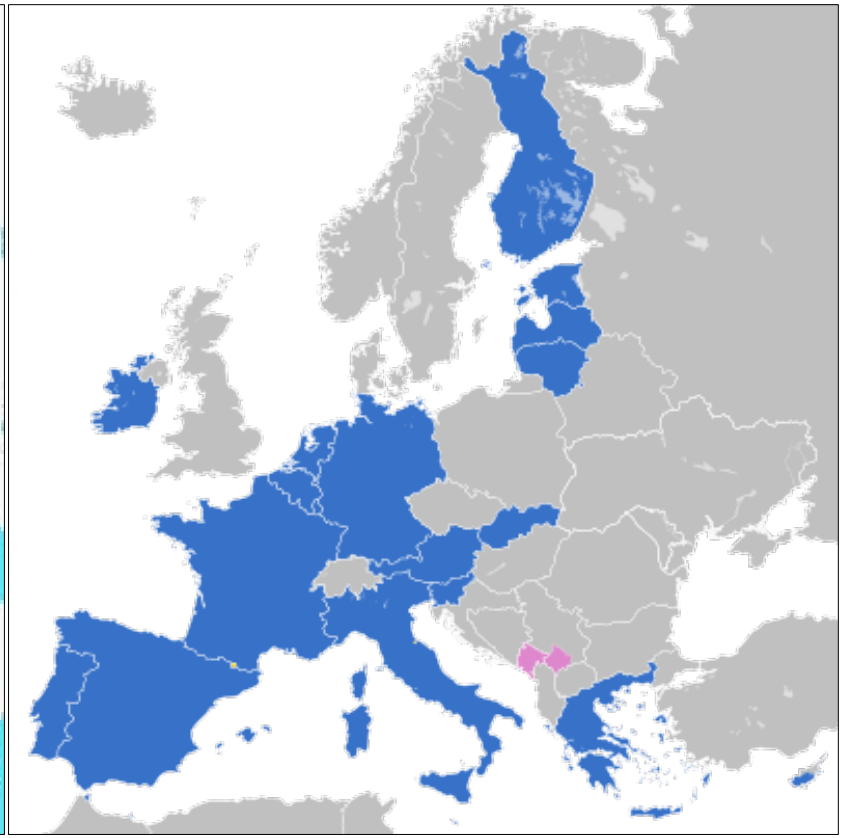
The Governing Council,
which is made up of the six members of the Executive Board
(which includes the president of the European Central Bank)
and the presidents of the National Central Banks,
makes the decision on monetary policy.

The Eurosystem, which was established under the terms of the Maastricht Treaty,
is even more independent than the Federal Reserve System
because its charter cannot be changed by legislation.

Central Banks

EU & Euro Zone









Central Banks

Turkish Central Bank

In the Ottoman Empire, gold coins were minted on behalf of the Sultan.
The empire put cash banknotes into circulation in 1840.
During the Crimean War, in 1854,
the Ottoman Empire, which borrowed from other nations for the first time in history,
needed a state bank to assume an intermediary function in the repayment of external debts.
As a result, the “Ottoman Bank”, headquartered in London, was established with English capital in 1856.
In 1863, the Ottoman Bank was dissolved and restructured as an English-French partnership
under the name “Imperial Ottoman Bank” and became a state bank.
The Imperial Ottoman Bank was granted the sole privilege of issuing banknotes for a period of thirty years.
The Bank, acting as Treasurer of the State, was assigned to collect State revenues,
make payments on behalf of the Treasury and discount Treasury bills,
as well as making interest and principal payments pertaining to domestic and foreign debts.

After the signing of Lausanne Treaty and the declaration of the Republic in 1923,
the preparations to establish a central bank of Turkey began in 1926,
but the organization was established on 3 October 1931
and opened officially on 1 January 1932.
The Bank had, originally, a privilege of issuing banknotes for a period of 30 years.
In 1955, this privilege was extended until 1999.
Finally it was prolonged indefinitely in 1994.

Central Banks

Turkey – CBRT / TCMB

The Law on the Central Bank of the Republic of Turkey, No. 1715 was enacted on 11 June 1930.
According to that law, the basic aim of the Bank was
to support economic development of the country.

In the recent past, however, the primary objective of the Bank was changed to be
to achieve and maintain price stability.
The Bank shall determine on its own discretion the monetary policy that it shall implement
and the monetary policy instruments that it is going to use
in order to achieve and maintain price stability.

Central Banks

Fundamental Powers of CBT (TCMB)

Fundamental powers of Central Bank of Turkey are:

- exclusive right to issue banknotes in Turkey
- right to determine of the inflation target
  and implement of monetary policy
- role of the lender of last resort for financial institutions
- power to request information from financial institutions

Money Supply

Players

The "cast of characters" in the money supply theory is as follows:

[1] The central bank,
the government agency that oversees the banking system
and is responsible for the conduct of monetary policy,

[2] Banks (depository institutions),
the financial intermediaries that accept deposits from individuals and institutions
and make loans,

[3] Depositors – individuals and institutions.

Of the three players, the central bank is the most important.
The central bank's conduct of monetary policy involves actions that affect its balance sheet.

Money Supply

Balance Sheet of the Central Bank

The Central Bank	
Assets	Liabilities
Securities	Curency in circulation
Loans to Financial Institutions	Reserves

Currency in circulation (C):

Currency held by the public; currency held by depository institutions is counted as part of the Reserves.

Reserves (R):

All banks have an account at the Central Bank in which they hold deposits.

Required reserves are those that the Central Bank requires banks to hold.

Any additional reserves the banks choose to hold are called excess reserves.

The Central Bank might require that for every dollar of deposits at a depository institution, a certain fraction (say, 10 cents) must be held as reserves.

This fraction (10%) is called the required reserve ratio.

The sum of the Currency in Circulation and the Reserves is called the monetary base: MB = C + R

Securities:

The Central Bank's holdings of securities issued by the Government most of the time, other securities in unusual circumstances.

Loans to Financial Institutions:

During normal times, the Central Bank makes loans only to banking institutions and the interest rate charged banks for these loans is called the discount rate.

Money Supply

Control of the Monetary Base

The primary way in which the Central Bank causes changes in the monetary base is through its open market operations;
the Central Bank buys/sells bonds from/to banks.
Buying expands MB (=C+R) by increasing reserves of banks held at the Central Bank.
Selling contracts MB.

Lending to financial institutions also expands MB by increasing reserves.

The Central Bank sets the discount rate (interest rate on loans to banks), and then banks make decisions about whether to borrow.
The component of Monetary Base which is created by loans from the Central Bank (called borrowed reserves)
is less tightly controlled
compared to the component that is created by open market operations
which is called the nonborrowed monetary base.

Money Supply

Multiple Deposit Creation

When the Central Bank supplies the banking system with 1$ of additional reserves,
deposits increase by a multiple of this amount,
a process called "multiple deposit creation".

For example, when the Central Bank buys 100 m$ worth bonds from the First National Bank, the bank finds that it has an increase in reserves of 100 m$.
Let us say the bank decides to make a loan
equal in amount to the 100 m$ rise in excess reserves.

The borrower deposits the borrowed amount of 100 m$ to Bank A.
If the required reserve ratio is 10%,
Bank A will now find itself with a 10 m$ increase in required reserves,
leaving it 90 m$ of excess reserves which it can lend.

The new borrower deposits the borrowed amount of 90 m$ to Bank B.
Since the required reserve ratio is 10%,
Bank B will now find itself with a 9 m$ increase in required reserves,
leaving it 81 m$ of excess reserves which it can lend.

The process goes on, eventually leading to a 1,000 m$ expansion in deposits.

Δ Deposits = Δ Reserves * Simple Deposit Multiplier

where Simple Deposit Multiplier = 1 / Required Reserve Ratio

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Money Supply

Multiple Deposit Creation

Central Bank's purchase of 100 m$ worth bonds
from the First National Bank
eventually leads to
a 400 m$ expansion in deposits
with a 25% Required Reserve Ratio.

Δ Deposits = Δ Reserves * Simple Deposit Multiplier

where
Simple Deposit Multiplier = 1 / Required Reserve Ratio

Required Reserve Ratio = 25%			
Bank	Increase in Deposits ($)	Increase in Loans ($)	Increase in Reserves ($)
First National	- - -	100.00	- - -
A	100.00	75.00	25.00
B	75.00	56.25	18.75
C	56.25	42.19	14.06
D	42.19	31.64	10.55
E	31.64	23.73	7.91
F	23.73	17.80	5.93
G	17.80	13.35	4.45
H	13.35	10.01	3.34
I	10.01	7.51	2.50
J	7.51	5.63	1.88
K	5.63	4.22	1.41
L	4.22	3.17	1.06
M	3.17	2.38	0.79
N	2.38	1.78	0.59
O	1.78	1.34	0.45
P	1.34	1.00	0.33
Q	1.00	0.75	0.25
R	0.75	0.56	0.19
S	0.56	0.42	0.14
T	0.42	0.32	0.11
U	0.32	0.24	0.08
V	0.24	0.18	0.06
W	0.18	0.13	0.04
X	0.13	0.10	0.03
Y	0.10	0.08	0.03
Z	0.08	0.06	0.02
Σ up to Z	399.77	399.83	99.94
. . .	. . .	. . .	. . .
Σ all banks	400	400	100

Money Supply

Overview

Player	Variable	Change in Variable	Money Supply Response	Reason
Central Bank	Nonborrowed Monetary Base	↑	↑	created by open market operations of the Central Bank there would be more monetary base for deposit creation
	Required Reserve Ratio	↑	↓	Simple Deposit Multiplier would decrease so less multiple deposit expansion
Banks	Borrowed Reserves	↑	↑	as banks borrow more from the Central Bank there would be more monetary base for deposit creation
	Excess Reserves	↑	↓	as banks willingly deposit more at the Central Bank there would be less loans and less deposit creation
Depositors	Currency holdings	↑	↓	as the public holds more currency and less bank deposits banks could lend less therefore less multiple deposit expansion

Note: Only increases (↑) in the variables are shown. The effects of decreases on the money supply would be the opposite of those indicated in the "Money Supply Response" column.

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Monetary Policy

Monetary Policy

Monetary Policy is the policy adopted
by the monetary authority of a country (the Central Bank)
that controls either the interest rate payable on very short-term borrowing or the money supply,
often targeting inflation or the interest rate
to ensure price stability and general trust in the currency.

Unlike Fiscal Policy which relies on government to spend its way out of recessions,
Monetary Policy aims to manipulate the money supply, i.e. 'printing' more money
or decreasing the money supply
by changing interest rates or removing excess reserves.

Monetary Policy

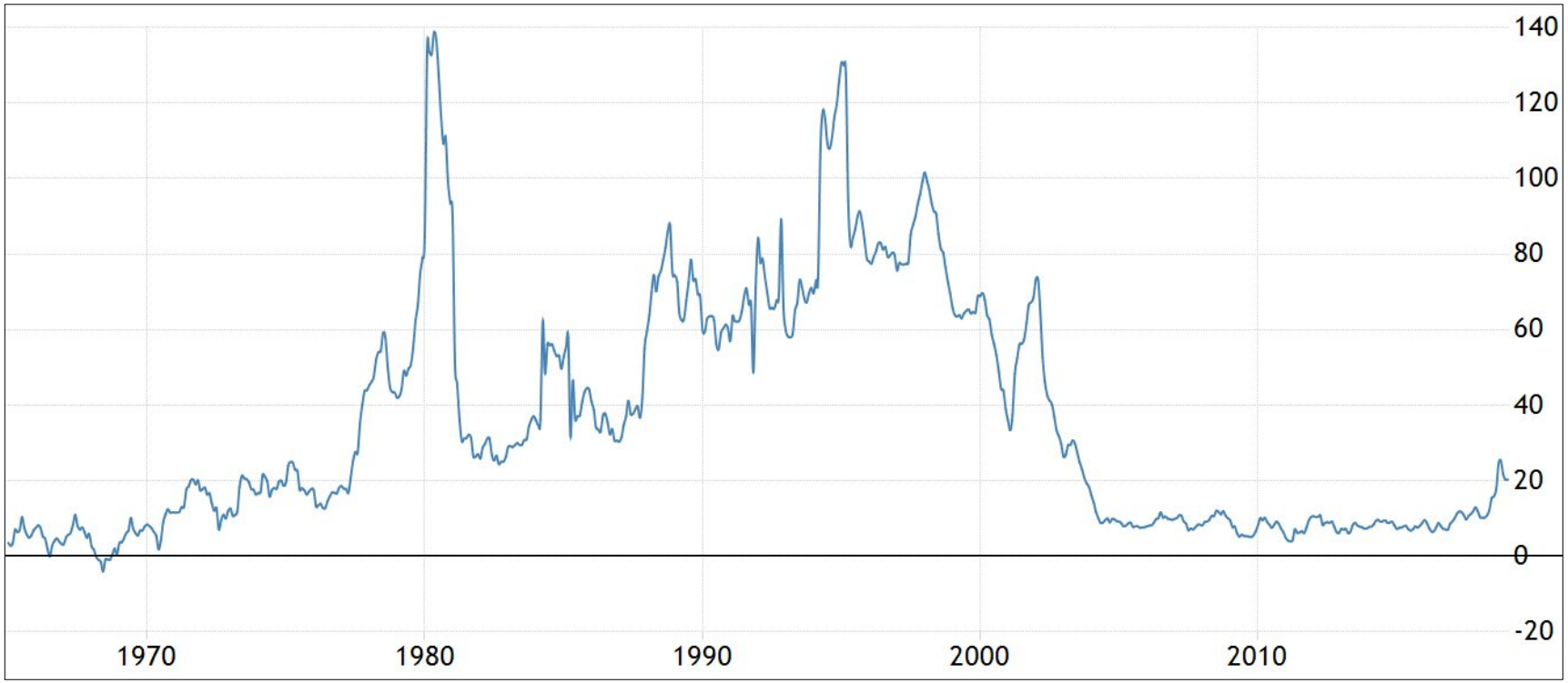
Price Stability

Getting monetary policy right is crucial to the health of the economy.
Overly expansionary monetary policy leads to high inflation,
which decreases the efficiency of the economy and hampers economic growth.
Monetary policy that is too tight can produce serious recessions
in which output falls and unemployment rises.
It can also lead to deflation, a fall in the price level,
such as occured in the US during the Great Depression (1933-1939)
and in Japan more recently.

Primary goal is the Price Stability which is defined as low and stable inflation.
It is desirable because a rising price level (inflation) creates uncertainty in the economy,
and that uncertainty might hamper economic growth.

Monetary Policy

Inflation in Turkey, Annual CPI



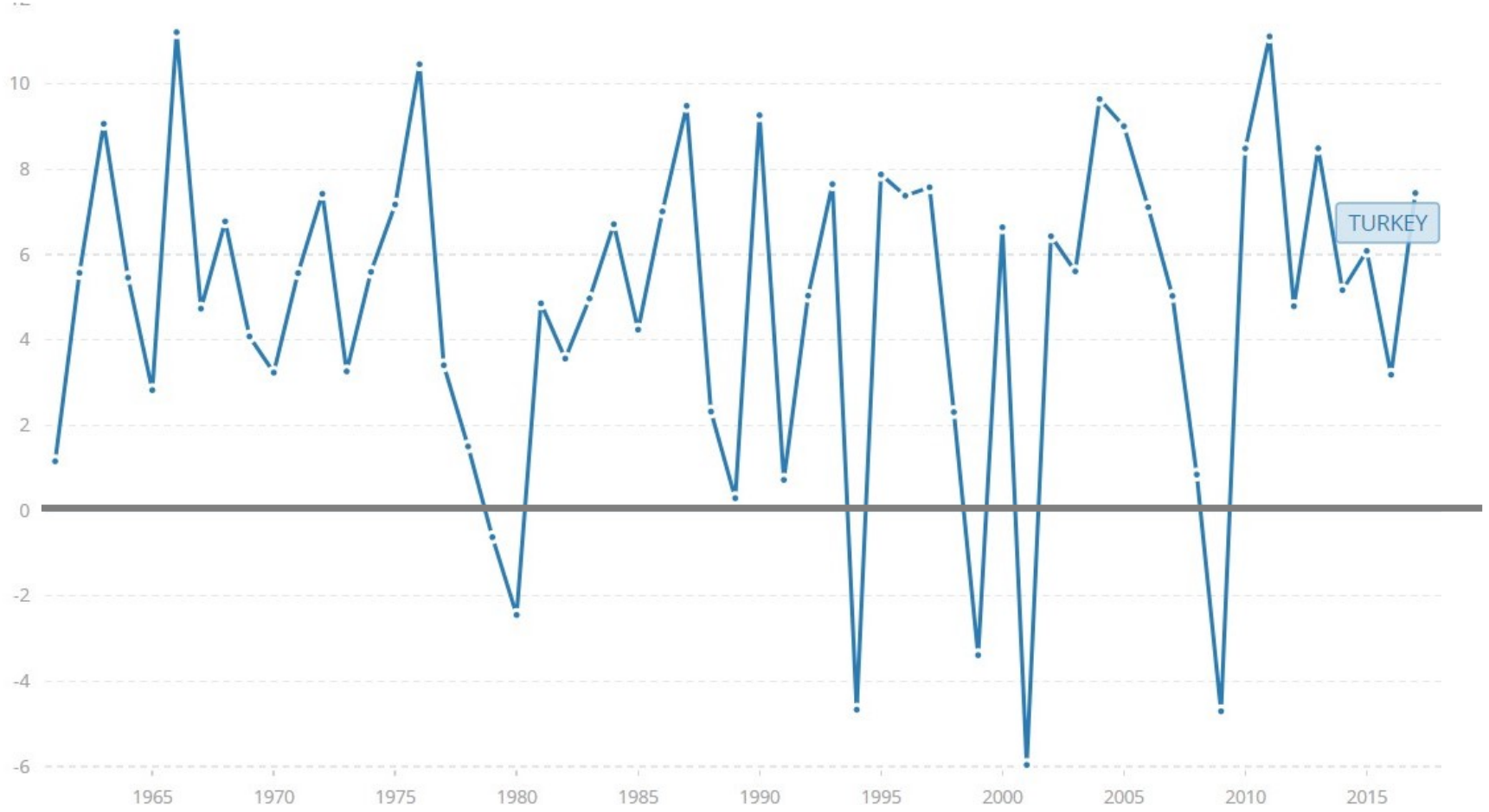
Monetary Policy

Other Goals

- High employment and economic growth
  - The goal is not zero unemplyment level but a level above zero
    that is consistent with full employment at which the demand for labor equals the supply of labor.
  - This level is called the natural rate of unemployment.
  - Since the level of unemployment is tied to the level of economic activity in the economy,
    a particular level of output is produced at the natural rate of unemployment,
    which is referred to as the natural rate of output or more commonly as the potential output.
- Stability of financial markets
  - Financial crises can interfere with the ability of financial markets
    to channel funds to people with productive investment opportunities
    and lead to sharp contraction in the economic activity.
- Interest-rate stability
  - is desirable because fluctuations in interest rates can create uncertainty in the economy
    and make it harder to plan for the future.
  - Fluctuations in interest rates affect consumers' willingness to buy houses, for example.
- Stability in foreign exchange markets
  - With the increasing importance of international trade to the economies, FX stability is crucial.

Monetary Policy

Economic Growth Rate in Turkey, Annual %



Monetary Policy

Nominal Anchor

A nominal variable such as the inflation rate or the money supply,
which ties down the price level to achieve price stability.

Having a strong nominal anchor is a key element in successful monetary policy.

Nominal anchor also limits the time-inconsistency problem,
in which monetary policymakers conduct monetary policy
in a discretionary way that focuses on short-term objectives
but produces poor long-run outcomes.

Actually, the time-inconsistency problem is something that we deal with
continually in our everyday life.

Monetary Policy

Inflation Targeting

Advantages:

- enables monetary policy to focus on domestic considerations
- readily understood by the public and is highly transparent
- increases accountability of the central bank
- appears to ameliorate the effects of inflatinary shocks

Disadvantages:

- inflation is not easily controlled by the monetary authorities,
  so that an inflation target is unable to send
  immediate signals to both the public and markets
- might impose a rigid rule on policy makers,
  although this has not been the case in practice
- a sole focus on inflation may lead to larger output fluctuations,
  although this has also not been the case in practice

In recent years, the Federal Reserve of US has had a strategy of having an implicit nominal anchor.
It has had a demonstrated success, producing low inflation
with the longest business cycle expansion in US history.

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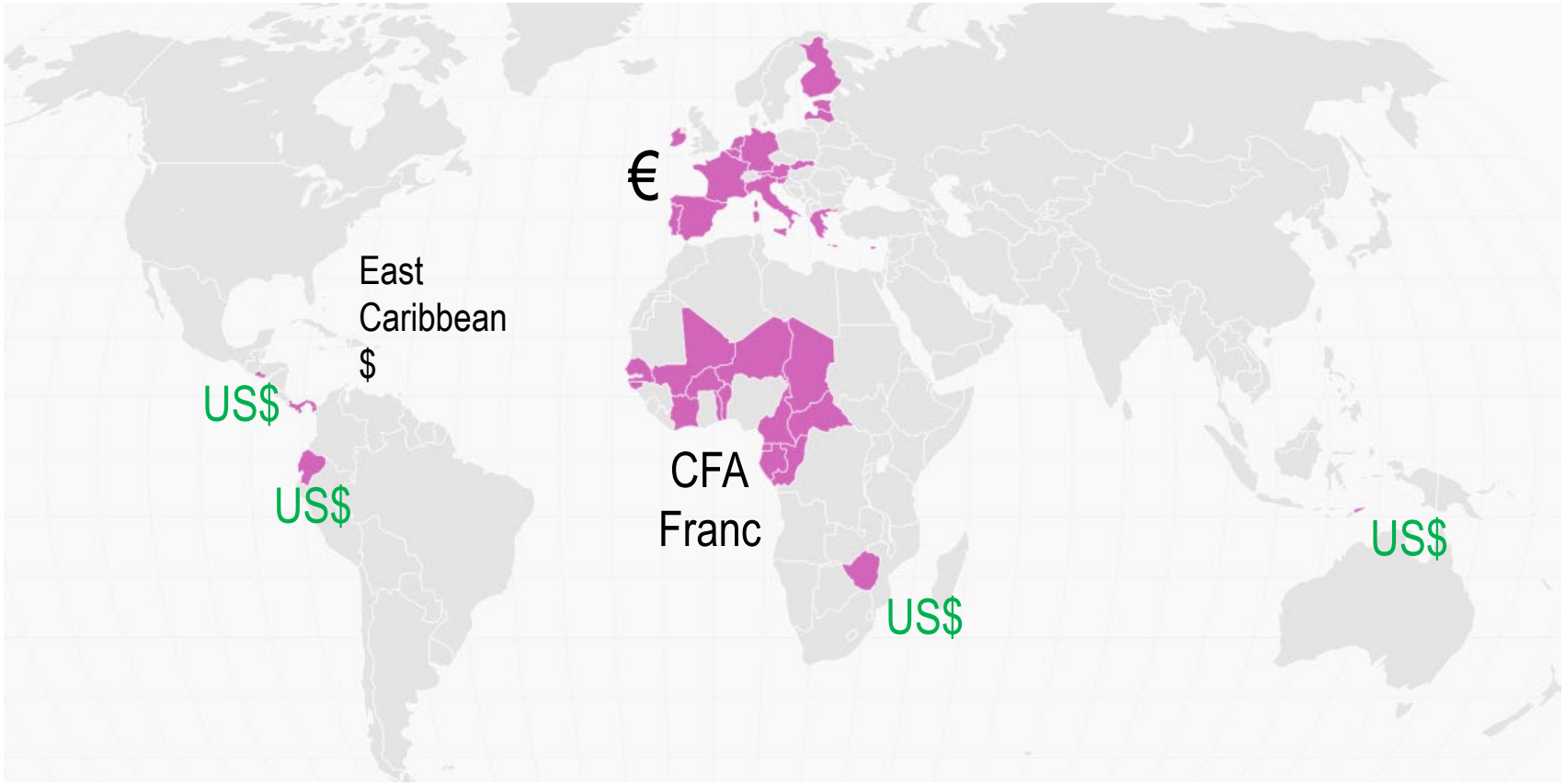
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Foreign Exchange Markets

Foreign Exchange Markets

Exchange Rate

The price of one currency in terms of another is called the Exchange Rate.
Fluctuations in the exchange rate also affect both inflation and output,
and are an important concern to monetary policymakers.
Most countries of the world have their own currencies; a few do not.



Foreign Exchange Markets

Transactions

Spot transactions involve the immediate (two-day) exchange of bank deposits at spot exchange rate.

Forward transactions involve the exchange of bank deposits at some specified future date at a predetermined forward exchange rate.

When a currency increases in value, it appreciates; when it falls in value it depreciates (or it undergoes devaluation).

Exchange rates are important because they affect the relative price of domestic and foreign goods.

When a country's currency appreciates (rises in value relative to other currencies), the country's goods become more expensive abroad and foreign goods in that country become cheaper

or vice versa.

Foreign Exchange Markets

FX Trade

FX market is organized as an over-the-counter market
in which several hundred dealers (mostly banks) stand ready to buy and sell deposits denominated in foreign currencies.

Because these dealers are in constant telephone and computer contact,
the market is very competitive;
in effect, it functions no differently from a centralized market.

The volume in this market is colossal, exceeding 4 trillion $ per day.
Sizes of individual transactions are in excess of 1 million $.

Foreign Exchange Markets

How Do FX Rates Come Out ?

Law of One Price:

If two countries produce an identical good,
and transportation costs and trade barriers are very low,
the price of the good should be the same throughout the world
no matter which country produces it.

Suppose that American steel costs 100 $ per ton
and identical Japanese steel costs 10,000 ¥ per ton.
For the Law of One Price to hold,
1 $ should be worth 100 ¥.

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Foreign Exchange Markets

How Do FX Rates Come Out ?

Theory of Purchasing Power Parity (PPP)

states that exchange rates between any two currencies
will adjust to reflect changes in the price levels of the two countries.

The theory of PPP is simply an application of the Law of One Price
to national price levels rather than to individual prices.

Suppose that the Yen price of Japanese steel rises 8% (to 10,800 ¥) due to inflation in Japan
while the Dollar price of the American steel rises 4% (to 104 $) due to inflation in the US;
then 1 $ should be worth 103.85 ¥ reflecting the inflation difference.

$$\frac{1+\text{ Japanese Inflation}}{1+\text{ American Inflation}} - 1 = \frac{1+\ 8\%}{1+\ 4\%} - 1 = \ 3.85\% \quad = \text{devaluation of Jap Yen against US\$}$$

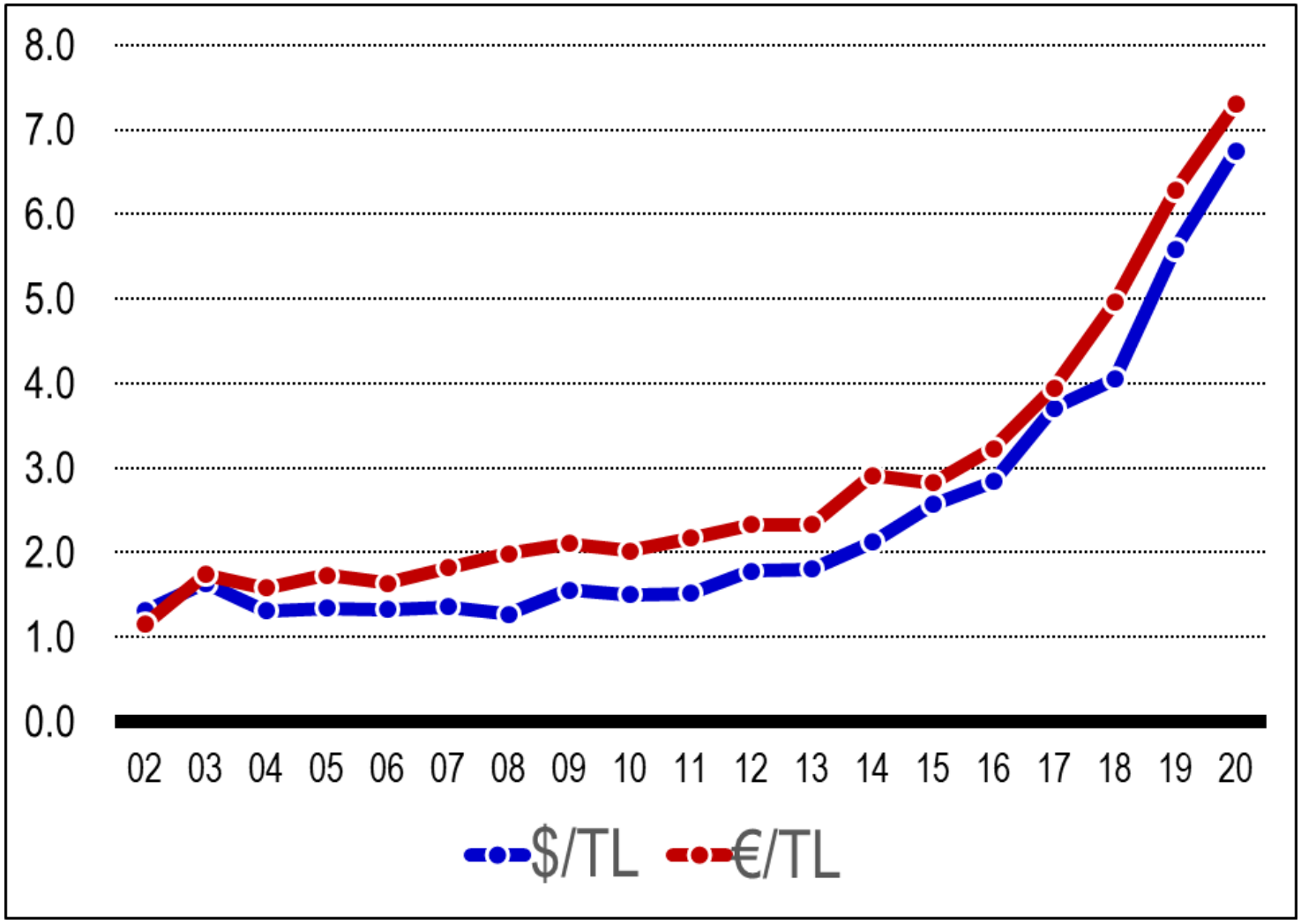
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Foreign Exchange Markets

$/TL and €/TL Exchange Rates

$ and € are the two most basic currencies in World trade and the Turkish foreign trade.





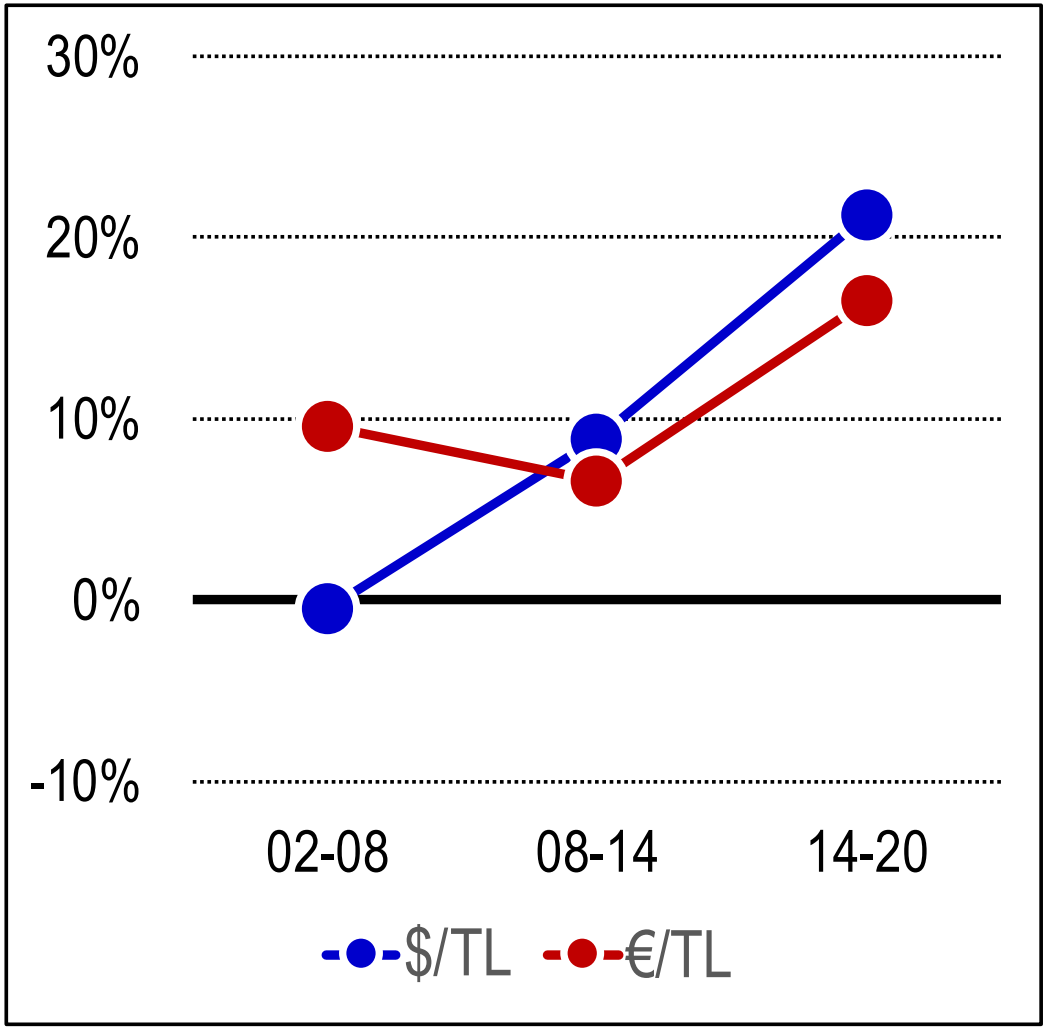
Foreign Exchange Markets



Average Annual Devaluation of TL against $ and €

April	$/TL	€/TL
2002	1.31	1.15
2008	1.27	1.99
2014	2.12	2.91
2020	6.75	7.30

	$/TL	€/TL
02-08	-1%	10%
08-14	9%	7%
14-20	21%	17%

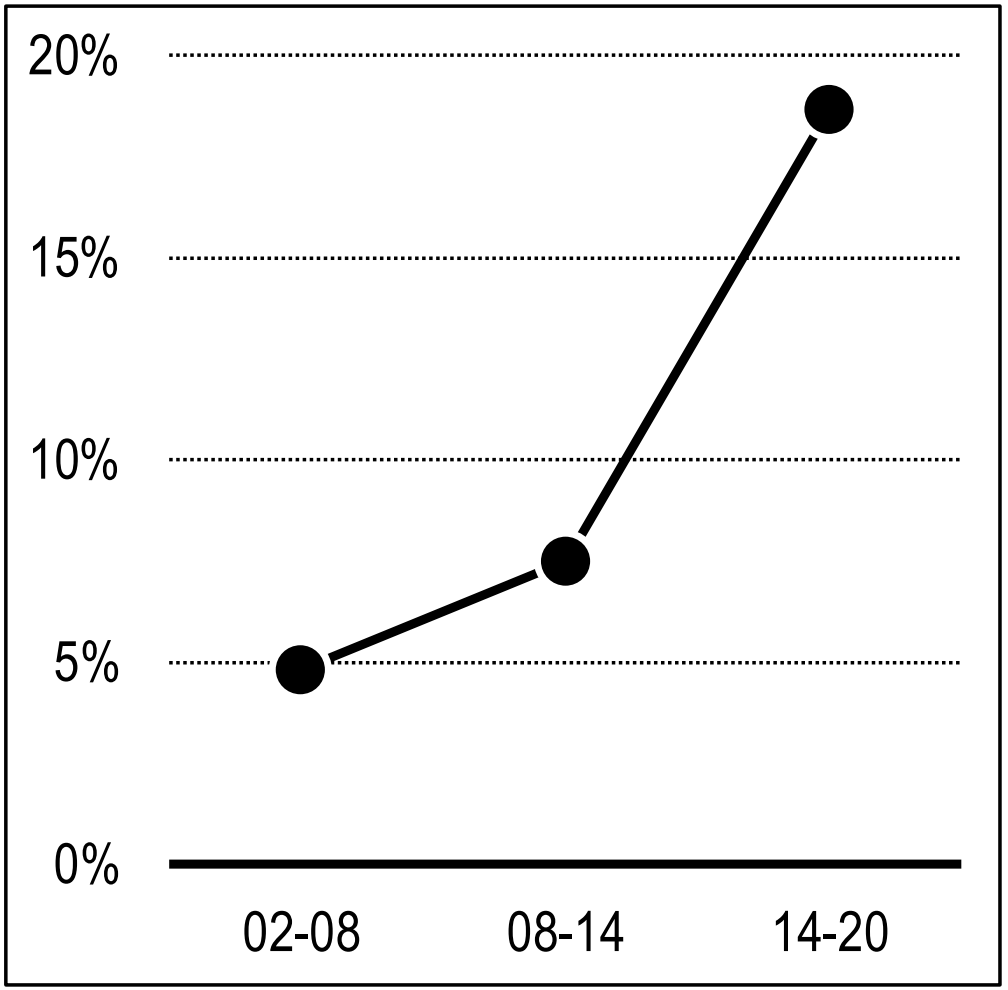


Foreign Exchange Markets

Average Annual Devaluation of TL against FX Basket of $ and €

$/TL	€/TL	April	$+€ Basket
1.31	1.15	2002	2.46
1.27	1.99	2008	3.26
2.12	2.91	2014	5.03
6.75	7.30	2020	14.05

$/TL	€/TL		$+€ Basket
-1%	10%	02-08	5%
9%	7%	08-14	7%
21%	17%	14-20	19%

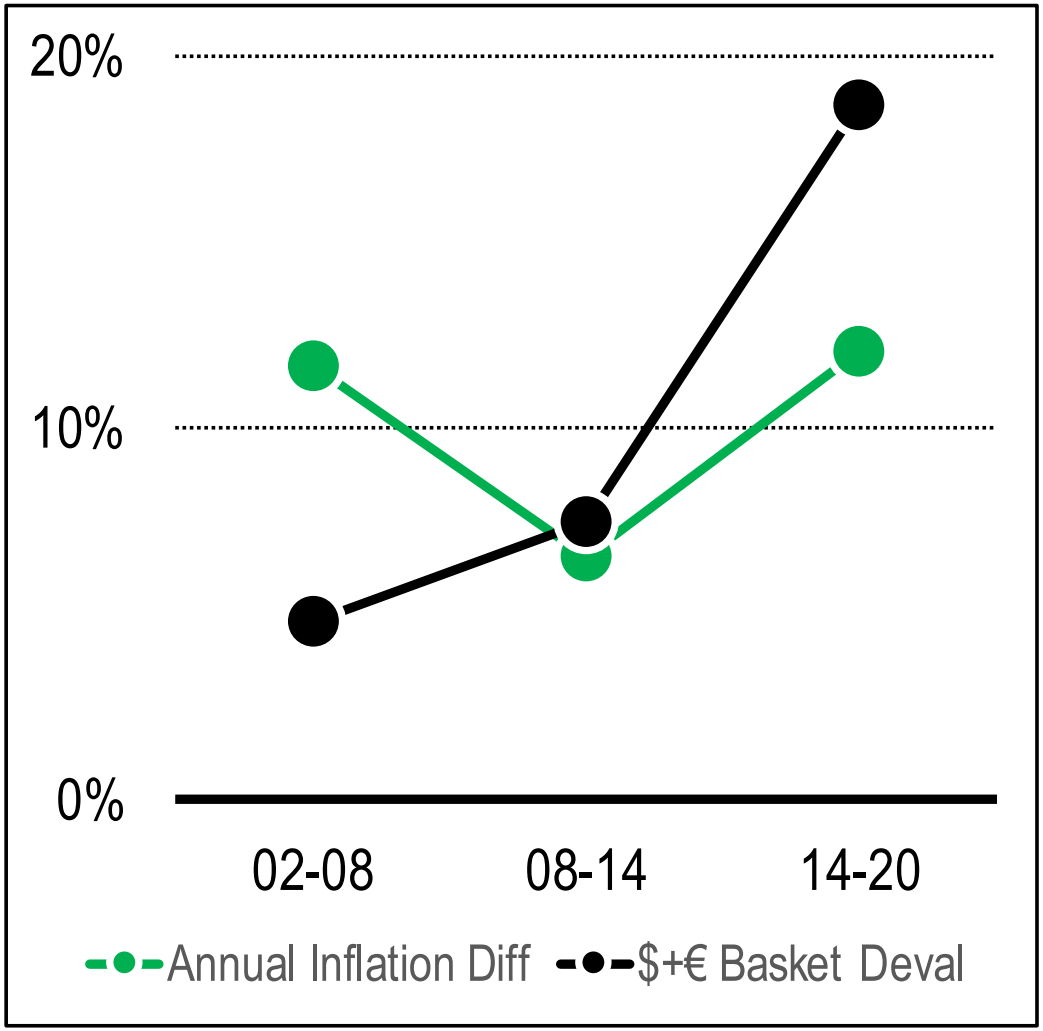


Foreign Exchange Markets

Avg Ann Devaluation of TL against $+€ Basket vs Avg Ann Inflation Difference

Inflation	TR	avg	US	D
02-08	14%	2.2%	2.7%	1.7%
08-14	8%	1.7%	2.0%	1.4%
14-20	13%	1.3%	1.5%	1.2%
avg	12%	1.7%	2.0%	1.4%

	Annual Inflation Diff	$+€ Basket Deval
02-08	12%	5%
08-14	7%	7%
14-20	12%	19%
avg	10%	10%



Foreign Exchange Markets

Interrelationship between FX Rates and Interest Rates

When domestic real interest rates rise,
the domestic currency appreciates
as demand for the domestic currency goes up.

On the other hand,
when domestic interest rates rise
due to to an unexpected increase in inflation (which effectively reduces real interest rates),
the domestic currency depreciates.

Foreign Exchange Markets

Calculating Forward FX Rate

Assume 1€ is bought for 6.45 TL and sold for 6.55 TL in the market today
and you ask a bank to quote €/TL rate for 1m € for 3 months from now
when you want to sell your export revenue.

The way the bank calculates the forward FX rate is as follows:
Since you will be bringing in 1m € to the bank 3 months from now,
with € borrowing interest at 4% per annum
the bank
[1] borrows 1m € / (1+(4%/4)) = 990,099 € today,
[2] sells those € for 990,099 *6.45 = 6,386,139 TL,
TL deposit rate at 12% per annum
[3] with those TL, makes a 3-month TL deposit
to get 6,386,139 *(1+(12%/4)) = 6,577,723 TL at the end of 3 months,
[4] 3 months later, gives those TL to you in exchange for 1m €
so the €/TL rate at that date would be
6,577,723 TL / 1m € = 6.58
for the bank purchasing your 1m € export revenue,
[5] with those €, pays back € that it borrowed 3 months ago.

Foreign Exchange Markets

Calculating Forward Rate for Purchasing FX

Assume 1€ is bought for 6.45 TL and sold for 6.55 TL in the market today
and you ask a bank to quote €/TL rate for purchasing 1m € for 3 months from now
when you want to pay for your imports.

The way the bank calculates the forward FX rate is as follows:
Since you will be taking 1m € out of the bank 3 months from now,
with € deposit rate at 2% per annum
the bank
[1] deposits 1m € / (1+(2%/4)) = 995,025 € today,
[2] in order to purchase those €, borrows 995,025 *6.55 = 6,517,414 TL,
TL loan rate at 18% per annum
[3] to be able to pay back that loan after 3-months
the bank will need 6,517,414 *(1+(18%/4)) = 6,810,698 TL at the end of 3 months,
[4] 3 months later, gives the 1m € deposit to you in exchange for 6,810,698 TL
so the €/TL rate at that date would be
6,810,698 TL / 1m € = 6.81
for the bank selling 1m € to you for your import payment,
[5] with those TL, pays back the TL loan that it took out 3 months ago.

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Foreign Exchange Markets

Calculating Forward FX Rate

	Bid	Offer
spot €/TL	6.45	6.55
€ funds (pa)	2%	4%
TL funds (pa)	12%	18%
term (months)	3	3
forward €/TL	6.58	6.81

Zero transaction costs are assumed.

$$\frac{1\text{m €}}{1+(\ 4\% / 4\ )} = 990{,}099\ € * 6.45 = 6{,}386{,}139\ TL * [1+(\ 12\% / 4\ )] = 6{,}577{,}723\ TL$$

$$\frac{6{,}577{,}723\ TL}{1\text{m €}} = 6.58\ €/TL$$

$$6.45 * \frac{1+(\ 12\% / 4\ )}{1+(\ 4\% / 4\ )} = 6.58\ €/TL$$

$$6.55 * \frac{1+(\ 18\% / 4\ )}{1+(\ 2\% / 4\ )} = 6.81\ €/TL$$

$$\text{Spot €/TL Bid} * \frac{1+\text{ TL funds Bid}}{1+\text{ € funds Offer}} = \text{Forward €/TL Bid}$$

$$\text{Spot €/TL Offer} * \frac{1+\text{ TL funds Offer}}{1+\text{ € funds Bid}} = \text{Forward €/TL Offer}$$

$$\frac{1\text{m €}}{1+(\ 2\% / 4\ )} = 995{,}025\ € * 6.55 = 6{,}517{,}413\ TL * [1+(\ 18\% / 4\ )] = 6{,}810{,}697\ TL$$

$$\frac{6{,}810{,}697\ TL}{1\text{m €}} = 6.81\ €/TL$$

$$\text{Spot €/TL} * \frac{1+\text{ TL int}}{1+\text{ € int}} = \text{Forward €/TL}$$

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week
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International Financial System

International Financial System

Balance of Payments

A bookkeeping system for recording all receipts and payments (private sector and government) that have a direct bearing on the movement of funds betweena nation and foreign countries.

The current account shows international transactions that involve currently produced goods and services.
The difference between merchandise exports and imports, the net receipts from trade, is called the trade balance.
Additional items included in the current account are the net receipts (cash flows received from abroad minus cash flows sent abroad) from three categories:

- investment income
- service transactions
- unilateral transfers (gifts, pensions, and foreign aid)

Another important item in the balance of payments is the capital account, the net receipts from capital transactions (e.g., purchases of stocks and bonds, bank loans, etc.).

The sum of the current account and the capital account equals the official reserve transactions balance (net change in government international reserves).

When economists refer to a surplus or deficit in the balance of payments, they actually mean a surplus or deficit in the offical reserve transactions balance.

International Financial System

Balance of Payments

Because the balance of payments must balance, the official transactions balance, which equals the current account plus the capital account, tells us the net amount of international reserves that must move between governments (as represented by their central banks) to finance international transactions.

Balance of Payments

Current Account,
international transactions
that involve currently produced
goods and services

- **Trade Balance,** difference between merchandise exports and imports, net receipts from trade
- **Investment Income** (interest and dividends paid in and out)
- **Service Transactions** (tourism, contractors, education, etc)
- **Unilateral Transfers** (gifts, pensions, and foreign aid)

+

Capital Account,
net receipts from capital transactions
(purchases of stocks and bonds,
bank loans, etc)

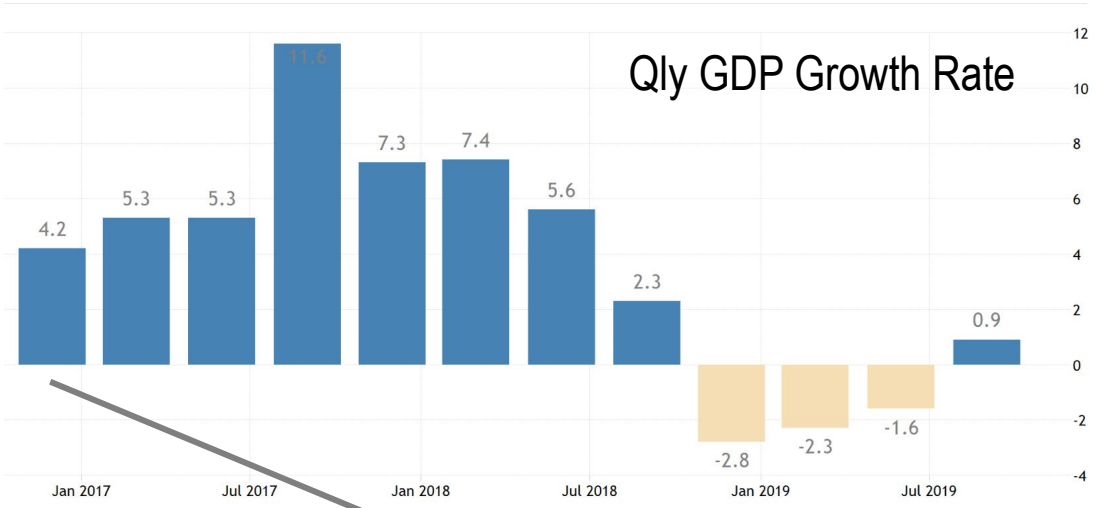
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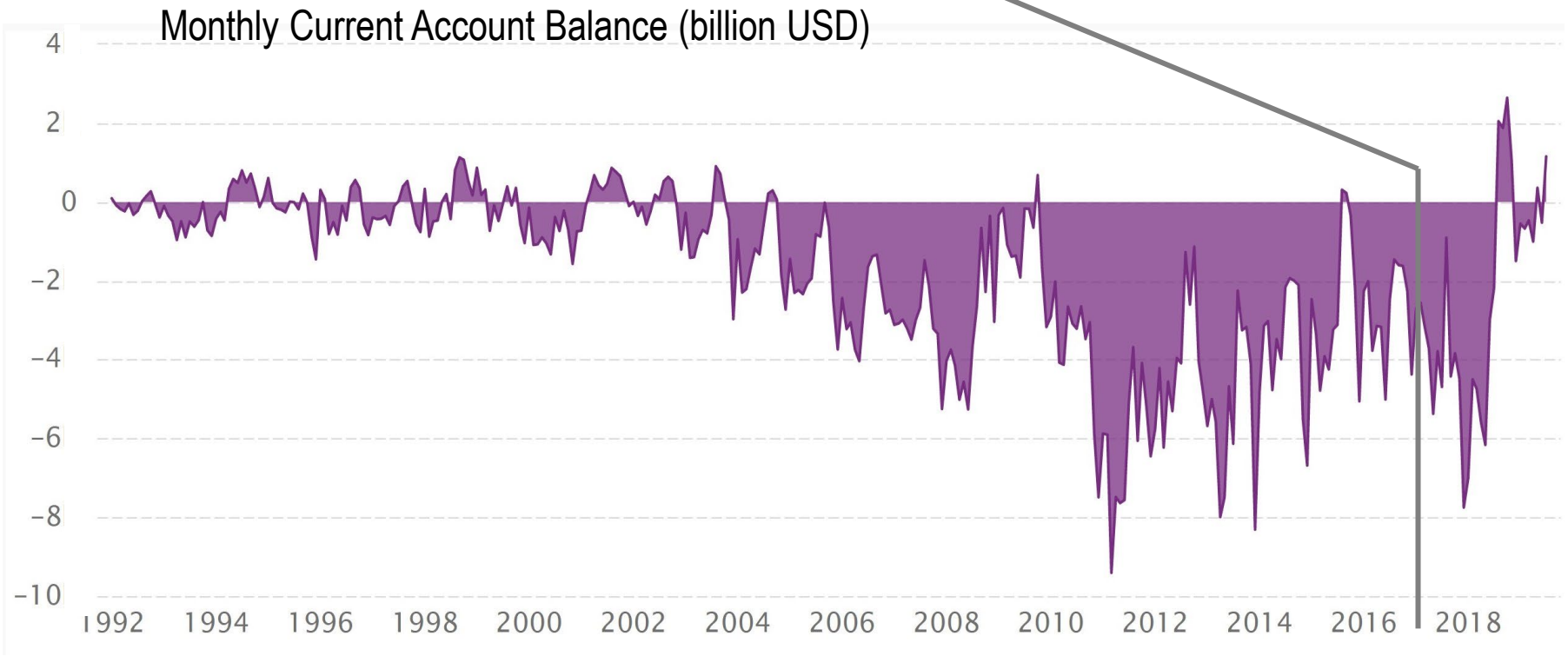
net change in
Government's
International Reserves

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International Financial System

Current Account Balance of Turkey



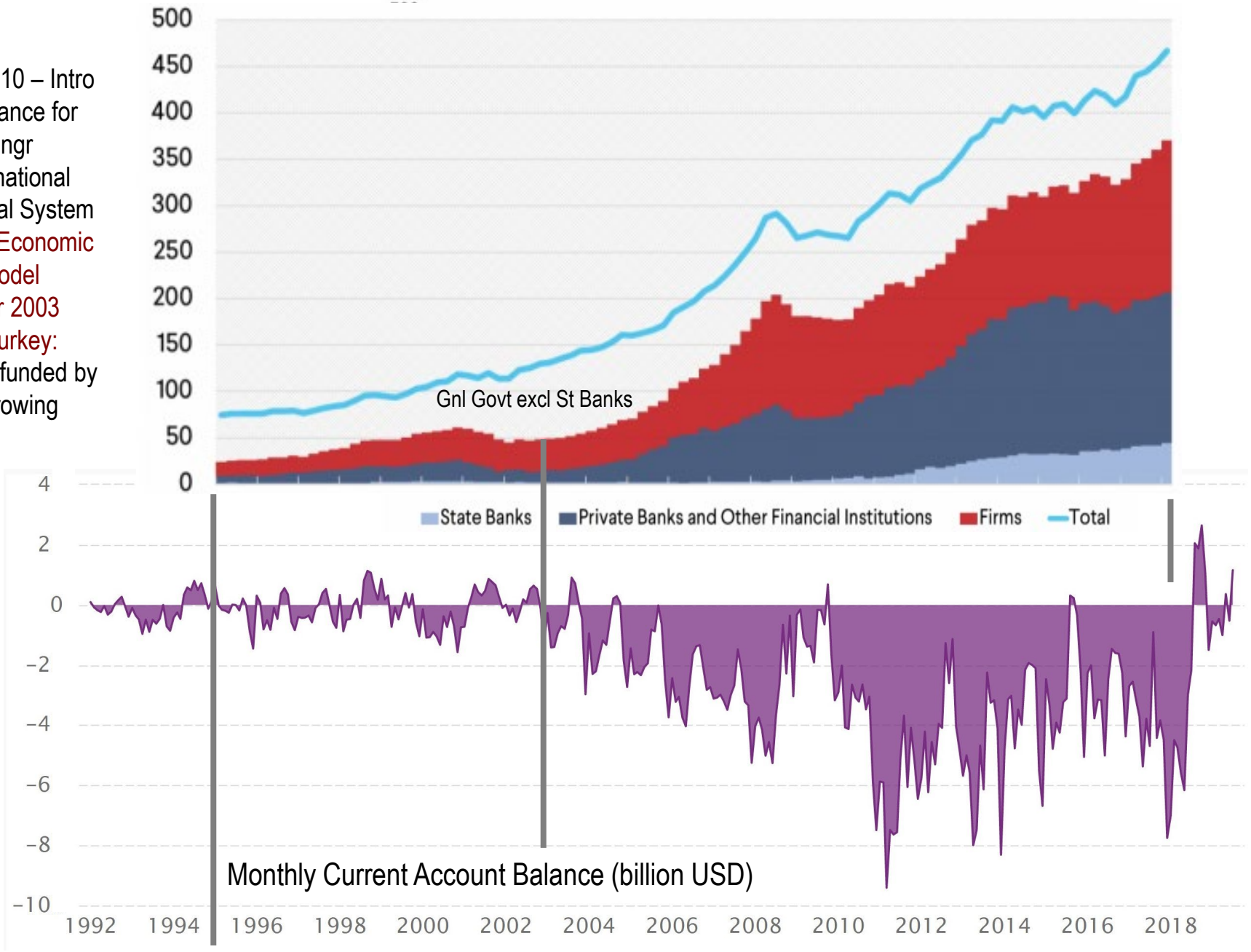


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International Financial System

A New Economic Model after 2003 in Turkey:

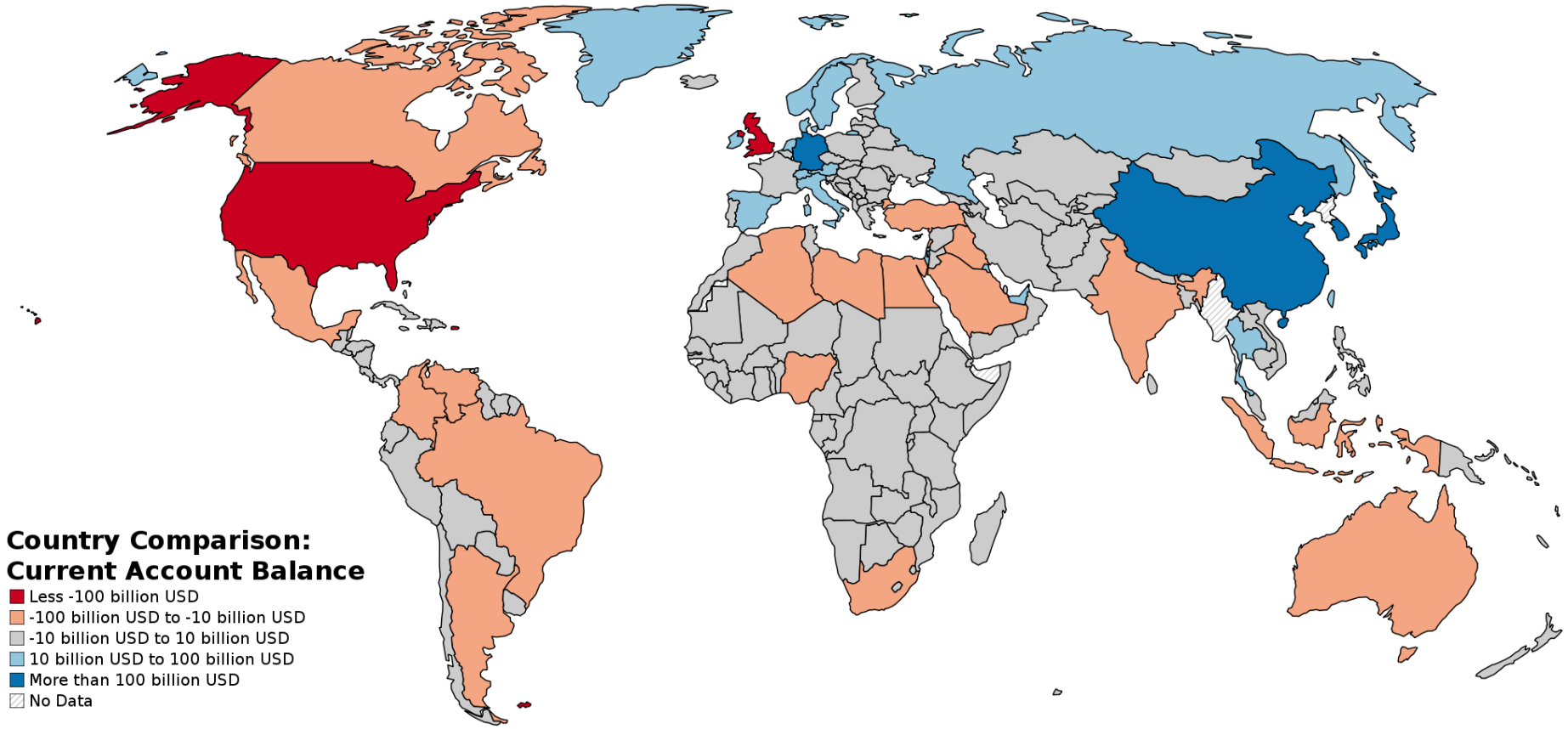
Growth funded by borrowing



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International Financial System

Current Account Balance in the World



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International Financial System

Exchange Rate Regimes

Classified into two basic types; «Fixed» and «Floating»,
and a third one which is more or less in between, namely, «Managed Float».

In a fixed exchange rate regime,
the value of a currency is pegged relative to
the value of another currency (called the anchor currency)
so that the exchange rate is fixed in terms of the anchor currency.

In a floating exchange rate regime,
the value of a currency is allowed to fluctuate against all other currencies.

When countries intervene in foreign exchange markets
in an attempt to influence their exchange rates
by buying and selling foreign assets,
the regime is referred to as a managed float regime (or a dirty float).

International Financial System

Past Exchange Rate Regimes

Before World War I,
the world economy operated under the gold standard,
a fixed exchange rate regime
in which the currency of most countries was convertible directly into gold at fixed rates,
so exchange rates between currencies were also fixed.

After World War II,
a fixed exchange rate system known as the Bretton Woods system was set up.
(Bretton Woods is a town in the US.)
As the US emerged from WWII as the world's largest economic power, with over half of the world's manufacturing capacity and the greater part of the world's gold, the Bretton Woods system of fixed exchange rates was based on the convertibility of US dollars into gold at 35$ per ounce.
Gold as well as US dollars were to kept by other countries as international reserves
with US guaranteeing to provide an ounce of gold for 35$ on demand.
As a result, US $ became the reserve currency of the world.

The Bretton Woods system was abondoned in 1971
with US declaring that it will no longer provide gold in return for US $ anymore.
Fixed exchange rate was abondoned in 1973.

From 1979 to 1990, the European Union instituted among its members its own fixed exchange rate system with some fluctuation allowed between narrow limits which was called the "snake".
In practice, currencies of all participating countries were pegged to German mark.
On January 1, 1999, Euro was launched with an initial value of 1.08 US $.

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Foreign Exchange Markets

Worth of € in $



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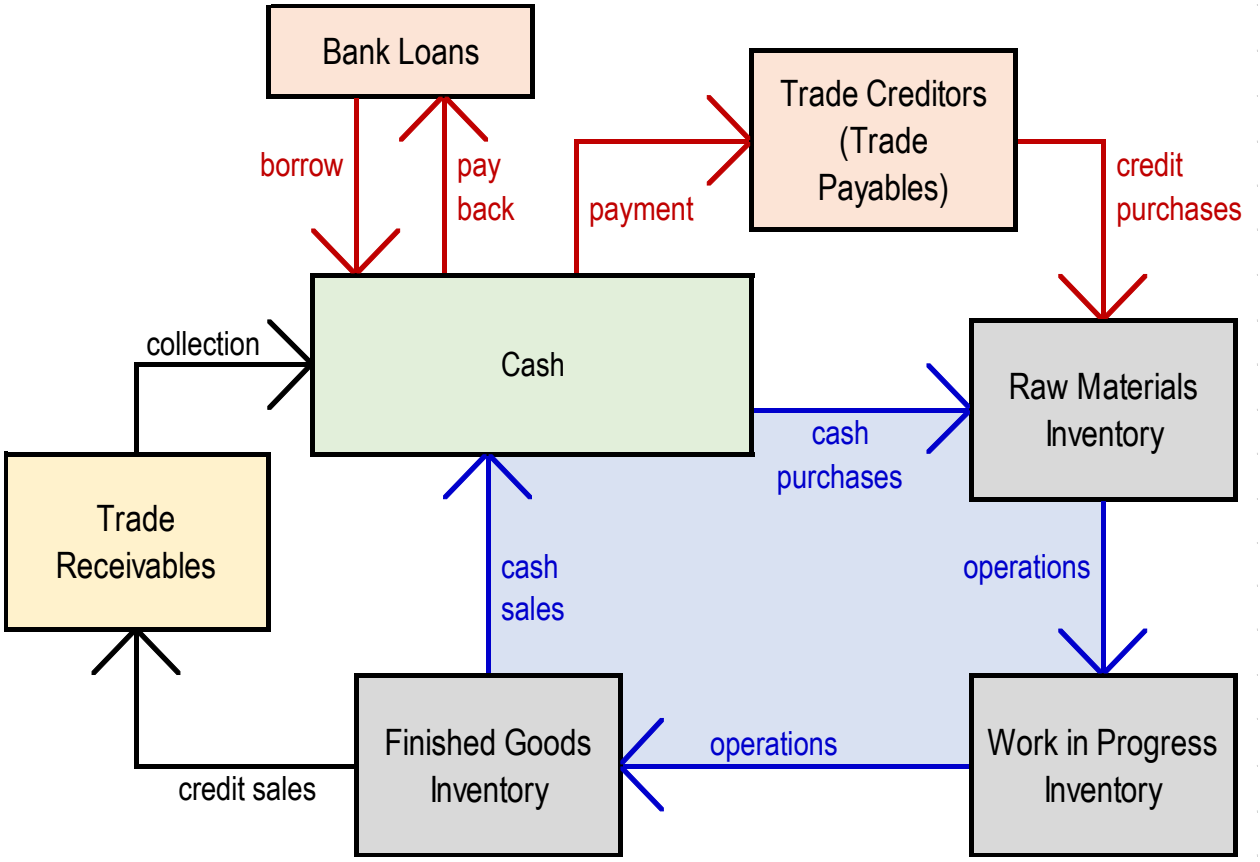
Financial Management

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Working Capital

Working Capital = Current Assets – Current Liabilities

Working Capital = +Cash and Equivalents +Trade Receivables +Inventories -Bank Loans -Trade Payables



Working Capital = Current Assets – Current Liabilities

Working Capital = +Cash and Equivalents +Trade Receivables +Inventories
-Bank Loans -Trade Payables

31.12.2018 Balance Sheet (mTL)

Cash and Equivalent	100		Bank Loans	200	
Trade Receivables	200		Trade Payables	300	
Inventories	300		Current Liabilities		500
Current Assets		600	Long-Term Liabilities		250
Long-Term/Fixed Assets		400	Owners' Equity		250
Total Assets		1.000	Total Liabilities & Owners' Equity		1.000

Working Capital, WC = 600 - 500 = 100 mTL

Working Capital, WC = +100 +200 +300 -200 -300 = 100 mTL

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Managing Working Capital

Inventories

Costs of holding inventories:

Lost interest, Storage cost, Insurance cost, Obsolescence.

Costs of not holding sufficient inventories:

Loss of sales and customer goodwill, Disruptions in production,
Loss of flexibility – cannot take advantage of opportunities,
Reorder costs – low inventories imply more frequent ordering.

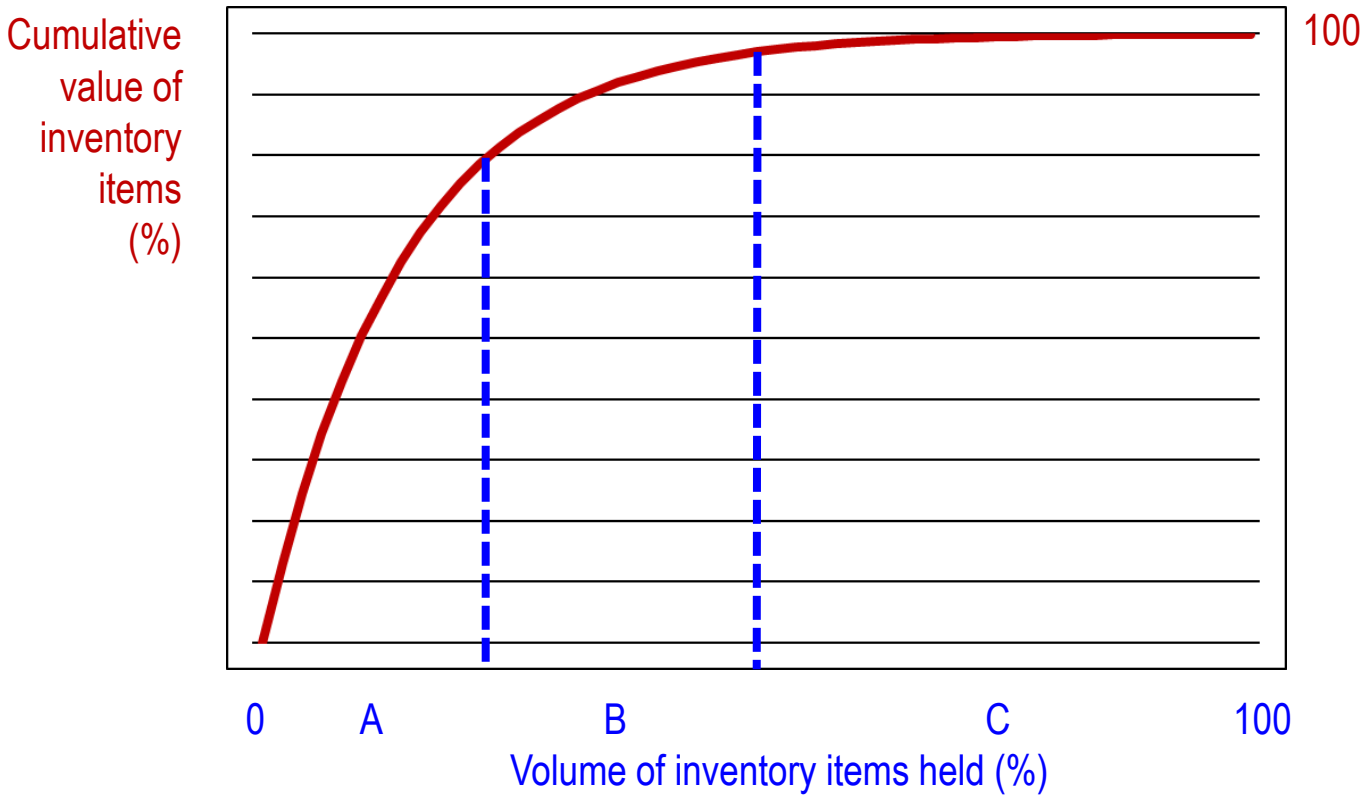
Inventory management guidelines:

- Identify optimum order size – models can help with this
- Set inventory reorder levels
- Consider Just-in-Time (JIT) inventories management
- Use budgets, use accounting ratios (e.g. inventories turnover period)
- Keep reliable inventory records

Inventories

ABC method of analysing and controlling inventories

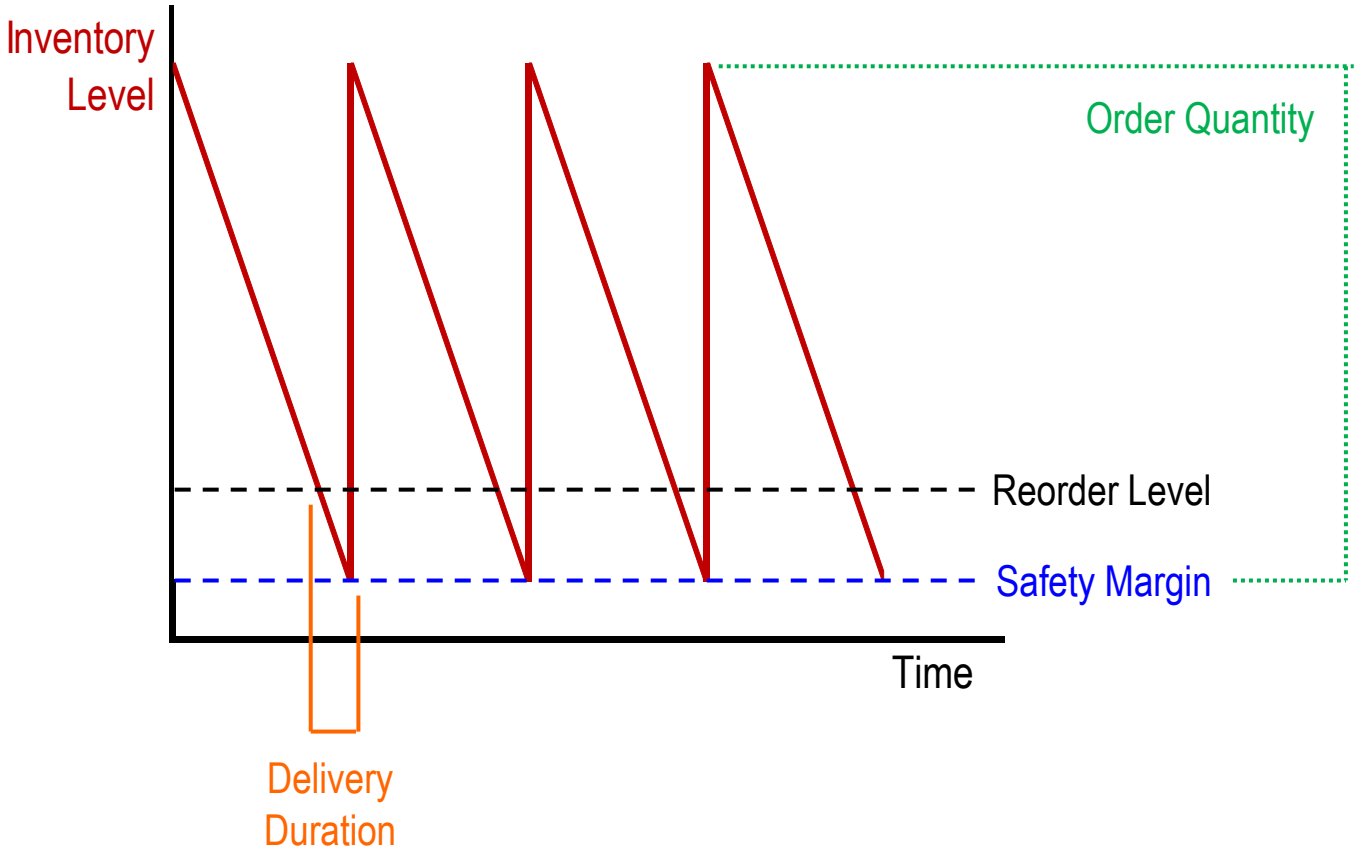
Category A inventories, few in quantity but a large proportion of the total value (20-80 "rule").
Category B, less valuable but more numerous. Category C, very numerous but relatively low in value.
Different inventory control rules would be applied to each category.



Inventories

Pattern of inventory movements over time

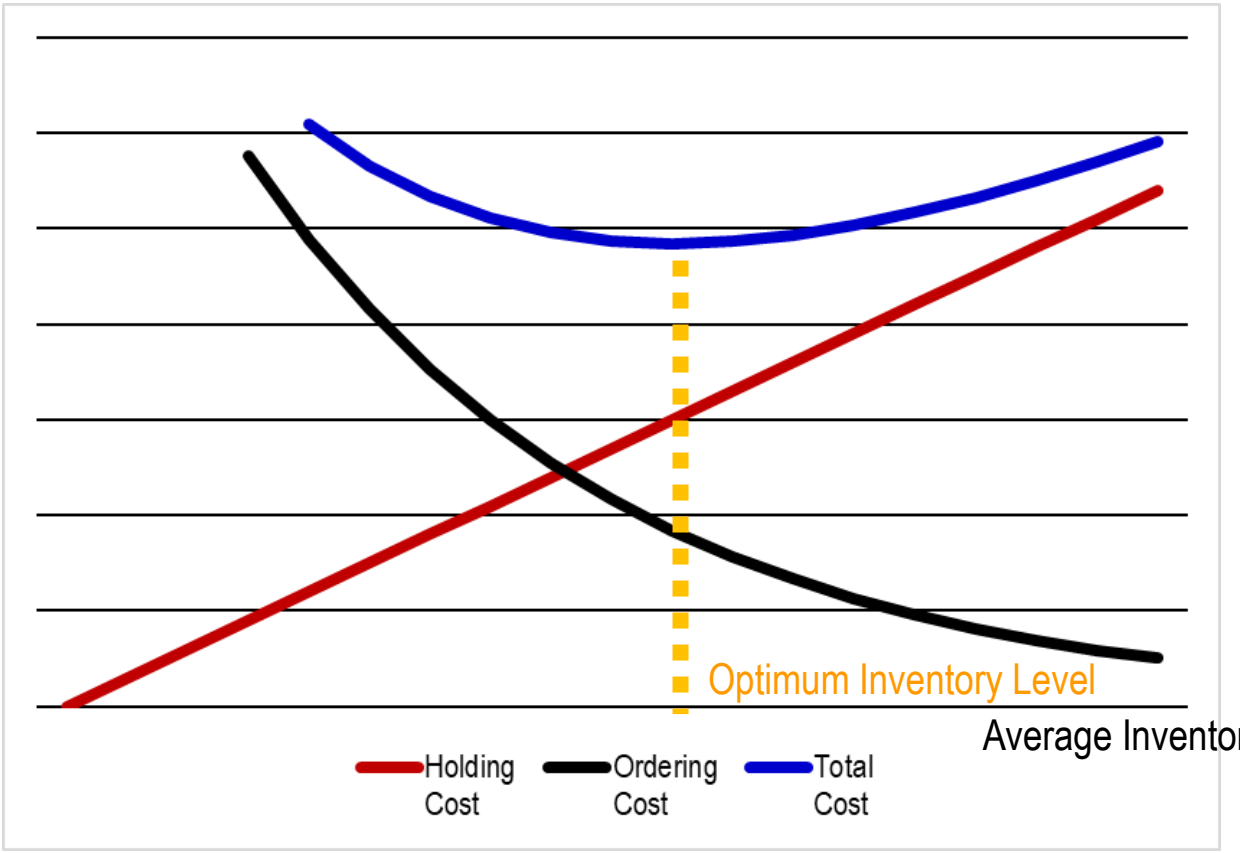
Constant rate of usage of the inventory items is assumed.



Inventories

Inventory holding and ordering costs

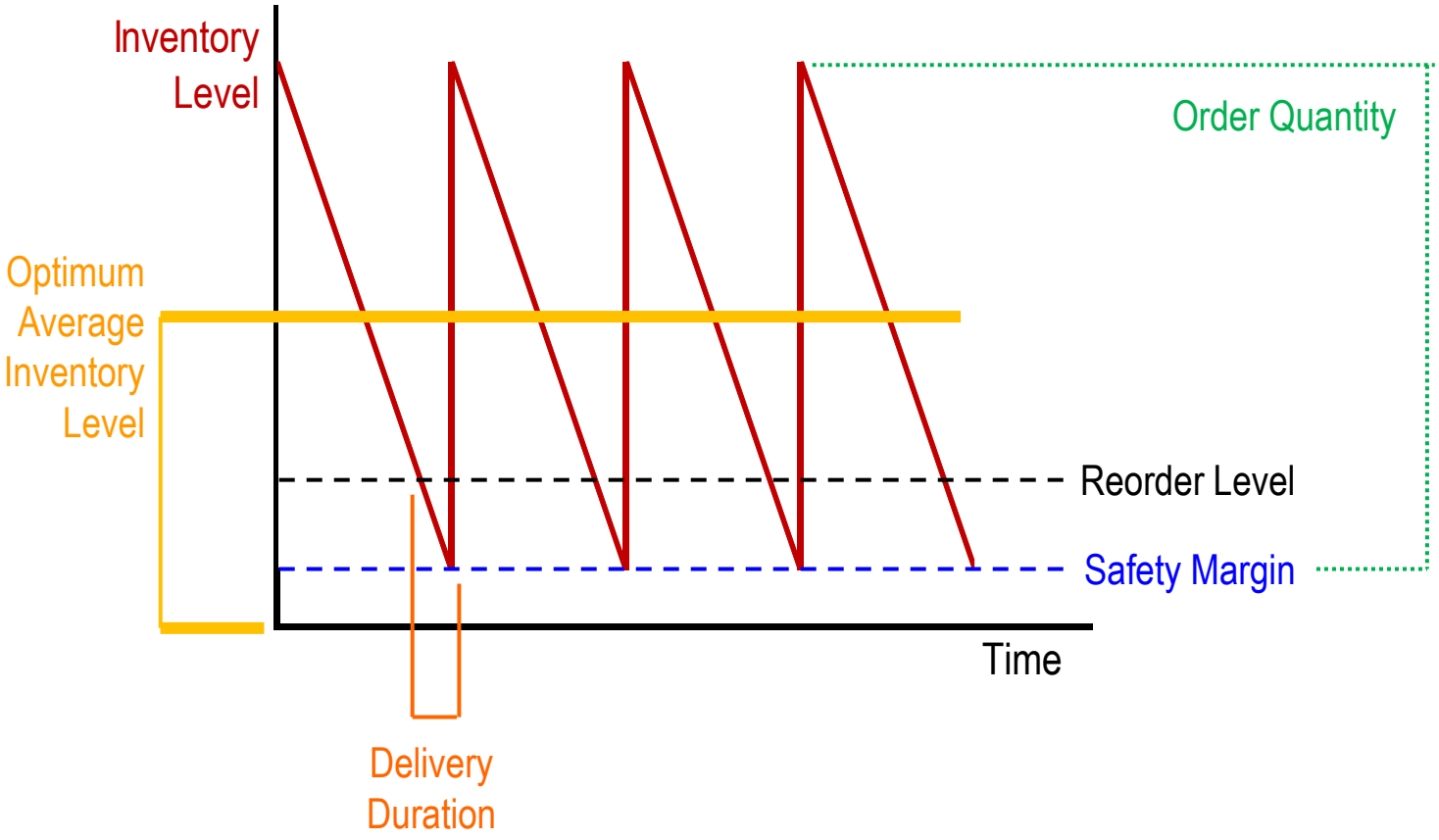
Low inventory levels imply frequent reordering and high annual ordering costs.
But they also imply relatively low inventory holding costs.
High inventory levels imply exactly the opposite.
So there is an optimum order size that will minimize the sum of ordering and holding costs (total costs).



Inventories

Optimum Inventory Level

Constant rate of usage of the inventory items is assumed.



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Managing Working Capital

Trade Receivables

When assessing which customers should be granted credit,
the five C's of credit can be used:
Capital – Capacity (to pay) – Collateral – Condition (of economy and industry) – Character.

Costs of allowing credit sales:

Lost interest, Lost purchasing power, Costs of assessing customer creditworthiness, Administration cost, Bad debts, Cash discounts (for prompt payment).

Cost of denying credit:

Loss of customer goodwill.

Trade receivables management guidelines:

- Establish a policy on granting credit as well as on bad debts
  - Assess and monitor customer creditworthiness
  - Establish an effective administration of receivables
    - Consider cash discounts
- Use financial ratios (e.g. average settlement period for receivables)
  - Use receivables aging tables

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Managing Working Capital

Cash

Cost of holding cash: Lost interest.

Cost of holding insufficient cash:

- Loss of supplier goodwill if unable to meet commitments on time
- Loss of opportunities
- Inability to claim cash discounts
- Cost of borrowing (if an obligation needs to be met at short notice)

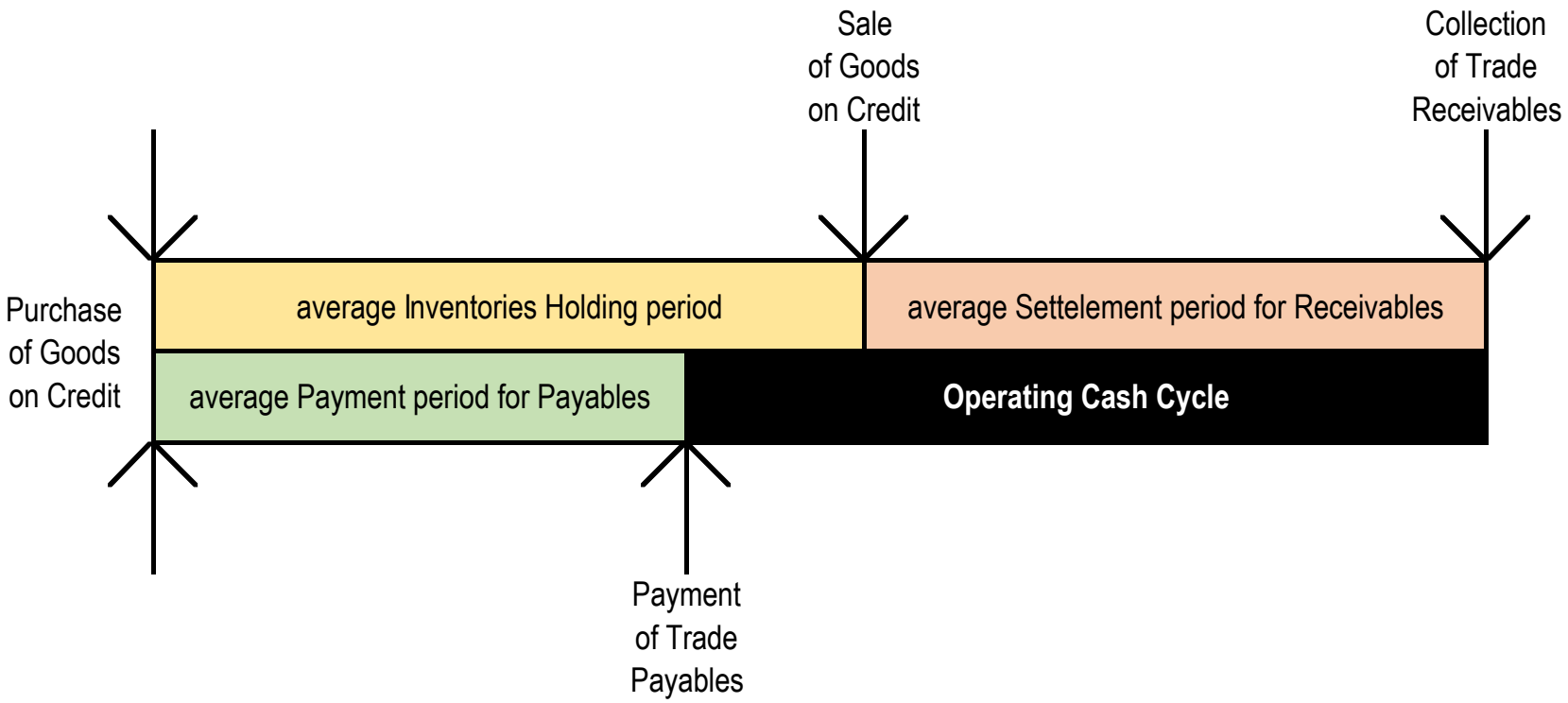
Cash management guidelines:

- Establish a policy with the view of Operating Cash Cycle
- Plan cash flows
- Make use of bank overdraft finance – it can be cheap and flexible
- Use short-term cash surpluses profitably
- Transmit cash promptly

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Managing Working Capital

Operating Cash Cycle



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Managing Working Capital

Measuring Operating Cash Cycle

average Inventories Holding period (days)

$$\frac{\text{average Inventories}}{\text{Cost of Goods Sold}} \times 365$$

avg Settelement period for Receivables (days)

$$\frac{\text{average Trade Receivables}}{\text{Credit Sales}} \times 365$$

avg Payment period for Payables (days)

$$\frac{\text{average Trade Payables}}{\text{Credit Purchases}} \times 365$$

Operating Cash Cycle (days)

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Managing Working Capital

Trade Payables

The cost of taking credit:

- Higher price than purchases for immediate cash settlement
- Administrative costs
- Restrictions imposed by the seller

The cost of not taking credit:

- Lost interest-free borrowing
- Lost purchaing power
- Paying at the time of purchase may be inconvenient

Payables management guidelines:

- Establish a policy
- Exploit free credit as much as possible
- Use accounting ratios (e.g. average settlement period)

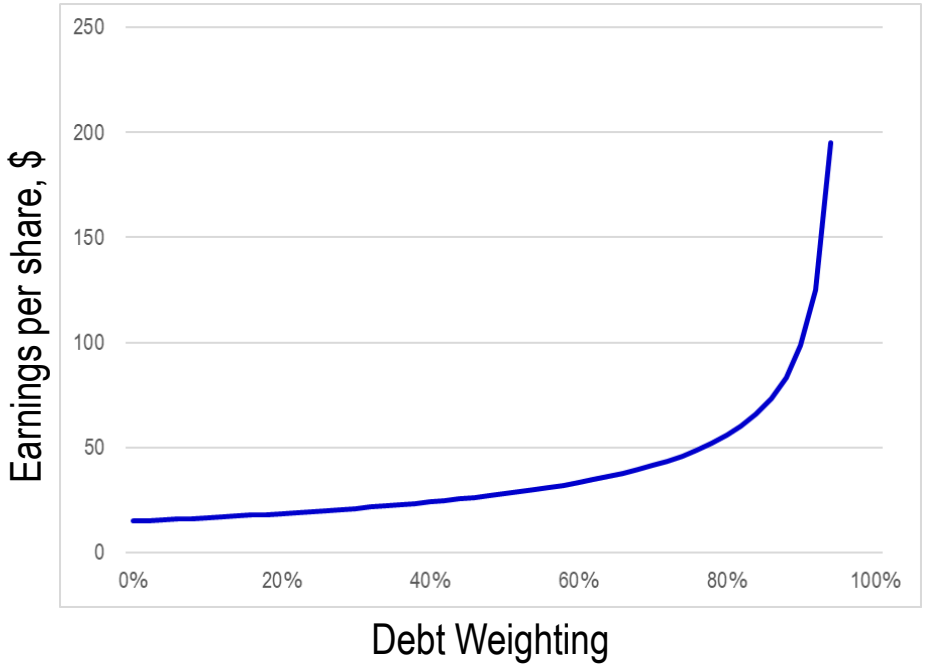
Weighted Average Cost of Capital, WACC

The cost of equity and the cost of debt are weighted by the proportion of equity and debt utilized. The weighted average cost of capital, WACC, is given by:

$$WACC = \left(\frac{E}{E+D}\right) R_E + \left(\frac{D}{E+D}\right) (1\text{-}T)\, R_{DBT}$$

where E is the amount of equity, D is the amount of debt
(E and D can be determined from their valus on the Balance Sheet),
R_E is the cost of equity, R_{DBT} is the before-tax cost of debt and T is the tax rate.

Debt is usually cheaper than equity for a profitable company.
As a result, there is an incentive to borrow in order to maximize value for shareholders.



Weighted Average Cost of Capital, WACC

$$WACC = \left(\frac{E}{E+D}\right) R_E + \left(\frac{D}{E+D}\right) (1\text{-}T)\, R_{DBT}$$

An examination of the above equation could lead to the false conclusion that the optimal solution would be to fund the company entirely from debt sources.
However, increasing debt makes the company more risky
due to increased fixed costs of servicing (making interest payments of) the debt.

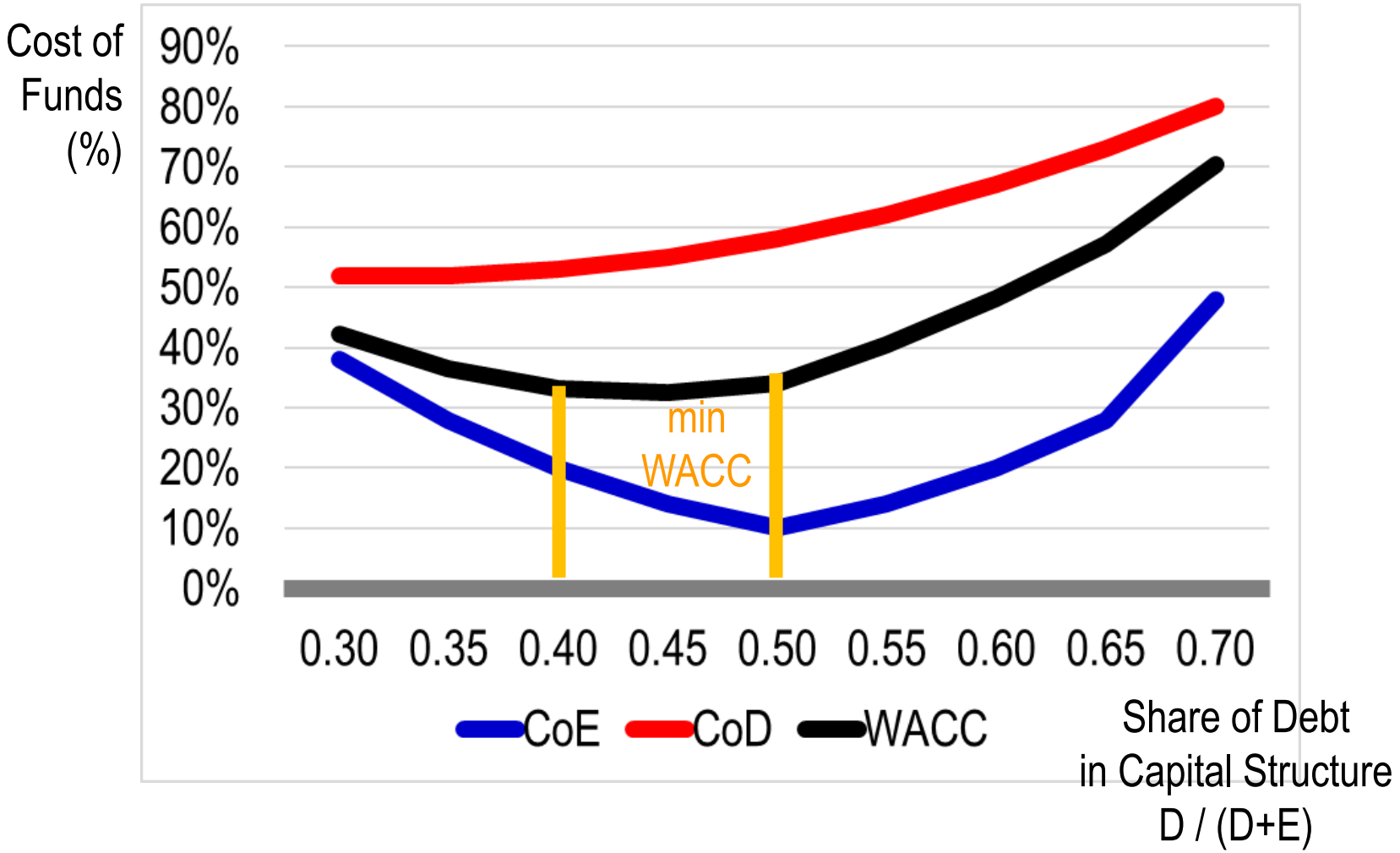




Investors are often more reluctant to invest in companies that have a high proportion of debt.
As a result, the cost of equity will increase for companies with an abnormally high proportion of debt.
Banks would also be more reluctant to lend to such companies.

Capital Structure

Optimizing



Dividend Policy

Affects Debt-Equity balance of the capital structure.

Is a way to let equity investors to realize their returns in cash,
from that perspective, ceteris paribus,
paying dividends have an effect of lowering cost of equity.

Nevertheless, it affects cash position of the company,
from that perspective,
it should also be considered
within the framework of Working Capital management.

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	Pazar 13:00	
MidTerm	7 Haziran	30 dk
Mazeret	14 Haziran	30 dk
Final	28 Haziran	90 dk
Bütünleme	12 Temmuz	90 dk

Tarih ve/veya saatlerdeki olası değişiklikleri lütfen "bys.marmara.edu.tr"den takip ediniz.

	30 dk	3Q	MidTerm	%20	=	%7	+	%7	+	%6			
Final %80	90 dk	7Q	Sınav	%74	=	%10	+	%10	+	%10	+	%10	+	%10	+	%12	+	%12
			HW-1	%6									

Sınav sorularını, sınavın başlangıç saatinde "ues.marmara.edu.tr" adresine
15. hafta dokümanı olarak
yükleyeceğim.

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Spring 2020

Cevaplarınız sınavın bitiş saatinde "turgayozbek58@gmail.com" adresime mail.
Attachment yapmayınız.
Cevap mailin içine, aşağıdaki formatta:

Marmara Üniversitesi
Mühendislik Fakültesi
NTE 1010 - Introduction to Finance for Engineers
2020 Bahar Dönemi
. . . . . Sınavı
Sınav tarihi ve saatini yazınız: . . . . . .
Adınızı Soyadınızı ve Öğrenci Numaranızı yazınız: . . . . . .

her soru için
Soru no. X cevabı: . . . . . .

Cevap mailinizdeki "time stamp" sınav bitiş saatini geçmemeli.

	gecikme süresi		
	≤5 dk	≤10 dk	10+ dk
MidTerm Mazeret	-%2	-%4	kabul edilmez

	gecikme süresi		
	≤10 dk	≤20 dk	20+ dk
Final Bütünleme	-%7	-%14	kabul edilmez

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Review

[1] Financial Accounting

Accounting vs Finance

Accounting is concerned with
collecting, analysing and communicating financial information.
Finance is concerned with
the ways in which funds for a business are raised and invested.

Financial Accounting versus Management Accounting

	Accounting	
	Financial	Management
seeks to meet the needs of	all other users	managers
nature of reports produced	general purpose	specific purpose; for decision or for business unit
level of detail	information is aggregated	considerable details, needed for operational decisions
regulations	subject to accounting regulations, mostly called GAAP "Generally Accepted Accounting Principles"	for internal use only, designed to meet needs of managers
reporting interval	mostly annual, also some half-yearly or quarterly	as required by managers; monthly, weekly or even daily
time horizon	for the past period; backward looking	past (actual) + generally forecast
range and quality of information	in monetary terms and verifiable	monetary + non-monetary figures; need overweights verifiability

[1] Financial Accounting

Accounting Principles and Conventions

Each country has a different variation of "Generally Accepted Accounting Principles".
For example, GAAP accepted in Turkey are called TR GAAP, in USA as US GAAP.

A version that is harmonized internationaly is called IFRS,
"International Financial Reporting Standards".

Although it is not necessary to have a detailed understanding of GAAP or IFRS
in order to understand the financial statements,
it is necessary to understand
some of the basic accounting principles and conventions:

- Business Entity

The business and its owners are treated separately for the purposes of accounting;
this differs from the legal treatment.

- Accrual Accounting
- Historical Cost

[1] Financial Accounting

Business Entity

The business and its owners are treated separately for the purposes of accounting;
this differs from the legal treatment.

The three main forms of legal structures of a business are

- Sole propriertorship (şahıs şirketi)

where owner is personally responsible for the debts of the business,

- Partnership (adi ortaklık)

where partners are "jointly and severally responsible"
meaning that creditors can attempt to recover bad debts
from all the owners, or each one separately, in their personal capacities,

- Corporation (şirket)

where the most that the owners can lose is their equity investment in the company.

[1] Financial Accounting

Accrual Accounting

Accrual accounting recognises a sale when it occurs, that is, when the sale documents are signed, not when the customer finally pays the cash in settlement.

If the transaction occurs before the cash transfer, the transaction is accrued,
and if the transaction is recognised after the cash transfer, it is deferred.
For example, deferred taxation is the recognition of taxes owed but not paid yet.

Because of the difference in timing between the recognition of a transaction and cash receipt or payment, there is the need to match the expenses to the correct period in which the revenue is recognised. Matching expenses and revenues allows for
the correct calculation of the profit that a company has made in a particular period.

On the other hand, the cost of an asset with a useful life of longer than a year
is spread out over its expected useful life.
This procedure is called depreciation.

[1] Financial Accounting

Historical Cost

The figures in the accounts represent historical values,
i.e. the cost of acquiring the assets or goods at the date of acquisition.
The value of an asset in the accounts, that is, its "book value" is unrelated to its value either within the business (replacement value) or to its value to others (market value).
Although this seems unrealistic, historical cost reporting removes the requirement for valuation based on subjective opinion,
which could result in manipulation of the financial position.

This convention also removes the need to adjust the figures for inflation.
Nevertheless, in high inflation environments (where cumulative inflation rate over three years exceeds 100%), "inflationary accounting" is allowed.

This is also in accord with the conservative position adopted by accounting. For example, if losses are probable, and can be reasonably estimated, they are recorded. However, if gains ocur, these are only recorded when they occur.

[1] Financial Accounting

Accounting Conventions

- Money Measurement

Resources not capable of being measured in monetary terms will be excluded. Accounting normally deals with only those items that are capable of being expressed in monetary terms. Money has the advantage that it is a useful common denominator with which to express the wide variety of resources held by a business.

- Going Concern

Financial statements are to be prepared on the assumption that the business will continue operations for the foreseeable future, unless this is known not to be true. In other words, it is assumed that there is no intention, or need, to sell off the non-current assets of the business. Based on this assumption, it is prudent to continue to show the value of non-current assets at their recorded values (that is, based on historic cost).

- Dual Aspect

Each transaction has two aspects, both of which will affect the balance sheet. The purchase of a car for cash results in an increase in one asset (vehicles) and a decrease in another (cash). The repayment of a bank loan results in the decrease in a liability (loans) and the decrease in an asset (cash / bank).

- Prudence

Financial statements should err on the side of caution. The convention represents an attempt to deal with the uncertainty surrounding many events reported in the financial statements, and evolved to counteract the excessive optimism of some managers and owners, which resulted in an overstatement of financial position.

- Objectivity

Objectivity convention seeks to reduce personal bias in financial statements. Financial statements should be based on objective verifiable evidence rather than on matters of opinion.

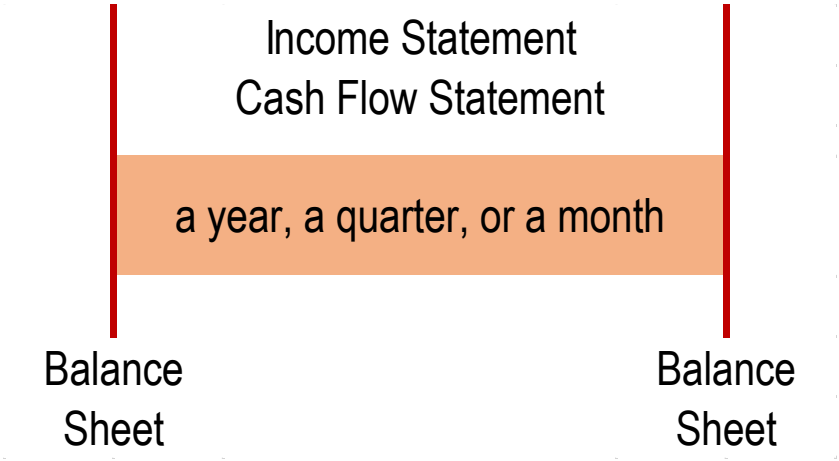
[2] Financial Statements

Major Financial Statements

Three major financial statements that form the core of financial accounting:

- Balance Sheet
  shows the accumulated wealth of the business at the end of the period,
- Income Statement (also known as Profit and Loss Account)
  shows the wealth (profit) generated or lost by the business during the period,
- Cash Flow Statement
  shows the cash in and cash out during the period.

Taken together, they provide an overall picture of the financial health of the business.



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[2] Financial Statements

Balance Sheet

Basic Structure of a Balance Sheet:
+ Assets (resources held by the business)
- Liabilities (obligation of the business to provide cash, or some other form of benefit to other individuals and organizations apart from the owners)
= Owner's Equity (accumulated wealth of the business)

Name of the Company
BALANCE SHEET (currency)
date

CURRENT ASSETS		CA=CA1+CA2+CA3+CA4+CA5	CURRENT LIABILITIES		CL=CL1+CL2+CL3+CL4
Cash & Bank Deposits		CA1	Bank Loans		CL1
Receivables		CA2	Payables		CL2
Inventories		CA3	Short Term Portion of Long Term Liabilities		CL3
Prepaid Expenses		CA4	Other Current Liabilities		CL4
Other Current Assets		CA5	LONG TERM LIABILITIES		LTL=LTL1+LTL2
FIXED ASSETS		FA=FA1+FA2+FA3	Long Term Bank Loans		LTL1
Plant & Equipment, net of Depreciation		FA1=P&Eg-Depr	Other Long Term Liabilities		LTL2
Plant & Equipment, gross	P&Eg		OWNER'S EQUITY		OE=OE1+OE2+OE3
less: Accumulated Depreciation	Depr		Paid Capital		OE1
Intangible Assets		FA2	Accumulated Profit (Loss)		OE2
Other Non-current Assets		FA3	Current Period's Profit (Loss)		OE3
TOTAL ASSETS		TA=CA+FA	TOTAL LIABILITIES & OWNER'S EQUITY		TL&OE=CL+LTL+OE

TA = TL&OE

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[2] Financial Statements

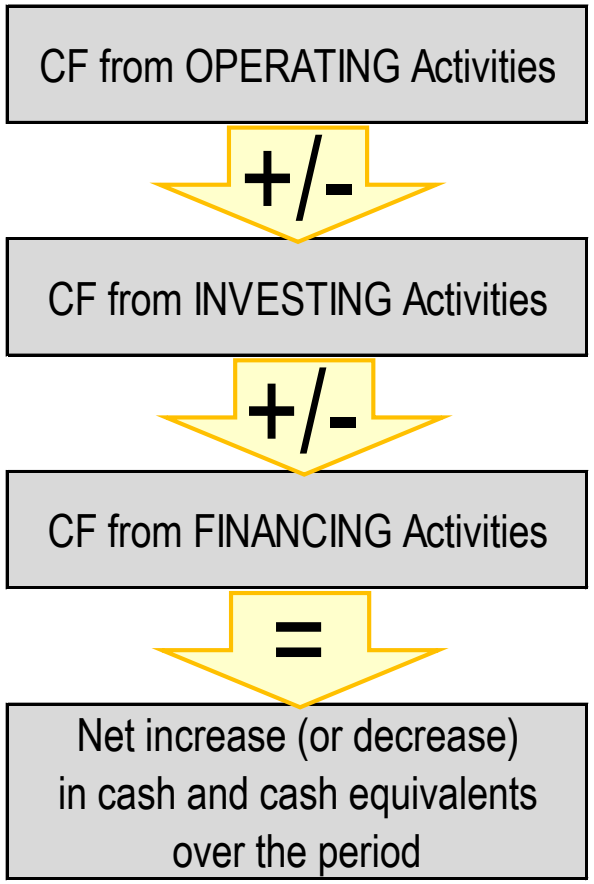
Income Statement

The purpose of the Income Statement,
also called Profit and Loss Account,
is to measure and report how much profit (wealth)
the business has generated
over a period
(most commonly a year, a quarter, a month).

My Shop Ltd Income Statement 1/1 - 31/12 /2017 ($m)
Sales Revenue
- Cost of Goods Sold
Gross Profit
- Selling, General and Admin Costs
- Other Operating Costs
- Depreciation Expense
Profit before Interest & Tax
- Interest Expense
- Uncollectible Receivables
+ Other Operational Income
- Other Operatinal Expense
+/- Net Investment Income (Expense)
Profit before Tax
- Corporate Tax on Profit
Current Period Profit (Profit after Tax)

[2] Financial Statements

Cash Flow Statement



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[3] Analysing & Interpreting Financial Statements

Ratio Analysis

Profitability

$$\text{RoE} = \frac{\text{Current Period Profit}}{\text{average Owner's Equity}}$$

$$\text{Net Profit Margin} = \frac{\text{Profit before Interest \& Tax}}{\text{Sales Revenue}}$$

$$\text{Gross Profit Margin} = \frac{\text{Gross Profit}}{\text{Sales Revenue}}$$

$$\text{P/E Ratio} = \frac{\text{Market Value per Share}}{\text{EPS}}$$

Leverage

$$\text{Leverage Ratio} = \frac{\text{Total Liabilities}}{\text{Owner's Equity}}$$

$$\text{Interest Cover Ratio} = \frac{\text{Profit before Interest \& Tax}}{\text{Interest Expense}}$$

$$\text{Dividend Payout Ratio} = \frac{\text{Dividends Declared}}{\text{Current Period Earnings available for Dividends}}$$

$$\text{Dividend Yield Ratio} = \frac{\text{Dividend per Share}}{\text{Market Value per Share}}$$

$$\text{EPS} = \frac{\text{Current Period Earnings available to Shareholders}}{\text{no of Shares}}$$

Efficiency

$$\text{Inventory Turnover Period (in days)} = \frac{\text{average Inventories held}}{\text{Cost of Goods Sold}} \times 365$$

$$\text{Avg Settlement Period of Receivables (in days)} = \frac{\text{average Trade Receivables}}{\text{Credit Sales Revenue}} \times 365$$

$$\text{Avg Settlement Period for Payables (in days)} = \frac{\text{average Trade Payables}}{\text{Credit Purchases}} \times 365$$

Liquidity

$$\text{Current Ratio} = \frac{\text{Current Assets}}{\text{Current Liabilities}}$$

$$\text{Acid Test Ratio} = \frac{\text{Current Assets excluding Inventories, Prepaid Exp, Other CA}}{\text{Current Liabilities}}$$

Investment

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[3] Analysing & Interpreting Financial Statements

Horizontal Analysis

Alex & Co ($m)	Balance Sheet 31-12 2015	31-12 2016	31-12 2017	Horizontal Analysis 31-12 2015	31-12 2016	31-12 2017
Current Assets	479	544	679	100	114	142
Cash, Cash Equivalents & Bank Deposits	30	4		100	13	0
Trade Receivables	208	240	273	100	115	131
Inventories	241	300	406	100	124	168
Prepaid Expenses						
Other Current Assets						
Non-Current Assets	361	510	587	100	141	163
Plant, Machinery & Equipment, gross						
less: Accumulated Depreciation						
equals: Plant, Machine & Equipm, net [f]	80	129	160	100	161	200
Land & Buildings	281	381	427	100	136	152
Other Non-Current Assets						
TOTAL ASSETS	**840**	**1.054**	**1.266**	**100**	**125**	**151**
Current Liabilities	288	291	432	100	101	150
Short-Term Bank Loans			76			
Trade Payables	240	221	314	100	92,1	131
Interest Payable						
Corporate Tax Payable	30	30	2	100	100	7
Accrued Expenses						
Dividends Payable (Unpaid Dividends) [d]	18	40	40	100	222	222
Other Current Liabilities						
Long-Term Liabilities	200	200	300	100	100	150
Long-Term Trade Payables						
Long-Term Bank Loans						
Other Long-Term Liabilities						
Long-Term Bonds [g]	200	200	300	100	100	150
Total Liabilities	**488**	**491**	**732**	**100**	**101**	**150**
Owner's Equity [g]	**352**	**563**	**534**	**100**	**160**	**152**
Paid-in Capital [a]	250	300	300	100	120	120
Equity Reserves						
Retained Earnings	60	138	263	100	230	438
Current Period Profit (Profit after Tax)	60	165	11	100	275	18
Declared Dividends [d]	-18	-40	-40	100	222	222
TOTAL LIABILITIES & OWNER'S EQUITY	**840**	**1.054**	**1.266**	**100**	**125**	**151**

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[3] Analysing & Interpreting Financial Statements

Vertical Analysis

Alex & Co ($m)	Income Statement year 2016	Income Statement year 2017	Vertical Analysis year 2016	Vertical Analysis year 2017
Sales Revenue [b]	2.240	2.681	100	100
- Cost of Goods Sold [c]	-1.745	-2.272	-78	-85
Gross Profit	495	409	22	15
- Selling, General and Admin Costs				
- Other Operating Costs	-252	-362	-11	-14
- Depreciation Expense				
Profit before Interest & Tax	243	47	11	2
- Interest Expense	-18	-32	-1	-1
- Uncollectible Receivables				
+ Other Operational Income				
- Other Operatinal Expense				
+/- Net Investment Income (Expense)				
Profit before Tax	225	15	10	1
- Corporate Tax on Profit	-60	-4	-3	0
Current Period Profit (Profit after Tax)	165	11	7	0

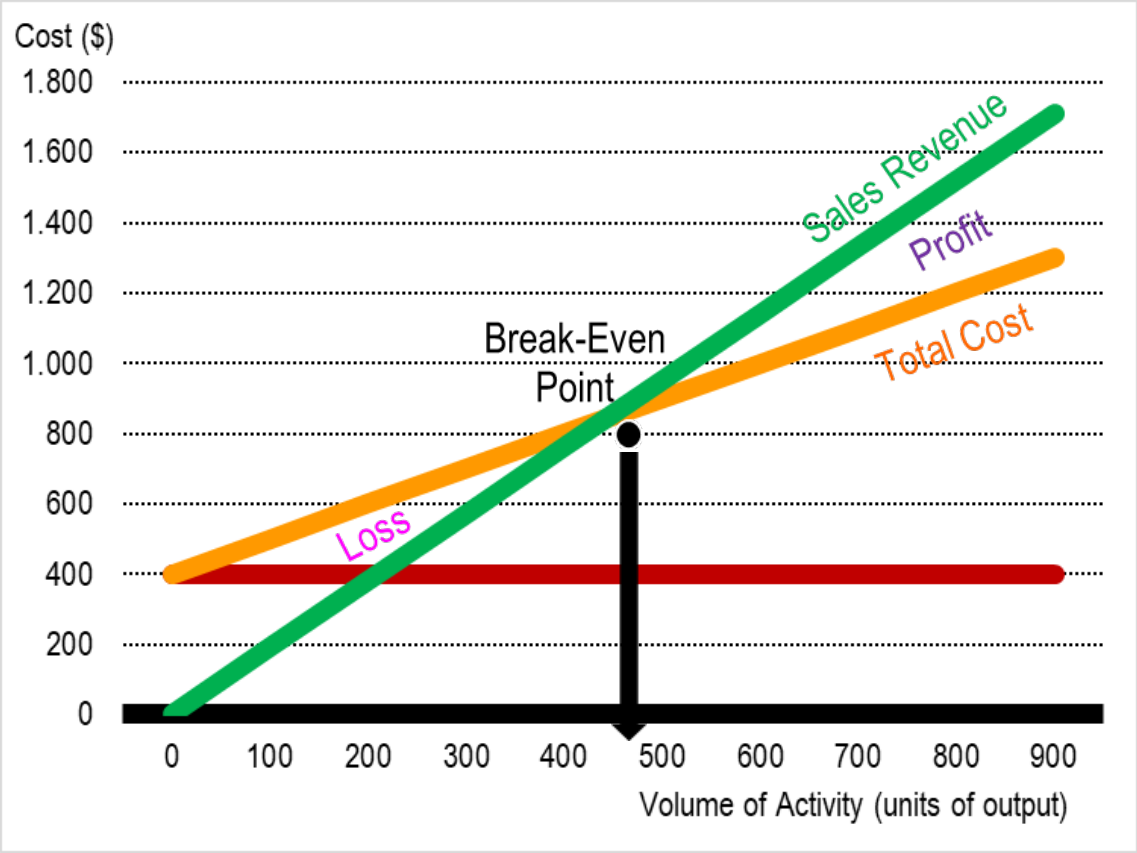
Alex & Co ($m)	Balance Sheet 31-12 2016	Balance Sheet 31-12 2017	Vertical Analysis 31-12 2016		Vertical Analysis 31-12 2017	
Current Assets	544	679	52	100	54	100
Cash, Cash Equivalents & Bank Deposits	4		0	1	0	0
Trade Receivables	240	273	23	44	22	40
Inventories	300	406	28	55	32	60
Prepaid Expenses						
Other Current Assets						
Non-Current Assets	510	587	48	100	46	100
Plant, Machinery & Equipment, gross						
less: Accumulated Depreciation						
equals: Plant, Machine & Equipm, net [f]	129	160	12	25	13	27
Land & Buildings	381	427	36	75	34	73
Other Non-Current Assets						
TOTAL ASSETS	**1.054**	**1.266**	**100**		**100**	
Current Liabilities	291	432	28	100	34	100
Short-Term Bank Loans		76				
Trade Payables	221	314	21	76	25	73
Interest Payable						
Corporate Tax Payable	30	2	3	10	0	0
Accrued Expenses						
Dividends Payable (Unpaid Dividends) [d]	40	40	4	14	3	9
Other Current Liabilities						
Long-Term Liabilities	200	300	19	100	24	100
Long-Term Trade Payables						
Long-Term Bank Loans						
Other Long-Term Liabilities						
Long-Term Bonds [g]	200	300	19	100	24	100
Total Liabilities	**491**	**732**	**47**		**58**	
Owner's Equity [g]	**563**	**534**	**53**	**100**	**42**	**100**
Paid-in Capital [a]	300	300	28	53	24	56
Equity Reserves						
Retained Earnings	138	263	13	25	21	49
Current Period Profit (Profit after Tax)	165	11	16	29	1	2
Declared Dividends [d]	-40	-40	-4	-7	-3	-7
TOTAL LIABILITIES & OWNER'S EQUITY	**1.054**	**1.266**	**100**		**100**	

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[4] Management Accounting

Break-Even Analysis

The Break-Even Point (BEP) is
the level of activity (in units of output or sales revenue) at which
Total Costs (fixed & variable) = Total Sales Revenue.
BEP = Fixed Costs / (Price – Unit Variable Costs)



[4] Management Accounting

Break-Even Analysis

Cottage Industries Ltd makes baskets.
The fixed costs of operating the workshop for a month total $500.
Each basket requires materials that cost $2.
Each basket takes one hour to make, and the business pays the basket makers $10 an hour.
The basket makers are all on contracts such that
if they do not work for any reason, they are not paid.
The baskets are sold to a wholesaler for $14 each.

What is the Break-Even Point (BEP) for basket making for the business?

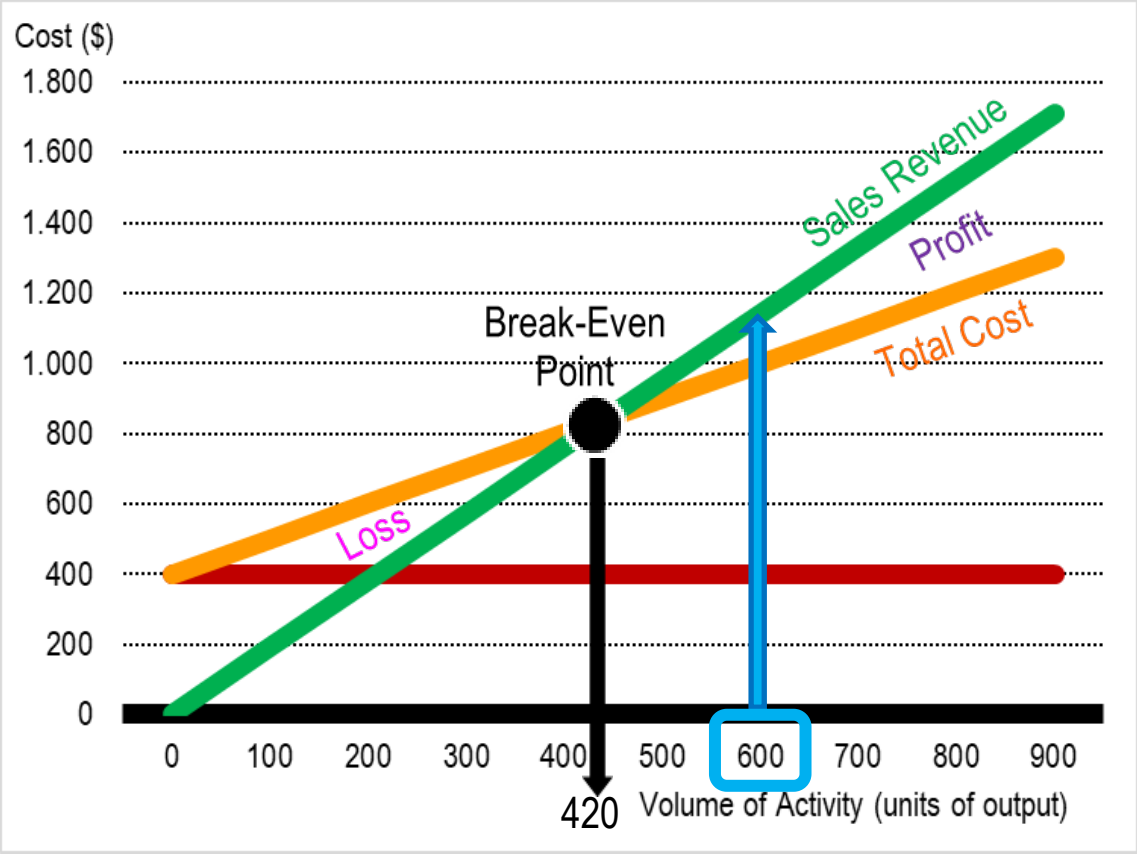
BEP (in number of baskets) = Fixed Costs / (Unit Price - Variable Costs per unit)
BEP = $500 per month / ($14 -$2 -$10) per basket
BEP = 500 / 2 = 250 baskets per month

[4] Management Accounting

Margin of Safety

Margin of Safety = excess of planned volume of activity over BEP / planned volume of activity

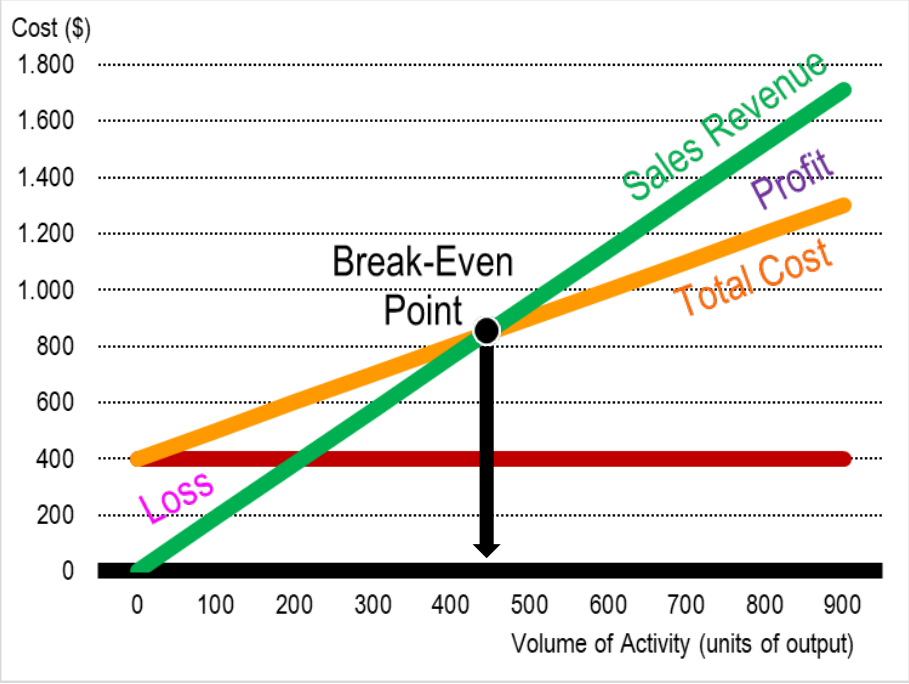
Margin of Safety = (600-420) / 600 = 30%

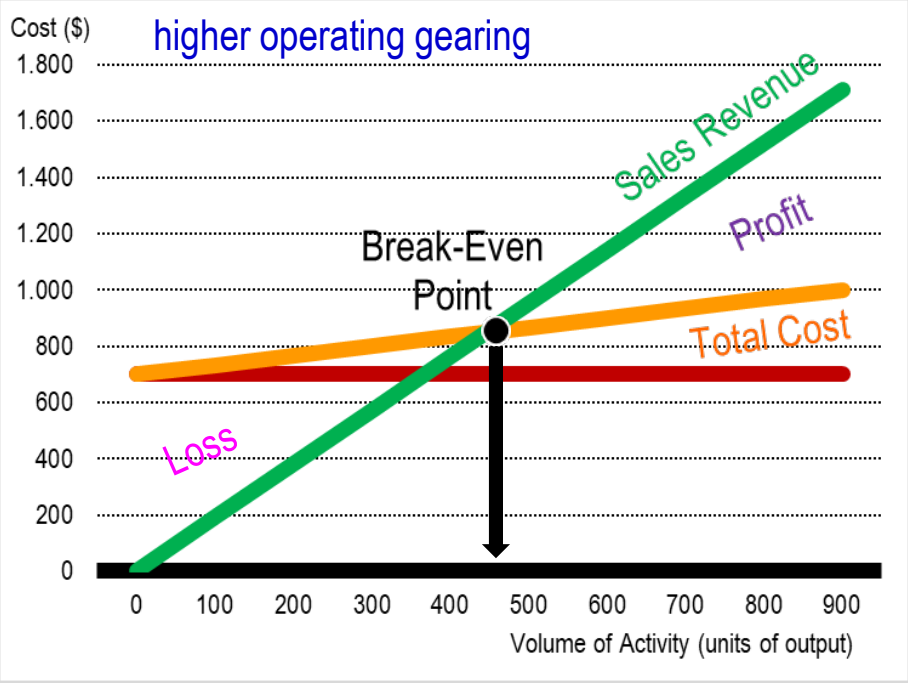


[4] Management Accounting

Operating Gearing

Operating Gearing = the extent to which the total costs of some activity are fixed rather than variable.
An activity with relatively high fixed costs compared with its variable costs
is said to have higher operating gearing.





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[4] Management Accounting

Activity-Based Costing

Modern Producers Ltd runs, like virtually all manufacturers, finished goods stores.
The business has two product lines, A and B.
Product A tends to be made in small batches and so low levels of finished goods inventories are held.
Product B is supllied in relatively large quantities instantly. As a consequence,
much of the finished goods stores is filled with Product B ready to be dispatched as an order is received.

Traditionally, the cost of of operationg the stores (an overhead cost) are allocated to products A and B
according to the number of direct labour hours worked on each one.
And, that did not seem fair to the new accountant Modern Producers Ltd.

He estimated that the costs of running the finished goods stores for the next year will be 90,000$.

It is also estimated that each unit of Product A will spend an average of 1 week in the stores before being sold.
With Product B, the equivalent period is 4 weeks.
Both products are of roughly similar size and have very similar storage needs.

It is estimated that next year 50,000 units of Product A and 25,000 units of Product B will pass through the stores.

The new accountant wants to allocate the overhead cost of running the finished goods stores to products A and B
according to Activity-Based Costing
with the quantity of products and the time spent in the stores as the cost drivers.

[4] Management Accounting

Activity-Based Costing

The total number of "product weeks" in the stores will be:

Product		estimated annual units for next year		average weeks spent in the stores before being sold		annual "product weeks"
A	:	50.000	x	1	=	50.000
B	:	25.000	x	4	=	100.000
						150.000

The finished goods store cost per "product week" is given by 90,000$ / 150,000 product weeks = 0.60$
Therefore each unit of product A and product B will be charged as follows:

Product		average weeks spent in the stores before being sold		cost ($) per "product week"		finished goods stores cost per unit ($)
A	:	1	x	0,60	=	0,60
B	:	4	x	0,60	=	2,40

[4] Management Accounting

Budgeting

Budget = a short-term business plan, mainly expressed in financial terms.

Budgets are the short-term means of working towards the business's objectives.

They are usually prepared for 12 months with subperiods of a month.

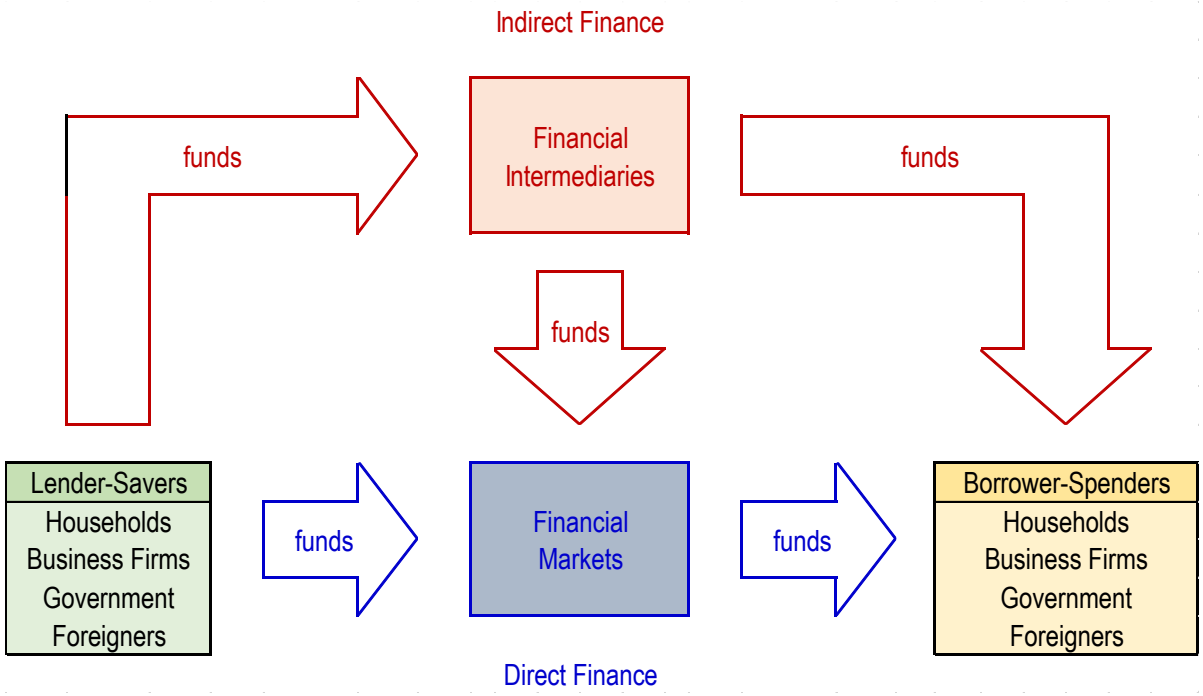
Uses are to:

- promote forward thinking
- help coordinate the various aspects
- motivate performance
- provide the basis of a system of control
- provide a system for authorisation

[5] Financial System

Flow of Funds through the Financial System

In Indirect Finance, borrowers borrow funds directly from Lenders in Financial Markets
by selling them securities (financial instruments)
which are claims on the borrower's future income or assets.



Financial markets perform the essential economic function of channeling funds
from households, firms, and governments
that have saved surplus funds by spending less than their income
to those that have a shortage of funds because they wish to spend more than their income.

[5] Financial System

Primary and Secondary Markets

Primary Market

is a financial market in which new issues of a Financial Security, such as a bond or stock,
are sold to initial buyers by the corporation or the government agency borrowing the funds.

Secondary Market

is a financial market in which securities that have been previously issued can be resold.
An important financial institution that assists in the initial sale of securities in the primary market
is the investment bank. It does this by underwriting securities:
It guarantees a price for a corporation's securities and then sells them to the public.
Securities brokers and dealers are crucial to a well-functioning secondary market.
Brokers are agents of investors who match buyers with sellers of securities;
dealers link buyers and sellers by buying and selling securities at stated prices.
Secondary markets serve two important functions:
First, they make it easier and quicker to sell financial instruments;
that is, they make the financial instruments more liquid.
Second, they determine the price of the security that the issuing firm sells in the primary market.
Secondary markets can be organized in two ways:
One method is to organize Exchanges, where buyers and sellers of securities
(or their agents or brokers) meet in one central location to conduct trades.
The other is to have an Over-the-Counter (OTC) market,
in which dealers at different locations who have an inventory of securities stand ready
to buy and sell securities "over the counter" to anyone who comes to them and is willing to accept their prices.

[5] Financial System

Equity

1. Ordinary Shares:

The investors provide the share capital for partial ownership of the business.
The return to the holders of ordinary shares is directly related to the performance of the business in two forms;

- The growth in the value of the company,
- The dividends paid by the company.

The holders of ordinary shares are exposed to the entire risk of the business.
If the company sells new shares to existing shareholders
who have pre-emptive right to subscribe to new shares, it is called a rights offer.
Current shareholders may also waive their right to subscribe to new shares.
In such a case, the company can raise equity capital
by selling new shares to new shareholders for cash; this called follow-on offer.

2. Preferred stock

Preferred Stock has the characteristics of both equity and debt.
Dividend offered by Preferred Stock is fixed, a percent of the face value of the shares.
These payments are similar to the interest payments on debt.
However, if there are insufficient funds, no payments are made
and this does not throw the company into bancruptcy.
Usually, these unpaid dividends accumulate and holders of Preferred Stock are paid in arrears.
Preference shares usually have no maturity date.
Unlike debt, the dividend payment to Preferred Stock is not tax-deductible.
In case of liquidation of the company,
the holders of Preferred Stock have a claim to to the assets prior to that of the ordinary shareholders.

[5] Financial System

Types of Financial Intermediaries

- Banks accept deposits from individuals and institutions and make loans.

- Life Insurance / Pension Companies insure people against financial hazards following a death and sell annuities (annual income payments upon retirement). They acquire funds from the premiums that people pay to keep their policies in force and use them to invest in financial markets.

- Mutual Funds acquire funds by selling their own shares to many individuals and use the proceeds to purchase diversified portfolios of stocks and bonds.

[5] Financial System

Money

Primary Functions of Money:

- Money in the form of currency, checks or electronic cards is a Medium of Exchange;
it is used to pay for goods and services.
- Money provides a Unit of Account;
it is used to measure value in the economy.
- Money also functions as a Store of Value;
it is a repository of purchasing power over time.

There are expanding levels of

money supply measurement:

M0 = physical money

(paper, coin, deposits of banks at Central Bank which the Central Bank has to provide cash on demand)

M1 = M0 +current deposits

M2 = M1 +deposits with maturity of less than 30 days

M3 = M2 +time deposits

[6] Interest Rates

Yield to Maturity

Present Value (PV) of the Debt Instrument may be formulated as follows:

$$PV = \sum_{t=1}^{n} \frac{CF_t}{(1+i)^t}$$

Yield to Maturity is the interest rate (i) that equates the present value of cash flow payments received from a debt instrument with its value today.
Yield to Maturity is also known as Internal Rate of Return (IRR).

If we know CF_t and the PV of the debt instrument,
then we can calculate i, that is, its yield to maturity.

If we know CF_t and the market interest rate relevant for the debt instrument,
then we can calculate its PV.

The effective cost of a corporate bond is determined by calculating the Yield to Maturity (also known as Internal Rate of Return (IRR)) of the cash flows arising from the issuance of bond.

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[6] Interest Rates

Effective Cost of a Corporate Bond

A company issues a bond repayable after 10 years.
The bond is issued with a face value of $ 100.000 and carries 7% coupon.
The lender provides the cash to the issuer by buying a corporate bond certificate.
Over the years, the lender receives coupon payments
(called coupon because the certificate contained coupons for each payment
which the lender have to submit in order to redeem that payment).
The coupon is a fixed proportion of the principal, in this case 7% .
At the end of the period, the lender is repaid the total face value of the bond by the issuer.
The company issued bonds worth $ 20 million.
The investment banking costs for arranging the bond amounted to 2,4% of the amount raised.
The bond was offered at a discount 1% to the face value to attract investors.
The tax rate of the company is 22% .
What is the before-tax and after-tax effective cost of this corporate bond?

(m $) time (years)	principal (face value)	2,4% investment banking fees	1,0% discount offered	7,0% coupon (interest) payments	principal payment	corporate bond cash flow	effective before-tax cost of the corporate bond (IRR) 7,50% discounting factors	discounted cash flows
0	20,0	-0,5	-0,2			19,3	1,0000	19,3
1				-1,4		-1,4	0,9302	-1,3
2				-1,4		-1,4	0,8653	-1,2
3				-1,4		-1,4	0,8050	-1,1
4				-1,4		-1,4	0,7488	-1,0
5				-1,4		-1,4	0,6966	-1,0
6				-1,4		-1,4	0,6480	-0,9
7				-1,4		-1,4	0,6028	-0,8
8				-1,4		-1,4	0,5607	-0,8
9				-1,4		-1,4	0,5216	-0,7
10				-1,4	-20,0	-21,4	0,4852	-10,4
								0,0 sum of disc CF

effective after-tax cost of the corporate bond = (1 -22% * 7,5%) = 5,9%

[6] Interest Rates

Real versus Nominal Interest Rate

When the effect of inflation is deducted from nominal interest rate, real interest rate is obtained.

Nominal Interest Rate	25%		
Inflation	19%		
		Today	A Year Later
Price of an Apple (TL)		1.00	1.19
Money that I have (TL)		1,000	1,250
No of Apples that I can Buy		1,000	1,050
Real Interest Rate	5%		

The above relationship may be formulated as follows:

$$1 + i_{real} = \frac{1 + i_{nominal}}{1 + inflation_{expected}} \qquad 1+5\% = \frac{1+25\%}{1+19\%}$$

Note that the expected inflation is used.

	Oct 2017	Oct 2018	Oct 2019	Oct 2020
[1-year deposit rate] nominal interest rate	10%	20%	11,5%	
[Δ CPI Last 12 Months] actual inflation	25%	9%	?	
real interest	-12%	10%	?	

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[6] Interest Rates

Interest-Rate Risk

Present Value (PV) of the Debt Instrument $= \sum_{t=1}^{n} \frac{CF_t}{(1+i)^t}$

For a given set of CF_t, when market interest rates (i) go up, the value (PV), that is, the price of the debt instrument drops. And, vice versa. Such volatility of prices of debt instruments is called Interest-Rate Risk. Prices of longer-maturity bonds respond more drastically to changes in interest rates. Dealing withinterest-rate risk is a major concern of managers of financial institutions and investors.

(m $) time (years)	principal (face value)	7,0% coupon (interest) payments	principal payment
0	20,0		
1		-1,4	
2		-1,4	
3		-1,4	
4		-1,4	
5		-1,4	
6		-1,4	
7		-1,4	
8		-1,4	
9		-1,4	
10		-1,4	-20,0

(m $) time (years)	market interest rate 10% discounting factors	discounted cash flows	
0			
1	0,9091	1,3	
2	0,8264	1,2	
3	0,7513	1,1	
4	0,6830	1,0	
5	0,6209	0,9	
6	0,5645	0,8	
7	0,5132	0,7	10,3
		17,1 m $ sum of disc CF	

market value of the corporate bond at market interest rate of 10%

(m $) time (years)	market interest rate 11% discounting factors	discounted cash flows	
0			
1	0,9009	1,3	
2	0,8116	1,1	
3	0,7312	1,0	
4	0,6587	0,9	
5	0,5935	0,8	
6	0,5346	0,7	
7	0,4817	0,7	9,6
		16,2 m $ sum of disc CF	

market value of the corporate bond at market interest rate of 11%

[6] Interest Rates

Yield Curve

A plot of the yields on bonds with differing terms to maturity
but the same risk, liquidity, and tax considerations is called a yield curve,
and it describes the term structure of interest rates for particular types of bonds.

Three empirical facts about term structure of interest rates are:

- Interest rates on bonds of different maturities move together over time.
- When short-term interest rates are low, yield curves are more likely
  to have an upward slope;
  when short-term interest rates are high,
  yield curves are more likely to slope downward (i.e. inverted).
- Yield curves almost always slope upward.

Three theories have been put forward to explain the term structure of interest rates,
i.e. shapes of yield curves:

- The Expectations theory
- The Segmented Markets theory
- The Liquidity Premium theory

[6] Interest Rates

Expectations Theory

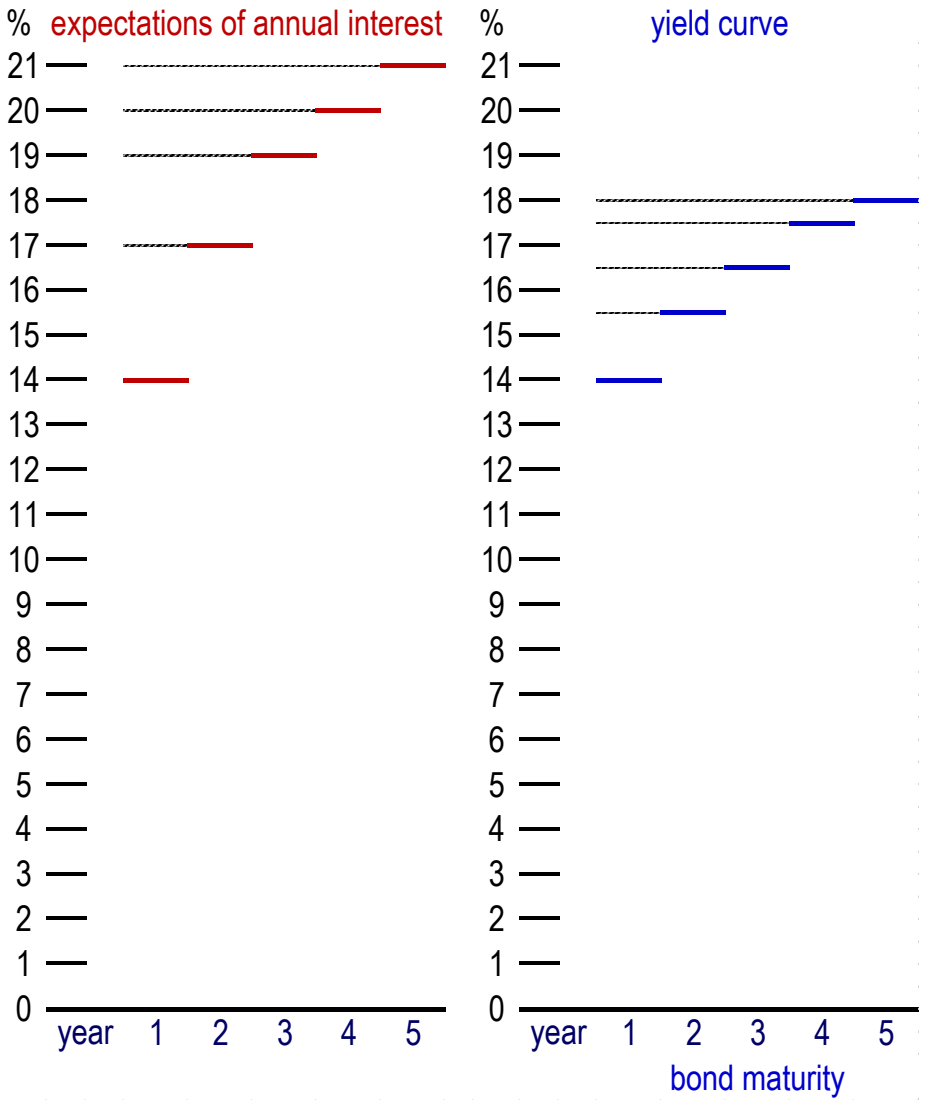
"The interest rate on a long-term bond will equal
an average of the short-term interest rates
that people expect to ocur
over the life of the long-term bond."
For a 5-year bond,
the yield formulation would be as follows:

$$(1 + i_{5yr})^5 = \prod_{t=1}^{5}(1 + i_t)$$

When the yield curve is upward-sloping,
the expectations theory suggests that
short-term interest rates are expected
to rise in the future.

If expectation of annual interest rates
for the next 5 years is as follows,
what would be the yield curve
according to the expectations theory?

year, t	interest rate, i	$1+i$	$\prod$	$\sqrt[t]{}$	interest rate	
1	14%	1,14	1,14	0,140	14,0%	on 1-year bond
2	17%	1,17	1,33	0,155	15,5%	on 2-year bond
3	19%	1,19	1,59	0,166	16,6%	on 3-year bond
4	20%	1,20	1,90	0,175	17,5%	on 4-year bond
5	21%	1,21	2,30	0,182	18,2%	on 5-year bond

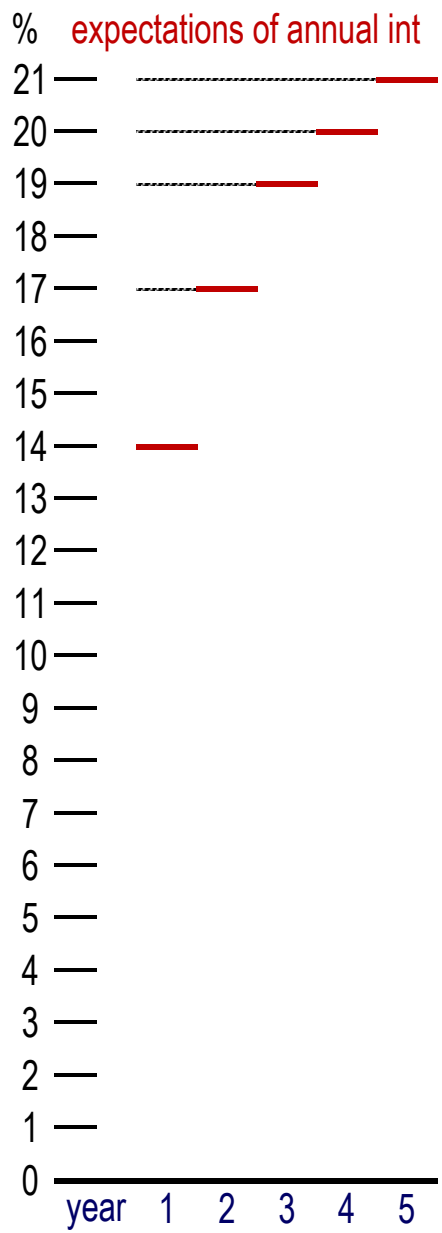


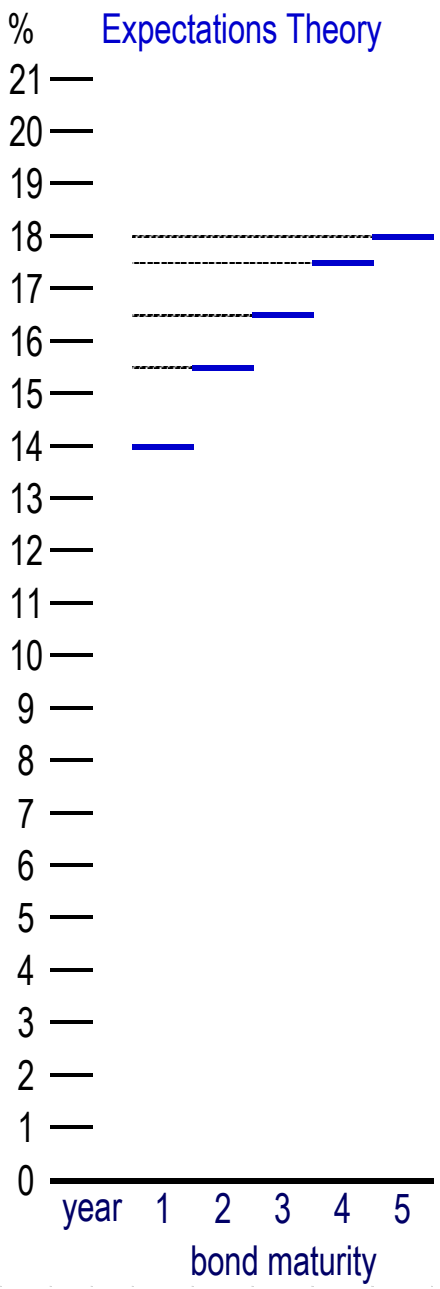
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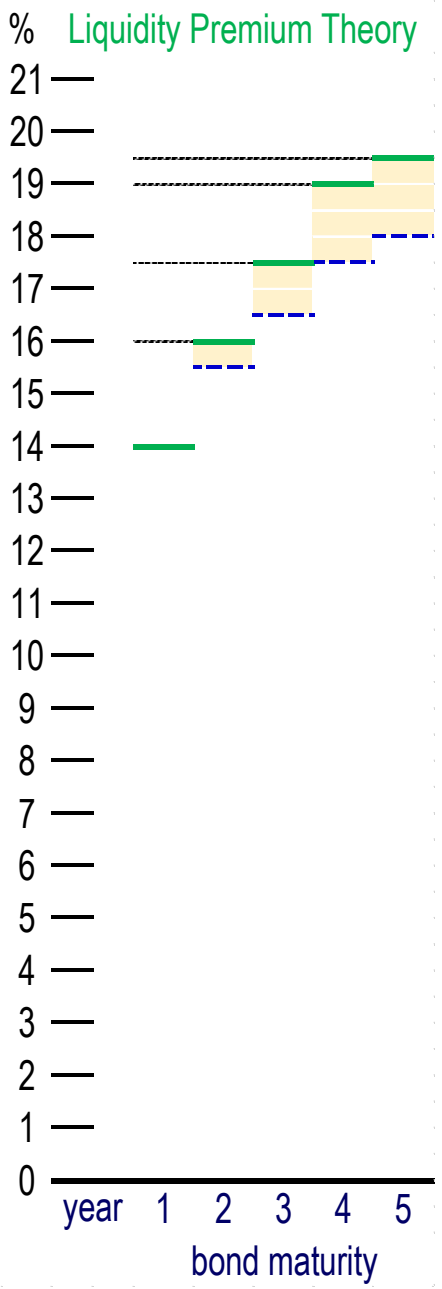
[6] Interest Rates

Liquidity Premium Theory

Because
the liquidity premium
is always positive
and grows
as the term to maturity
increases,
the yield curve implied
by the liquidity
premium theory
is always above
the yield curve implied
by the expectations
theory and
has a steeper slope.







[7] Stock Markets

Computing the Price of Common Stock

Common stock is the principal way that corporations raise equity capital.
Stockholders – those who hold stock in a Corporation –
own an interest in the Corporation proportional to the percentage outstanding shares they own.
This ownership interest gives them the right to vote
and to be the residual claimant of all funds flowing into the firm (known as cash flows),
meaning that the stockholder receives whatever remains
after all other claims against the firm's assets have been satisfied.

Stockholders may receive dividends from the net earnings of the Corporation.
Dividends are payments made periodically,
usually every quarter in the U.S. and annually in Turkey, to stockholders.

In addition, the stocholder has the right to sell the stock.
One basic principle of finance is that the value of any investment is found by computing
the present value of all cash flows the investment will generate over its life.

The cash flows a stockholder might earn from stock are dividends, the sales price, or both.

[7] Stock Markets

The One-Period Valuation Model

To value the stock today,
you need to find the present discounted value of the expected cash flows (future payments).
The cash flows consist of one dividend payment plus a final sales price.

P_0 = the curent price of the stock
(the zero subscript refers to time period zero, or the present)

Div_1 = the dividend paid at the end of year 1

k_e = the required return on investments in equity

P_1 = the price at the end of the first year; predicted sales price of the stock

$$P_0 = \frac{Div_1}{(1+k_e)} + \frac{P_1}{(1+k_e)}$$

[7] Stock Markets

The Generalized Dividend Valuation Model

Using the present value concept,
we can extend the one-period dividend valuation model to any number of periods:
The value of a stock today is the present value of all future cash flows.

P_0 = the curent price of the stock

D_t = the dividend paid at the end of year t

k_e = the required return on investments in equity

P_N = the price at the end of year N; predicted sales price of the stock

$$P_0 = \frac{D_1}{(1+k_e)^1} + \frac{D_2}{(1+k_e)^2} + \ldots + \frac{D_t}{(1+k_e)^t} + \ldots + \frac{D_N}{(1+k_e)^N} + \frac{P_N}{(1+k_e)^N}$$

[7] Stock Markets

The Generalized Dividend Valuation Model

The generalized dividend model may be rewritten without the final sales price.

P_0 = the curent price of the stock
D_t = the dividend paid at the end of year t
k_e = the required return on investments in equity

$$P_0 = \sum_{t=1}^{\infty} \frac{D_t}{(1+k_e)^t}$$

This tells us that the price of stock is determined only by the present value of the dividends and nothing else matters.
In other words, buyers of the stock expect that the firm will pay dividends someday.
Most of the time a firm institutes dividends
as soon as it has completed the rapid growth phase of its life cycle.
However, this model requires that we compute the present value of an infinite stream of dividends, a process that could be difficult, to say the least.

[7] Stock Markets

Valuation - Gordon Growth Model

The Gordon growth model is a simplified method of computing stock value that depends on the assumption that the dividends are growing at a constant rate forever. Given our uncertainty regarding future dividends, this assumption is often the best we can do. Also, the growth rate is assumed to be less than the required return on equity which is a reasonable assumption.

P_0 = the curent price of the stock
D_0 = the most recent paid dividend
D_1 = the dividend expected to be paid at the end of year 1
g = the expected average constant growth rate in dividends
k_e = the required return on investments in equity

$$P_0 = \frac{D_0 \times (1+g)^1}{(1+k_e)^1} + \frac{D_0 \times (1+g)^2}{(1+k_e)^2} + \ldots + \frac{D_0 \times (1+g)^t}{(1+k_e)^t} + \ldots + \frac{D_0 \times (1+g)^\infty}{(1+k_e)^\infty}$$

$$P_0 = \frac{D_0 \times (1+g)}{(k_e - g)}$$

$$P_0 = \frac{D_1}{(k_e - g)}$$

[8] Financial Crises

2007-2008 Crisis

It began in 2007 with a crisis in the subprime mortgage market in the United States,
and developed into a full-blown international banking crisis
with the collapse of the investment bank Lehman Brothers on September 15, 2008.
Excessive risk-taking by banks such as Lehman Brothers
helped to magnify the financial impact globally.
Massive bail-outs of financial institutions
and other palliative monetary ("quantitative easing")
and fiscal policies
were employed to prevent a possible collapse of the world financial system.

The crisis was nonetheless followed by a global economic downturn, the Great Recession.

The Europen debt crisis,
a crisis in the banking system of the European countries using the euro, followed later.

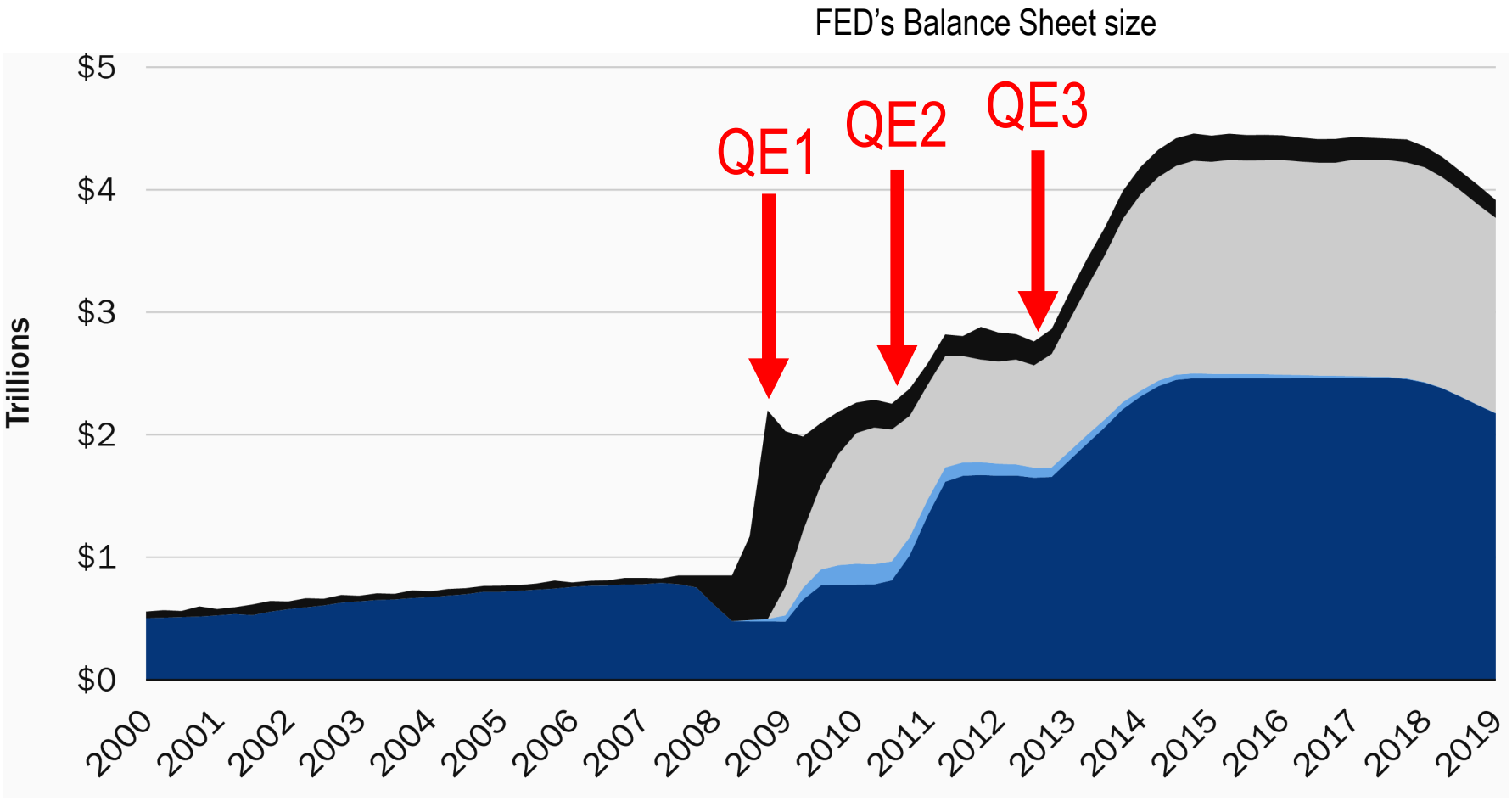
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[8] Financial Crises

2007-2008 Crisis

Quantitative Easing

Quantitative easing (QE) is a monetary policy, implemented by the Central Bank, primarily to energize the economy.
The Central Bank will create money to buy government securities from the market in order to lower interest rates and increase the money supply.
These economic conditions will then trigger financial institutions to promote increased lending and to make money supply more liquid.



[9] Banks

Asset Transformation

Banks make profits through the process of asset transformation:
They borrow short (accept short-term deposits)
and lend long (make long-term loans).

Primary concerns of Bank Management:

1. To make sure that the bank has enough ready cash to pay its depositors
when there are deposit outflows
the bank must engage in Liquidity Management.

2. Pursuing an acceptably low level of risk
by acquiring assets that have a low rate default
and by diversifying asset holdings (Asset Management).

3. Acquring funds at low cost (Liability Management).

4. Deciding the amount of capital the bank should maintain
and then acquiring the needed capital (Capital Adequacy Management).

[9] Banks

Primary risks in banking
are as follows.

1. Credit Risk

The risk arising because borrowers may default.

2. Interest-rate Risk

The riskiness of earnings and returns on bank assets
that results from interest-rate changes.

Credit risk management principles
involving loan activities are the following:

1. screening and monitoring
2. long-term customer relationships
3. loan commitments
4. collateral and compensating balances
5. credit rationing

[9] Banks

Estimating the Size of the Interest-Rate Risk

Gap and duration analyses tell a financial institution
if it has more rate-sensitive liabilities than assets
["rate-sensitive" = with interest rates that change frequently (at least once a year)
"fixed-rate" = with interest rates that remain unchanged for a long period (over a year)]
in which case a rise in interest rates will reduce profits
and a fall in interest rates will raise profits.

Gap = rate-sensitive Liabilities – rate-sensitive Assets
By multiplying the Gap times change in interest rate,
we can immediately obtain the effect on bank profits.

Duration = the average lifetime of a security's stream of payments
By multiplying the weighted average duration of a bank's assets and of its liabilities
times the amount of assets and of liabilities
times change in interest rate,
we can immediately obtain the approximate effect on bank's total asset value.

Financial institutions manage their interest-rate risk by modifying their balance sheets
but can also use strategies involving financial derivatives.

[9] Banks

Gap Analysis

The Bank

Assets	m$	Liabilities	m$
Rate-sensitive Assets	20	Rate-sensitive Liabilities	50
Variable-rate and short-term loans		Variable-rate Certificates of Deposits	
Short-term securities		Money market deposit accounts	
Fixed-rate Assets	80	Fixed-rate Liabilities	50
Reserves		Current account deposits	
Long-term loans		Savings deposits	
Long-term securities		Equity Capital	

Gap = rate-sensitive Liabilities – rate-sensitive Assets

Gap = 50 -20 = 30 million $

For example, if interest rates fall by 5%, then the bank's profits increase by

30 x 5% = 1.5 million $

[9] Banks

Duration Analysis

Duration = the average lifetime of a security's stream of payments

"Duration" concept approximates
the sensitivity of a security's market value (*MV*) to a change in its interest rate (i).

% Δ in *MV* of security ~ -% Δ in i * duration in years

average weighted duration (years)	The Bank	m$		m$	years
3	Assets	100	Liabilities	90	2
			Equity Capital	10	

For example, if interest rates fall by 5%, then the change in bank's net worth would be

100 m$ * 5% * 3 years = 15 m$
-90 m$ * 5% * 2 years = -9 m$

15 -9 = 6 m$ increase.

[10] Central Banks

Fundamental Powers of Central Bank of Turkey (TCMB)

- exclusive right to issue banknotes in Turkey
- right to determine of the inflation target and implement of monetary policy
- role of the lender of last resort for financial institutions
- power to request information from financial institutions

[10] Money Supply

Control of the Monetary Base

The primary way in which the Central Bank causes changes in the monetary base
is through its open market operations;
the Central Bank buys/sells bonds from/to banks.
Buying expands MB (=C+R) by increasing reserves of banks held at the Central Bank.
Selling contracts MB.

Lending to financial institutions also expands MB by increasing reserves.

The Central Bank sets the discount rate (interest rate on loans to banks), and then banks make decisions about whether to borrow.
The component of Monetary Base which is created by loans from the Central Bank
(called borrowed reserves)
is less tightly controlled
compared to the component that is created by open market operations
which is called the nonborrowed monetary base.

[10] Money Supply

Multiple Deposit Creation

When the Central Bank supplies the banking system with 1$ of additional reserves,
deposits increase by a multiple of this amount,
a process called "multiple deposit creation".

For example, when the Central Bank buys 100 m$ worth bonds from the First National Bank,
the bank finds that it has an increase in reserves of 100 m$.
Let us say the bank decides to make a loan
equal in amount to the 100 m$ rise in excess reserves.

The borrower deposits the borrowed amount of 100 m$ to Bank A.
If the required reserve ratio is 10%,
Bank A will now find itself with a 10 m$ increase in required reserves,
leaving it 90 m$ of excess reserves which it can lend.

The new borrower deposits the borrowed amount of 90 m$ to Bank B.
Since the required reserve ratio is 10%,
Bank B will now find itself with a 9 m$ increase in required reserves,
leaving it 81 m$ of excess reserves which it can lend.

The process goes on, eventually leading to a 1,000 m$ expansion in deposits.
Δ Deposits = Δ Reserves * Simple Deposit Multiplier
where Simple Deposit Multiplier = 1 / Required Reserve Ratio

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[11] Monetary Policy

Goals

1. Price stability
2. High employment and economic growth
3. Stability of financial markets
4. Interest-rate stability
5. Stability in foreign exchange markets

[12] Foreign Exchange Markets

How Do FX Rates Come Out ?

Law of One Price:

If two countries produce an identical good,
and transportation costs and trade barriers are very low,
the price of the good should be the same throughout the world
no matter which country produces it.

Theory of Purchasing Power Parity (PPP)

states that exchange rates between any two currencies
will adjust to reflect changes in the price levels of the two countries.

The theory of PPP is simply an application of the Law of One Price
to national price levels rather than to individual prices.

Suppose that the Yen price of Japanese steel rises 8% (to 10,800 ¥) due to inflation in Japan
while the Dollar price of the American steel rises 4% (to 104 $) due to inflation in the US;
then 1 $ should be worth 103.85 ¥ reflecting the inflation difference.

$$\frac{1+\text{Japanese Inflation}}{1+\text{American Inflation}} - 1 = \frac{1+8\%}{1+4\%} - 1 = 3.85\% \quad = \text{devaluation of Jap Yen against US\$}$$

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[12] Foreign Exchange Markets

Calculating Forward Rate for Selling FX

[Forward Bid Rate]

Assume 1€ is bought for 6.45 TL and sold for 6.55 TL in the market today
and you ask a bank to quote €/TL rate for 1m € for 3 months from now
when you want to sell your export revenue.

Calculating Forward Rate for Purchasing FX

[Forward Offer Rate]

Assume 1€ is bought for 6.45 TL and sold for 6.55 TL in the market today
and you ask a bank to quote €/TL rate for purchasing 1m € for 3 months from now
when you want to pay for your imports.

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[12] Foreign Exchange Markets

Calculating Forward FX Rate

	Bid	Offer
spot €/TL	6.45	6.55
€ funds (pa)	2%	4%
TL funds (pa)	12%	18%
term (months)	3	3
forward €/TL	6.58	6.81

Zero transaction costs are assumed.

$$\frac{1\text{m €}}{1+(\,4\%\,/\,4\,)} = 990{,}099\text{ €} * 6.45 = 6{,}386{,}139\text{ TL} * [1+(\,12\%\,/\,4\,)] = \frac{6{,}577{,}723\text{ TL}}{1\text{m €}} = 6.58\text{ €/TL}$$

$$6.45 * \frac{1+(\,12\%\,/\,4\,)}{1+(\,4\%\,/\,4\,)} = 6.58\text{ €/TL}$$

$$\text{Spot €/TL Bid} * \frac{1+\text{ TL funds Bid}}{1+\text{ € funds Offer}} = \text{Forward €/TL Bid}$$

$$\frac{1\text{m €}}{1+(\,2\%\,/\,4\,)} = 995{,}025\text{ €} * 6.55 = 6{,}517{,}413\text{ TL} * [1+(\,18\%\,/\,4\,)] = \frac{6{,}810{,}697\text{ TL}}{1\text{m €}} = 6.81\text{ €/TL}$$

$$6.55 * \frac{1+(\,18\%\,/\,4\,)}{1+(\,2\%\,/\,4\,)} = 6.81\text{ €/TL}$$

$$\text{Spot €/TL Offer} * \frac{1+\text{ TL funds Offer}}{1+\text{ € funds Bid}} = \text{Forward €/TL Offer}$$

[13] International Financial System

Balance of Payments

Because the balance of payments must balance, the official transactions balance, which equals the current account plus the capital account, tells us the net amount of international reserves that must move between governments (as represented by their central banks) to finance international transactions.

Balance of Payments

Current Account,
international transactions
that involve currently produced
goods and services

- **Trade Balance,** difference between merchandise exports and imports, net receipts from trade
- **Investment Income** (interest and dividends paid in and out)
- **Service Transactions** (tourism, contractors, education, etc)
- **Unilateral Transfers** (gifts, pensions, and foreign aid)

+

Capital Account,
net receipts from capital transactions
(purchases of stocks and bonds,
bank loans, etc)

=

net change in
Government's
International Reserves

[13] International Financial System

Exchange Rate Regimes

Classified into two basic types; «Fixed» and «Floating»,
and a third one which is more or less in between, namely, «Managed Float».

In a fixed exchange rate regime,
the value of a currency is pegged relative to
the value of another currency (called the anchor currency)
so that the exchange rate is fixed in terms of the anchor currency.

In a floating exchange rate regime,
the value of a currency is allowed to fluctuate against all other currencies.

When countries intervene in foreign exchange markets
in an attempt to influence their exchange rates
by buying and selling foreign assets,
the regime is referred to as a managed float regime (or a dirty float).

[14] Managing Working Capital

Working Capital = Current Assets – Current Liabilities

Working Capital = +Cash and Equivalents +Trade Receivables +Inventories
-Bank Loans -Trade Payables

31.12.2018 Balance Sheet (mTL)

Cash and Equivalent	100		Bank Loans	200	
Trade Receivables	200		Trade Payables	300	
Inventories	300		Current Liabilities		500
Current Assets		600	Long-Term Liabilities		250
Long-Term/Fixed Assets		400	Owners' Equity		250
Total Assets		1.000	Total Liabilities & Owners' Equity		1.000

Working Capital, WC = 600 - 500 = 100 mTL

Working Capital, WC = +100 +200 +300 -200 -300 = 100 mTL

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[14] Managing Working Capital

Measuring Operating Cash Cycle

average Inventories Holding period (days)

$$\frac{\text{average Inventories}}{\text{Cost of Goods Sold}} \times 365$$

avg Settelement period for Receivables (days)

$$\frac{\text{average Trade Receivables}}{\text{Credit Sales}} \times 365$$

avg Payment period for Payables (days)

$$\frac{\text{average Trade Payables}}{\text{Credit Purchases}} \times 365$$

Operating Cash Cycle (days)

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[14] Managing Working Capital

Weighted Average Cost of Capital, WACC

$$WACC = \left(\frac{E}{E+D}\right) R_E + \left(\frac{D}{E+D}\right) (1\text{-}T)\, R_{DBT}$$

An examination of the above equation could lead to the false conclusion that the optimal solution would be to fund the company entirely from debt sources.
However, increasing debt makes the company more risky
due to increased fixed costs of servicing (making interest payments of) the debt.





Investors are often more reluctant to invest in companies that have a high proportion of debt.
As a result, the cost of equity will increase for companies with an abnormally high proportion of debt.
Banks would also be more reluctant to lend to such companies.